Electronic Batch Manufacturing Record (eBMR) Regulatory Requirements

Tablets edition and Injectables edition
India (Revised Schedule M) · US FDA · EU GMP

241
controls the system will implement

678
clause-level requirements behind them

587
requirements checked against official texts

Version	0.9 (draft for review)
Date	27 September 2026
Purpose	Independent verification of the rules the eBMR will follow, before any software is built

All regulatory content is paraphrased. Clause references point to the official texts; they are not quotations.

How to use this document

This pack lists every rule the eBMR software will follow, where each rule comes from, and how the software will satisfy and test it. Please review each control and mark a decision. We will then build only what you confirm.

What to do for each control

1. **Read "What the system does".** This is the behaviour we intend to build.
2. **Check the "Regulatory basis".** Each clause reference points to the official text. Confirm that the clause says what we claim and that nothing is missing.
3. **Mark a decision:** *Follow* (build as described), *Don't follow* (drop it), or *Modify* (write the change in Comments).
4. **Pay most attention** to controls marked partly checked or search-based, and to all Design choice items. Design choices are our own proposals and are not required by any regulation.

Legend

Enforced	A hard rule inside the software. No customer, site or administrator setting can switch it off.
Configurable	The customer sets a value within safe limits. A setting may be made stricter, never looser than the regulation.
Design choice	Good practice that we propose. No regulation requires it. You may decide to drop it.
Both editions Tablets Injectables	Which product the control applies to. The two editions are separate products that share one compliance core (audit trail, e-signatures, access control).
All checked	Every requirement behind the control was compared with the official text you supplied.
Partly checked Search-based	Some or all source requirements come from web-search summaries because the official text was not available. Verify these first.
Supporting	Comes only from supporting documents (EU Annex 15/16, EU Chapter 1, EMA Q&A, ICH). Lower priority for batch records.
(s) · (u) · (draft)	Markers inside citations: (s) secondary source only, (u) unverified, (draft) July 2025 EU draft that is not yet in force.
P · P~ · S · U · [chk] · [gk]	India part only: P = re-read in the Gazette text; P~ = in the text but the clause number is inferred; S = secondary source; U = unconfirmed detail. [chk] = FDA/EU comparison value checked in the official files; [gk] = general knowledge.
n123	Reference to a clause-level requirement in the Appendix (click to jump in the PDF).

How the markets combine

Every batch is made in India, so Revised Schedule M always applies. Each product also declares its export markets (US, EU). The software applies every rule of every market the batch serves. Where two rules set different values, the stricter value wins (for example the longest record-retention period or the tightest time limit). Part D lists each topic where India differs from US or EU and the value applied for every market combination.

Scope and sources

The pack covers the rules that govern batch manufacturing records and the software that holds them. Documents are grouped as **Core** (needed for any compliant batch record) and **Supporting** (only a small part applies, mostly the release step or the quality system).

Document	Market	Version used	Status	Obtained from	Tier	Req.	Checked
21 CFR Part 211	US	eCFR, current as of 24 Sep 2026	Binding regulation	Uploaded official text	Core	61	59
21 CFR Part 11	US	eCFR, current as of 24 Sep 2026	Binding regulation	Uploaded official text	Core	20	19
FDA Data Integrity Q&A	US	December 2018	Guidance	Uploaded official text	Core	34	34
FDA Aseptic Processing Guidance	US	September 2004	Guidance	Uploaded official text	Core (Injectables)	122	117
Other US FDA law / guidance	US	Various (DSCSA, Part 201, Part 210, CSA 2025, PV 2011)	Law / guidance	Search results	Core / supporting	12	1
EU GMP Chapter 4	EU	2011 (in force 30 Jun 2011); July 2025 draft	Binding; draft pending	Uploaded official text (2011)	Core	42	31
EU GMP Chapter 5	EU	Revision in force 1 Mar 2015	Binding	Uploaded official text	Core	40	40
EU GMP Annex 11	EU	2011 (in force 30 Jun 2011); July 2025 draft	Binding; draft pending	Uploaded official text (2011)	Core	39	24
EU GMP Annex 1	EU	C(2022) 5938, in force 25 Aug 2023 (8.123 from 25 Aug 2024)	Binding	Uploaded official text	Core (Injectables)	201	201
EU GMP Annex 16	EU	2016	Binding (EU release only)	Search results	Supporting	8	0
EU GMP Annex 15	EU	2015	Binding (validation)	Search results	Supporting	5	0
EU GMP Chapter 1	EU	2013	Binding (quality system)	Search results	Supporting	3	0
EU GMP Annex 22 (draft)	EU	July 2025 draft	Draft	Search results	Supporting	1	0
EMA GMP Q&A	EU	Q&A pages (e.g. data integrity, Aug 2016)	Guidance	Search results	Supporting	10	0
ICH Q9(R1) / Q10 / Q12	ICH	Q9(R1) 2023; Q10 2008; Q12 2019	Guidance adopted by FDA and EU	Search results	Supporting	15	0
Revised Schedule M (India)	IN	G.S.R. 922(E), 28 Dec 2023	Binding (Drugs Rules 1945)	Uploaded official text (compressed copy, some pages missing)	Core	62	61
Other Indian law	IN	Drugs Rules 1945; IT Act 2000; Bharatiya Sakshya Adhiniyam 2023	Binding law	Search results	Core / supporting	3	0

Not included

- **EU GMP Annex 2** (biological medicinal products) and **Annex 13** (investigational products): not applicable to tablets or standard injectables. Tell us if your customers make biological injectables such as vaccines or insulin.
- **WHO-GMP** and other export markets: largely covered by meeting US FDA and EU GMP; not mapped separately.
- **Pharmacopoeial test limits** (USP, Ph. Eur., IP): handled as configurable master data per market; listed in the build outline rather than here.

Known gaps in the source material

- The uploaded copy of Revised Schedule M was compressed and is missing some pages (see Part D, section F). Those items are marked for verification.
- EU Annex 15, Annex 16, EU Chapter 1, EMA Q&A pages and ICH texts were not supplied; their items are search-based.
- The July 2025 EU drafts of Annex 11 and Chapter 4 were not supplied and are not yet in force. Where we adopt a stricter draft expectation, it is marked (draft).

At a glance

Part	Topic	Applies to	Controls	Enforced	Configurable	Design choice	Source req.	Checked
A	Data integrity and computerised systems	Shared compliance kernel · both editions	50	50	22	24	130	93
B	Production, documentation and batch release	Shared compliance kernel · both editions	60	52	32	16	144	105
C	Sterile manufacturing controls	Injectables edition	79	76	46	4	328	319
D	India layer: Revised Schedule M and Indian law	Mandatory for every batch made in India	52	49	28	5	65	61

A control can be both enforced and configurable (for example a fixed minimum with a configurable stricter value), so type counts can exceed the number of controls.

Contents

1. **Part A · Data integrity and computerised systems** (50 controls)
   - (a) ALCOA+ principles mapped to controls, and which document defines each term
   - (b) Revision notes: what is binding, what is draft, and how the kernel treats drafts
   - (c) Claims made in earlier research that are NOT actually required
   - (d) Conflicts between rows and how they were resolved
   - (e) Rows that are not an eBMR data-integrity concern
   - (f) Open questions
2. **Part B · Production, documentation and batch release** (60 controls)
   - A. Master records (MBR / master formula, processing and packaging instructions)
   - B. Batch record issuance and execution
   - C. Materials: receipt, status, dispensing and charge-in
   - D. Equipment, premises, cleaning and logs
   - E. In-process controls, time limits and yields
   - F. Packaging and labeling
   - G. Material and product status: quarantine, rejected, reprocessed, recovered and returned
   - H. Deviations, investigations and CAPA
   - I. Batch record review and release
   - J. Traceability, product quality review and records retention
   - (a) Required content: merged US + EU checklists
   - (b) Release models by market, and one batch destined for several markets
   - (c) Claims made by earlier research that are NOT actually required
   - (d) Conflicts between rows and how they were resolved (with official-text evidence)
   - (e) "Not an eBMR concern": rows with reasons
   - (f) Open questions
3. **Part C · Sterile manufacturing controls** (79 controls)
   - Theme 0: Cross-cutting rules (apply to every theme)
   - Theme 1: Personnel qualification and gowning gate
   - Theme 2: Cleanroom/area status and environmental and personnel monitoring linked to the batch window
   - Theme 3: Utilities (WFI, clean steam, gases)
   - Theme 4: Aseptic process simulation (media fill): status and failure consequences
   - Theme 5: Interventions catalogue and log
   - Theme 6: Time limits driven by recorded events
   - Theme 7: Components, container-closures, depyrogenation, sterilisation cycles and load traceability
   - Theme 8: Sterilising filtration, filter integrity (PUPSIT/post-use) and bioburden
   - Theme 9: Filling, fill weight/volume checks, stoppering and capping
   - Theme 10: Lyophilisation
   - Theme 11: Terminal sterilisation
   - Theme 12: 100% visual inspection, AQL and container-closure integrity
   - Theme 13: Sterility, endotoxin and particulate testing, and release
   - Theme 14: Batch record review dossier and release gate
   - Theme 15: Technology profiles (isolator, RABS, BFS, FFS, closed/SUS)
   - What the FDA aseptic guidance (2004) does NOT cover, and where those rules come from
   - Conflicts between rows and how they were resolved (with evidence)
   - "Not an eBMR concern"
   - Open questions
4. **Part D · India layer: Revised Schedule M and Indian law** (52 controls)
   - A. Controls
   - B. Market-profile resolution table
   - C. India-specific fields
   - D. Electronic records and signatures in India
   - E. Status, in-force dates and deadlines (as of 27 Sep 2026)
   - F. Gaps: text that could not be verified in the uploaded Gazette copy
   - G. Rows not treated as eBMR concerns, and open questions
5. **Appendix · Clause-level requirements register**
6. **Reviewer sign-off**

PART A

Data integrity and computerised systems

Shared compliance kernel · both editions · 50 controls

Scope: the shared compliance kernel used by both the TABLETS and INJECTABLES editions. Every control here is kernel (applies to both editions) unless a line says "Tablets:" or "Injectables:". India Revised Schedule M is mandatory for every product and lives in the separate India catalog. The kernel combines the two catalogs using the same stricter-wins rule described below.

DI-01 Record-type, official-record and raw-data register

Enforced | Configurable | Both editions | 4 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- Each tenant keeps a controlled register (versioned, approved by QA e-signature) of every GMP record type the eBMR holds or references: batch/packaging record, logbooks, forms, attachments, device printouts, interface-sourced data. For each type it records:
- official medium (electronic, paper or hybrid);
- which item is the original and which are copies;
- raw / derived / informational class;
- owner, retained metadata, retention rule ID and review rule;
- where the record is stored.
- Raw-data floor: every data item a quality decision relies on is raw data. This covers release, IPC acceptance, yield and reconciliation evaluation, and deviation decisions. The engine derives "decision use" from limits, gates and review rules, and blocks approval of an MBR or register that classes such a field as anything other than raw.
- For anything executed in the eBMR, the electronic record is the official record. PDFs and printouts are copies and are marked as copies (DI-40). Each hybrid point must be declared with its linkage method (DI-41, DI-43).
- All eBMR records are treated as Part 11 records. The design does not rely on the 2003 enforcement-discretion policy, because the eBMR replaces paper and users create and modify GMP data in normal operation.

TYPE DETAIL

ENFORCED: the requirements list must exist, and the raw-data floor is checked at MBR/register approval. Register content is CONFIGURABLE per tenant.

REGULATORY BASIS

US: 11.1(b), 11.1(e), 11.2(a), 11.3(b)(4),(6),(9); FDA DI Q&A Q10, Q12; FDA Part 11 Scope and Application guidance (2003) sec. III (s) | EU: Ch.4 Principle (records include raw data; for electronic records the user defines raw data, at minimum all data behind quality decisions), 4.1, 4.10 | ICH/EMA/draft: draft Ch.4 data governance / ALCOA++ (draft)(u)

INDIA / US / EU DIFFERENCES

EU Ch.4 is explicit on defining raw data. FDA gets to the same place through Q10/Q12 and the completeness wording in 211.188/211.194. There is no conflict, and the EU wording sets the floor.

HOW IT WILL BE TESTED

Classify an IPC result that feeds an acceptance limit as "informational" and try to approve the MBR. It is blocked. The register shows the official medium and storage location for each record type.

Source requirements: n96, n616, n59, n648, n227, n43

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-02 Metadata bound to every value across the data lifecycle

Enforced | Design choice | Both editions | 3 of 7 source requirements checked against official text

WHAT THE SYSTEM DOES

- Every stored GMP value carries its metadata, and the two cannot be separated. The metadata is:
- unit;
- acting user ID;
- server timestamp;
- source (device, interface or manual);
- instrument/equipment ID;
- material identity and status where relevant;
- MBR version and step;
- formula version for derived values;
- a link to the value's audit-trail history.
- Metadata and audit trail travel with the value into every copy, export, backup, archive and migration. A copy without them is not a true copy (DI-41).
- Each record has a tracked lifecycle state: captured, verified, reviewed, used for decision, archived, eligible for disposal, disposed. Disposal happens only through DI-47.
- Values are stored locale-neutral (canonical number, unit code, UTC). Language and locale affect display only, never limits, units or meaning.

TYPE DETAIL

ENFORCED (metadata binding). DESIGN-CHOICE (explicit lifecycle-state model and canonical storage): this makes the draft Ch.4 lifecycle expectations demonstrable, and the engine enforces it.

REGULATORY BASIS

US: 11.10(e); 211.188, 211.194(a); FDA DI Q&A Q1a (life cycle through disposition), Q1b (metadata includes audit trails; data-metadata relationship kept secure and traceable), Q1e | EU: Ch.4 4.1, 4.10 | ICH/EMA/draft: draft Ch.4 draft-4.12 data lifecycle (draft)(s); draft Ch.4 metadata/audit-trail and ALCOA++ provisions (draft)(s)

INDIA / US / EU DIFFERENCES

FDA Q1b says outright that audit trails are metadata. The 2011 EU texts only imply it, and the EU draft states it. No conflict.

HOW IT WILL BE TESTED

Export a batch and restore it into an isolated environment. For 20 sampled values, confirm that unit, user, time, source ID, MBR/formula version and full change history are all present.

Source requirements: n60, n640, n111, n181, n108, n110, n88

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-03 Computerised-system validation (application, infrastructure, life cycle)

Enforced | Design choice | Both editions | 6 of 8 source requirements checked against official text

WHAT THE SYSTEM DOES

- The vendor supplies a validation package for each release: risk-based requirements, design/configuration specs, test evidence including negative and boundary tests on parameter limits, data limits and error handling, validation deviations and how each was resolved, release notes, known issues, and SDLC/QMS evidence. The customer performs risk-based verification of its configured use.
- Infrastructure is qualified: servers, DB, network, cloud tenancy and peripherals.
- Validation scope follows FDA's notion of the "system": hardware, software, peripherals (balances, scanners, label printers, terminals), network, cloud hosting, personnel roles and training, and the SOPs/manuals.
- A production tenant accepts a new software release for GMP use only after the customer records an approval e-signature on the validation summary for that release (staged release: validation tenant, then production). Every batch stamps the release it ran on (DI-10).
- The system can detect invalid or altered records. A scheduled integrity job checks record and signature hashes and raises an alarm if anything was changed outside the application (the hash mechanism is a design choice).
- High-impact functions get the deepest testing: limit checks, calculations, signature binding, status/release interlocks, exception reports, and any automated check that replaces a human check.
- Customer-specific scripts, reports and interfaces are treated as bespoke, with their own specifications, tests and quality metrics.
- Automated test tools and test environments carry documented adequacy assessments.
- A paper-to-electronic control map shows that no paper control was lost without a risk justification.
- Retirement is planned, so records stay readable after decommissioning (DI-46).

TYPE DETAIL
ENFORCED (release-approval gate per tenant; integrity job). Package content is a vendor DESIGN-CHOICE within the listed minimum.

REGULATORY BASIS
US: 11.10(a); 211.68(a)-(b); FDA DI Q&A Q1f, Q3 | EU: Annex 11 Principle, §1, §3.3, §4.1, §4.2, §4.5, §4.6, §4.7 | ICH/EMA/draft: ICH Q9(R1) sec. 1 (s); draft Annex 11 §9 qualification/validation incl. cloud documentation access (draft)(s)

INDIA / US / EU DIFFERENCES
Annex 11 is more prescriptive on documentation: supplier assessment, COTS documentation review, bespoke metrics, test-tool assessment. 11.10(a) adds the explicit duty to be able to detect altered records. Both apply.

HOW IT WILL BE TESTED
In a test environment, change a DB row outside the application. The integrity job alarms and names the record. Try to execute a GMP batch on a release that has no approved validation summary in that tenant. It is refused.

Source requirements: n44, n147, n628, n200, n194, n209, n642, n167

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-04 Verification of each MBR version and configured workflow before it becomes Effective

Enforced | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- QA cannot set Effective status until a verification record is linked. This applies to every MBR version and every configured GMP workflow, report, calculation, interface mapping and review-by-exception rule set. The record shows that the steps, requirements, limits, calculations and signature points work as intended.
- The verification method is documented review plus test. A dry-run/simulation mode is one acceptable method, not a requirement (see DI-05 if one is offered).
- MBR steps name the computerised systems they rely on: PLC/SCADA recipe and version, weighing system, checkweigher, vision/serialisation system, autoclave or lyophiliser controller, filter-integrity tester.
- PAT steps define the method, the model version and the acceptance criteria. An MBR containing a PAT step without acceptance criteria cannot be approved.
- At execution, the record captures the system/recipe version actually used and flags any mismatch with the MBR.
- The MPCR carries author and independent-checker e-signatures from two different people, then QU approval (the signature types are in DI-35; MBR content minima are in the production catalog).

REGULATORY BASIS
US: 211.68(b), 211.100(a), 211.186(a); FDA DI Q&A Q3, Q6, Q11 | EU: Ch.4 preamble "instructions" type (starting materials, equipment and computerised systems to be named; PAT with acceptance criteria), 4.3; Annex 11 §4.1, §10 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES
FDA Q3 says explicitly that qualifying the platform does not prove that a given MPCR calculates correctly. The EU reaches the same result through Ch.4 4.3 approval plus Annex 11 validation. A verification record for every version satisfies both.

HOW IT WILL BE TESTED
Try to make a new MBR version Effective without a verification record. It is blocked. Run a batch whose PLC reports a different recipe version from the MBR. It is flagged as an exception.

Source requirements: n61, n36, n617

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-05 Segregation of non-GMP functions (training, simulation, sandbox, analytics)

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- Non-GMP functions keep their data in segregated stores. Examples: training tenant, MBR simulation, planning/scheduling, analytics.
- They cannot:
- write to GMP records, inventory, equipment status or genealogy;
- change GMP configuration;
- use real lot or batch identifiers (a reserved namespace is used instead);
- have a non-GMP record converted into, or merged with, a GMP record.
- Their outputs carry a visible NOT-GMP watermark.
- A GMP record can only come from an Effective MBR.
- The validation package contains a documented impact assessment for each non-GMP function.

TYPE DETAIL
ENFORCED (segregation). Offering a simulation mode at all is a DESIGN-CHOICE: FDA does not expect one.

REGULATORY BASIS	US: FDA DI Q&A Q3 (assess and mitigate influence of non-CGMP functions on CGMP operations); 211.68(b) | EU: no dedicated clause; Annex 11 §1 (risk management) applies | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US guidance is explicit, and the EU covers it through general risk management. No conflict.
HOW IT WILL BE TESTED	From the training tenant, try to post to a production batch or reuse a real lot number. It is blocked. Simulation output carries the watermark.

Source requirements: n643, n61

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-06 System inventory, system description and requirements traceability

Enforced · Design choice · Both editions · 1 of 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• The system exports its own configuration inventory: modules, each enabled interface with endpoint and version, connected devices (balances, scanners, PLC/SCADA, printers, vision systems), and data flows. • The export is compared automatically with the approved system description. An undocumented interface or device raises an alarm to the system owner and becomes a periodic-review item. • The customer inventory lists the eBMR and each connected system with its GMP function. • User requirements are based on documented risk assessment and GMP impact, and they are traceable to configuration items and tests. The traceability matrix is generated from linked IDs and kept current with each release and configuration change. • Validation evidence includes boundary tests (just inside, on and just outside) for every critical limit, and negative tests of error handling.
TYPE DETAIL	ENFORCED (inventory export and drift alarm). DESIGN-CHOICE (generated traceability matrix): this makes the draft's mandatory traceability matrix cheap to keep current, and it is enforced.
REGULATORY BASIS	US: 211.68(b); FDA DI Q&A Q1f | EU: Annex 11 §4.3, §4.4, §4.7 | ICH/EMA/draft: draft Annex 11 §6 requirements, traceability matrix (draft)(s)
INDIA / US / EU DIFFERENCES	Only the EU texts explicitly require an inventory and a system description for critical systems. That requirement is adopted for all markets.
HOW IT WILL BE TESTED	Enable an interface that is not in the approved description. The drift alarm fires. Every critical limit shows boundary-test evidence in the matrix.

Source requirements: n149, n193

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-07 Risk-based data-criticality classification (raises controls, never removes floors)

Enforced · Configurable · Both editions · 1 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every MBR field and step carries a criticality class (e.g., CQA/CPP-linked, GMP-supporting, informational) with a reference to an approved risk assessment. • The class drives three things: whether manual entry needs a second-person or validated-electronic check (DI-30), whether limits are hard or soft, and highlighting in review (DI-25/26). • Floors apply to all GMP data regardless of class, and a class cannot remove them: audit trail, attribution, mandatory reason for change, and inclusion of every post-entry change in audit-trail review. • A new field defaults to "critical" until it is classified. Downgrading a class requires a QA e-signature, a linked risk assessment and change control. • The data-risk assessment covers the whole business process, including manual hand-offs at interfaces, and prefers automated capture for critical values. • Injectables: sterility-assurance data (sterilisation cycles, filter integrity, EM excursions, interventions) is critical by default. • Tablets: blend uniformity, weight, hardness and similar IPCs are critical by default.
TYPE DETAIL	CONFIGURABLE (class per field; default critical; downgrade only under QA-approved change control). Floors are ENFORCED.
REGULATORY BASIS	US: FDA DI Q&A Introduction (risk-based strategies), Q8 (risk assessment considers data criticality, controls, product impact) | EU: Annex 11 §1, §6 | ICH/EMA/draft: EMA GMP DI Q&A on data criticality and data risk (s); ICH Q9(R1) sec. 1 (s)
INDIA / US / EU DIFFERENCES	Both markets allow a risk-based extent of controls. Neither lets risk classification remove the audit trail or attribution, which is why the floors exist.
HOW IT WILL BE TESTED	Try to downgrade a field without QA approval. It is blocked. A change to a field classed "informational" still appears in the audit trail and in the batch audit-trail review list.

Source requirements: n148, n202, n203, n183, n209

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-08 Closed-system classification, hosting and supplier agreements, open-system data flows

Enforced · Both editions · 4 of 7 source requirements checked against official text

WHAT THE SYSTEM DOES	• The deployment is documented as a closed system. The regulated company, not the vendor, controls access provisioning and approves privileged access. • Vendor or support access is individually named, time-boxed, approved by the customer, protected by MFA, logged and reviewed. If customer control cannot be demonstrated, the tenant is treated as open (11.30): the 11.10 controls plus extra measures (e.g., encryption, digital signatures) as the circumstances need. • Data flows that cross organisational boundaries protect authenticity, integrity and (where appropriate) confidentiality from creation to receipt. Examples: data exchanged with a CMO or contract giver, imported supplier CoAs, trading-partner messages, remote support. The protection is TLS in transit plus signed payloads or an equivalent. The importer records source metadata and the supplier's qualification status. • A formal written agreement exists with every third party, internal and group IT included: vendor, cloud host, validation contractor, remote-maintenance provider, archive provider. It covers: • responsibilities and data ownership; • backup/restore and retention;

	• audit-trail availability; • notification of changes and security incidents; • support during inspections and customer audit rights; • exit/data return as a full export with audit trails and signatures in readable form. • Supplier QMS and audit information can be produced for inspectors. The need for a supplier audit is decided by risk. • The MIA holder / contract giver keeps access to complete batch records for the full retention period, including after the contract ends. • System documentation and the computer systems themselves are readily available for FDA inspection.
TYPE DETAIL	ENFORCED for the product capabilities: customer-controlled access, the vendor-access grant workflow and log, and full export. The agreements are organisational and are tracked in a go-live checklist.
REGULATORY BASIS	US: 11.1(e), 11.3(b)(4),(9), 11.10 introductory text (authenticity, integrity, confidentiality when appropriate, non-repudiation), 11.30 | EU: Annex 11 §2, §3.1, §3.2, §3.4 | ICH/EMA/draft: EMA GMP DI Q&A on outsourced activities / chain of contracts (s); ICH Q10 sec. 2.7 (s); draft Annex 11 §7 supplier and service management incl. exit strategy (draft)(s)
INDIA / US / EU DIFFERENCES	Annex 11 §3.1 requires agreements even with internal IT departments, which is broader in scope. 11.30 adds a separate control set when access is not controlled by the record owners. Both apply.
HOW IT WILL BE TESTED	A support engineer tries to log in without a customer-approved time-boxed grant. Refused. A full tenant export re-opens in the stand-alone viewer with audit trails and signature manifestations intact.

Source requirements: n43, n53, n166, n179, n186, n225, n615

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-09 Change and configuration management (including systems documentation)

Enforced | Both editions | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every configuration object is versioned through Draft, Review, Approved (e-signature) and Effective. This covers MBRs, workflows, calculations, limits, roles and permission sets, signature policies, interface settings, exception/RBE rules, retention rules and report templates. • Each change carries a change-record ID. The configuration audit trail stores diffs, and an exportable baseline is kept per environment. Drift from the validated baseline raises an alarm. • New batches can use only one Effective version per product plus batch size (processing), or per product, pack size and pack type (packaging). A batch in progress stays on the version it was issued from unless an approved change says otherwise. • There is no direct DB editing in production, and no admin-console path that changes specifications, process parameters or MBR content outside the change workflow. • Only authorised people can change master records or other records. A change without a change-record ID is refused. • Systems documentation (URS, configuration specs, SOPs, admin manuals) is held under controlled distribution, access and use, with a time-ordered revision history. • Software releases are tested in a non-production environment before production.
REGULATORY BASIS	US: 211.68(b), 211.100(a); 11.10(k)(1)-(2); FDA DI Q&A Q4 | EU: Annex 11 §10, §4.7 (test environments); Ch.4 4.3, 4.29 (change-control records) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Both markets require controlled change. 11.10(k) additionally requires documentation control with a time-sequenced audit trail. Both apply.
HOW IT WILL BE TESTED	Try to change a specification limit through the admin console. Impossible. Submit a configuration change without a change-record ID. Rejected. A configuration diff can be retrieved for any past date.

Source requirements: n157, n52, n8, n62

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-10 Version-in-force stamping (executed-context snapshot)

Enforced | Design choice | Both editions | 1 of 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each executed step stores the version or value of everything it depended on: • software release and configuration baseline ID; • MBR version and hash, and formula version; • specification limits and SOP version; • equipment status and calibration due date; • material status; • device and recipe versions. • A later change to master data never alters how a completed record is displayed or how its results were evaluated.
TYPE DETAIL	DESIGN-CHOICE, engine-enforced. No rule names this mechanism, but without it 211.188(a), 11.10(e) and Ch.4 4.20 could not be shown to hold after master data changes.
REGULATORY BASIS	US: 211.188(a) (accurate reproduction of the master), 11.10(e) | EU: Ch.4 4.20 (record based on the currently approved instructions), Ch.4 preamble (instructions name computerised systems); Annex 11 §10 | ICH/EMA/draft: draft Annex 11 §9 context (draft)(s): recording the version in force per batch is a design means from the requirement, not reported draft text
INDIA / US / EU DIFFERENCES	No conflict.
HOW IT WILL BE TESTED	Tighten a specification limit after a batch completes. The completed record still shows the original limit version and its original pass/fail evaluation.

Source requirements: n194, n617

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-11 Periodic evaluation and routine checks of the eBMR itself

Enforced | Configurable | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• For a chosen period the system generates a periodic-review pack covering: • software releases and configuration changes; • open validation gaps; • incidents and interface failures; • access changes and privileged activity; • audit-trail review completion; • backup/restore and DR test results; • archive fixity and readability results; • time-sync events; • alarm statistics; • supplier SLA performance; • training status of active users. • The approved periodic-review report and its actions are stored in the system or linked to it. • A written routine-check program for the eBMR as a "computer or related system" covers interface health, backup success, time sync and integrity-job results, and its check records are kept.
TYPE DETAIL	CONFIGURABLE for review frequency. No rule sets a number. DESIGN-CHOICE default: 12 months; shorter allowed; longer only with a QA-approved risk rationale. Pack content is ENFORCED.
REGULATORY BASIS	US: 211.68(a) (routine checks under a written program, records kept) | EU: Annex 11 §11 | ICH/EMA/draft: draft Annex 11 §14 periodic review (draft)(u)
INDIA / US / EU DIFFERENCES	211.68(a) applies a routine calibration/inspection/check program to computers as well as other equipment. Annex 11 §11 requires periodic evaluation of validated state. Both apply.
HOW IT WILL BE TESTED	Generate the pack for a test period and trace 5 sampled entries per topic back to their source logs.

Source requirements: n158, n178, n7

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-12 Incident management linked to batches

Enforced · Design choice · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• The system automatically raises incidents for: • interface failures; • loss of connection during a step; • client or server crashes; • repeated signature failures; • time-sync faults; • print failures; • integrity-check failures. • Users can also raise incidents manually. • Each incident is linked to the affected batches and steps. An open incident linked to a batch is shown on the review and disposition screens and blocks disposition until it has been assessed. • Every incident carries a criticality rating. A critical incident cannot be closed without a root cause and a CAPA link.
TYPE DETAIL	ENFORCED for logging, batch linkage, and root cause plus CAPA on critical incidents. The disposition block is a DESIGN-CHOICE, engine-enforced by default: it puts the Ch.4 4.27 "special observations" indication into effect.
REGULATORY BASIS	US: 211.192 (unexplained discrepancies investigated) | EU: Annex 11 §13; Ch.4 4.27 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Only the EU text explicitly requires incident management for computerised systems. It is adopted for all markets.
HOW IT WILL BE TESTED	Kill the balance interface in the middle of a weighing. An incident is created, linked to the batch and shown as open on the disposition screen. Try to close it as critical without a root cause and CAPA. Blocked.

Source requirements: n161

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-13 Security baseline, confidentiality and non-repudiation evidence

Enforced · Configurable · Design choice · Both editions · 1 of 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Encryption: • TLS on all client and interface connections; plain HTTP is refused. • Encryption at rest for the DB, attachments, backups and archives. • Access to confidential data (formula composition, personal data) is restricted by role and site. • Shop-floor clients are hardened: kiosk mode, removable storage disabled. • Platform hygiene: • malware protection and patch SLAs; • segmentation between the OT line network and the IT network; • isolation of unsupported platforms. • MFA is required for all remote access and all privileged access. • Internet-exposed or SaaS deployments are penetration-tested periodically. • RTO and RPO are documented per module. The execution module has the tightest RPO (Injectables: continuous aseptic filling). • Each signature has a non-repudiation evidence package: identity-verification record, authentication event, manifestation and audit trail.
TYPE DETAIL	ENFORCED for TLS, encryption at rest, and MFA on remote/privileged access (designed to the stricter draft; no conflict). CONFIGURABLE for RTO/RPO values and pen-test frequency (DESIGN-CHOICE defaults: annual pen test; RPO set by risk).

REGULATORY BASIS	US: 11.10 introductory text (authenticity, integrity, confidentiality when appropriate, non-repudiation), 11.30; 211.68(b) (backup data secure from alteration, erasure, loss) | EU: Annex 11 §7.1, §12.1, §12.2 | ICH/EMA/draft: draft Annex 11 §15 security incl. reported 15.19 penetration testing (draft)(s); draft Annex 11 §10 encryption, §11 MFA for remote access (draft)(s)
INDIA / US / EU DIFFERENCES	The 2011 texts set general protection duties. The draft Annex 11 is prescriptive, and the kernel designs to it.
HOW IT WILL BE TESTED	A non-TLS connection is refused. USB storage on a kiosk is blocked. A user without formula permission cannot open the composition. For a sampled signature, the evidence package can be produced.

Source requirements: n176, n615

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-14 Unique personal accounts; no shared write-capable accounts

Enforced · Both editions · 3 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every GMP action requires an individually assigned, authenticated account. GMP actions are create, change, verify, review, approve, sign and acknowledge (including alarm acknowledgment). • An account flagged generic or shared cannot be given any write, sign, verify or acknowledge permission. The engine checks this when a role is assigned, not only by policy. • A shared read-only view (e.g., a line-status display) is allowed only if it is technically unable to create, change, sign, verify or acknowledge anything. • Interface and service accounts are non-interactive and cannot sign. Their entries are attributed to the interface or device. • On shared terminals, each entry is attributed to the person who authenticated for that entry. Re-authentication is fast (e.g., badge plus PIN). • Injectables: cleanroom terminals use a method that works with gloves (badge/RFID plus PIN, or biometrics), so a shared login is never needed.
REGULATORY BASIS	US: 11.10(d), 11.10(g), 11.100(a), 11.300(a); 211.68(b), 211.188(b)(11), 211.194(a)(7)-(8); FDA DI Q&A Q5 | EU: Annex 11 §12.1, §12.4 | ICH/EMA/draft: draft Annex 11 §11 identity and access management (shared accounts read-only only) (draft)(s)(u)
INDIA / US / EU DIFFERENCES	Both markets allow shared read-only viewing (FDA Q5; draft). Neither allows shared accounts for any action, including second-person review.
HOW IT WILL BE TESTED	Assign a write permission to an account flagged generic. Refused. A service account attempts a signature. Blocked. A directory scan finds no generic account holding GMP permissions.

Source requirements: n63, n173, n226, n47, n159

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-15 Identity verification, credential issuance and permanent identifier uniqueness

Enforced · Design choice · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Before any signature component is issued, account activation requires a recorded identity verification: who verified, method, document type and date. • User IDs are never reused or reassigned, even after deactivation. A permanent registry spans sites and tenants. Accounts are disabled, never deleted, so historical attribution survives. • There is no proxy or delegated signing. • Initial and admin-reset passwords are single-use and must be changed before the account can sign. This guarantees that misusing another person's signature would need two or more people to collude. • US market: a go-live checklist per legal entity records submission of the FDA non-repudiation certification letter (11.100(c)). This is an organisational duty. The system flags a missing letter for US-market tenants but does not block execution.
TYPE DETAIL	ENFORCED. NRL tracking is a DESIGN-CHOICE, because it is a firm's submission to FDA, not a system function.
REGULATORY BASIS	US: 11.100(a)-(c) (as amended 2 March 2023), 11.200(a)(2)-(3), 11.300(a) | EU: Annex 11 §12.1, §12.4 | ICH/EMA/draft: draft Annex 11 §11 (draft)(s)
INDIA / US / EU DIFFERENCES	The NRL certification is US-only, with no EU equivalent. Identity verification before issuing credentials is explicit only in the US text and is adopted for all markets.
HOW IT WILL BE TESTED	Create an account with a deactivated user's ID. Blocked. Activate an account with no identity-verification record. Blocked. Sign with an admin-reset password before changing it. Refused.

Source requirements: n56

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-16 Access lifecycle: recorded grants, point-in-time access lists, recertification, prompt removal

Enforced · Configurable · Both editions · 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every grant, change and revocation of access rights is recorded in a protected log with requester, approver, date and reason. The requester and the approver must be different people (design). • The system reports the current and point-in-time (any past date) list of users, roles, privileges and sites for each system. This is the "authorised users and privileges" list. • A leaver event from HR or the IdP disables the account. DESIGN-CHOICE default: the same business day. No rule sets a time. • Periodic access recertification by the area owners records the reviewer, the outcome and each user's last activity. DESIGN-CHOICE default: every 6 months; allowed 3-12 months.
TYPE DETAIL	ENFORCED for recording and point-in-time reports. CONFIGURABLE for timings (no regulatory numbers exist).
REGULATORY BASIS	US: 11.10(d); 211.68(b); FDA DI Q&A Q4 (documented list of authorised personnel and privileges) | EU: Annex 11 §12.3 | ICH/EMA/draft: draft Annex 11 §11 (timely handling and review of access rights) (draft)(s)
INDIA / US / EU DIFFERENCES	EU §12.3 explicitly requires recording of access changes. FDA Q4 recommends a documented privilege list. Both apply.

HOW IT WILL BE TESTED	Produce the role list as of a past date and reconcile it with the access-request records. Disable a leaver, then try to log in and to sign. Both fail.

Source requirements: n644, n159, n62, n47, n173

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-17 Authority checks per action, least privilege and segregation of duties

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• At every action the engine checks that the user holds a role permitted for that action on that object. Objects are scoped by legal entity, site, area, product, line and step type. Actions include: • execute, verify, supervise; • QA review and audit-trail review; • disposition / QP certification; • MBR author, checker, approver; • master-data change, configure, administer. • A failed check blocks the action (it does not just warn) and is logged. • Signature meanings are bound to roles. For example, only the QU-disposition role can sign "Approved for release" or "Rejected". • Segregation of duties is enforced by the engine: • the performer of an entry is not its verifier; • the MBR author, checker and QU approver are three different users; • a user who executed or verified any step of batch X cannot do the audit-trail review of batch X or sign its disposition. • Data is isolated between sites by default. • Status changes on materials, intermediates and batches are QU-only e-signatures; the details are in the production catalog.
TYPE DETAIL	ENFORCED for the checks, performer-not-verifier and author-not-checker. The executor-not-reviewer/disposer rule is a DESIGN-CHOICE, engine-enforced: it is good practice supporting QU independence and the draft's independent-reviewer expectation, not explicit rule text.
REGULATORY BASIS	US: 11.10(g); 211.22(a),(c); 211.68(b); 211.101(c)-(d), 211.103, 211.186(a), 211.188(b)(11) | EU: Annex 11 §2 (level of access), §12.1; Ch.4 4.20(c), 4.21(c) | ICH/EMA/draft: draft Annex 11 §11 least privilege (draft)(s); draft Annex 11 §12 independent audit-trail reviewer (draft)(s)
INDIA / US / EU DIFFERENCES	US second-person rules (211.101, 211.103, 211.186(a), 211.188(b)(11)) require the checker to be someone other than the performer. EU Ch.4 asks for a checker "where appropriate". The stricter US rule applies.
HOW IT WILL BE TESTED	Run a permission-matrix suite with positive and negative tests for every role and action. The performer tries to verify their own entry. Refused. A batch executor tries to sign disposition of that batch. Refused. A role without the QU function tries to sign "Approved for release". Refused.

Source requirements: n50, n2, n173

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-18 System-administration segregation (administrators independent of record content)

Enforced | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• No account in a tenant can hold both a system-administration permission and a GMP data permission. The engine rejects the assignment. • System-administration permissions: user/role management, configuration, interface settings, audit-related settings, file or DB access. • GMP data permissions: execute, verify, review, approve, release, MBR authoring. • Administrators cannot change GMP data values, cannot sign GMP records, and cannot grant themselves any role. Every role grant needs a second approver. • Break-glass accounts are sealed and time-limited, need dual authorisation, raise an alarm when used, and are reviewed afterwards. They cannot perform GMP actions. • Privileged actions go to a protected system log. That log is covered by periodic review (DI-11) and by the configuration audit trail. • The product has no in-built exception for small sites: FDA Q4 offers no fallback. Any compensating arrangement is owned by the firm (see open questions).
TYPE DETAIL	ENFORCED (designed to the stricter draft; FDA text says "should").
REGULATORY BASIS	US: 11.10(d), 11.10(g); 211.68(b); FDA DI Q&A Q4 (administrator rights assigned to people independent of record content) | EU: Annex 11 §12.1, §12.2 | ICH/EMA/draft: draft Annex 11 §11 segregation of duties, §12 privileged functions only for administrators independent of GMP work (draft)(s)
INDIA / US / EU DIFFERENCES	FDA recommends. The 2011 EU text is silent. The EU draft is explicit, so the kernel follows the draft.
HOW IT WILL BE TESTED	Assign an administrator role to a user who holds a QA role. Refused. An administrator tries to edit a GMP value through the UI or API. Refused and logged. A break-glass login raises an alarm and cannot sign.

Source requirements: n645, n62, n168

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-19 Credential protection: password policy, lockout, misuse detection, token and badge management

Enforced | Configurable | Both editions | 1 of 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each ID-plus-password combination belongs to one individual. This follows from unique IDs (DI-15). • Issued credentials are checked, recalled or revised periodically. At least one of these must be active, and the engine refuses a configuration with neither: • password aging (DESIGN-CHOICE default: 90 days); • a documented risk-based equivalent, i.e. compromised-password screening plus access recertification. • Lockout after N consecutive failed login or signing attempts. DESIGN-CHOICE default N = 5; allowed 3-5. • Transaction safeguards: failed attempts, signing attempts on locked or disabled accounts, and anomalous use raise an immediate alert to the security unit (and to management where defined). Acknowledgment is tracked and unanswered alerts escalate.

	• A lost, stolen or possibly compromised badge or token is de-authorised electronically at once. A replacement is issued only after identity is re-verified. • Badges and tokens are tested when issued and periodically, and the results are recorded. • There is no "remember password", and administrators can never see passwords.
TYPE DETAIL	ENFORCED (mechanisms). CONFIGURABLE (numeric values). Neither Part 11 nor Annex 11 (2011) sets any number for password length, aging, lockout count or alert time.
REGULATORY BASIS	US: 11.300(a)-(e); 11.200(a)(2)-(3) | EU: Annex 11 §12.1, §12.2 | ICH/EMA/draft: draft Annex 11 §11 automatic lock-out after repeated failed log-ins (draft) (s)
INDIA / US / EU DIFFERENCES	11.300(d) requires immediate and urgent reporting of attempted misuse, which is stricter than the 2011 EU text. The EU draft adds lockout. Both apply.
HOW IT WILL BE TESTED	Make 5 failed signing attempts. The account locks and the security unit receives the alert within the configured time. A revoked badge cannot authenticate. Badge test records exist.

Source requirements: n58, n173

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-20 Device and data-source checks

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every device has a registered identity and an assigned room or line: terminals, tablets, balances, scanners, instruments, PLC/SCADA connectors, printers. • Every value stores the ID of the device it came from. • For steps where the MBR declares device binding, data is accepted only from a registered device assigned to that step's location. Recipe downloads go only to the requirements listed equipment for the step. • A measuring device can be used only if it is within calibration and has passed any required pre-use check. The calibration details are in the production catalog.
TYPE DETAIL	CONFIGURABLE for device-to-location binding per step. The rule applies it "as appropriate". Default ON for weighing, dispensing and critical-parameter capture; it can be turned OFF only for non-critical steps, with a rationale. The device registry and the source-ID stamp are ENFORCED.
REGULATORY BASIS	US: 11.10(h); 211.68(a) | EU: Annex 11 §5 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	11.10(h) is the only explicit device-check rule. The EU reaches a similar result through §5 built-in checks.
HOW IT WILL BE TESTED	Data arriving from an unregistered or relocated scale is rejected with an alert. A stored weight shows its source balance ID.

Source requirements: n51, n7

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-21 Training of system users, e-signature accountability policy and signature register

Enforced | Configurable | Design choice | Both editions | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Before a user can execute, verify, review or approve an action, the engine checks that the user holds a current qualification for that operation, SOP version and MBR (via LMS integration or an internal curriculum). A lapse is a hard stop that names the missing item. A supervisor cannot sign on the user's behalf. • The qualification evidence as it stood at signing time is snapshotted with each executed signature. • Required training content: using the eBMR; preventing and detecting data-integrity problems; audit-trail review (for reviewers). • Training also covers administrators, key users and the vendor's developers and maintainers (evidenced through supplier qualification). • A written e-signature accountability policy exists. Before first signing, a user acknowledges it electronically, and the acknowledgment is recorded (design). • The signature register maps each user ID to full name, initials (if used), roles, permitted signature meanings and validity dates. It is kept at least as long as any record that user signed, and it can resolve any signer as of the signing date. • Injectables: aseptic roles also need current gowning and APS qualification. That check lives in the sterile catalog.
TYPE DETAIL	ENFORCED (signature register; acknowledgment before first signing is an engine-enforced DESIGN-CHOICE). CONFIGURABLE for training gating: default hard block, which may be relaxed to a flag-and-review mode only by QA-approved configuration. The rules require trained staff; they do not require system gating.
REGULATORY BASIS	US: 11.10(i), 11.10(j); 211.25(a); FDA DI Q&A Q16 | EU: Annex 11 §2; Ch.4 4.29 (personnel records incl. signature lists and training) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	11.10(i) explicitly extends training duties to developers and maintainers. EU Ch.4 4.29 explicitly requires signature lists. Both apply.
HOW IT WILL BE TESTED	Expire a user's SOP training. Execution and verification are both blocked. A new user tries to sign before acknowledging the policy. Blocked. An archived signer resolves to the name and role valid on the signing date.

Source requirements: n3, n52, n655, n624

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-22 Audit trail: capture scope and entry content

Enforced | Both editions | 5 of 8 source requirements checked against official text

WHAT THE SYSTEM DOES	• The server generates the audit trail independently of the UI, and it is append-only. • It records every create, modify, delete/void, confirm/verify, sign, status change and configuration change of GMP-relevant data: execution data, MBRs, master data, roles, limits, review and release. • Each entry holds: • user ID, printed full name and role at the time;

- action type;
- object, record and field;
- old and new value;
- server date-time (UTC plus site offset);
- reason, chosen from a controlled list with free text. A reason is mandatory for any modification and any delete/void of GMP data after initial commit. It is not required for initial creation.
- source terminal or device ID (design).
- A separate protected system log records: logins, failed logins, lockouts, permission changes, configuration changes, interface and equipment-communication events, and alert records.
- The audit trail is kept at least as long as its subject record and travels with it (DI-02).
- It can be viewed and exported in plain language (not raw codes) and filtered by batch, step, user, data class and date.

REGULATORY BASIS	US: 11.10(e); FDA DI Q&A Q1b, Q1c (data-level and record/system-level audit trails; reprocessing justified) | EU: Annex 11 §9 (reason for change or deletion of GMP-relevant data; available in intelligible form), §12.4 (identity and date/time for entering, changing, confirming, deleting); Ch.4 4.9 | ICH/EMA/draft: draft Annex 11 §12 (creation included; who/what/when/why; role; time zone; reason prompted) (draft)(s); draft Ch.4 metadata/audit-trail provisions (draft)(s); draft summary (draft)(u)
INDIA / US / EU DIFFERENCES	• The US makes the audit trail mandatory and covers creation, but 11.10(e) does not require a reason. • EU 2011 makes the audit trail risk-based ("consideration should be given"), but it requires a documented reason for any change or deletion and captures confirmations. • Combined: the trail is mandatory and covers creation, changes, deletions and confirmations, and a reason is required for changes and deletions. • The draft's role and time-zone fields are adopted.
HOW IT WILL BE TESTED	As each role, create, modify and void a value. Every entry carries all fields. Saving a modification without a reason is blocked. A failed login, a role change and a configuration change each appear in the system log.

Source requirements: n48, n156, n168, n111, n640, n160, n88, n226

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-23 Audit trail: always on, immutable, tamper-evident

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• No tenant, site, MBR, role or administrator setting can disable, pause, narrow at capture time, edit or purge the GMP audit trail. • Audit-related settings (e.g., adding event types) can be changed only by independent administrators (DI-18) under change control. The change itself goes to the protected system log and is flagged for QA. • No UI or API function deletes audit entries during retention. • The trail is tamper-evident: it is hash-chained, a scheduled verification job checks the chain, and a break raises a critical alarm. DB-level activity monitoring detects direct changes.
TYPE DETAIL	ENFORCED. The hash chain is the DESIGN-CHOICE used to meet "secure" and "ability to discern altered records".
REGULATORY BASIS	US: 11.10(e) (secure), 11.10(a) (discern altered records); FDA DI Q&A Q4, Q12 (inference; the Q&A does not itself say who may switch audit trails off) | EU: Annex 11 §9 | ICH/EMA/draft: draft Annex 11 §12 (should not be possible to turn off, edit or delete) (draft)(s)
INDIA / US / EU DIFFERENCES	Neither the 2011 EU text nor the FDA Q&A explicitly forbids deactivation. 11.10(e) requires a secure trail, and the draft forbids deactivation outright, so the kernel follows the draft.
HOW IT WILL BE TESTED	Search the UI, the API and the admin console for any disable or purge function. None exists. In a test environment, change an audit row directly in the DB. The verification job alarms and names the entry.

Source requirements: n645, n168, n44

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-24 Corrections, voids and deletions: the original is always preserved

Enforced | Design choice | Both editions | 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• After initial commit, a GMP value can only be corrected by creating a new version linked to the original. • The original stays visible in the record, in review views and on printouts (DI-40). • A correction requires the corrector to re-authenticate (signature meaning "Corrected by"). It records user, date-time and a mandatory reason. • "Delete" of GMP data is implemented as void/invalidate with a reason. The data is retained and shown as voided. • Correcting signed content supersedes the signature (DI-37). • Physical purge is possible only after retention ends, through DI-47. Deletion at DB level is prevented or detected.
TYPE DETAIL	ENFORCED (original preserved, reason, no deletion within retention). DESIGN-CHOICE, engine-enforced: a correction needs re-authentication because EU Ch.4 4.9 asks for alterations to be signed and dated, and an e-signature ceremony is the conservative reading. Physical-deletion ban during retention: 11.10(e) tolerates audited deletion; the ban is stricter.
REGULATORY BASIS	US: 11.10(e); FDA DI Q&A Q1c, Q12, Q13; Background (data secure until disposition) | EU: Ch.4 4.9; Annex 11 §9, §12.4 | ICH/EMA/draft: EMA GMP DI Q&A on data risk (s)
INDIA / US / EU DIFFERENCES	US: prior information must not be obscured. EU: alteration signed and dated, original readable, reason (Annex 11 §9: for all GMP-relevant changes and deletions). The EU is stricter on the reason and the signature.
HOW IT WILL BE TESTED	Try to delete an entry through the UI and the API. Only void-with-reason is offered. Correct a value. The original is visible in the record, in the review list and in the PDF, alongside the corrector's signature and reason.

Source requirements: n658, n145, n203, n48

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-25 Audit-trail review within batch review (step level and before disposition)

Enforced | Both editions | 1 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Step level.** When a checker or verifier signs a significant step, the system shows inline every post-entry change, void and signature supersession on that step's data, like paper cross-outs: old and new value, who, when, why. The checker's signature covers that review. • **Batch level.** Before QU disposition (US) and before QP certification (EU, where recorded in the system), someone must complete and e-sign a batch-scoped audit-trail review task ("Audit trail reviewed"). • The reviewer must not have executed or verified any step in that batch. • Disposition/certification is blocked until the review is signed and every discrepancy raised from it is closed or impact-assessed. • **Scope.** Every change to any data in the batch record is in scope. Criticality may order or highlight entries but never excludes them. Scope also includes batch-linked system events: interface failures, signature failures, lockouts, time-sync faults and alarms. • **No deferral.** There is no option to defer audit-trail review until after release. • **Discrepancies.** Any entry can be escalated to a deviation directly. • **QU/QP view.** The QU and QP can open the full batch audit trail from the disposition/certification screen. That screen shows review status, reviewer, date and findings.
REGULATORY BASIS	US: 211.22(a), 211.188(b), 211.192; FDA DI Q&A Q7 (reviewers of a record review the audit trails of its data; QU review includes audit trails), Q8 (same frequency as the data: after each significant step and before release) | EU: Annex 11 §9 (regular review); Ch.4 4.27 (indication of changes to critical data; records available to QP) | ICH/EMA/draft: draft Annex 11 §12 review (written procedure, independent reviewer, before release, QP access) (draft)(s); draft summary (draft)(u); draft removal of 2011 §15 with QP audit-trail access (draft)(s)
INDIA / US / EU DIFFERENCES	• The US is stricter on timing: every significant step, plus before release. • EU 2011 says only "regularly". • The draft moves to "before release" but is reported to allow a justified later review. That option conflicts with the US before-release rule and is not offered.
HOW IT WILL BE TESTED	Change a value after entry. It appears in the checker's step view and in the batch review list. Disposition stays blocked until the review is signed. A batch executor tries to sign the review. Refused. No configuration offers deferral.

Source requirements: n65, n169, n172, n170, n226

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-26 Review by exception, error detection and risk-based review of system logs

Enforced | Configurable | Both editions | 3 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• The engine builds a batch exception report from a mandatory set of exception types: • out-of-limit values; • deviations and overrides; • forced, skipped or NA steps; • post-entry changes and voids/invalidations; • repeats and aborted attempts; • late or back-entered data; • manual entry where automatic capture was configured; • interface fallbacks and failures; • signature failures; • alarm events; • time-sync events; • open incidents. • Customers may add types or raise severity. They cannot remove a mandatory type. • A clean batch report states "no exceptions" and shows a completeness summary. Every item drills down to the full record. Reviewers work on the electronic record, not on printouts. • The RBE rule set is version-controlled configuration and changes only under QA-approved change control. It is validated by seeded-exception tests with zero false negatives, and challenged periodically by comparing sampled full audit trails with the filtered view. • A new product or process, or one with a poor recent history, defaults to full review until a configured confidence criterion is met (design). • Instrument-status and communication logs and alert records are reviewed at a frequency set by a documented risk assessment. Any such event tied to a batch surfaces in that batch's review. • Detection is built in: range checks, completeness checks before sign-off, out-of-sequence/late-entry detection, and interface reconciliation.
TYPE DETAIL	ENFORCED for the mandatory type set, validation, drill-down and change control. CONFIGURABLE for whether an MBR uses RBE: default full review until the confidence criterion is met. The QU still approves the complete record (DI-39).
REGULATORY BASIS	US: 211.68(b), 211.192; FDA DI Q&A Q1a (design for easy detection of errors, omissions, aberrant results), Q3 (validate the workflow), Q8 and footnote 13 (risk-based review of instrument status/communication logs and alerts; reliability of review proven) | EU: Annex 11 §9; Ch.4 4.20 Note (compliance summaries and exception reports for continuously monitored validated processes) | ICH/EMA/draft: EMA GMP DI Q&A on review of electronic data / review by exception (s)
INDIA / US / EU DIFFERENCES	The EMA Q&A explicitly accepts review by exception when it is scientifically justified and validated. The FDA DI Q&A does not mention RBE: it requires the QU to review all records, audit trails included, and asks that review reliability be proven. So RBE is used only as a validated way of doing the full QU review, never as a reduction of what the QU approves.
HOW IT WILL BE TESTED	Seed a batch with one instance of every exception type. All are listed. A clean batch shows "no exceptions" with its summary. Try to edit an RBE rule without a change record. Blocked.

Source requirements: n184, n201, n646, n639, n61

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-27 Trusted time

Enforced | Configurable | Design choice | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every GMP timestamp comes from the synchronised server clock: entries, audit trail, signatures, status changes, step start and completion, and significant intermediate stages. • Client-device clocks and user-typed times are never used as the time of an action.

	• Timestamps are stored in UTC with the site time zone and offset, and displayed as site-local time with the offset. Ordering stays correct across daylight-saving changes. • Event times a user enters from another source (e.g., cycle start read off a printout) are separate, flagged data fields. They are kept distinct from the time the entry was made. • Server-to-reference offset is monitored: • Beyond the threshold, an alarm and an incident are raised, and entries made during the fault are flagged for review. • Capture is never simply blocked without the controlled contingency path of DI-43, because staff must never fall back to scrap-paper notes.
TYPE DETAIL	ENFORCED for server time and UTC storage. CONFIGURABLE for the drift threshold: DESIGN-CHOICE default 1 s, allowed 0.1-5 s. No rule sets a clock tolerance.
REGULATORY BASIS	US: 11.10(e) (computer-generated, time-stamped), 11.50(a)(2); 211.188(b)(1) (dates) | EU: Annex 11 §12.4 (date and time), §14 (time and date of signature); Ch.4 4.20(b), 4.21(b) (dates and times) | ICH/EMA/draft: draft Annex 11 §12 and §13 (time zone where relevant) (draft)(s)
INDIA / US / EU DIFFERENCES	211.188(b)(1) asks only for dates in the batch record. EU 4.20(b)/4.21(b) require dates and times, so the stricter EU rule applies. Time-zone display comes from the draft. The DI Q&A says nothing about clocks: NTP, UTC and the drift threshold are design choices.
HOW IT WILL BE TESTED	Change a terminal clock, then record a step. The stored time is unaffected. Run a DST-transition test. Simulate loss of the time reference. An alarm and an incident are raised.

Source requirements: n160, n88, n10

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-28 Contemporaneous capture, immediate durable commit, late entries and offline buffering

Enforced | Design choice | Both editions | 5 of 7 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each GMP entry is committed to durable, audit-trailed server storage the moment the user confirms it (auto-save per entry). • Executed GMP data has no editable client-side draft. • There is no end-of-shift bulk save. • Limit evaluation (pass/fail, OOS warning) is shown only after the value is committed, so a value cannot be re-keyed before it is saved. • Free-text or scratch fields can never be the source of a GMP field. • A late entry (recorded after the event) goes only through an explicit late-entry path: • a mandatory reason; • both the claimed event time and the entry time are stored; • the entry is flagged for review. • Back-dating a batch, packaging or logbook entry is impossible. • Batch and packaging records identify who performed each significant step and who checked it, who performed each IPC, and dates and times. Attribution comes from DI-22 and DI-35. • Logbook entries are chronological and attributed, and are filled in automatically from execution wherever possible. • An SOP pairs these controls with the procedural duty to enter data at the time of the work. • Offline/edge mode, if offered: • entries are signed and hashed locally at capture against a trusted clock; • they are immutable, uploaded in order and flagged "offline-captured"; • they never overwrite server data; a conflict becomes an exception; • a status check that needs server data either uses a pre-cached value with a validity window or blocks the step.
TYPE DETAIL	ENFORCED. Offline mode is a DESIGN-CHOICE; if enabled, its rules are ENFORCED.
REGULATORY BASIS	US: 211.100(b), 211.160(a), 211.182 (chronological equipment log), 211.188; FDA DI Q&A Q12 | EU: Ch.4 4.8, 4.20(b),(c),(f), 4.21(b),(c), 4.31 | ICH/EMA/draft: EMA GMP DI Q&A on temporary memory and on OOS warnings before saving (s)
INDIA / US / EU DIFFERENCES	• The final FDA Q&A says no storage method may allow data to be manipulated before a permanent record exists. • The "temporary memory" and "warning before save" wording comes from the EMA Q&A, not from the FDA final text. • Commit-before-evaluate satisfies both.
HOW IT WILL BE TESTED	Enter a value, then kill the browser. The value persists with its timestamp. Enter an out-of-limit value. The audit trail shows the value committed before any warning appeared. Submit a late entry without a reason. Blocked. Try to insert a back-dated logbook entry. Blocked.

Source requirements: n10, n66, n183, n221, n106, n92, n94

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-29 Complete data: every attempt, repeat, aborted operation and invalidated result is retained; invalidation needs an investigation

Enforced | Configurable | Both editions | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every capture attempt on product is stored with who, when and reason, and is shown in review. This includes: • repeat weighings and re-reads; • failing or suspect IPC results; • aborted weighings, cancelled label prints and aborted equipment cycles; • restarted steps. • There is no unrecorded trial, practice or check mode on product samples. Set-up and suitability checks use registered standards under their own IDs, never a product sample ID. • Correcting an entry error (DI-24) is distinct from invalidating a genuine result. • Invalidation requires a link to a documented investigation (deviation or OOS-type) with a scientifically sound justification. • The invalidated data and the investigation report are part of the record the QU sees at disposition. • A repeat-until-pass pattern is flagged. DESIGN-CHOICE default: any new attempt after a failing result is flagged. • For IPCs, all individual values are stored (not just means), with: • the raw instrument output; • a link to the material and lot tested; • the method reference and version; • the sample quantity;

	• calculations with units and conversion factors. • Where the SOP classes a shop-floor IPC as a laboratory test, second-person review of the original record is mandatory (211.194(a)(8)).
TYPE DETAIL	ENFORCED. CONFIGURABLE for flag thresholds, stricter only.
REGULATORY BASIS	US: 211.160(b), 211.188, 211.192, 211.194(a)(3)-(8); FDA DI Q&A Q2, Q12 (aborted/incomplete runs captured and justified), Q13 (all data including errors, failing, passing and suspect retained and reviewed) | EU: Ch.4 Principle (raw data), 4.20(f); Annex 11 §9 | ICH/EMA/draft: draft Ch.4 ALCOA++ "complete" (draft)(s)
INDIA / US / EU DIFFERENCES	FDA is explicit, through Q2 and Q13. The EU reaches the same result through its raw-data definition. No conflict. Applying Q12/Q13 to production IPCs is by analogy with the laboratory examples.
HOW IT WILL BE TESTED	Take three readings on one sample where the first two fail. All three are stored and flagged. Try to invalidate a value without an investigation link. Blocked. Abort a weighing. A reason is required and the attempt appears in review.

Source requirements: n69, n651, n652, n41, n17

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-30 Manually entered critical data: additional accuracy check

Enforced | Configurable | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• A manually entered value classed critical (DI-07) must pass an additional accuracy check before the step can complete. The check is one of: • an independent second person using their own authentication, who is not the performer; • a validated electronic means: barcode match against the expected item/lot, or device capture that replaces manual keying (the interface then falls under DI-31). • The engine refuses to approve an MBR in which a critical manual field has no check method. • Every manual field gets plausibility checks: type, range, format, unit, expected lot or material. • A value transcribed by hand is marked "manual/transcribed" and always needs second-person verification. This covers interface bypass and back-entry from paper. • Manual readings (plate counts, visual checks, gauge readings) are entered by the reader at the time of reading and verified by a different person. The record shows the reader, the verifier and both times. • US-market floors (details in the production catalog): • every container of a dispensed component is checked by a second person (211.101(c)); • each charge-in is added by one person and verified by a second (211.101(d)); • each yield calculation is verified by a second person (211.103); • each significant step identifies the person performing it and the person directly supervising or checking it (211.188(b)(11)). • One-person checking is allowed only where the operation itself is performed by automated equipment under 211.68 and one person checks that the equipment did it correctly (211.68(c)).
TYPE DETAIL	ENFORCED (floor). CONFIGURABLE for the check method per field (second person vs validated electronic check), within the floors.
REGULATORY BASIS	US: 211.68(c), 211.101(c)-(d), 211.103, 211.188(b)(11), 211.194(a)(8); FDA DI Q&A Q10 | EU: Annex 11 §6 | ICH/EMA/draft: draft Annex 11 §10 handling of data (plausibility checks; prefer validated interfaces to manual transfer) (draft)(s)
INDIA / US / EU DIFFERENCES	EU §6 lets a validated electronic check replace the second operator for manual data. The US second-person rules for dispensing, charge-in and yield drop to one person only when the operation itself is performed by automated equipment. For US-market products, device capture on its own does not remove the second-person check at dispensing and charge-in unless QA documents that the operation qualifies under 211.68(c) (see open questions).
HOW IT WILL BE TESTED	Configure a critical manual field and try to complete the step without verification. Blocked. The same user tries to verify. Refused. Take the interface down. The manual value is flagged "manual" and needs verification.

Source requirements: n151, n175, n649

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-31 Interface data integrity

Enforced | Configurable | Design choice | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each interface checks what it receives. In scope: ERP/WMS, LIMS, balances, PLC/SCADA/historian, EMS, label printers, vision/serialisation, QMS. The checks are: • schema and completeness validation; • identity matching (material, lot, batch, equipment); • unit and range checks; • duplicate and sequence detection; • acknowledgment and retry; • reconciliation by counts or checksums. • Check depth is set by the interface's complexity and reliability. • A failed or rejected transfer raises an alarm. It never silently drops a value or substitutes a default. A partial transfer is never committed. • Each received value keeps its provenance: source system, source record ID, source timestamp, raw value and unit. • An interface-captured value can be overwritten only by an audit-trailed correction with a reason. A manual override appears in the exception report (DI-26). • Transport is authenticated and encrypted (DI-13).
TYPE DETAIL	ENFORCED (no silent loss or defaulting; provenance; alarms). CONFIGURABLE for check depth per interface risk. Idempotency keys and checksums on every interface are a DESIGN-CHOICE: 211.68(b) scales checking to complexity and reliability.
REGULATORY BASIS	US: 211.68(b) (input/output checked for accuracy; degree and frequency by complexity and reliability) | EU: Annex 11 §5 | ICH/EMA/draft: draft Annex 11 §10 data transfer (draft)(s)
INDIA / US / EU DIFFERENCES	Consistent across markets. Annex 11 §5 adds "secure", which leads to the transport security.
HOW IT WILL BE TESTED	Inject malformed, duplicate, out-of-order and wrong-batch messages. Each is rejected or de-duplicated, with an alarm. Cut the link mid-transfer. No partial value lands in the record.

Source requirements: n155, n8, n88

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-32 Instrument and device data; original records from simple devices; dynamic data

Enforced | Configurable | Both editions | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Where a device can transmit, capture its data directly: value, raw device output, device ID, unit and device timestamp. The raw output is stored and must match the displayed value. The electronic capture is then the original record. • Manual keying is allowed only in a documented fallback mode, flagged and verified (DI-30). • Where a balance or pH-meter printout is the original (no electronic capture), the printout or a verified true copy is retained and linked by reference to the eBMR entry. Thermal slips are true-copied promptly before they fade (DI-41). • Dynamic device data are kept in native form with a viewer: trends, cycle data, spectra, chromatograms, PAT outputs. A static rendering alone never replaces them.
TYPE DETAIL	ENFORCED (an original is defined per data source; dynamic data kept native). CONFIGURABLE for direct capture per step: default ON wherever a validated device interface exists.
REGULATORY BASIS	US: 211.68(a)-(b), 211.180(d), 211.194(a)(4); FDA DI Q&A Q1d, Q9, Q10 | EU: Ch.4 Principle (raw data), 4.20 Note; Annex 11 §5 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	FDA Q10 is the only text that explicitly says a simple device's printout can be the original and that a printout of dynamic data is not enough. It is adopted for all markets.
HOW IT WILL BE TESTED	A weighing with direct capture stores a raw string that matches the display. A weighing with a printed ticket shows the ticket linked. Trend data from a restored archive can be re-rendered interactively.

Source requirements: n7, n648, n67, n41, n17

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-33 Calculations and reprocessing

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• GMP calculations run only in validated, version-controlled engine logic. Examples: yield, reconciliation %, potency-adjusted quantities, means/RSD, unit conversions. • The logic is a sandboxed expression language with a closed function library, and GMP masters have no custom-script escape hatch (design). • Each result stores its inputs, formula version, units and conversion factors, and the result. • Automated yield calculations are independently verified by one person. Manual calculations are done by one person and verified by a second (211.103). • Recalculation or reprocessing is allowed only in SOP-defined cases (e.g., after an input correction, or re-evaluated PAT data). Every result version is kept with its inputs, parameters, user, time and justification, and reviewers see the full history. • A spreadsheet used for a GMP calculation must be a controlled template version, stored in native form with its formulas intact (a dynamic record). It is never the unlogged source of a GMP result.
TYPE DETAIL	ENFORCED. The sandboxed expression language is an engine-enforced DESIGN-CHOICE.
REGULATORY BASIS	US: 211.68(b), 211.103, 211.188(b)(7), 211.194(a)(5); FDA DI Q&A Q1d, Q3, Q6, Q14 | EU: Annex 11 §4.7 | ICH/EMA/draft: draft Ch.4 draft-4.12 (derived data reconstructable from raw data) (draft)(s)
INDIA / US / EU DIFFERENCES	US text is explicit about verifying yield. The EU reaches the same result through validation. Applying Q14 to production calculations is by analogy with laboratory reprocessing.
HOW IT WILL BE TESTED	The yield step shows inputs, formula version, result and a verification signature. Correct an input and recalculate. Both results are kept, with their reasons.

Source requirements: n641, n653, n108

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-34 Operational sequencing checks

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• An execution state machine enforces: • MBR step order and declared parallel branches; • prerequisites: line clearance, equipment and material status, IPC due, timers, QA gates; • guarded batch lifecycle transitions: Issued, In Execution, Production Complete, Under Review, Released/Rejected, with Hold and Cancelled. • Out-of-sequence execution is possible only through a deviation path approved in the system. It is recorded and appears in review (design).
TYPE DETAIL	ENFORCED wherever the MBR declares sequence (the rule applies sequencing "as appropriate").
REGULATORY BASIS	US: 11.10(f); 211.100(b) | EU: Ch.4 4.20(h) (signed authorisation for departures); Annex 11 Principle (no loss of process control) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Only the US has an explicit sequencing-check rule, and it is adopted for all markets.
HOW IT WILL BE TESTED	Try step N+1 before step N. Blocked. Try disposition before review is complete. Blocked.

Source requirements: n49

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-35 Electronic-signature catalogue: every predicate-rule signature point

Enforced | Both editions | 4 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES	• A vendor-defined catalogue of signature types covers every point where a rule asks for a signature, initials, a check or an approval: • MPCR/MBR prepared, independently checked, QU-approved; • batch record issued (accurate reproduction checked); • step performed; step checked or directly supervised; • component container checked at dispensing; charge-in verified; • yield verified; • IPC performed and verified; • deviation authorised; correction; • labeling specimen approved; • logbook entry performed and double-checked; • production and packaging record approved; • audit trail reviewed; • QU review and disposition; • QP certification (if used); • true copy verified; • void or reprint authorised. • Each type defines: • the permitted roles and the meaning text; • the independence rule (which earlier signers on the same object it must differ from); • the re-authentication rule (DI-38). • Every type is a full Part 11 / Annex 11 e-signature, never a click-acknowledgment. • A handwritten (stylus) signature on an electronic record, if offered, is still a handwritten signature: it must show the 11.50 elements and be linked under 11.70. • The company signature policy (who may sign which meaning) is configured in the system. • E-signing is the default everywhere. There is no print-to-sign path. • Customers may add signature points but cannot remove the minimum set.
REGULATORY BASIS	US: 11.1(c)-(d), 11.2(a), 11.3(b)(7)-(8); 211.101(c)-(d), 211.103, 211.182, 211.186(a), 211.188(a),(b)(11), 211.194(a)(7)-(8); FDA DI Q&A Q11 (e-signature satisfies "full signature, handwritten"; document signer-identification controls) | EU: Ch.4 4.3, 4.9, 4.13, 4.20(c),(f),(h),(i), 4.21(c),(g),(i), 4.31; Annex 11 §14 | ICH/EMA/draft: draft Ch.4 signatures and signature policy (draft)(s); draft Ch.4 hybrid systems and recognition of e-signatures (draft)(s)
INDIA / US / EU DIFFERENCES	US signature points are more numerous (second-person rules). EU adds approval of the completed processing and packaging records by the responsible person. The catalogue is the union.
HOW IT WILL BE TESTED	An inventory maps every predicate signature point to a catalogue type. Try to complete a cleaning-log entry with a plain acknowledgment. Impossible.

Source requirements: n614, n145, n36, n68, n113, n182

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-36 Signature manifestation

Enforced | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every human-readable form of a signed record shows, for each signature: • the printed full name as it was at signing; • user ID and role; • date and time with time zone / UTC offset; • the meaning. • The human-readable forms are screen, print, PDF and export. • The manifestation is stored immutably with the signature when it is applied, not re-derived later from the user table. It is controlled like the record itself.
REGULATORY BASIS	US: 11.50(a)-(b) | EU: Annex 11 §14(c) (time and date) | ICH/EMA/draft: draft Annex 11 §13 (name, role, meaning, date/time with time zone where applicable) (draft)(s); draft Ch.4 signatures (draft)(s)
INDIA / US / EU DIFFERENCES	The printed name and meaning are US requirements (11.50). EU 2011 requires only date and time. The draft adds role and time zone. All are adopted.
HOW IT WILL BE TESTED	Sign a step, then view it on screen, in the PDF and in the export. All elements appear. Rename the user. Historic signatures keep the original name.

Source requirements: n54, n162, n171

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-37 Signature-record linking and invalidation on change

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• A signature record binds the signer ID, the manifestation, the meaning, the timestamp and a cryptographic hash of the exact content version it signs. • No UI or API path can cut, copy or transfer a signature to another record. An integrity job detects any DB-level transplant. • A permitted change to signed content automatically: • marks the signature "superseded by change"; • shows the element as unsigned; • requires re-signing by the relevant signer(s). The history stays in the audit trail. • Wet-ink signing on printouts is avoided. If it cannot be avoided, the scanned signed page is indexed to the record ID and version hash under an SOP. A wet-ink signature never substitutes for an e-signature the MBR requires.
TYPE DETAIL	ENFORCED (linking). DESIGN-CHOICE, engine-enforced: invalidation on change goes beyond the 11.70 "ordinary means" standard and follows the draft's hybrid-signature expectation.

REGULATORY BASIS	US: 11.70; 11.10(e) | EU: Annex 11 §14(b) (permanently linked) | ICH/EMA/draft: draft Annex 11 §13 (hybrid: later change must invalidate the signature) (draft)(s)
INDIA / US / EU DIFFERENCES	11.70 targets falsification by ordinary means. Annex 11 §14 asks for permanent linkage. Automatic supersession after a change goes further than both and matches the draft.
HOW IT WILL BE TESTED	Correct a signed value. The signature shows as superseded and re-signing is required. Copy a signature row to another record in a test DB. The integrity check fails.

Source requirements: n55, n162, n171

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-38 Signature components, re-authentication and session rules

Enforced | Configurable | Both editions | 2 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Non-biometric signatures use two distinct components: ID plus password/PIN, or badge plus PIN.

TYPE DETAIL

CONFIGURABLE (default = strictest practical). Floors are ENFORCED.

DEFAULTS

every signing requires all components.

- **Allowed relaxation** (CONFIGURABLE, QA-approved): within one continuous controlled session, routine signings other than disposition, certification or MBR approval may use only the private component (password/PIN). A signing is never completed with zero components.
- The continuous session ends at lock, inactivity timeout, logout or a terminal switch. The next signing then uses all components.
- Shop-floor terminals lock after inactivity. DESIGN-CHOICE default: 5 min; allowed 1-15 min.
- A verifier signs with their own full credentials in the same dialog. The performer's session can never supply the verifier's signature.
- The signing dialog states the meaning explicitly and shows what is being signed.
- Biometric signatures must be usable only by their owner, with evidence of liveness/anti-spoofing.
- MFA at signing can be switched on (optional, stricter).
- There is no remember-password and no pre-signing.

REGULATORY BASIS	US: 11.200(a)(1)(i)-(ii), (a)(2)-(3), (b); 11.10(d) | EU: Annex 11 §12.1, §14 | ICH/EMA/draft: draft Annex 11 §13 (re-authentication at signing) (draft)(s); draft IAM/e-signature summary incl. reported MFA for signatures (draft)(u)
INDIA / US / EU DIFFERENCES	11.200 allows a single private component for later signings within one continuous session. The EU draft expects re-authentication at each signing, and typing the password/PIN each time satisfies that. The default of all components every time meets both.
HOW IT WILL BE TESTED	Sign twice in one session. The second signing needs at least the password. After a timeout, both components are needed. Try to apply the verifier's signature through the performer's session. Refused.

Source requirements: n57, n47, n171, n226

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-39 QU disposition and QP certification by e-signature; quarantine interlock

Enforced | Configurable | Both editions | 3 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- **US.** Batch disposition (Released/Rejected) is a QU-role e-signature with an explicit meaning.
- No other role, administrators included, can set a released status.
- The QU is shown the complete record: invalidated data with their investigation reports (DI-29), deviations, exceptions and the audit-trail review (DI-25).
- The release record lists the criteria evaluated. The criteria themselves are in the production catalog.
- **EU.** If the system records certification/release (QP-certification feature ON), only accounts registered as a QP named on the authorisation can certify, by e-signature, and the certifier's identity is recorded.
- This stays in force even though the July 2025 draft reportedly drops the stand-alone clause.
- The QP screen links to the full audit trail and its review record.
- **Quarantine interlock.**
- An uncertified or unreleased batch cannot be allocated or shipped in the integrated ERP/WMS.
- Status is carried across inter-site transfers.
- Notice of certification to the site that moves stock to saleable status is a structured, signed, audit-trailed message with acknowledgment, as defined in the technical agreement.
- Any electronic safeguard used as the quarantine control is validated (DI-03).
- A release approval sent by email or chat is not the record (DI-40).

TYPE DETAIL	ENFORCED. The QP-certification feature is CONFIGURABLE ON/OFF per tenant; when ON its rules are ENFORCED. An Indian exporter's own system usually records QU disposition only.
REGULATORY BASIS	US: 211.22(a), 211.165(a), 211.192; FDA DI Q&A Q2 | EU: Annex 11 §15; Annex 16 sec. 4 (s) | ICH/EMA/draft: draft Annex 11 (removal of 2011 clause 15; QP access to audit trail) (draft)(s)
INDIA / US / EU DIFFERENCES	US: QU approval before release or distribution. EU: QP certification (Annex 16), which is recorded by QP-only e-signature only if the system is used for it. 2011 §15 stays binding until a final revision replaces it.
HOW IT WILL BE TESTED	A QA reviewer, a supervisor and an admin each try to certify. All fail. The QP certifies, and the record shows name, date/time and meaning. ERP stock status changes only after certification.

Source requirements: n163, n170, n191, n2, n31

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-40 Copies for inspection, printouts that flag changes, GMP communications

Enforced | Both editions | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• One action produces an accurate and complete copy of any batch record in two forms: • **Human-readable** (PDF/A): all values, corrections showing both the original and the new value, exceptions, deviations, attachments, signature manifestations and the audit trail. • **Electronic**: structured JSON/XML/CSV with metadata, native attachments, and dynamic data with a viewer. • Report configuration cannot trim a copy below this minimum. • A printout of a record supporting batch release flags every value changed since original entry: old value, new value, who, when and why, or a reference to the audit-trail entry. Unchanged values are not flagged. • Every printout or export carries: • a "copy" mark with print time, user and record status; • the record ID and "page x of y" (good practice). • Every export is itself logged. • Records can be retrieved at once from every site where the activity happened (central hosting is acceptable if retrieval is immediate). • Read-only, audited, time-limited viewing access can be granted for inspections (design option). • GMP decisions are made in the system by e-signature: release, deviations, holds. Comments and messages linked to a batch are kept with an audit trail. An approval that arrives by email is attached and kept with the batch record, and all of these appear in inspection copies.
REGULATORY BASIS	US: 11.10(b), 11.10(e) (audit trail available for review and copying); 211.180(c); FDA DI Q&A Q17 and footnote 14 | EU: Annex 11 §8.1, §8.2; Ch.4 4.27 (all records available to the QP) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Only the EU (§8.2) explicitly requires printouts that indicate changed data. The US (11.10(b)) requires copies in both human-readable and electronic form. The union applies.
HOW IT WILL BE TESTED	Export a batch and compare it field by field with the database. Correct a value, then print. The release printout flags the change and shows the original. A records package for 5 batches includes their audit trails.

Source requirements: n45, n70, n656, n154, n32

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-41 True copies (paper to electronic, electronic to electronic)

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• The true-copy function keeps the content and meaning of the original: all metadata needed to reconstruct the activity, and the original's static or dynamic nature. • Each true copy carries the source ID, a checksum and capture metadata. • A verifier e-signature confirms that the copy is complete and legible. • A true copy of a dynamic record is kept in the original format or in one that preserves content and meaning, and a suitable reader stays available for the whole retention period. • Fading or short-lived media are true-copied promptly: thermal balance, pH or checkweigher slips, and chart-recorder output (e.g., autoclave or lyophiliser charts in the Injectables edition). • Paper originals are kept unless QA approves destruction under an SOP after a verified true copy exists (see open questions).
TYPE DETAIL	ENFORCED (preservation of content, meaning and metadata). DESIGN-CHOICE, engine-enforced: the verifier signature is not described in the FDA Q&A; it follows the EU draft's documented-verification expectation, which does not conflict.
REGULATORY BASIS	US: 211.180(d), 211.68(b); FDA DI Q&A Q1e, Q9, Q10 | EU: Ch.4 4.1 (official copies), 4.10 | ICH/EMA/draft: draft Ch.4 retention/copies and glossary (true/verified copy) (draft)(s)
INDIA / US / EU DIFFERENCES	FDA says "true copy". "Certified" or "verified copy" is EU/PIC/S terminology. The requirements match, and the EU draft adds documented verification.
HOW IT WILL BE TESTED	Scan a thermal printout. The copy carries a checksum, a verifier signature and a link to its step. Alter the file. The checksum check fails.

Source requirements: n67, n114, n647

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-42 Controlled issuance of electronic record instances and templates

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Issuing records: • Batch record instances, logbooks and ad-hoc GMP forms are issued only from Effective, approved template versions. • Each issued instance stores the master version ID and hash, and is an exact instantiation of it plus the batch header. • Instances are uniquely numbered, and a later master edit never changes an issued instance. • Reconciliation of issuance (issued, in execution, completed, voided) shows no gaps. • Voided or re-issued instances are kept and retrievable, each with a written reason and a signer. • A spreadsheet or electronic worksheet used to create CGMP data is a controlled form. An uncontrolled attachment is allowed only as labelled supporting information, never as the source of a GMP result.
TYPE DETAIL	ENFORCED. CONFIGURABLE for QA approval of each void: default ON. Q6 asks for a written justification under QU or other document control, not QA sign-off on every void.
REGULATORY BASIS	US: 211.100(a), 211.186, 211.188(a); FDA DI Q&A Q3, Q6 | EU: Ch.4 4.1 (controls for electronic templates, forms, masters), 4.2 (reproduction without error), 4.3 | ICH/EMA/draft: EMA GMP DI Q&A on template control (s)
INDIA / US / EU DIFFERENCES	No conflict.
HOW IT WILL BE TESTED	Change the MBR after issuing a batch. The issued instance is unchanged and its hash still matches its source version. Void an instance. It stays retrievable with its reason. The reconciliation report has no gaps.

Source requirements: n38, n64

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-43 Paper in the loop: hybrid steps, contingency (downtime) forms and business continuity

Enforced | Configurable | Design choice | Both editions | 3 of 7 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Hybrid points.**
- Each hybrid point is declared in the requirements list (DI-01) with its official record. Examples: supplier CoA, device printouts, chart recorders, downtime forms.
- No GMP step may have a required signature that exists only on paper.
- **Paper forms.**
- A paper form can be printed only through controlled issuance. Each page carries a unique serial number, batch ID, MBR and template version, "page x of y" and issue time. Issuer, quantity and destination are logged.
- User roles cannot print a template outside this control.
- A reprint needs a reason and QA approval, is marked as a reprint and is linked to the original serial number.
- **Reconciliation.** A batch cannot close until every issued form is reconciled as returned/used or voided with a reason. A form reported missing raises a deviation automatically.
- **Business continuity.**
- The downtime procedure is documented and tested.
- Contingency forms are pre-issued, generated from the current master. Entries are made on them at the time of the work.
- After recovery, the data is back-entered and flagged "transcribed from paper". Each back-entered value gets second-person verification (DI-30), and both the original execution time and the entry time are kept.
- The paper originals are true-copied (DI-41) and kept, and the contingency period is reviewed. There are no ad-hoc notes.
- **Fallback timing.** How quickly the fallback must be available is set by risk.
- Injectables: aseptic filling needs an immediate, pre-staged fallback.
- Tablets: may pause at defined hold points.

TYPE DETAIL

ENFORCED for issuance, reconciliation and back-entry. CONFIGURABLE for drill frequency and fallback time: DESIGN-CHOICE default is an annual downtime drill per site. §16 requires the arrangements to be documented and tested, not "validated".

REGULATORY BASIS

US: 211.100(b), 211.180(d), 211.186, 211.188(a); FDA DI Q&A Q6, Q12 (no scrap-paper transcription) | EU: Ch.4 4.1 (hybrid relationships and controls stated); Annex 11 §6, §16 | ICH/EMA/draft: EMA GMP DI Q&A on blank/template forms (s); draft Ch.4 hybrid systems, reported as draft 4.82-4.85 (draft)(s); draft Ch.4 hybrid provisions (draft)(u)

INDIA / US / EU DIFFERENCES

Both markets allow controlled paper. The EU draft pushes firms to avoid hybrid signing wherever possible, and the kernel follows it.

HOW IT WILL BE TESTED

Issue 3 contingency sheets and return 2. Batch closure is blocked and a deviation is raised. In a downtime drill, back-entered values are flagged, verified and linked to scans, with both timestamps.

Source requirements: n164, n650, n185, n222, n112, n182, n96

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-44 Protection and integrity checking of stored data

Enforced | Configurable | Design choice | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- Finalised records sit on protected, redundant storage, and completed records are immutable (design: WORM/object lock).
- A scheduled integrity check runs over hashes and DB consistency.
- A scheduled readability check opens and renders sampled old records, including their audit trails and signatures.
- Results are recorded, and any failure opens an incident (DI-12).
- Access to the data is maintained for the whole retention period, including across upgrades.

TYPE DETAIL

ENFORCED (the checks exist and alarm). CONFIGURABLE for frequency: DESIGN-CHOICE default is quarterly for readability sampling and weekly for integrity jobs. Neither rule sets a frequency.

REGULATORY BASIS

US: 11.10(a), 11.10(c); 211.68(b) | EU: Annex 11 §7.1 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

EU §7.1 explicitly requires stored data to be checked for accessibility, readability and accuracy. It is adopted for all markets.

HOW IT WILL BE TESTED

The periodic readability report lists the sampled records, each opened with its audit trail. Deliberately corrupt a record in a test environment. The check detects it.

Source requirements: n46, n152, n44

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-45 Backup and restore

Enforced | Configurable | Design choice | Both editions | 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- Regular automated backups cover all relevant data as one consistent set:
- database and attachments;
- configuration (MBRs, roles);
- audit trails and signature records;
- system logs.
- There are two tiers:
- (1) **Retention-grade backup.** A true copy, including metadata, in the original or a compatible format, secured against alteration, erasure and loss, and kept for the full retention period. This is FDA's meaning of the 211.68(b) backup, which FDA equates with an archive.
- (2) **Operational/DR snapshots.** These never count toward (1). A replica that mirrors deletions or corruption does not qualify.
- Backups are kept physically and logically apart from primary data (separate site, account and credentials) and are encrypted. Immutable storage is preferred: production administrators cannot delete backups.
- Each backup job's success or failure is logged, and a failure raises an alarm.
- Restores are tested during validation and then periodically. A restore test compares record counts and hashes and opens sample records with their audit trails, and its results are recorded.

	• RPO and RTO are set by risk; the Injectables execution module has the tightest.
TYPE DETAIL	ENFORCED (both tiers, separation, alarms, periodic restore testing). CONFIGURABLE for frequencies: DESIGN-CHOICE default is a quarterly restore test.
REGULATORY BASIS	US: 211.68(b); FDA DI Q&A Q1e (backup is a true copy kept for the retention period; temporary crash copies do not satisfy 211.68(b)) | EU: Annex 11 §7.2 | ICH/EMA/draft: draft Annex 11 §16 backup (physically and logically separate; restore tested regularly) (draft)(s)
INDIA / US / EU DIFFERENCES	For the US, the backup must last the full retention period. EU §7.2 requires the integrity and restorability of backups to be checked during validation and monitored periodically. The draft adds separation. All are adopted.
HOW IT WILL BE TESTED	Restore into an isolated environment and hash-compare records, audit trails and signatures with production. Make a backup job fail. An alarm is raised. Try to delete a backup with production-admin credentials. Refused.

Source requirements: n153, n177, n8, n60, n88

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-46 Archiving

Enforced | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Completed batch records are archived to read-only storage with the full audit trail, signatures, attachments and metadata. "Completed" includes released, rejected and never-released batches. • Fixity is checked periodically. • Retrieval and re-rendering are tested before and after any relevant system change (hardware, software upgrade, format change). Upgrade validation includes retrieving archived records created under earlier versions. • For dynamic data, a validated viewer is kept for the full retention period, or the data is migrated (DI-48) to a format that preserves content and meaning. • Retiring the system requires a documented migration plan or a read-only retention plan.
TYPE DETAIL	ENFORCED. §17 sets no fixity frequency; the default follows DI-44.
REGULATORY BASIS	US: 11.10(c); 211.68(b), 211.180(d); FDA DI Q&A Q1e, Q9 | EU: Annex 11 §17 | ICH/EMA/draft: draft Annex 11 §17 archiving (draft)(s)
INDIA / US / EU DIFFERENCES	Consistent across markets.
HOW IT WILL BE TESTED	Archive a batch and run a fixity check. Upgrade the test environment, then retrieve and render the record. Content, audit trail and signatures are intact and the hashes match.

Source requirements: n165, n647, n177

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-47 Retention periods, holds and controlled disposal

Enforced | Configurable | Both editions | 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• The retention engine computes each record's retention end from genealogy. The floor is the maximum over every rule that applies to the product's markets, plus the India catalog value: • **US batch records:** at least 1 year after batch expiry. OTC products exempt from expiry dating: 3 years after distribution of the batch. • **US component, container, closure and labeling records:** at least 1 year after expiry (read as the expiry of the last lot that used them). Exempt OTC: 3 years after distribution of that last lot. • **EU batch documentation:** 1 year after expiry or at least 5 years after QP certification, whichever is longer. IMPs: at least 5 years after the last clinical trial using the batch is completed or formally discontinued. • **EU critical documentation supporting the marketing authorisation** (e.g., validation and stability raw data): kept while the authorisation is in force. Process-validation raw data: at least as long as the records of every batch released on the strength of that validation. • **Audit trails and signature records:** at least as long as their subject record. • **Shared and master objects:** extended to the latest retention end of any record that depends on them. This covers MBR and formula versions, validation-linked configuration, component lot records, logbooks, the signature register and training evidence. • If the QP certification date of an exported batch is unknown, its retention end stays open (no disposal) until the date is received. • Tenants may lengthen retention, never shorten it. • Legal holds and DI-investigation holds (DI-50) override disposal. • After retention ends, disposal happens only through a QA-approved workflow that writes a disposal record: what, when and who approved. No user or configuration can delete a record during retention.
TYPE DETAIL	ENFORCED (floors). CONFIGURABLE for longer periods.
REGULATORY BASIS	US: 211.180(a)-(b); 11.10(c), 11.10(e); FDA DI Q&A Q1a (life cycle through disposition), Background | EU: Ch.4 4.10, 4.11, 4.12; Annex 11 §7.1 | ICH/EMA/draft: draft Ch.4 data lifecycle with controlled destruction (draft)(s)
INDIA / US / EU DIFFERENCES	The EU is usually stricter for EU-certified batches because of the "5 years after certification" limb. Example: a 24-month shelf life certified 2 months after manufacture gives US = manufacture + 36 months, EU = manufacture + 62 months, so the EU period wins. The US is the only source of the component-record and OTC rules. The kernel takes the maximum.
HOW IT WILL BE TESTED	Try to delete a batch record before its retention end. Blocked. A component lot used in two batches with different expiries follows the later one. An EU batch whose certification date has not been received shows retention "open".

Source requirements: n32, n104, n152, n658, n181

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-48 Data migration

Enforced | Design choice | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Any move of data to another format or system is a validated migration. This includes schema upgrades, archive exports, legacy imports, platform changes and tenant exit.

- The migration verifies that neither values nor meaning change. It checks:
- record counts and per-record checksums;
- field-level values, units and precision;
- timestamps and time zones;
- the audit trail;
- signature manifestations and signature-record linkage.
- The migration report is approved before go-live.

TYPE DETAIL	ENFORCED (the checks). Automating them is a DESIGN-CHOICE: §4.8 requires the checks, not automation.
REGULATORY BASIS	US: 11.10(a), 11.10(c); 211.68(b), 211.180(d); FDA DI Q&A Q9 | EU: Annex 11 §4.8, §14(b) (linkage survives) | ICH/EMA/draft: draft Annex 11 §10 data migration (draft)(s)
INDIA / US / EU DIFFERENCES	Only the EU (§4.8) has an explicit migration rule. The US reaches the same result through 11.10(c) and the true-copy rules.
HOW IT WILL BE TESTED	In a migration dry run, counts and hashes match, and spot-checked audit trails, e-signature meaning and date-time, and units are unchanged.

Source requirements: n150, n175, n647

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-49 Alarms and equipment events in the batch context

Enforced | Configurable | Both editions | Search-based only (0 of 1 checked against official text)

WHAT THE SYSTEM DOES

- GMP-relevant alarms that occur during a batch appear in the batch record. They may come from the eBMR itself or from connected SCADA/EMS. Each shows:
- the time raised and the time acknowledged;
- who acknowledged it;
- the comment and action taken;
- a deviation link where one is required.
- Acknowledging a GMP-critical alarm requires a comment.
- The alarm log cannot be erased or deactivated.
- Alarm limits are controlled configuration (DI-09).
- Each batch has an alarm log that is part of review (DI-26), and alarms are trended across batches.
- Injectables: aseptic-area alarms (differential pressure, particles, glove breach, filter-integrity failure) are visible to reviewers and the QP.
- Tablets: examples are compression-force excursions and checkweigher reject bursts.

TYPE DETAIL	CONFIGURABLE for the integration scope, which depends on the connected systems. Where alarms are integrated, the rules above are ENFORCED. The kernel is designed to the draft (no conflict).
REGULATORY BASIS	US: 211.68(a); FDA DI Q&A Q8 footnote 13 (alert records reviewed on a risk basis) | EU: Ch.4 4.20(e),(h) (relevant events, special problems) | ICH/EMA/draft: draft Annex 11 §8 alarms (draft)(s)
INDIA / US / EU DIFFERENCES	The 2011 texts have no alarm clause. The draft adds one, and the kernel adopts it.
HOW IT WILL BE TESTED	Simulate an excursion. It appears with raise and acknowledgment times, acknowledger identity and comment. Try to acknowledge without a comment. Refused. Try to delete the alarm log. Impossible.

Source requirements: n174

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

DI-50 Data-integrity investigations, remediation and scoping support

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- A DI concern can be raised as an investigation linked to users, time windows and batches.
- Linked records and audit trails go on hold until the investigation closes: no archive purge and no disposal.
- Scoping reports list, per user and per period, all records created or changed, across batches and sites.
- A user's GMP rights can be revoked immediately across all sites.
- Tighter controls (e.g., added verifications) can be rolled out as CAPA through configuration under change control.
- The documented DI risk assessment of eBMR workflows is reviewed periodically.
- Tips and suspected falsification go into the QMS investigation route and are never handled informally.

TYPE DETAIL	ENFORCED for holds, scoping reports and immediate revocation. The tooling scope is a DESIGN-CHOICE.
REGULATORY BASIS	US: 211.22(a), 211.192; FDA DI Q&A Q15, Q18, Introduction | EU: Annex 11 §13 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	FDA is explicit (Q15, Q18). The EU covers it through incident management and CAPA.
HOW IT WILL BE TESTED	Open a DI investigation for a user and date range. The system lists every record that user created or changed. The linked batches cannot be disposed of until the investigation closes.

Source requirements: n654, n657

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

(a) ALCOA+ principles mapped to controls, and which document defines each term

What each source actually does (row basis: n60, n110, n227 and the ALCOA verification notes):

- **FDA DI Q&A (Dec 2018), Q1a.**
  - It defines *data integrity* as the completeness, consistency and accuracy of data.
  - It then *lists* the ALCOA traits (attributable, legible, contemporaneously recorded, original or a true copy, accurate) without defining any of them.
  - Footnote 5 ties each letter to Part 211: attributable to 211.101(d), 211.122, 211.186, 211.188(b)(11); legible to 211.180(e); contemporaneous to 211.100(b), 211.160(a); original/true copy to 211.180, 211.194(a); accurate to 211.22(a), 211.68, 211.188.
  - FDA does not use "enduring", "available" or "traceable" as ALCOA terms. It uses "traceable" only for the data-metadata relationship (Q1b). "Complete" and "consistent" appear in its data-integrity definition, not as ALCOA letters. (Primary.)
- **EU GMP Ch.4 (2011, binding).** It uses none of the ALCOA labels, but its text carries the ideas (primary):
  - controls for the accuracy, integrity, availability and legibility of documents (Principle);
  - handwritten entries clear, legible and indelible (4.7);
  - records made when each action is taken, with significant activities traceable (4.8);
  - alterations signed and dated, with the original still readable (4.9);
  - record integrity kept for the whole retention period (4.10).
- **EU Annex 11 (2011, binding).** Stored data checked for accessibility, readability and accuracy (§7.1). Identity and date-time recorded for entering, changing, confirming or deleting data (§12.4). (Primary.)
- **EU Ch.4 July 2025 draft.** Defines ALCOA++ as ten attributes: attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring, available, traceable. "Traceable" is its addition to ALCOA+. The definition wording was confirmed only through secondary excerpts and could not be checked against a local official text (draft)(s)/(u).
- **EMA GMP DI Q&A (2016)** uses ALCOA terminology (s). The EMA GCP computerised-systems guideline (2023) defines several "+" terms, but it is a GCP document and is cited only as evidence of EU usage (s). Neither is in the local files.

Principle	Defined by	eBMR meaning used here (paraphrase)	Kernel controls
Attributable	Listed: FDA Q1a. Defined: EU Ch.4 draft (draft)(s). Binding hooks: 211.188(b)(11), Annex 11 §12.4	Every value, change and confirmation is tied to one authenticated person (or registered device/interface) and a time	DI-14, DI-15, DI-17, DI-20, DI-22, DI-35, DI-36, DI-37, DI-38
Legible	Listed: FDA Q1a. Defined: EU Ch.4 draft (draft)(s). Binding hooks: Ch.4 Principle, 4.7, 4.9; 211.180(e)	Readable in plain language for the whole retention period; corrections never hide the original	DI-22, DI-24, DI-36, DI-40, DI-41, DI-44
Contemporaneous	Listed: FDA Q1a ("contemporaneously recorded"). Defined: EU Ch.4 draft (draft)(s). Binding hooks: 211.100(b), 211.160(a); Ch.4 4.8	Recorded and durably committed when the activity happens, on trusted server time; late entries explicit and flagged	DI-27, DI-28
Original (or true copy)	Listed: FDA Q1a ("original or a true copy"). Defined: EU Ch.4 draft (draft)(s). Binding hooks: 211.180(d), 211.194(a); Ch.4 Principle (raw data)	First capture kept (including dynamic form and metadata), or a verified true copy keeping content and meaning	DI-01, DI-32, DI-41, DI-45
Accurate	Listed: FDA Q1a. EU Ch.4 draft wording not verified (draft)(u). Binding hooks: 211.68(b), 211.188; Annex 11 §5, §6; Ch.4 Principle	Correct values from validated capture, interfaces and calculations; manual critical data double-checked	DI-03, DI-04, DI-30, DI-31, DI-33
Complete	FDA data-integrity definition (Q1a), not an ALCOA letter; EU Ch.4 draft (draft)(u)	All attempts, repeats, aborted and invalidated data plus metadata and audit trail retained; no silent omissions	DI-02, DI-26, DI-29, DI-42, DI-43
Consistent	FDA data-integrity definition (Q1a), not an ALCOA letter; EU Ch.4 draft (draft)(u)	Uniform units, formats and time order; sequence enforced; out-of-order entries stand out	DI-02, DI-10, DI-27, DI-34
Enduring	EU Ch.4 draft only (draft)(u); not an FDA term. Binding hook: Ch.4 4.10, Annex 11 §7.1	Survives intact on protected media through backups, archives and migrations	DI-44, DI-45, DI-46, DI-48
Available	EU Ch.4 draft only (draft)(u); not an FDA term. Binding hooks: Ch.4 Principle; Annex 11 §7.1; 211.180(c)	Retrievable in readable form for review, QP/QU decisions and inspection throughout retention	DI-08, DI-40, DI-46, DI-47
Traceable	EU Ch.4 draft (the "++" addition) (draft)(s); FDA uses the word only for data-metadata links (Q1b). Binding hook: Ch.4 4.8	The whole history (creation, change, movement, location) can be reconstructed from identifiers and audit trail	DI-02, DI-10, DI-22

(b) Revision notes: what is binding, what is draft, and how the kernel treats drafts

- **Binding today (checked against local texts).**
  - EU Annex 11 rev. 1 and EU Ch.4 rev., both in operation since 30 June 2011.
  - 21 CFR Part 11: eCFR as of 24 Sept 2026, including the 2 March 2023 amendment to 11.100(c)(1).
  - 21 CFR Part 211: eCFR as of 24 Sept 2026.
  - FDA DI Q&A: final guidance of December 2018 (non-binding, but checked as primary).
- **Pending.** The July 2025 consultation drafts of a revised Annex 11 and a revised Chapter 4 (EC/EMA with PIC/S) are not in force. Their final wording and effective dates were not known when this research was done. Their content here comes only from secondary summaries (a few items unverified). The draft section numbers used in this catalog (Annex 11 draft §6-§17; Ch.4 draft 4.12 and 4.82-4.85) are *reported* numbers.
- **Design rule applied.** Adopt the draft wherever it is stricter and conflicts with no binding text. Draft-derived elements adopted:
  - audit trail covers creation, is never disableable, and prompts for a reason (DI-22, DI-23);
  - administrators independent of GMP work (DI-18);
  - audit-trail review before release by an independent reviewer, with QP access (DI-25);
  - signature manifestation shows role and time zone (DI-36);
  - re-authentication at every signing (DI-38);
  - supersession of signatures after change and avoidance of hybrid signing (DI-37, DI-43);
  - IAM: least privilege, lockout, MFA for remote/privileged access (DI-16, DI-17, DI-19, DI-13);
  - encryption, patching, pen-testing and DR objectives (DI-13);
  - backups separated from primary data (DI-45);
  - expanded periodic review (DI-11);
  - supplier exit strategy and access to cloud validation documentation (DI-08, DI-03);
  - traceability matrix (DI-06);
  - alarms (DI-49);
  - ALCOA++, data lifecycle and verified true copies (DI-02, DI-41, DI-47).
- **Draft elements deliberately not followed.**
  - (1) The reported removal of the stand-alone QP e-signature clause: 2011 §15 stays binding, and removal would be less strict (DI-39).
  - (2) A reported allowance for justified audit-trail review after release: it conflicts with 211.22/211.192 as FDA reads them in Q8, so it is not offered (DI-25).
  - (3) MFA for every signature: unconfirmed, so it is offered as an option, not as the default (DI-38).
- **FDA side.**
  - The 2003 Part 11 Scope and Application guidance (s) grants enforcement discretion on validation, audit trails, retention and copying. The kernel does not rely on it (DI-01).
  - Earlier research carried some wording that is **not** in the final Dec 2018 Q&A and is therefore not treated as FDA text:
    - "review critical data audit trails with each record, on a schedule set by system complexity";
    - "temporary memory" (that is EMA wording);
    - a small-site fallback for administrator/QA overlap (n62, n65, n66 verification notes).
- **On finalisation of the drafts:** re-run a gap assessment clause by clause against DI-02, DI-08, DI-11, DI-13, DI-14 to DI-19, DI-22, DI-23, DI-25, DI-36 to DI-39, DI-41, DI-43, DI-45 to DI-47 and DI-49, and correct the reported section numbers.

(c) Claims made in earlier research that are NOT actually required

The kernel still implements many of these as design choices. The point is that nobody should cite them as regulatory text.

1. **Numeric floors for passwords, lockout or session timeout.** Guardrail 25 speaks of "regulatory floors". Neither 11.300 nor Annex 11 §12 sets any number, and the draft reportedly requires lockout only qualitatively. Every number in DI-19 and DI-38 is a design default.
2. **Mandatory password aging.** 11.300(b) gives aging only as an example of periodic checking. A documented risk-based equivalent is acceptable (n58).
3. **"11.10(e) requires a reason for change."** It does not. The binding reason duty comes from Annex 11 §9 (all changes and deletions of GMP-relevant data) and Ch.4 4.9 ("where appropriate") (n48 scope note). The reason is still mandatory in the kernel.
4. **"Annex 11 (2011) makes an audit trail mandatory."** §9 says building one in should be *considered* on the basis of risk (n156, corrected). The audit trail is mandatory in the kernel because of 11.10(e) and because any risk assessment of an eBMR would conclude it is needed.
5. **"Annex 11 §14 requires printed name, meaning and re-authentication."** §14 requires equal effect to handwritten signatures inside the company, permanent linkage, and date and time. Printed name, meaning and signing components come from 11.50 and 11.200 (n145, n154, n162).
6. **Specific technologies.** Hash chains, cryptographic content hashes, WORM/object lock, encryption at rest, DB activity monitoring, idempotency keys and checksums on *every* interface are design choices serving 11.10(a), 11.10(e) and 211.68(b). No rule names them (n8, n44, n48, n55, n88).
7. **Clock rules.** NTP synchronisation, UTC storage and drift thresholds appear in no 2011 EU or FDA text, and the DI Q&A says nothing about clocks (n88). Only the draft adds time zone to audit and signature records (s).
8. **A ban on physical deletion.** 11.10(e) assumes deletions may happen and must be audit-trailed. The ban during retention is a stricter design choice (n48, n658).
9. **Automatic invalidation and re-signing of a signature after its content changes.** This goes beyond the 11.70 "ordinary means" standard. It is a design choice that matches the draft's hybrid-signature expectation (n55, n171).
10. **Electronic acknowledgment of the accountability policy before first signing.** 11.10(j) requires written policies, not a system acknowledgment (n52).
11. **Inspector accounts, retrieval SLAs, and FD&C 704(a)(4) / remote regulatory assessments.** None is in the DI Q&A. They are other-source or design items (n45, n70).
12. **Verifier signature on scanned true copies; conditions for destroying paper originals.** Neither is described in the FDA Q&A. Documented verification is an EU draft expectation (s) (n67, n114).
13. **QA approval of every void or reprint.** A firm choice. Q6 asks for a written justification under document control (n64, n222).
14. **An MBR simulation/dry-run mode.** Not an FDA expectation. Q3 asks for verification of each MPCR, by any sound method (n61).
15. **A small-site exception letting one person be both administrator and QA "per the DI Q&A".** The final Q&A contains no such fallback (n50, n62).
16. **"Audit-trail review of critical data only" or "frequency set by system complexity".** This is draft-era wording, not the final Q&A. The final text ties review frequency to the data's own review frequency or to a documented risk assessment (n65).
17. **"Temporary memory" as FDA text.** The phrase is EMA Q&A wording. The final FDA Q&A bans any storage that allows manipulation before a permanent record exists (n66).
18. **A "validated" downtime mode.** Annex 11 §16 requires the arrangements to be documented and tested (n164).
19. **Per-field criticality tagging.** A design means for evidencing risk management, not required by Annex 11 §1 (n148).
20. **Training-gated hard stops and qualification snapshots.** 211.25 and Annex 11 §2 require qualified staff, not system gating (n3, n655, n624).
21. **Automated migration reconciliation; a fixity-check frequency; backup alerts and RPO values.** §4.8, §17 and §7.2 require the checks, not automation or specific frequencies (n150, n153, n165).
22. **Binding every device to a location.** 11.10(h) applies device checks "as appropriate" (n51).
23. **"Executor cannot release".** Good practice supporting QU independence. It is not 211.22 text (n2, n50).
24. **Requester, approver and reason on every access change, and point-in-time role history.** Annex 11 §12.3 only requires that access changes be recorded (n159).
25. **System-generated periodic-review evidence.** §11 does not require the evidence to be system-generated (n158).
26. **MFA for every signature under the draft.** Unconfirmed (n226).
27. **Serial-number reconciliation as the Ch.4 4.21(h) "electronic control" equivalent.** An interpretation. Serialisation comes from other EU legislation (n95).
28. **QP certification recorded in every tenant.** Annex 11 §15 applies only *when* the system is used to record certification. An Indian exporter's own eBMR normally records QU disposition, and the EU importer's QP certifies elsewhere (n163).
29. **A 10+ year retention horizon and training-record retention under 211.180.** These come from policy or other rules, not 211.180 (n32).
30. **Scanned or vision images as the Ch.4 4.21(f) printed-material specimen.** The literal expectation is physical samples whenever possible. An image is an interpretation, so both are supported (n94).

(d) Conflicts between rows and how they were resolved

1. **Scope of audit-trail review.**
   - Conflict: n111, n148, n172 and n202 imply that review can focus on GMP-critical entries; n65 (corrected) says every change in the batch record is in scope.
   - Evidence: FDA Q7 says the people who review a record review the audit trails of changes to that record's data. 211.192 requires QU review of *all* production and control records, and Q7 counts audit trails as part of them.
   - Resolution: every post-entry change is in scope; criticality only orders and highlights (DI-25).
2. **Timing of audit-trail review; deferral after release.**
   - Conflict: n172 (draft, s) reports that a justified later review is allowed; n156 (2011) says only "regularly"; n65 says after each significant step and before release.
   - Evidence: FDA Q8 applies the data's own review frequency to its audit trail, which means 211.188(b) at each significant step and 211.22 before release.
   - Resolution: no deferral option; the stricter US timing applies (DI-25).
3. **Audit trail mandatory vs risk-based.**
   - Conflict: n156 (§9 "consideration ... based on risk") vs n48 and n168.
   - Evidence: 11.10(e) uses mandatory wording for audit trails in closed systems.
   - Resolution: mandatory (DI-22).
4. **Is creation of data audited?**
   - Conflict: n156 (2011 EU: changes and deletions) vs n48 and n168.
   - Evidence: 11.10(e) covers operator entries that create, modify or delete records; the draft reportedly adds creation.
   - Resolution: creation is audited (DI-22).
5. **Reason for change.**
   - Conflict: Ch.4 4.9 says "where appropriate"; 11.10(e) is silent; n48 says always.
   - Evidence: Annex 11 §9 requires the reason for any change or deletion of GMP-relevant data to be documented.
   - Resolution: mandatory for every change or void of GMP data; not required at creation (DI-22, DI-24).
6. **Void approval.**
   - Conflict: n64's system text calls QA approval per void a firm choice, but n64's own test demands QA approval; n38 requires "an approver"; n222 requires QA approval for reprints.
   - Evidence: Q6 asks for written justification under QU or other document control.
   - Resolution: a reason and an attributable signature are ENFORCED; QA approval is CONFIGURABLE, default ON (DI-42, DI-43).
7. **Re-authentication at signing.**
   - Conflict: n57 (Part 11 minimum: private component only within a continuous session; recommends both components for critical signatures) vs n171 (draft: re-authenticate at each signing).
   - Resolution: all components at every signing by default. Relaxation is allowed only down to the private component within a continuous session, for routine signings, which still meets both texts (DI-38).
8. **MBR dry-run.**
   - Conflict: n61's test makes a dry-run verification record a precondition; n61's corrected text calls the dry-run mode a design choice.
   - Evidence: Q3 requires the workflow to be verified, not a specific method.
   - Resolution: a verification record is mandatory and the method is open (DI-04, DI-05).
9. **Administrator/GMP-user exception.**
   - Conflict: n50 cites an "approved exception with a second reviewer (per the DI Q&A)"; n62 says the final Q&A has no fallback; n173 (draft) says no admin rights for GMP workers.
   - Evidence: the Q4 text recommends administrator independence and has no exception clause.
   - Resolution: no product-level exception (DI-18). The staffing issue is an open question.
10. **Verification signature on true copies.**

    `- Conflict: n67's test requires it; n67's corrected text says the Q&A does not describe it; n114 (draft) expects documented verification.`

    `- Resolution: design to the draft, since there is no conflict with FDA (DI-41).`
11. **QP e-signature clause.** n170 (draft reportedly drops 2011 §15) vs n163 (binding §15). The binding text prevails; the clause is kept (DI-39).
12. **Physical deletion.** n48's system text bans it, while its scope note says 11.10(e) allows audited deletion; n658 allows purge after retention. Resolution: no deletion during retention; controlled, recorded disposal afterwards (DI-24, DI-47).
13. **Training gating.** n3: hard stop with no override. n655: blocked *or flagged* per SOP. n624: good practice. Resolution: CONFIGURABLE, default hard block (DI-21).
14. **Packaging reconciliation exemption.**

    `- Conflict: n95 (EU Ch.4 4.21(h): reconciliation data may be omitted where robust electronic controls operate and the omission is justified)`

    `- Resolution: for US-market products only the 211.125(c) waiver is available. The EU exemption applies only to EU-only products and never rem`
15. **Retention period.** n32 and n647 give the US figure only (1 year after expiry); n152 gives the EU figure.

    `- Evidence: Ch.4 4.11 says 1 year after expiry or at least 5 years after QP certification, whichever is longer; 211.180(a) says at least 1 ye`

    `- Resolution: the maximum across the product's markets (DI-47).`
16. **Attribution of "temporary memory".**

    `- Conflict: n66 (FDA row; verification note says the phrase is not in the final Q&A) vs n183 and n221 (EMA Q&A).`

    `- Resolution: attributed to EMA. The control (commit before evaluation) satisfies both texts (DI-28).`

17. **Blocking capture on clock drift.**

    `- Conflict: n88's test says capture stops on edge devices past the drift threshold; n88's correction warns that blocking without a fallback p`

    `- Resolution: server timestamps, an alarm and an incident, and the controlled contingency path of DI-43 (DI-27).`

18. **Review by exception.**

    `- Conflict: n184 and n201 (EMA: RBE acceptable when justified and validated) vs n65 (FDA: the QU reviews all records, audit trails included).`

    `- Resolution: RBE is only a validated way of doing the full QU review. The QU approves the complete record, with full drill-down (DI-26).`

19. **Signature elements in EU rows.** n162 and n145 list printed name and meaning as if they came from Annex 11. The check notes show they come from 11.50. Resolution: kept for all markets as the union (DI-36).
20. **Printed-material specimen.** n94 offers a vision image; the Ch.4 4.21(f) literal text asks for samples whenever possible. Resolution: both are supported, with a physical specimen by default for EU-market packaging (routed to the packaging catalog).

(e) Rows that are not an eBMR data-integrity concern

(e1) Not an eBMR concern: another system or organisational process owns the requirement

- **n83** DSCSA package-level tracing, EPCIS and exemption timeline (secondary). Owned by serialisation level 3/4 and trading-partner exchange. At most, the eBMR reads an aggregation-completeness status at release (packaging catalog). If a DI hook is needed, the exchange falls under DI-08 (open-system flows) and DI-31 (interfaces).
- **n596** 211.65 equipment contact surfaces and lubricants. This is equipment design and qualification. The eBMR only reads the equipment's qualification status (production catalog).
- **n600** 211.84(d)-(e) component testing and CoA reliance. This is QC/LIMS release testing. The eBMR only reads the resulting lot status.
- **n602** 211.94 container-closure suitability. This is design and qualification. The eBMR only enforces the approved bill of materials (production catalog).
- **n603** 211.142 warehousing quarantine and storage conditions. Owned by the WMS. The eBMR reads quarantine status through the DI-39 interlock.
- **n612** 211.204 returned products. Owned by QMS/WMS.
- **n613** 211.208 drug product salvaging. Owned by QMS/WMS.

(e2) eBMR functions outside this topic: routed to the production, packaging and materials catalogs (DI hooks noted)

- **n22** 211.122 labeling and packaging material controls and the special controls for cut labels (packaging). Hooks: device challenge records (DI-20, DI-32); e-signatures (DI-35).
- **n23** 211.125 labeling issuance and reconciliation (packaging). See conflict (d)14.
- **n27** 211.137 expiration dating (production/packaging). Hook: calculate the date once, with no manual entry (DI-33).
- **n28** 211.84(a), 211.87, 211.89 material quarantine, retest and rejection (materials). Hook: status changes are QU-only e-signatures (DI-17).
- **n35** 211.184 component, container, closure and labeling records (materials). Hook: append-only inventory ledger (DI-24).
- **n37** 211.186(b) required MBR content (production). Hook: MBR verification (DI-04).
- **n95** Ch.4 4.21(h) reconciliation exemption (packaging). See conflict (d)14.
- **n124** Ch.5 5.52 checks at arrival in the packaging area (packaging).
- **n128** Ch.5 5.57-5.58 code readers, counters and print durability (packaging). Hook: device checks (DI-20).
- **n141** Ch.5 5.32-5.34 starting-material labelling and status (materials). Hook: when labels carry only a barcode, the system is the authoritative source and must be validated for that (DI-03).
- **n635** Ch.5 5.19 and 5.21 cleaning records and status labels (production/equipment). Hook: chronological, attributed logbook entries (DI-28).

(f) Open questions

1. **Final EU drafts.** Final text and effective dates of the July 2025 Annex 11 and Chapter 4 revisions. Confirm:
   - the reported section numbers;
   - whether audit-trail review after release is ever allowed;
   - MFA for signatures;
   - IAM, alarm and hybrid details;
   - the ALCOA++ definitions. Then re-run the gap assessment listed in (b).
2. **EU retention for exported batches.** "5 years after QP certification" needs the certification date, which the EU importer's QP holds. A technical-agreement data feed is needed. Until it exists, retention stays open (DI-47). The kernel must also take the India Schedule M figure from the India catalog when computing the maximum.
3. **211.101(c)/(d) and 211.68(c).** Does a validated balance interface with barcode identity checking count as "automated equipment", so that dispensing or charge-in can drop to one person? This needs a regulatory-affairs position. The default keeps the second person for US-market products (DI-30).
4. **Small sites.** Staffing for administrator segregation (DI-18): vendor-managed or group-IT administration, and any firm-owned compensating control, need a decision per customer.
5. **EU Ch.4 4.9 "signed and dated".** Is an audit-trail entry by an authenticated user enough, or is a signature ceremony needed? The kernel takes the conservative route (re-authentication for corrections, DI-24). Confirm with QA before any relaxation.
6. **SaaS classification.** Can a vendor-hosted, multi-tenant deployment be shown to be a closed system (customer-controlled access, vendor access under customer approval), or must 11.30 open-system measures apply? Data-residency constraints also need confirming (DI-08).
7. **What counts as "remote access" for MFA.** Does access from the shop floor to a SaaS tenant count? The default treats badge plus PIN as two factors on site and requires MFA off-network and for all privileged/vendor access (DI-13, DI-38).
8. **Destroying paper originals after a verified true copy.** 211.180(d) allows true copies, and the 2011 EU texts are silent. The default is to keep the paper; destruction only under a QA-approved SOP (DI-41).
9. **FDA and review by exception.** The FDA DI Q&A neither endorses nor forbids RBE; it requires proven review reliability. The default is full review until a validated confidence criterion is met. The criterion needs defining per customer (DI-26).
10. **Design defaults with no regulatory number.** Customer risk assessments must confirm them:

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    - lockout 5 attempts;

    - inactivity lock 5 min;

    - clock drift 1 s;

    - periodic review 12 months;

    - access recertification 6 months;

    - restore test quarterly;

    - downtime drill annual;

    - password aging 90 days (or its equivalent).
    ```

11. **Auditor-facing citations.** The EMA GMP DI Q&A, PIC/S PI 041, Annex 16 and ICH Q9(R1)/Q10 were not in the local files (all (s)). Verify them against official text before citing them in customer validation documents.
12. **IPC records and 211.194.** Are shop-floor IPCs "laboratory records" under 211.194? If so, second-person review of original records is mandatory (DI-29). This is a documented firm decision per product.
13. **Independence of the audit-trail reviewer.** Does a QA person who performed in-process QA checks on the batch count as "involved"? The default blocks anyone who executed or verified any step (DI-17, DI-25).
14. **Alarm scope.** Which SCADA/EMS alarms are GMP-relevant per line needs defining per site before DI-49 can be configured (Injectables edition especially).
15. **Other-source claims.** The DSCSA timeline (n83), the FDA 2025 remote-assessment guidance (n70) and FD&C 704(a)(4) are outside the local texts and need their own verification. They belong to other catalogs.

PART B

Production, documentation and batch release

Shared compliance kernel · both editions · 60 controls

Shared compliance kernel for the TABLETS and INJECTABLES eBMR editions. India Revised Schedule M is handled in the separate India catalog; the kernel merges catalogs with a stricter-wins rule, so any Schedule M obligation that is stricter than what is written here overrides it.

Citation markers: no mark means the clause was checked against the official text in the official text (s) means secondary source only, not in the uploaded texts. (u) means the row is unverified and clause numbering is approximate. (draft) means a draft that is not binding.

Types: ENFORCED means kernel behaviour that no tenant, site, MBR or admin setting can weaken. CONFIGURABLE means a safe default plus an allowed range. DESIGN-CHOICE means product behaviour that goes beyond the regulatory text. It is recommended but not required by the rule, unless the control says so.

Market set: every batch (and every packaging order) carries its destination markets, e.g. {US}, {EU} or {US, EU}. "Stricter wins" means the engine applies the union of the obligations of every market in the set. Where a control says "(EU in set)", the behaviour switches on only when EU is a destination.

Edition tags: [T] = Tablets, [I] = Injectables. Untagged controls apply to both editions.

A. Master records (MBR / master formula, processing and packaging instructions)

PR-01 Master record scope and keying

Enforced · Both editions · All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	A processing master (manufacturing formula plus processing instructions) is keyed by product code, strength and batch size. A packaging master is keyed by product, pack size and pack type. A batch can only be created for a batch size that an approved master lists explicitly, either as one size or as an approved set of discrete sizes. There is no free scaling at batch creation. Packaging masters are versioned and approved separately and reference the bulk product code.
REGULATORY BASIS	US: 211.186(a) (each product and each batch size) | EU: Ch.4 lead-in above 4.17 (each product and batch size); 4.19 (each product, pack size and type) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	The two are equivalent for processing. EU states the packaging keying (pack size and type) explicitly, and the kernel applies it to every market.
HOW IT WILL BE TESTED	Creating a batch at a size that no approved master lists is refused. A packaging order for a pack type with no approved packaging master is refused.

Source requirements: n89, n91

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-02 Master preparation, independent check and quality-unit approval

Enforced · Configurable · Both editions · All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	Master lifecycle is Draft → Checked → Approved → Effective. The minimum route is: • the preparer's dated e-signature; • an independent checker's dated e-signature (a different individual from the preparer); • a quality-unit approval e-signature. Before the checker can sign, automated consistency checks run: BOM total against batch size, unit consistency, formula syntax, limits within registered values, and a typed entry field for every data point. The effective date is mandatory, and approval does not make the master executable before that date. Every master and every rendered output carries a unique document ID and version. Customers may add signers but cannot remove the minimum. A "skip review" switch exists only in non-GMP test mode.
TYPE DETAIL	ENFORCED (minimum route) + CONFIGURABLE (extra reviewers). Default: the checker and the QCU approver must be different users. Allowed: the checker may also be the QCU approver.
REGULATORY BASIS	US: 211.186(a); 211.100(a); 211.22(c); 11.1(c); 11.100(c) | EU: Ch.4 Principle (instructions error-free and in writing); 4.1; 4.2; 4.3; 4.4 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US explicitly requires an independent check of the master by a second person, so US is stricter here. EU requires approval, signature and dating by authorised persons, plus a defined effective date; EU is stricter on the effective date. 211.186(a) asks for a "full signature, handwritten". A Part 11-compliant e-signature counts as its equivalent (11.1(c)), provided the firm has filed the 11.100(c) certification with FDA.
HOW IT WILL BE TESTED	When the same user tries to prepare and check a master, the check is refused. A master with no effective date cannot be approved. A seeded unit error is caught before the check signature.

Source requirements: n97, n618

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-03 MBR content completeness gate

Enforced · Both editions · All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	The MBR designer refuses submission for approval unless every item in Section (a)-1 (processing) or (a)-2 (packaging) is present, for the markets in the product's market set. Specifically: • every data-entry point has a typed field (unit, precision, range or list); • every CPP and IPC has limits;

- every yield checkpoint has a theoretical basis and min/max limits;
- every printed component has a specimen (see D-1). An item may be marked "not applicable" only if the rule itself makes it conditional ("where applicable", "where appropriate", "where necessary"), and only with a signed justification.

REGULATORY BASIS	US: 211.186(b)(1)-(9); 211.100(a); 211.101(a); 211.110(a)-(b); 211.111; 211.125(c) | EU: Ch.4 4.6; 4.17(a)-(d); 4.18(a)-(g); 4.19(a)-(i) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	See Section (a). US is stricter on labeling specimens, active content per dosage unit, calculated excess and yield limits. EU is stricter on equipment preparation, pre-start clearance checks, bulk storage requirements and packaging-instruction detail.
HOW IT WILL BE TESTED	Starting from a complete test master, delete one mandatory item at a time. Approval is refused each time, and the refusal names the missing item.

Source requirements: n90, n89, n91, n99, n37

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-04 Registration (MA / application) conformity and market validity of masters

Configurable | Both editions | 1 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each master version holds the registration references for each target market (EU MA number and variation; US application number and supplement). • Any master element can be flagged as registered or as an Established Condition (EC), with its market and reporting category. • Approval requires a regulatory-conformity sign-off. • A version becomes effective for a market only when all required approvals or notifications for that market are recorded. • A batch can be created only from a version valid for every market in its market set. • The batch record shows the master version and the registration references used. • A change linked to a PACMP stays open until the protocol's acceptance criteria are met on the eBMR batches it names.
TYPE DETAIL	CONFIGURABLE. Default: hard block when a registered-flagged limit is wider than the requirements listed value, or when a registered element changes without a recorded regulatory action for that market. Allowed: warning only, for elements not flagged as registered or Established Condition.
REGULATORY BASIS	US: 211.100(a); 211.186(b)(9) | EU: Ch.5 Principle; Ch.4 4.2 | ICH/EMA/draft: Annex 16 section 1 (s); ICH Q12 ch.3-5 (ECs, PACMP, PLCM) (s); EU Variations Guidelines applying from 15 Jan 2026 (s)
INDIA / US / EU DIFFERENCES	EU text links production directly to the MA (Ch.5 Principle, 4.2). US reaches the application through 211.100(a) and FD&C Act filing rules (s). For a multi-market batch, the stricter rule is that both registrations must be satisfied at once, i.e. the process must run inside the intersection of the requirements listed ranges.
HOW IT WILL BE TESTED	Widening a registered CPP range with no variation reference is blocked. Creating a batch for a market not linked to the master is refused.

Source requirements: n629, n187, n218, n219

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-05 Master versioning, supersession, periodic review, retirement and document inventory

Enforced | Configurable | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Once version n+1 becomes effective, no new batch can start on version n. • Batches already in progress follow a configured rule. Default: finish on the issued version; migration to the new version only under change control. The record shows which version governed each step. • Every master carries a periodic-review due date and an overdue alert. • A retired master blocks new batches, but every executed record stays retrievable and printable. • The system exports an inventory of its controlled documents (masters, templates, forms, label templates, embedded instructions) with ID, version, status, effective date and review date.
TYPE DETAIL	ENFORCED (supersession and retirement) + CONFIGURABLE (review interval, set by customer SOP; the texts give no number)
REGULATORY BASIS	US: 211.100(a); 211.186(a); 211.68(b) | EU: Ch.4 4.3; 4.5; 4.20; 4.21; 4.32; Annex 11 section 10 | ICH/EMA/draft: ICH Q10 1.6.1 and 3.1 (s)
INDIA / US / EU DIFFERENCES	EU requires periodic review (4.5) and a document inventory (4.32) explicitly, so EU is stricter. Blocking superseded versions comes from 4.3 (effective date) and 4.20/4.21 (currently approved instructions), not from 4.5 (see D-11).
HOW IT WILL BE TESTED	After v2 is effective, starting a batch on v1 is blocked. An overdue review appears on the dashboard. The inventory export reconciles with the site document list.

Source requirements: n98, n627, n217

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-06 Change control for masters and linked GMP objects

Enforced | Both editions | 2 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES	An Effective master cannot be edited. A new version needs a linked, approved change record that holds: • description and rationale; • a risk assessment whose formality matches the risk; • an impact assessment covering specifications, equipment, validation, cleaning, training, labels, products sharing the line, and other sites holding copies; • a regulatory-impact decision for each market; • a validation reference, for process-impacting changes (CPP ranges, equipment class, materials or suppliers, batch size, sequence, test method). A diff report lists the changed fields. Implementation is tracked per site. A post-implementation effectiveness review is scheduled automatically. Shop-floor test methods can only be changed through this route, with a reason and comparative data.

REGULATORY BASIS	US: 211.100(a); 211.22(c); 211.68(b); 211.194(b) | EU: Ch.5 5.24; 5.25; 5.26; Ch.4 4.29 (change control); Annex 11 section 10 | ICH/EMA/draft: ICH Q10 3.2.3 (s); ICH Q9(R1) 5.1 (s); ICH Q12 ch.6 (s); EU Annex 15 section 11 (u)
INDIA / US / EU DIFFERENCES	The intent is equivalent. EU requires validation of significant process changes explicitly (5.25), and US requires QCU approval of changes to procedures (211.100(a)). The kernel applies both.
HOW IT WILL BE TESTED	A direct edit of an Effective master is impossible. A process-impacting change with no validation link cannot be approved. The diff report lists exactly the changed fields.

Source requirements: n215, n198, n143, n220, n606, n205

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-07 Quality-risk-management linkage for risk-based settings

Configurable | Both editions | Search-based only (0 of 7 checked against official text)

WHAT THE SYSTEM DOES	• Scope: this applies to risk-based settings in a master, namely criticality, verification mode, IPC frequency and sample size, hard vs soft limits, review-by-exception scope, and campaign or concurrency rules. Each cites a risk assessment by ID and version. • Scoring scales are controlled master data. • Assessments store the formality level and its reasons, and keep hazard, harm, probability, severity and detectability as separate fields. • Linked events raise a re-review task, and review-by-exception for the affected step pauses until QA confirms. Linked events are repeated deviations on a step, a PQR finding, or a change. • Reducing IPC frequency requires change control that cites trend data.
TYPE DETAIL	CONFIGURABLE. Default: every master element classified critical must cite an approved, versioned risk assessment. Allowed: extend the requirement to all elements. Not allowed: removing it for critical elements.
REGULATORY BASIS	US: none binding (Q9 is FDA guidance (s)) | EU: Annex 11 section 6 (criticality via risk management); Ch.5 5.20-5.21 | ICH/EMA/draft: ICH Q9(R1) sections 3, 4.3, 4.6, 5.1-5.3, Annex II (s); EU Ch.4 draft 4.13 (draft)
INDIA / US / EU DIFFERENCES	This is guidance level in both regions. EU GMP Part III incorporates Q9 (s). The default is a recommended configuration, not a legal minimum.
HOW IT WILL BE TESTED	Submitting a master with a critical parameter and no linked assessment is blocked. Three deviations on a low-risk step open a re-review task and pause review-by-exception for that step.

Source requirements: n204, n205, n206, n207, n211, n210, n109

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-08 CPP and IPC limit model (validated, registered and internal tiers)

Enforced | Configurable | Both editions | 1 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	Each CPP and IPC field stores its unit, precision, limits and source (validation report or registration). Limits are tiered: registered, internal action, and optional alert. A result outside a registered limit automatically creates a deviation flagged "registration-relevant", which appears on the QCU/QP release screen. Out-of-limit values are stored exactly as acquired. The operator cannot suppress the flag, and re-entry is a correction, not a replacement.
TYPE DETAIL	ENFORCED (units, limits and source on every CPP/IPC; out-of-limit values stored and flagged) + CONFIGURABLE (limit tiers and breach action). Default: breaching a registered or action limit stops the next dependent step until a deviation is opened. Allowed: for internal (non-registered) limits, the step may continue with a deviation.
REGULATORY BASIS	US: 211.110(b); 211.186(b)(9); 211.100(b) | EU: Ch.4 4.18(d)-(e); Ch.5 5.43 | ICH/EMA/draft: EU Annex 15 section 5 (u); Annex 16 section 3 and EMA Q&A on Annex 16 (s)
INDIA / US / EU DIFFERENCES	US is more explicit that in-process specifications must match final specifications and be derived statistically where possible (211.110(b)). Tagging registered limits serves the EU Annex 16 route (s), which has no US counterpart.
HOW IT WILL BE TESTED	A value outside a registered limit creates a deviation with the registration flag, and it is visible on the release screen. The operator cannot clear the flag.

Source requirements: n195, n189, n90

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-09 Process validation status and validation-batch flags

Configurable | Both editions | 1 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• A master records its validation basis (report IDs). • Batches under a protocol carry a visible flag and an enhanced sampling plan. • The release screen shows the protocol and report status. • CPP and IPC data can be exported as structured data (with record counts or checksums) for CPV/OPV trending.
TYPE DETAIL	CONFIGURABLE. Default: a batch made under a PPQ or concurrent-validation protocol is flagged, and cannot be certified or released until its validation outcome is recorded. Allowed: US-only batches may be released per an approved protocol that permits concurrent release. For EU certification there is no exception (Annex 15 (u)).
REGULATORY BASIS	US: 211.100(a); 211.110(a) | EU: Ch.5 5.23; 5.24; 5.25; 5.26; 5.42 | ICH/EMA/draft: FDA Process Validation guidance (2011) stages 2-3 (s); EU Annex 15 section 5 (OPV, concurrent validation) (u)
INDIA / US / EU DIFFERENCES	EU expects the QP to have concurrent-validation results before certifying (Annex 15, (u)). For US this is at guidance level. EU is stricter.
HOW IT WILL BE TESTED	A PPQ-flagged batch cannot be released or certified until the validation outcome is entered. CPP data exported for N batches and recomputed from the source records match exactly.

Source requirements: n143, n197, n80, n196

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

B. Batch record issuance and execution

PR-10 Batch record issuance and batch numbering

Enforced | Both editions | 1 of 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- The batch record is generated programmatically from the Effective master version, with no retyping.
- The system verifies that the instance matches the master (hash or field comparison).
- An authorised issuer e-signs and dates the issued record, with signature meaning "issued, reproduction verified".
- The batch number comes from a controlled sequence. It is unique per site and product and has a globally unique internal key.
- The destination-market set is recorded at issue. Packaging orders may narrow it later under control.
- Later master edits never change an issued record.
- Test, simulation and training runs are segregated, watermarked and cannot become GMP records.

REGULATORY BASIS

US: 211.188(a); 211.186; 211.130(c); 210.3(b) lot-number definition (s) | EU: Ch.4 4.2 (working copies from masters must not introduce errors); 4.20 and 4.21 (records based on approved instructions) | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

US explicitly requires that the reproduction be checked for accuracy, dated and signed (211.188(a)), so US is stricter: the issuance signature is mandatory for all markets. EU requires only that the reproduction method cannot introduce errors.

HOW IT WILL BE TESTED

An issued record matches the master hash and carries the issuance e-signature. A duplicate batch number is rejected. A record cannot be issued from a Draft master.

Source requirements: n97, n1

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-11 Contemporaneous recording, time stamps, late entries and paper fallback

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- Action times come from a synchronised server clock (stored as UTC plus the site offset). A user can never set the time of an action.
- Late entries go only through an explicit late-entry path. It records the actual activity time as a data field, the entry time and a reason, and the entry is flagged in the exception report.
- Downtime fallback: printable instructions and forms are generated from the Effective master. Each carries a unique form ID, the batch ID, "page x of y" and a dedicated space for every entry.
- Issued fallback forms are counted and reconciled on return. Their data are transcribed back with verification, and a scan of the form is linked to the record.

TYPE DETAIL

ENFORCED (server-side time; no backdating) + CONFIGURABLE (what counts as late). Default: any entry made after the step's planned window has closed is flagged.

REGULATORY BASIS

US: 211.100(b) (documented at time of performance); 211.188; 11.10(e) | EU: Ch.4 4.1 (hybrid systems); 4.6; 4.7; 4.8; Annex 11 section 16 | ICH/EMA/draft: EMA data-integrity Q&A / PIC/S PI 041 for fallback-form reconciliation (s)

INDIA / US / EU DIFFERENCES

Equivalent.

HOW IT WILL BE TESTED

A backdating attempt is impossible. A late entry shows both times, the reason and an exception flag. Issued fallback forms reconcile to returned forms.

Source requirements: n100, n99, n618

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-12 Corrections to recorded data

Enforced | Configurable | Design choice | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

A correction keeps the original value visible in the record and in review views. The audit trail captures the old and new values, the user, the server time and the reason (mandatory). A correction after step sign-off requires re-signature. Corrections to critical data (quantities, CPP values, IPC results, yields) need the configured approval. Every correction appears in the exception report.

TYPE DETAIL

ENFORCED (never overwrite; reason mandatory) + CONFIGURABLE (approval level for critical-data corrections). Default: second person or QA approves. The approval requirement cannot be removed; this is a design choice beyond the text.

REGULATORY BASIS

US: 11.10(e); 211.100(b) | EU: Ch.4 4.9; Annex 11 section 9 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

4.9 asks for a reason only "where appropriate", but Annex 11 section 9 requires one for any change to GMP-relevant data. EU is therefore stricter, and the kernel always requires a reason.

HOW IT WILL BE TESTED

Correct a signed weight. The original stays visible, a reason is required, the step must be re-signed, and the correction is listed in the exception report. A correction without a reason is refused.

Source requirements: n101

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-13 Step-sequence enforcement and the in-line deviation path

Enforced | Configurable | Design choice | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

The engine enforces step order, prerequisites and branch conditions. A user cannot skip or reorder a mandatory step, or use equipment or material the master does not list. The only way around the defined flow is an in-line deviation record (description, immediate action, impact) approved by the e-signature of a user designated competent for that deviation class. Planned departures are approved before execution; unplanned events are approved after the fact. Quality-relevant classes are routed to QC/QA. The deviation ID is linked to the step and shown in the record.

TYPE DETAIL

ENFORCED + CONFIGURABLE (deviation classes and routing). Default: the operator cannot approve their own deviation (a design choice).

REGULATORY BASIS	US: 211.100(b); 11.10(f) | EU: Ch.5 5.15; Ch.4 4.20(h); 4.21(g) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU requires written approval by a competent person, with QC involved when appropriate. US requires the deviation to be recorded and justified. EU is stricter on approval, and the kernel applies it for all markets.
HOW IT WILL BE TESTED	Skipping a mandatory step is blocked until a deviation is approved by an authorised role. A user cannot approve their own deviation.

Source requirements: n119

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-14 Significant steps: performer plus checker, and the limited scope of single-person automated checking

Enforced | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	The master flags steps as significant. Each significant step records a performer and a checker or supervisor, who must be a different authenticated individual. • Where "automated performance" may apply: the mode may cut the humans involved to one checker only for step types mapped to 211.101(c), 211.101(d), 211.103, 211.182 and 211.188(b)(11). • Device conditions: the device must be current in its calibration/check programme (211.68(a)), and the automated function must be validated (211.68(b)). • What is recorded in that mode: the device ID and raw data, plus one checker's e-signature whose stated meaning is "automated operation checked". • If automation is unavailable, the step escalates automatically to two-person mode and the fallback appears as an exception. "Unavailable" means the device is offline, a value is keyed manually, or a barcode is overridden. • What the flag can never relax: any other two-person rule, such as the independent check of the master (211.186(a)), the second-person verification of the 100% visual check of hand-applied labeling (211.122(g)(3)), and second-person review of lab records (211.194(a)(8)).
REGULATORY BASIS	US: 211.68(a)-(c); 211.188(b)(11); 211.101(c)-(d); 211.103; 211.182 | EU: Ch.4 4.20(c); 4.21(c) (checker "where appropriate"); Annex 11 section 6 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US requires a checker on every significant step. EU requires one only where appropriate. US is stricter, and the kernel applies it to all markets.
HOW IT WILL BE TESTED	With the scale disconnected during an automated dispense, the step demands a second, different user. Enabling automated mode on a 211.122(g)(3) step is refused at configuration time.

Source requirements: n9, n39

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-15 Competence, role and area authorisation

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	A user can perform, verify or approve only if they hold a role permitted for that action, step type, MBR and site. Performer and verifier are separate permissions. Optional interlocks: • current training on the governing SOP or master version; • authorisation for the restricted area where the step runs. [I] Gowning and APS qualification for aseptic areas come from Annex 1 and the FDA 2004 aseptic guidance, not from 211.28 (see sterile catalog).
TYPE DETAIL	ENFORCED (role/authority check at the moment of each action) + CONFIGURABLE (training-currency interlock and restricted-area interlock). Default: both interlocks ON. Allowed: OFF only where the evidence is controlled in another validated system that the eBMR queries.
REGULATORY BASIS	US: 211.25(a)-(c); 211.28(c); 11.10(g) | EU: Ch.5 5.1; 5.2; 5.16; EU Ch.2 (s) | ICH/EMA/draft: EU Annex 1 (gowning) (s)
INDIA / US / EU DIFFERENCES	EU 5.16 restricts access to all production premises. US 211.28(c) covers only areas designated as limited-access. EU is broader. Hygiene and illness rules in 211.28(a),(b),(d) are organisational controls, not eBMR controls.
HOW IT WILL BE TESTED	A user whose training has lapsed is refused on the step. A user without area authorisation cannot sign a step in the filling suite.

Source requirements: n142, n4, n634

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-16 Record completeness gate (processing complete / packaging complete)

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	"Processing complete" and "Packaging complete" are allowed only when two conditions hold. • Every step is in one of three states: executed; marked N/A by the system (branch not taken); or marked N/A by an authorised user with a reason and signature. • Every applicable item of Section (a)-3 or (a)-4 is present. The items are: dates and times; equipment unit IDs; component lots and quantities; IPC and lab results; packaging-area checks before and after use; yields; labeling specimens; container/closure description; sampling; performer and checker IDs; investigations; 211.134 results; the responsible production person's approval. Category checks depend on step type; for example, packaging-area checks apply only to packaging orders.
REGULATORY BASIS	US: 211.188(b)(1)-(13) | EU: Ch.4 4.20(a)-(i); 4.21(a)-(i) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	See Section (a).
HOW IT WILL BE TESTED	Removing one label specimen or one checker signature blocks completion, and the message names the gap.

Source requirements: n39

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

C. Materials: receipt, status, dispensing and charge-in

PR-17 Receipt, lot identity and approved source

Enforced | Design choice | Both editions | All 11 source requirements checked against official text

WHAT THE SYSTEM DOES

- Lot identity:
  - Each lot of each shipment gets a unique internal receiving/control code. The same supplier lot arriving in two shipments gets two codes.
  - Each supplier batch in a multi-batch delivery becomes a separate internal lot.
  - The receipt record holds: name on the delivery note and containers; in-house name/code; date received; supplier and manufacturer; manufacturer batch number; total quantity and number of containers; internal lot number; remarks.
- Receiving checks and quarantine:
  - Each container (or container group) is visually examined for labeling, damage, seal integrity (including any tamper-evident seal) and contamination. The examiner's identity is recorded.
  - If documents or labels disagree with the approved manufacturer/supplier list, the lot is blocked and routed to QC.
  - New lots default to Quarantine.
- Problems after receipt: a problem with a container found at storage, dispensing or charging raises a record to QC and blocks that container.
- Sampling:
  - Sampled containers are marked as sampled.
  - Sample labels identify the material, lot, source container, date and sampler.
  - Stratified sub-samples cannot be composited.
- Other rules:
  - Bought-in intermediates and bulk follow the same flow as starting materials.
  - The approved-source check is repeated at dispensing (design choice).

TYPE DETAIL

ENFORCED at the eBMR boundary: a lot with an incomplete receipt record cannot be dispensed or issued. The receipt functions themselves are a DESIGN-CHOICE, used only if the eBMR/MES suite performs receipt; otherwise ERP/WMS does, over an interface.

REGULATORY BASIS

US: 211.80(a),(d); 211.82(a)-(b); 211.84(a)-(c); 211.184(a)-(b) | EU: Ch.4 4.22; 4.23; Ch.5 5.3; 5.4; 5.6; 5.27; 5.29; 5.30; 5.31; 5.33; 5.45 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

EU requires a check against approved manufacturer/supplier data at every delivery (5.30) and API supply-chain traceability (5.29). US requires a distinct code for each lot in each shipment (211.80(d)). The kernel applies both.

HOW IT WILL BE TESTED

One delivery containing three supplier batches creates three lots. A supplier label naming an unapproved manufacturer blocks the lot. A lot with no complete receipt record cannot be issued.

Source requirements: n597, n598, n621, n630, n638, n632, n637, n631, n599, n35, n141

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-18 Material status and usability gate at dispensing, issue and charge-in

Enforced | Configurable | Both editions | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- What can be used: only lots in Released status, within their retest or expiry date, and of the exact item called for by the master (plus grade and approved source where the master specifies them). This applies to dispensing, issue and charge-in.
- Hard blocks: quarantined, on-test, rejected, expired, retest-overdue and blocked lots, with no operator override.
- Identity: confirmed by scanning system-generated labels.
- Status and labels: the status on screen and on the physical label must agree, so a status change triggers relabelling.
- Stock rotation: the system proposes the oldest approved lot first. Picking another lot needs a reason, which appears in a rotation-exception report.
- Storage conditions: storage-condition attributes stop a lot from being staged in an incompatible location.

TYPE DETAIL

ENFORCED (status, identity and retest/expiry block) + CONFIGURABLE (rotation rule). Default: oldest approved first. Allowed: first-expiry-first-out with a documented SOP rationale.

REGULATORY BASIS

US: 211.80(d); 211.84(a); 211.86; 211.87; 211.89 | EU: Ch.4 4.14(e); 4.24; Ch.5 5.5; 5.7; 5.32; 5.34 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

EU 5.34 allows only QC-released material within its retest period. US 211.87 requires retesting "as necessary". EU is stricter, so the date-based block applies to all markets (D-2).

HOW IT WILL BE TESTED

Scanning a quarantined lot is blocked. A lot past its retest date is blocked. Picking a newer lot demands a reason.

Source requirements: n13, n622, n620, n601, n633, n28

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-19 Dispensing targets and dispensing execution

Enforced | Configurable | Both editions | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Who dispenses: only users holding the dispenser designation, following the procedure embedded in the step.
- Targets:
  - Targets come from the master formula, including any stated calculated excess.
  - Where the master calls for it, the target is adjusted for potency or assay. The adjustment uses the approved result of the released lot (never a typed value), and a compensating excipient quantity is calculated.
  - A master or target that would deliver less than 100% of the labeled amount of active ingredient is refused.

- Weighing: quantities are captured from a calibrated device and checked against tolerance. An out-of-tolerance quantity cannot be confirmed without a deviation.
- Container labels: each dispensed container is labeled automatically with:
  - component name or item code;
  - receiving/control number;
  - quantity in the container;
  - the destination batch (product name, strength, lot number);
  - a unique container ID (a design choice).
- Reprints need a reason and void the earlier label.

TYPE DETAIL	ENFORCED (label content, tolerance, designated dispensers, 100% intent check) + CONFIGURABLE (potency-adjustment method, per master)
REGULATORY BASIS	US: 211.101(a)-(b); 211.186(b)(4)-(5) | EU: Ch.5 5.12; 5.13; 5.37 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	The at-least-100% intent rule and the four label elements are explicit US requirements. EU requires designated dispensers and clean, correctly labelled containers. The kernel applies both. Potency adjustment is one way to meet 211.101(a), not a mandate.
HOW IT WILL BE TESTED	An API lot assayed at 98.5% raises the target and lowers the filler, with an audit trail. The label shows all four required elements. A non-designated user is refused.

Source requirements: n11, n12, n120, n118

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-20 Independent check of each dispensed material

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	In manual weighing, a second, different individual confirms three things for each dispensed container: the component was released by the QCU; the quantity matches the batch record; the container is correctly identified. All three outcomes are recorded with both identities and timestamps. In automated mode (validated scale, barcode and status interlocks, device in its calibration programme), one human verifier signs. If the device goes offline, the step reverts to two-person mode.
TYPE DETAIL	ENFORCED (manual mode) + CONFIGURABLE (automated mode). Default in automated mode: one human verifier who is not the user who started the dispense. Allowed: the same user, but only when the dispense ran on a validated automated device under 211.68 (see F-1). No mode allows zero human verification.
REGULATORY BASIS	US: 211.101(c); 211.68(c) | EU: Ch.5 5.38; Annex 11 section 6 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US requires a second person unless the operation is automated, and then still one person. EU requires an independent, recorded check but does not require a second person. US is stricter because it always requires a human verification (D-3).
HOW IT WILL BE TESTED	In manual mode, the same user dispensing and verifying is blocked. With the scale offline, the step switches to two-person mode.

Source requirements: n13, n120

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-21 Batch kit and charge-in verification

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	Materials dispensed for a batch are grouped into a kit with a conspicuous batch label. A kit cannot be issued while a component is missing. Charge-in requires scanning every dispensed container, and each must match the batch, step and material, at the process point the master defines. A missing container blocks step completion, and an unplanned container is rejected. Every addition needs a performer and a different verifier, or one verifier when the addition is automated under 211.68.
TYPE DETAIL	ENFORCED + DESIGN-CHOICE. The design choice (default off) requires the verifier to sign within a set time window from a terminal registered to that room.
REGULATORY BASIS	US: 211.101(d); 211.68(c); 211.188(b)(3)-(4) | EU: Ch.5 5.39; Ch.4 4.18(d) (sequence of addition); 4.20(d) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US explicitly requires second-person verification of each addition. EU requires kits kept together and labelled, and a defined order of addition. US is stricter on the witness, and the kernel applies it to all markets.
HOW IT WILL BE TESTED	Omitting one container prevents completion. A wrong container is rejected. A verifier who is also the performer is blocked.

Source requirements: n14, n120

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

D. Equipment, premises, cleaning and logs

PR-22 Equipment and container identification

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- Every major equipment item has a distinctive ID. The ID is recorded automatically in the batch record at unit level (not just the equipment class) and checked against the classes and units the master allows.
- The following show product or material, strength, batch number and stage, in the company's approved label format: every compounding and storage container, bulk container, processing line and major equipment item, and rooms where appropriate. A physical label or a validated electronic display may be used.
- Identification is updated on every state change.
- Colour coding is optional.

TYPE DETAIL	ENFORCED (unit ID in the record; contents and status identification) + DESIGN-CHOICE (barcode/RFID; electronic status displays in place of paper labels)
REGULATORY BASIS	US: 211.105(a)-(b); 211.188(b)(2) | EU: Ch.5 5.12; 5.13; Ch.4 4.20(e); 4.21(e) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU adds strength and, where appropriate, rooms. US adds the processing phase where necessary. The kernel applies the union of both.
HOW IT WILL BE TESTED	A step with a unit the master does not allow is rejected. The batch report shows unit IDs. The status display matches the system state after each transition.

Source requirements: n16, n118

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-23 Equipment and instrument status gate before use

Enforced | Configurable | Both editions | 3 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- At step start, the engine checks that the scanned unit:
  - belongs to the required equipment class and capacity;
  - is qualified;
  - has no overdue calibration or preventive maintenance;
  - is Clean at the cleaning level the changeover matrix requires, and within its validated clean-hold time;
  - is not allocated to another batch.
- The pre-use cleanliness inspection is recorded.
- Measuring instruments are blocked when their calibration is overdue or failed. Examples: balances, pH meters, checkweighers, gauges and integrity testers.
- A failed as-found calibration lists all batches and steps that used the device after its most recent passing calibration, for impact assessment.
- End of use sets the unit to Dirty and starts the dirty-hold timer.

TYPE DETAIL	ENFORCED (checks run and block) + CONFIGURABLE (clean-hold and dirty-hold limits, campaign length and changeover matrix; values come from cleaning validation, not from the rule)
REGULATORY BASIS	US: 211.63; 211.67(a)-(c) (incl. (b)(6) pre-use cleanliness inspection); 211.68(a); 211.160(b)(4); 211.194(d) | EU: Ch.4 4.18(b)-(c); 4.30; 4.31 | ICH/EMA/draft: EU Annex 15 section 10 (cleaning hold times) and section 4 (re-qualification) (u)
INDIA / US / EU DIFFERENCES	Hold-time limits come from cleaning validation (Annex 15, (u)), not from 211.67 or Ch.5, so their values are configurable. The block itself is enforced once a value exists. US states explicitly that out-of-specification instruments must not be used (211.160(b)(4)).
HOW IT WILL BE TESTED	A Dirty unit is blocked. An expired clean-hold is blocked. A balance with overdue calibration is blocked. A failed as-found calibration produces the list of affected batches.

Source requirements: n6, n605, n198, n199, n90

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-24 Equipment and area logbooks

Enforced | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- Scope: electronic logbooks for each major production equipment item, each major or critical analytical instrument used on the shop floor, and (EU in set) each area where product is processed.
- Content: logbooks are filled automatically from eBMR events:
  - use start and stop, with product, lot, date and time;
  - cleaning, with a performer plus a second person verifying, or a single verifier when a validated automated CIP/SIP did the cleaning;
  - maintenance other than routine lubrication and adjustment (from a CMMS link);
  - calibration;
  - repairs.
- Order: entries are kept in chronological order. Late or out-of-sequence entries are flagged with a reason, and the original chronology is preserved.
- US dedicated-equipment option: dedicated equipment may skip an individual log only when EU is not in the market set. In that case its cleaning, maintenance and use records are embedded in each batch record, and the lot-number sequence is checked.

REGULATORY BASIS	US: 211.67(c); 211.182; 211.180 | EU: Ch.4 4.29 (maintenance, cleaning); 4.31 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU 4.31 reaches further: logbooks for areas and critical analytical equipment, including repairs and calibrations, with no dedicated-equipment exemption. US 211.182 is stricter on double-checked cleaning and maintenance signatures. The kernel applies the union of both (D-14).
HOW IT WILL BE TESTED	Each batch appears in the logbook with its date, time, product and lot, consistent with the batch records. The same user performing and verifying a manual cleaning is blocked. A room logbook shows use and cleaning.

Source requirements: n34, n6, n635

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-25 Room and area suitability, occupancy and sanitation status

Enforced | Configurable | Design choice | Both editions | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Room master data: area type; permitted activities and material statuses; cleaning/sanitation status and schedule; maintenance status.
- Step binding: each step is bound to permitted area types. Execution is blocked in an unapproved area, a room under maintenance, or a room overdue for cleaning.
- Occupancy: a room occupied by one product refuses a second product unless it is configured for concurrent use. Consecutive use by another product requires cleaning and clearance first.

	• Room conditions: where the master states conditions (temperature/RH, pressure cascade, dust extraction), the engine reads BMS/EMS status at step start and during the step. A breach blocks the step or raises a deviation, as the master rule says. Readings for the step window are attached to the record.
TYPE DETAIL	ENFORCED (occupancy, permitted area type, maintenance and cleaning status) + CONFIGURABLE (room-condition checks). Default: ON wherever the master states conditions. Concurrent use of a room needs a configuration that cites a risk assessment; QA approval of that configuration is a design choice.
REGULATORY BASIS	US: 211.42(b)-(c); 211.46(a)-(c); 211.56(a)-(d); 211.58 | EU: Ch.5 5.9; 5.10; EU Ch.3 (s) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US permits "other control systems" instead of separate areas. EU permits simultaneous or consecutive work only where it creates no mix-up or cross-contamination risk. The effect is equivalent.
HOW IT WILL BE TESTED	Starting product Y in a room occupied by product X is blocked. RH above the master limit blocks the step or raises a deviation. A room overdue for cleaning is blocked.

Source requirements: n117, n592, n595, n593

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-26 Processing pre-start clearance and transfer-route checks

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Before each processing operation, the system checks electronically that: ◦ the previous batch is closed in the room and equipment logs; ◦ the equipment is Clean; ◦ no materials of other orders are allocated to the room. • The operator then confirms that no foreign materials, products, residues or documents are physically present. The confirmation is e-signed into the record. • Before a product transfer through pipework, hoses or transfer panels, a connection check is recorded. • Design choice for the route check: scan the source, route components and destination and compare them with the master route. A mismatch blocks the transfer.
TYPE DETAIL	ENFORCED (clearance before each processing operation; route check before each transfer) + CONFIGURABLE (checklist format for processing clearance: default ON, allowed OFF; second verifier: default ON, allowed OFF; repeat clearance after an interruption: default ON)
REGULATORY BASIS	US: 211.67(b)(6); 211.100(a) | EU: Ch.4 4.18(c); Ch.5 5.14; 5.40 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU explicitly requires processing-area clearance (5.40, 4.18(c)) and transfer-connection checks (5.14). US requires only a pre-use cleanliness inspection of equipment. EU is stricter. A checklist is not required for processing clearance (D-5).
HOW IT WILL BE TESTED	Starting compounding while the prior batch is not cleared is blocked. Scanning a hose to the wrong destination blocks the transfer.

Source requirements: n116, n90, n140

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-27 Cross-contamination controls: dedication, changeover, campaigns, penicillin, dust and events

Enforced | Configurable | Both editions | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• QRM outputs on the product master: ◦ reference to HBEL-based cleaning limits; ◦ dedication flags for facility, area, equipment and product-contact parts (e.g. punches and dies [T]; filters and filling needles [I]); ◦ allowed rooms and equipment, and campaign rules; ◦ the cleaning level required for each changeover; ◦ whether cleaning verification is needed after a campaign, for higher-risk products. The engine enforces all of these. • Penicillin: ◦ Penicillin products and components can never be assigned to non-penicillin facilities or equipment, and vice versa. ◦ A non-penicillin lot flagged as potentially exposed needs a penicillin test showing none detected before release. • Dust [T]: open handling of potent or sensitising dry materials requires containment or extraction prerequisites (a design choice). • Events: ◦ An event log (spill, breakage, containment breach) is linked to the batch and to the room and equipment logbooks. ◦ Logging an event sets the room or equipment to "cleaning required". ◦ Events feed the deviation process and the periodic review of cross-contamination measures. • Non-medicinal products cannot be scheduled on medicinal equipment without a recorded justification.
TYPE DETAIL	ENFORCED (penicillin segregation; enforcement of whatever dedication and changeover rules are configured) + CONFIGURABLE (the QRM-derived values themselves)
REGULATORY BASIS	US: 211.42(d); 211.46(d); 211.176; 211.67(a) | EU: Ch.5 5.10; 5.11; 5.17; 5.18; 5.19; 5.20; 5.21; 5.22 | ICH/EMA/draft: EMA guideline on health-based exposure limits (s)
INDIA / US / EU DIFFERENCES	EU requires a documented QRM, including a toxicological evaluation, to set dedication and control measures (5.20), and periodic review of them (5.22). US sets hard penicillin segregation and testing (211.42(d), 211.176). The kernel applies both. Cleaning verification at every changeover is not required (see C-7).
HOW IT WILL BE TESTED	A potent product on non-dedicated equipment outside its allowed list is blocked. A penicillin component in a non-penicillin facility is blocked. A spill event sets the room to "cleaning required".

Source requirements: n121, n139, n594, n636, n635

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-28 Microbiological-control and sterility-assurance prerequisites (kernel hook)

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• [I] Sterilised items: the engine blocks consumption of sterilised or depyrogenated items unless the cycle ran a validated recipe/version for that load and passed. • [I] Aseptic steps: the engine blocks them unless the line and its operators hold current APS qualification and the items are within validated hold times. The detailed rules are in the sterile catalog. • [I] Monitoring data: environmental and personnel monitoring results for the rooms and time window are linked to the batch. Open excursions appear at QA review and QP certification. • [T] Masters may define microbial controls, such as aqueous granulation hold limits and water-system status. Rooms and times used stay traceable so EM data can be retrieved.
TYPE DETAIL	ENFORCED [I] / CONFIGURABLE [T]
REGULATORY BASIS	US: 211.113(a)-(b); 211.94(c) | EU: Ch.4 4.27; 4.29 (environmental monitoring); Ch.5 5.10 | ICH/EMA/draft: EU Annex 1 (2022) (s); FDA aseptic processing guidance (2004) (s)
INDIA / US / EU DIFFERENCES	211.113(b) requires validated aseptic and sterilisation processes. The specific gates come from Annex 1 and the FDA 2004 guidance, not from the text of 211.113.
HOW IT WILL BE TESTED	A failed autoclave cycle prevents issue of the load's items. An open EM excursion appears on the QP screen.

Source requirements: n20, n626

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

E. In-process controls, time limits and yields

PR-29 In-process controls and sampling

Enforced | Configurable | Both editions | 4 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES	• IPC plans and samples: ◦ Every IPC or sample is created from a plan defined in the master. The plan sets the sampling point, method, frequency, sample size or units, limits, procedure version, tool and precautions. ◦ Each sample gets a unique ID and a label (product, lot, sampling point, date/time, sampler). ◦ Sample quantities count in yield and reconciliation. ◦ Taking fewer units than the plan requires, or missing an IPC in its interval, creates an exception or deviation. • Results: ◦ Each result is recorded with its performer and method reference. ◦ A repeat test is added as a new result and never overwrites the first. ◦ Reagents, buffers and reference standards used in shop-floor tests are scanned by lot and checked for expiry and standardisation status. • Typical IPCs: [T] weight variation, hardness, thickness, disintegration, blend uniformity. [I] clarity, completeness or pH of solutions, bioburden, fill weight or volume. • [T] Failed IPC: a failed or missed periodic IPC holds, at container level, all output made after the most recent passing check. • PAT and models: ◦ PAT or model results are accepted as IPC records carrying the model ID and version and the validity diagnostics. ◦ A model output outside its validity domain requires a confirmatory test. This follows the FDA draft position (draft).
TYPE DETAIL	ENFORCED (plan execution, sample identity, result recording) + CONFIGURABLE (frequency and sample size per master, each citing a QRM reference)
REGULATORY BASIS	US: 211.110(a)-(c); 211.160(b)(2)-(3); 211.165(c); 211.194(b)-(c); 211.188(b)(5),(10) | EU: Ch.4 4.18(e); 4.25; 4.26; Ch.5 5.43 | ICH/EMA/draft: FDA draft guidance "Considerations for Complying With 21 CFR 211.110" (Jan 2025) (draft)(s); ICH Q9(R1) Annex II (s)
INDIA / US / EU DIFFERENCES	US is more prescriptive. It lists IPC categories that apply "where appropriate" (211.110(a)(1)-(6)) and wants in-process specifications derived from process averages and variability. EU requires IPC instructions with limits, and IPCs that are carried out and recorded.
HOW IT WILL BE TESTED	Taking fewer units than the plan requires creates a deviation. Both results of a repeated IPC are visible. An expired pH buffer is blocked. [T] A skipped hardness check puts output since the last passing check on hold.

Source requirements: n604, n623, n133, n606, n18, n210

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-30 Time limits and hold times

Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Named timers have start and end events. They run server-side from persisted timestamps. • A timer warns before expiry. On breach it creates a deviation that needs justification and a QA assessment. • Holds of intermediates and bulk record the container, location, storage condition and maximum hold time. • A maximum filling-to-labelling interval for unlabelled units is a design choice.
TYPE DETAIL	CONFIGURABLE. The master defines timers where appropriate, each with a validation or registration reference. Default on breach: a deviation is created and release is blocked until it is dispositioned. Timers are never evaluated from client clocks.
REGULATORY BASIS	US: 211.111; 211.110(c) | EU: Ch.4 4.18(f); Ch.5 5.41; 5.54 | ICH/EMA/draft: EU Annex 1 (s), for sterile-process timers
INDIA / US / EU DIFFERENCES	US explicitly requires time limits where appropriate, and departures must be justified and documented. EU requires appropriate holding conditions and bulk storage requirements. US is stricter, and the kernel applies it to all markets.
HOW IT WILL BE TESTED	A simulated breach creates a deviation automatically and flags the batch. After a client and server restart, the elapsed time is still correct.

Source requirements: n19, n125

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-31 Yield calculation, verification and investigation

Enforced | Design choice | Both editions | 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Checkpoints: the master defines a yield checkpoint at each appropriate phase. Each has a theoretical basis (excluding materials flagged as lost in processing), a formula, a rounding rule and min/max limits. • Calculation: the system calculates actual yield and percent of theoretical from the captured quantities (inputs, outputs, samples, rejects, waste). Users cannot type a yield percentage. • Verification: every automated calculation is independently verified by one person. A manual calculation needs a performer and a second verifier. • Out of limits: a result outside the limits automatically creates an investigation and blocks release. • Corrections: changing any input quantity triggers an audit-trailed recalculation of the yield.
TYPE DETAIL	ENFORCED + DESIGN-CHOICE. Design choices: the verifier must not be the person who entered the quantities; optional alert limits.
REGULATORY BASIS	US: 211.103; 211.186(b)(6)-(7); 211.188(b)(7); 211.192; 210.3(b) yield definitions (s) | EU: Ch.4 4.17(c)-(d); 4.20(g); Ch.5 5.8; 5.44 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US requires independent verification of every yield calculation and ties investigation to the master's min/max limits. EU requires an expected final yield with limits, and investigation of significant yield deviations. US is stricter (D-9).
HOW IT WILL BE TESTED	A yield of 94.0% against limits of 95.0-101.0% creates an investigation automatically and blocks release. Without a verifier the step cannot complete. An independent recalculation matches the displayed yield.

Source requirements: n15, n1, n115, n89

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

F. Packaging and labeling

PR-32 Receipt, examination, segregated storage and access control for labeling and packaging materials

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Receipt and examination: ◦ Every incoming lot of primary or printed packaging gets its own reference number. ◦ Each shipment of labeling or packaging material is sampled representatively and examined or tested against its specification before use, including the text/artwork version. ◦ A record shows the receipt, the examination and whether the shipment was accepted or rejected. • Storage and access: ◦ Labels and labeling are stored separately for each product, strength, dosage form and quantity of contents, with identification, in access-controlled locations. ◦ Cut labels and loose printed items are stored and moved in closed containers whose IDs are recorded. • Obsolete material: obsolete or outdated labels, labeling and packaging materials are blocked and routed to destruction with a record. This covers superseded artwork and material past its expiry or retest date. • Gang printing: gang-printed labeling for different products, strengths or net contents is refused unless it is differentiated by size, shape or colour.
TYPE DETAIL	ENFORCED at the eBMR boundary (usable only when complete) + DESIGN-CHOICE (if the suite performs receipt itself)
REGULATORY BASIS	US: 211.122(a)-(f); 211.184(a),(b),(d),(e) | EU: Ch.4 4.14 (specimen of printed materials in the specification); 4.22; Ch.5 5.46; 5.47; 5.48 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US is explicit on separate storage by product, strength, form and quantity, and on the gang-printing prohibition. EU is explicit on closed containers for cut labels and on recording destruction. The kernel applies both.
HOW IT WILL BE TESTED	Issuing a label lot with obsolete artwork is blocked. An unauthorised user cannot open a label-store transaction.

Source requirements: n126, n22

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-33 Labeling issuance and packaging-material verification at line start

Enforced | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Only an authorised role can issue packaging materials of any kind, following an approved procedure. • Labeling issued for a batch is scanned by its reference number. It is examined for identity and for conformity with the labeling the master specifies (artwork version, text, NDC/GTIN where applicable). • Issued quantities are recorded for reconciliation. • At line start, every packaging component and bulk container is scanned against the packaging master BOM (quantity, identity, conformity). The result is recorded in the packaging record before packaging can start.
REGULATORY BASIS	US: 211.125(a),(b),(f); 211.130(d); 211.184(d) | EU: Ch.5 5.46; 5.52; Ch.4 4.21(d),(h) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Equivalent.
HOW IT WILL BE TESTED	Scanning a wrong carton code is blocked. Packaging cannot start until the issued quantity is recorded.

Source requirements: n24, n23, n124

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-34 Packaging line clearance before and after use

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Pre-use clearance: ◦ The first step of every packaging order is line clearance against a line-specific checklist defined in the master. The checklist covers the work area, lines, printers and coders, vision recipes, reject bins and documents. ◦ Clearance cannot be completed while the previous order on the line is still open or unreconciled, or while components of another order remain allocated to the line. ◦ No packaging step, print job or line start is possible until clearance is signed. • Post-use inspection: at order end, an inspection of the packaging and labeling area after use is recorded in the packaging record.
TYPE DETAIL	ENFORCED (pre-use clearance against a checklist; post-use inspection) + CONFIGURABLE (second-person verifier). Default: ON. Allowed: OFF, unless the customer designates clearance a significant step under 211.188(b)(11), in which case it is ON.
REGULATORY BASIS	US: 211.130(e); 211.188(b)(6) | EU: Ch.4 4.19(f)-(g); Ch.5 5.50 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU requires the clearance to follow a checklist (5.50). US requires an inspection before use, and documentation of inspections both before and after use (211.188(b)(6)). The kernel applies the union: checklist, before and after (D-4, D-5).
HOW IT WILL BE TESTED	Starting packaging without a signed clearance is blocked. Closing an order without the post-use inspection is blocked.

Source requirements: n123, n24, n39

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-35 Packaging segregation and line identification

Enforced | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	A line runs only one active packaging order (one product, strength and lot) at a time. A different product cannot start on an adjacent line that is not physically segregated; segregation is an attribute of lines and rooms. Every packaging station or line shows the product name and batch number of the active order (on an HMI or a line board). The display clears only after order close and line clearance.
REGULATORY BASIS	US: 211.130(a); 211.42(c)(6) | EU: Ch.5 5.49; 5.51 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US accepts physical or spatial separation. EU requires physical segregation for different products packaged close to each other. EU is stricter.
HOW IT WILL BE TESTED	Starting product Y on an unsegregated adjacent line while product X runs is blocked. The station display shows the product and batch.

Source requirements: n122, n24

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-36 Printing, coding, variable data and cut-label controls

Enforced | Configurable | Both editions | 2 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Variable data: ◦ Variable data are sent electronically from the batch record to printers and coders: lot, expiry, NDC/GTIN, serial number. ◦ Manual keying is blocked, or detected by vision. ◦ A print job is invalidated if batch data change after set-up. • Print checks: ◦ Printing devices are monitored so every imprint matches the print specified in the record. ◦ A start-up check is recorded before the line enters production mode, followed by periodic checks with specimen images. ◦ Hand printing is re-checked at a configured interval. • Durability: printed and embossed information must be clear and durable against fading or erasure. This is confirmed when the artwork or specification is approved. • Cut labels on immediate containers, unit cartons or qualifying multi-unit cartons: the packaging master cannot be approved without one of these four controls: ◦ a line dedicated to each strength of each product; ◦ electronic or electromechanical 100% examination for correct labeling; ◦ 100% visual examination of hand-applied labeling by one person, verified independently by a second (never reducible under 211.68(c)); ◦ an automated technique that physically prevents incorrect labeling. Off-line overprinting is flagged for extra controls. • US barcode: at least the NDC, in a linear or 2D symbology to a recognised standard. The NDC master holds 10- and 12-digit formats with effective dates, and the NDC recorded on a historical lot is never rewritten.
TYPE DETAIL	ENFORCED (variable data from the record; print checks; one of the 211.122(g) options whenever cut labeling is used) + CONFIGURABLE (print-check method). Default: 100% vision inspection. Allowed: start-up plus periodic checks with specimens.
REGULATORY BASIS	US: 211.122(g)(1)-(4), (h); 211.130(c); 201.18 (s); 201.25 as revised by the March 5, 2026 NDC final rule (12-digit NDC effective March 7, 2033; transition to March 6, 2036) (s) | EU: Ch.5 5.55; 5.56; 5.57; 5.58; Ch.4 4.21(f) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US mandates one control from a closed list whenever cut labels are used, and requires monitoring of printing devices. EU requires every printing operation to be checked and recorded, prefers roll-fed labels, and asks for special care with cut labels. US is stricter on cut labels. The kernel applies the union.
HOW IT WILL BE TESTED	Changing batch data after print set-up invalidates the job. A wrong expiry printed on a test unit is rejected by vision and stops the line. A packaging master using cut labels with no 211.122(g) option cannot be approved.

Source requirements: n127, n84, n85, n128

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-37 On-line packaging controls, label examinations and unusual events

Enforced | Design choice | Both editions | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Minimum IPC plan:
  - At a defined frequency, with pass/fail and sample counts, the packaging IPC plan checks at least: pack appearance; completeness; correct product and materials; correct overprint; correct functioning of line monitors (code readers, label counters, vision systems).
  - A missed IPC raises an alert and an exception.
- Label examinations:
  - During finishing, packages are examined for the correct label.
  - At completion, a representative sample is visually examined for correct labeling. The sample size is set by SOP; AQL is optional.
  - Both results are recorded.
- Samples: withdrawn samples are recorded by quantity and purpose. No transaction returns sampled units to the line.
- Unusual events:
  - Units involved in an unusual event (jam, drop, suspected mix-up) are counted into a set-aside bucket linked to an event record.
  - They can be reintroduced only after a special inspection, a linked investigation and approval by an authorised role; every part of this is recorded in detail.
  - Set-aside quantities flow into reconciliation.

TYPE DETAIL

ENFORCED + DESIGN-CHOICE. Design choices: the person who raises an event cannot approve its reintroduction; units not reintroduced default to reject.

REGULATORY BASIS

US: 211.134(a)-(c); 211.188(b)(13); 211.100(b); 211.192 | EU: Ch.5 5.57; 5.59; 5.60; Ch.4 4.19(i); 4.21(d),(g) | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

US requires the end-of-run representative label examination. EU requires the minimum on-line checks, checks that devices work, and controlled reintroduction after unusual events. The kernel applies the union.

HOW IT WILL BE TESTED

Skipping the end-of-run examination prevents order completion. Adding samples back to good output is impossible. Reintroducing 40 jammed units without approval is blocked.

Source requirements: n129, n26, n130, n128

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-38 Filled but unlabelled containers ("bright stock")

Enforced | Design choice | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

Filled, unlabelled containers held for later labelling are tracked as quarantined intermediates at tray or container level. Their identification must be enough to determine name, strength, quantity and lot; each unit does not need its own label. Location and hold start are also recorded. A labelling order can consume only bright stock of the matching batch, verified by scan, and reconciles it at labelling. Containers must be clean before filling, and foreign-matter checks run where the master defines them.

TYPE DETAIL

ENFORCED + DESIGN-CHOICE (maximum unlabelled hold time, with an alert)

REGULATORY BASIS

US: 211.130(b); 211.94(c) | EU: Ch.5 5.53; 5.54 | ICH/EMA/draft: EU Annex 1 (s) for depyrogenation and sterile holds [I]

INDIA / US / EU DIFFERENCES

Equivalent. EU adds that labelling should normally follow filling as quickly as possible.

HOW IT WILL BE TESTED

Labelling trays of batch A under an order for batch B is rejected. The unlabelled inventory reconciles to the labelling run.

Source requirements: n24, n125

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-39 Reconciliation of bulk, labeling, printed packaging materials and serials

Enforced | Configurable | Both editions | 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Balance: at order close, for bulk and for each printed or labeling item, issued = used + destroyed + returned + samples + rejects + set-aside. This is reconciled against the units produced.
- Limits: limits are narrow, preset in the master, and based on historical operating data.
- Discrepancies: an out-of-limit or unusual discrepancy requires a linked investigation. Order close and release or certification stay blocked until it is satisfactorily explained.
- Component lots: use of each component lot is reconciled and feeds the inventory records.
- Serialised products: commissioned serials = packed + decommissioned, with zero unexplained serials.
- Labeling waiver: labeling reconciliation may be switched off only when all of these hold:
  - the labeling is cut or roll labeling;
  - a validated 100% electronic examination under 211.122(g)(2) is active;
  - (EU in set) robust electronic controls are documented. Reconciliation of bulk and batch-coded printed material is never waived when EU is in the set.

TYPE DETAIL

ENFORCED (reconciliation, and investigation of discrepancies before release) + CONFIGURABLE (limits; labeling waiver). Default: no waiver.

REGULATORY BASIS

US: 211.125(c); 211.184(c); 211.192; DSCSA 581/582 (s) | EU: Ch.4 4.21(h); Ch.5 5.8; 5.61 | ICH/EMA/draft: none

INDIA / US / EU DIFFERENCES

The US waiver is narrow (cut or roll labeling plus 211.122(g)(2)). EU 4.21(h) allows quantities to be left out of the record when electronic controls are robust, but 5.61 still requires discrepancies to be explained before release. The kernel applies the stricter combination (D-6).

HOW IT WILL BE TESTED

12 labels unaccounted for against a limit of 0 creates a deviation and blocks close and release. Enabling the waiver without a 211.122(g)(2) device is refused.

Source requirements: n131, n81, n23, n95

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-40 Packaging close-out: batch-coded leftovers and returns to stock

Enforced | Design choice | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	At order close, unused labeling and packaging materials that carry the batch code are destroyed, with quantity, method, date and performer recorded, and cannot be returned to stock. Uncoded printed leftovers can return only through a counted return-to-stock transaction, following the documented procedure, and are identified and stored to prevent mix-ups.
TYPE DETAIL	ENFORCED + DESIGN-CHOICE (witness signature on destruction)
REGULATORY BASIS	US: 211.125(d)-(e) | EU: Ch.5 5.62; Ch.4 4.21(h) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Equivalent. EU requires the destruction to be recorded; US requires destruction of excess labeling that bears lot or control numbers.
HOW IT WILL BE TESTED	No transaction exists to return batch-coded cartons to stock. A return of uncoded labels updates stock only after a count verification.

Source requirements: n132

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-41 OTC tamper-evident packaging [T, US OTC retail only]

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Required checks: the packaging master cannot be approved without steps that check the tamper-evident features (e.g. seal integrity, band presence). • Labeling-statement check: the statement must name every feature and any capsule-sealing technology, and refer to the identifying characteristic. • Capsules: two-piece hard capsules require verification of the sealing. • Failures: a failed seal check triggers segregation and a deviation.
TYPE DETAIL	ENFORCED when the product master flag "US OTC retail" is true; not applicable otherwise
REGULATORY BASIS	US: 211.132(a)-(c) (exemptions and related procedures in (d)-(f)) | EU: none in Ch.4/5 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US-only. The EU anti-tampering device for prescription products is a separate obligation (see PR-42).
HOW IT WILL BE TESTED	An OTC packaging master without a tamper-evident check fails approval.

Source requirements: n25

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-42 Serialisation, product identifiers and safety features

Configurable | Both editions | Search-based only (0 of 3 checked against official text)

WHAT THE SYSTEM DOES	• US prescription packs (DSCSA): ◦ Content: the product identifier is the NDC plus a serial of up to 20 characters, the lot and the expiry, in human- and machine-readable form. ◦ Carriers: a 2D data matrix on packages; linear or 2D on homogeneous cases. ◦ Before release, an automated check confirms that the lot and expiry encoded equal the record values. ◦ Serial events are recorded: allocated, commissioned, packed, decommissioned, sampled, destroyed. ◦ A serial → lot → record lookup answers verification requests within 24 h (s). • EU prescription products in scope: a unique identifier and an anti-tampering device, per Directive 2011/62/EU and Delegated Regulation (EU) 2016/161 (s). • Release gate: release is blocked if serial reconciliation fails or serial events are missing.
TYPE DETAIL	CONFIGURABLE per market (ON automatically when a market requires it)
REGULATORY BASIS	US: FD&C Act 581(14), 582(a)(9), 582(b) (s); FDA product-identifier Q&A (June 2021) (s); 201.25 (s) | EU: Directive 2011/62/EU; Delegated Regulation (EU) 2016/161 (s) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	The two are separate obligations, one per market. A batch going to both markets needs separate market-specific packaging orders, each with its own identifier scheme.
HOW IT WILL BE TESTED	A wrong expiry encoded in a test code is rejected. Serial reconciliation shows zero unexplained serials. A query on a commissioned serial returns its lot, expiry and status.

Source requirements: n81, n82, n85

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-43 Expiry dating

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Calculation: the printed and recorded expiry is derived from the shelf life stated in the currently effective finished-product specification or product master, using the defined start-of-shelf-life rule. It is never typed. • Stability basis: the shelf life must reference approved stability data. • Labeling: storage conditions on the labeling are linked to the shelf life. Products reconstituted at dispensing carry both expiries. • Tentative dating: a shelf life based on accelerated data is flagged, so the batches made under it can be found. • Stability samples: samples drawn from a batch are recorded with quantity, container-closure, storage condition and study ID.

TYPE DETAIL	ENFORCED + CONFIGURABLE (start-of-shelf-life rule, set per product with QA approval; see F-9)
REGULATORY BASIS	US: 211.137(a)-(d); 211.166(a)-(b); 211.194(e) | EU: Ch.4 4.16 (shelf life in finished-product specification); 4.19(e) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Equivalent.
HOW IT WILL BE TESTED	Changing the shelf life without a stability reference is blocked. A new shelf life applies only to batches made after its effective date.

Source requirements: n607, n620, n27

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-44 Reserve samples

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Product reserve samples (per lot): ◦ The filling or packaging master includes a reserve-sample step for each lot. ◦ Quantity: at least double the quantity required for all release tests, sterility and pyrogen tests excluded. ◦ Container and storage: the marketed immediate container-closure (or one with essentially the same characteristics), labelled and stored as the labeling requires. ◦ The reserve units count in yield and reconciliation. • Annual examination: the system schedules an annual visual examination of statistically selected lots. Deterioration found there opens an investigation. • Retention: disposal is blocked until 1 year after expiry (OTC products exempt from expiry dating: 3 years from distribution). • API reserve, where the suite handles receipt: the same 2x rule. It is kept until 1 year after the expiry of the last product lot containing that API.
TYPE DETAIL	ENFORCED (US in set) / CONFIGURABLE (EU rules; see F-8)
REGULATORY BASIS	US: 211.170(a)-(b); 211.84(b) | EU: EU GMP Annex 19 (not in uploaded texts) (s) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US quantities and periods are verified here. EU reference and retention sample rules (Annex 19) were not verified; see F-8.
HOW IT WILL BE TESTED	Packaging cannot complete without a reserve sample of at least the computed quantity. Disposal before 1 year after expiry is blocked.

Source requirements: n609, n610

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

G. Material and product status: quarantine, rejected, reprocessed, recovered and returned

PR-45 Quarantine and release status model

Enforced | Both editions | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every received lot starts in Quarantine at receipt. Every produced lot (intermediate, bulk, finished) starts in Quarantine when its output is posted. • A lot moves to Released only through a release transaction by an authorised QC/QCU role. EU finished product also needs QP certification (PR-55). • Interfaces to ERP and WMS cannot allocate or ship quarantined or rejected stock. • In-process materials are approved or rejected by the QCU at the significant phases the master defines. • Electronic (administrative) quarantine is accepted where the validated system enforces it, and physical labels show the same status. • Status changes are e-signed and audit-trailed. • Finished-lot status is held per market (PR-56).
REGULATORY BASIS	US: 211.42(c)(1),(7); 211.82(b); 211.110(c); 211.142(a) | EU: Ch.4 4.24; Ch.5 5.5; 5.63; 5.64; 5.65 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Both accept non-physical quarantine: US through "other control systems", EU through administrative quarantine. Equivalent.
HOW IT WILL BE TESTED	Shipping a quarantined lot through the ERP interface is rejected. Using a quarantined intermediate in the next step is blocked.

Source requirements: n138, n592, n622, n28, n141

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-46 Rejected materials and products

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Blocking: "Rejected" is a hard status. It blocks every consumption, issue, packaging and shipping transaction and appears on the label. • Segregation: rejected stock must be moved to a restricted, separate location. The system blocks moves to normal locations. • Disposition: return to supplier, reprocess or destroy. It needs approval from an authorised role; the action actually carried out and its quantities are recorded. • Rejects arising in-process (for example checkweigher or visual-inspection rejects) go into identified reject containers, are counted, and are handled the same way.
REGULATORY BASIS	US: 211.89; 211.110(d); 211.165(f); 211.184(e); 211.42(c)(2) | EU: Ch.5 5.66 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US accepts a quarantine system that prevents use. EU requires separate storage in restricted areas. EU is stricter, so the kernel requires both the electronic status and a physical restricted location (D-10).

HOW IT WILL BE TESTED	Dispensing a rejected lot is blocked. Moving rejected stock to a released-stock location is blocked.

Source requirements: n134

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-47 Reprocessing and recovery of earlier batches

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Reprocessing: ◦ It runs only from a QCU-approved reprocessing procedure or master. That procedure is linked to the non-conforming batch, its investigation and a documented risk evaluation. ◦ It creates a child execution record and genealogy edges. ◦ The product must meet its specifications before acceptance. ◦ It is flagged for review and PQR. • Recovery (adding conforming earlier batches of the same product at a defined stage): ◦ The step type exists only where the master defines it, and only for the same product. ◦ Prior authorisation is required, with a referenced risk evaluation that includes the effect on shelf life. ◦ The source batch IDs and quantities are recorded, which also meets 4.20(d). • Additional testing: before release of any reprocessed or recovered-material batch, a QC decision on the need for additional testing is a mandatory field.
TYPE DETAIL	ENFORCED + CONFIGURABLE (per product, reprocessing can be disabled, e.g. where the registration forbids it for a sterile product [I])
REGULATORY BASIS	US: 211.115(a)-(b); 211.165(f) | EU: Ch.5 5.67; 5.68; 5.69; Ch.4 4.20(d) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	EU calls reprocessing exceptional and requires a risk evaluation and a QC decision on extra testing. US requires QCU review and approval. The kernel applies both. "Rework" is not a Part 211 term.
HOW IT WILL BE TESTED	Starting reprocessing without QCU approval or a risk evaluation is blocked. Recovering material from a different product is blocked. A missing QC decision on additional testing blocks release.

Source requirements: n21, n135, n136

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-48 Returned products

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Receipt: booking in returned goods creates a quarantined lot of type "returned", linked to its original batch. • Assessment: disposition requires a QC assessment record with a QC e-signature. It considers the product's nature, its storage requirements, its condition and history, and the time since issue. • Disposition rules: the default is destroy. If any doubt is recorded, the only options are destruction or chemical recovery of the active substance. • Record content: the record holds name, label potency, dosage form, lot, reason for return, quantity, disposition date and final disposition. • Investigation: if the reason for return implicates other batches, a 211.192 investigation is linked.
REGULATORY BASIS	US: 211.204; 211.208 | EU: Ch.5 5.70; Ch.4 4.29 (returns) | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US lets examination or testing resolve doubts about storage conditions. EU does not allow re-issue or re-use once any doubt exists. EU is stricter, and the kernel applies it to all markets (D-12).
HOW IT WILL BE TESTED	Repackaging returned units without a QC assessment is blocked. After a "doubt" assessment, only destroy or chemical recovery is offered.

Source requirements: n137

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

H. Deviations, investigations and CAPA

PR-49 Deviation record, classification and approval

Enforced | Configurable | Design choice | Both editions | 1 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Creation: a deviation raised from a step pre-fills the batch, step, master version, equipment and materials. • Classification: planned or unplanned; impact on a registered detail, on GMP, or neither; severity. • Approval and holds: a competent person approves it, with QC involved for quality-relevant classes. Linked batches are placed on hold. • Annex 16 section 3 fields (EU in set), for certifying a batch with an unexpected deviation: ◦ confirmation that registered specifications are met; ◦ a QRM impact assessment; ◦ root cause and actions; ◦ an assessment of whether a variation is needed. • Recurrence and register: a counter warns when the same deviation recurs. A register of batches certified under this route is kept for the PQR. • Availability impact (design choice): an optional field for shortage-sensitive products. Availability can never override a quality disposition.
TYPE DETAIL	ENFORCED (record, approval, batch link, hold of linked batches) + CONFIGURABLE (deviation classes) + DESIGN-CHOICE (availability-impact field)
REGULATORY BASIS	US: 211.100(b); 211.192 | EU: Ch.5 5.15; Ch.4 4.20(h); 4.21(g); 4.29 | ICH/EMA/draft: Annex 16 section 3 and EMA Q&A on Annex 16 (s); ICH Q9(R1) 6.1 (s); EU GMP Ch.1 (s)

INDIA / US / EU DIFFERENCES	The Annex 16 section 3 route has no US equivalent. In the US, a batch failing a specification is rejected (211.165(f)) and the failure is investigated (211.192).
HOW IT WILL BE TESTED	A failure of a registered limit creates a deviation with the MA flag. The third recurrence shows a warning on the certification screen.

Source requirements: n119, n214, n208, n189, n224

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-50 Investigation of discrepancies and failures

Enforced | Both editions | 1 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Triggers: an investigation opens for any unexplained discrepancy or specification failure, even after distribution. Unexplained discrepancies include yields outside master limits, reconciliation discrepancies and time-limit breaches.
- Impact extension (mandatory step):
  - The system offers queries for other batches of the same product and for other products that shared component lots, intermediates, equipment (previous and next product), rooms, personnel, utilities or time windows.
  - Linked batches are held.
- Closure: the written record requires conclusions and follow-up.
- Root cause (major and critical investigations):
  - A root-cause category and the method used are required.
  - Concluding "human error" requires a documented assessment of process, procedure and system factors.
  - "Not determined" requires a most-probable cause and actions.

REGULATORY BASIS	US: 211.192; 211.188(b)(12); 211.198(b)(2); 211.204 | EU: Ch.4 4.29 (investigations); Ch.5 5.44; 5.61 | ICH/EMA/draft: EU GMP Ch.1 1.4(xiv) (s); ICH Q9(R1) 4.3 (s); ICH Q10 3.2.2 (s)
INDIA / US / EU DIFFERENCES	US explicitly extends investigations to other batches and products, including after distribution. EU Ch.1 (s) is explicit about the depth of root-cause analysis and about justifying a human-error conclusion. The kernel applies both.
HOW IT WILL BE TESTED	An investigation cannot close without conclusions and follow-up. The impact query returns the expected batch set, checked against an independent query. "Human error" without the system-factor assessment is blocked.

Source requirements: n40, n211, n228, n214

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-51 CAPA linkage and effectiveness

Configurable | Design choice | Both editions | Supporting | Search-based only (0 of 1 checked against official text)

WHAT THE SYSTEM DOES

- CAPAs are created from, or linked to, deviations and investigations, with links in both directions: the batch lists its deviations and CAPAs, and each CAPA lists the batches it affects.
- An effectiveness check is scheduled, with criteria that can be measured from eBMR data.
- A CAPA that needs a master change cannot close until the new master version is effective.

TYPE DETAIL	CONFIGURABLE. Default: CAPA completion does not block release, but the investigation's follow-up does. Allowed: QA may make a CAPA release-blocking. The automated effectiveness checks are a DESIGN-CHOICE.
REGULATORY BASIS	US: 211.192 (follow-up) | EU: EU GMP Ch.1 (s) | ICH/EMA/draft: ICH Q10 3.2.2 (s)
INDIA / US / EU DIFFERENCES	"CAPA" is ICH Q10 / EU Ch.1 wording. 211.192 requires conclusions and follow-up. Guidance level in both regions.
HOW IT WILL BE TESTED	Closing a CAPA that needs a master change is blocked while the change is not yet effective.

Source requirements: n213

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

I. Batch record review and release

PR-52 Production approval and quality-unit review (including review by exception)

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- Production approval: after processing and after packaging, the person responsible for that operation approves the processing record and the packaging record by e-signature. The approval covers the exceptions within their scope.
- QA workbench and exception report:
  - The QA review workbench shows an exception report generated from a mandatory taxonomy: limit breaches, deviations, overrides, corrections of critical data, late entries, interface fallbacks, timer breaches, open verifications, alarms, audit-trail events on critical data, and special observations or comments.
  - Each exception is dispositioned individually. The full record can always be viewed.
- Reduced reports:
  - Automatically generated reports may be cut down to compliance summaries plus exception/OOS reports only for steps configured, under change control with a validation reference, as validated and continuously monitored.
  - The underlying data are kept and can be drilled into.
- QCU approval: the QCU approval e-signature states that the complete record, including the packaging and labeling records, was reviewed.

TYPE DETAIL	ENFORCED (full record available; every exception dispositioned; QCU approval of the complete record) + CONFIGURABLE (production-responsible approval: default ON, mandatory when EU is in the set; review-by-exception scope)

REGULATORY BASIS	US: 211.192; 211.22(a); 11.10(e) | EU: Ch.4 4.20(i); 4.21(i); 4.20 Note; 4.27; Annex 11 sections 8.2 and 9 | ICH/EMA/draft: FDA data-integrity Q&A (2018) on audit-trail review (s)
INDIA / US / EU DIFFERENCES	EU requires approval of the records by the person responsible for production (4.20(i), 4.21(i)); US has no explicit equivalent. US requires QCU review and approval of all production and control records, packaging and labeling included, before release. The 4.20 Note permits shorter reports but does not remove that US duty. Stricter combination: production approval, then QCU approval of the complete record.
HOW IT WILL BE TESTED	Seed one exception of each type. All appear, and approval stays blocked until each is dispositioned. Enabling review by exception on a step with no validation reference is refused.

Source requirements: n40, n93, n105

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-53 Release gating on open deviations and investigations

Configurable | Both editions | 1 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	The gate reads live status from the eBMR and the linked QMS/LIMS. Batches linked as "potentially affected" are held. CAPA completion is tracked separately (PR-51).
TYPE DETAIL	CONFIGURABLE. Default: release or certification is blocked while any linked deviation or investigation is open. Allowed: release with an investigation still open, only when the QCU (US) or QP (EU) records an e-signed justification that the investigation is advanced enough to support the decision. The justification appears on the release record and the certificate data. Not allowed: this exception for reconciliation discrepancies (5.61 requires them explained before release) or for any registered-specification failure.
REGULATORY BASIS	US: 211.192; 211.165(f) | EU: Ch.5 5.61 | ICH/EMA/draft: Annex 16 section 1.7 (investigations complete enough to support certification) (s); ICH Q10 3.2.2 / EU Ch.1 (s)
INDIA / US / EU DIFFERENCES	Neither 211.192 nor Annex 16 1.7 literally requires every investigation to be closed before release. Annex 16 (s) requires it to be complete to a level sufficient to support certification. The guardrail's unconditional block is stricter than both texts (D-8).
HOW IT WILL BE TESTED	A batch with an open deviation cannot be released. With the e-signed justification it can, and the exception is printed on the release record. An open reconciliation discrepancy cannot be justified away.

Source requirements: n214, n188, n223, n115

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-54 US batch disposition by the quality control unit

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Only users with the QCU disposition role can release a finished lot for the US market. • Prerequisites: ◦ QCU record review approved (PR-52) and investigations handled per PR-53; ◦ lab conformance to final specifications, including identity and strength of each active ingredient; ◦ microbiological testing where the product must be free of objectionable organisms; ◦ [I] sterility and pyrogen testing; ◦ [T] a release-rate result for each active ingredient of a controlled-release product; ◦ reconciliation passed; ◦ QC decisions on reprocessed or recovered material recorded. • Failing product is rejected. • Contract manufacture: when the site makes product under contract, the record also supports approval or rejection by the contract giver's QCU (211.22(a)).
TYPE DETAIL	ENFORCED (US in set)
REGULATORY BASIS	US: 211.22(a),(c); 211.142(a); 211.165(a),(b),(d),(f); 211.167(a),(c); 211.192 | EU: n/a | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	US only. The QCU is an organisational unit; no named individual is required, unlike the EU QP.
HOW IT WILL BE TESTED	An extended-release tablet lot missing one active's release-rate result cannot be released.

Source requirements: n608

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-55 EU QP certification

Enforced | Both editions | 2 of 8 source requirements checked against official text

WHAT THE SYSTEM DOES	• Who certifies: where certification is recorded in the system, only users holding the QP role can certify. They do so by e-signature, and the system identifies the certifying QP. • Certification cockpit (live status): ◦ record completeness and the production and QA approvals; ◦ deviations and changes, with their approval status and regulatory status for the EU MA; ◦ OOS/OOT results and investigations, and validation status (concurrent-validation results must be available); ◦ a partial-manufacturing confirmation, with Appendix I content, for each upstream stage in the genealogy performed at another site; ◦ open complaints or recalls on the product; ◦ flags for special observations and for changes to critical data. • Blocking: certification is blocked while a mandatory item is red, unless the QP records a justified decision. • Certification register: each certification goes into an append-only, exportable register (batch, product/strength/pack, market, QP, date/time, confirmations relied on), kept for at least 5 years (s).

- Real-time release (RTRT) products: certification may rest on validated summary and exception reports of PAT or parameter data, but only where the MA allows it.
- QP outside the eBMR: typically the EU importer's QP for an Indian site. The eBMR then records the external certification reference, and the lot's EU status cannot change until that reference is recorded.

TYPE DETAIL	ENFORCED (EU in set)
REGULATORY BASIS	US: n/a | EU: Ch.4 4.27; Ch.4 preamble footnote 1 (RTRT); Annex 11 section 15 | ICH/EMA/draft: Directive 2001/83/EC Art. 51 (s); Annex 16 section 1, 1.7 and Appendix I (s); EU Annex 15 concurrent validation (u)
INDIA / US / EU DIFFERENCES	EU only; there is no US counterpart. Only the certification step must be an e-signature (Annex 11 section 15).
HOW IT WILL BE TESTED	A non-QP attempt to certify is refused. An open OOS blocks certification. A register entry is created at signature and cannot be edited.

Source requirements: n105, n187, n188, n223, n190, n192, n619, n197

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-56 Market-specific disposition for multi-market batches

Enforced | Both editions | Supporting | Search-based only (0 of 2 checked against official text)

WHAT THE SYSTEM DOES

- Disposition status is held per finished lot and per market.
- A lot is releasable for a market only when:
  - it meets that market's registration (ECs, specifications, process);
  - its market-specific packaging record is complete;
  - that market's release authority has signed (US: QCU; EU: QP).
- A deviation found acceptable for one market is assessed separately for the other.
- A rejection in one market opens an investigation link visible to all markets.
- See Section (b).

REGULATORY BASIS	US: 211.22(a); 211.192 | EU: Ch.5 Principle | ICH/EMA/draft: Annex 16 (s); ICH Q12 ch.3 (s)
INDIA / US / EU DIFFERENCES	Each market's release authority is independent. Release in one market implies nothing about the other.
HOW IT WILL BE TESTED	A batch certified for the EU but not yet released by the US QCU cannot be shipped to the US, and the reverse also holds.

Source requirements: n218, n187

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

J. Traceability, product quality review and records retention

PR-57 Genealogy and lot traceability (including distribution and complaint lookups)

Enforced | Design choice | Both editions | 6 of 8 source requirements checked against official text

WHAT THE SYSTEM DOES

- A lot number resolves to its complete history:
  - master version, executed records and every signature;
  - component and intermediate lots, with supplier, manufacturer and manufacturer batch;
  - equipment and rooms;
  - IPC and QC results;
  - deviations;
  - packaging, serialisation and disposition;
  - distribution to consignees.
- Traversal: backward and forward traversal works at any depth, including splits, merges, reprocessing, recovery and transfers between sites.
- Integrity: each genealogy edge is committed atomically with the business event that creates it. Any graph database is only a derived, rebuildable and reconciled projection.
- Distribution records, linked from ERP, hold product name and strength, dosage form, consignee name and address, date, quantity and lot.
- Lookups: a complaint-to-lot lookup opens the record, genealogy, deviations and shipments. An API supply-chain traceability report can be produced for each finished batch.

TYPE DETAIL	ENFORCED (backward and forward traceability) + DESIGN-CHOICE (graph-database projection; bitemporal storage)
REGULATORY BASIS	US: 211.130(c); 211.184(c); 211.188(b)(2)-(3); 211.105(b); 211.182; 211.150(b); 211.196; 211.192; 211.198; 210.3(b) (s); 201.18 (s) | EU: Ch.4 Records definition (batch history includes distribution); 4.8; 4.23; 4.28; Ch.5 5.29 | ICH/EMA/draft: none
INDIA / US / EU DIFFERENCES	Equivalent. EU explicitly includes distribution in the batch history and requires API supply-chain traceability. Room- and person-level forward tracing are not explicit requirements anywhere.
HOW IT WILL BE TESTED	A seeded split/merge/rework scenario returns exactly the expected nodes. A forward trace from a component lot lists every consignee, matching an ERP extract.

Source requirements: n87, n1, n84, n621, n637, n30, n146, n42

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-58 Product quality review / annual product review data

Enforced | Design choice | Both editions | 2 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- Data types: CPPs, IPCs, yields, deviations, reprocessing, rejections, reconciliation results and release decisions are stored as typed data with units, limits and specification version.

- Extracts: extracts per product and period are reproducible and carry source-record links. They cover:
  - a representative number of approved and rejected batches;
  - complaints, recalls and returned or salvaged products;
  - investigations, changes and CAPA effectiveness.
- Checks: extract counts reconcile to the requirements lists.

TYPE DETAIL	ENFORCED (structured, queryable, reproducible data) + DESIGN-CHOICE (SPC dashboards, trend rules)
REGULATORY BASIS	US: 211.180(e)(1)-(2) | EU: Ch.4 4.29 (summaries such as PQR) | ICH/EMA/draft: EU GMP Ch.1 1.10 (s); ICH Q10 3.2.1 and 3.2.4 (s); FDA Process Validation guidance (2011) (s); EU Annex 15 OPV (u)
INDIA / US / EU DIFFERENCES	US requires an evaluation at least once a year. The EU PQR scope (Ch.1 1.10, (s)) is broader: significant deviations, CAPA effectiveness and changes.
HOW IT WILL BE TESTED	A product extract for a period reconciles to the batch list and the deviation and CAPA registers. Re-running it gives identical results.

Source requirements: n33, n216, n212, n80, n196, n625

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-59 Record retention, record-location register and archiving

Enforced | Configurable | Both editions | 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- Retention date for each batch record: the latest of all that apply:
  - US: expiry + 1 year (for OTC products exempt from expiry dating: distribution + 3 years);
  - EU in set: the later of expiry + 1 year and QP certification + 5 years;
  - DSCSA transaction and serial data: at least 6 years (s);
  - IMP: 5 years after the last trial ends;
  - longer national or product rules, and legal holds.
- Recalculation and edge cases:
  - The date is recomputed if the expiry changes.
  - For batches never certified, the default is the later of expiry + 1 year and disposition + 5 years (F-2).
  - Component, container, closure and labeling records are kept for 1 year after the expiry of the last lot that used them (211.180(b); see F-10).
  - Validation raw data that supported release are kept at least as long as the batches they support (4.12).
- Deletion: impossible before the retention date. Afterwards it needs a QA-approved, audit-trailed disposition.
- Scope: the full record, including audit trails, attachments and linked historian extracts.
- Location register: maps each activity to its record and its storage location.
- Archive:
  - checksums, periodic readability checks and tested migration;
  - a complete human-readable rendering on demand;
  - records stay available at the site, or can be retrieved electronically at once, for inspection.

TYPE DETAIL	ENFORCED (retention floors; no deletion before the retention date) + CONFIGURABLE (longer national, product or legal-hold rules)
REGULATORY BASIS	US: 211.180(a)-(d); 211.198(b); 211.68(b); 11.10(b)-(c) | EU: Ch.4 Principle; 4.10; 4.11; 4.12; Annex 11 sections 7, 8 and 17 | ICH/EMA/draft: DSCSA 582(b) (s)
INDIA / US / EU DIFFERENCES	EU is usually longer because of the 5 years after certification; US is tied to expiry. Stricter wins: the maximum (D-7).
HOW IT WILL BE TESTED	A batch with a 24-month shelf life certified on 2026-01-10 gets a retention date of 2031-01-10. With a 60-month shelf life, expiry + 1 year governs. Deleting before the date is refused and logged.

Source requirements: n103, n102, n82, n42, n146

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

PR-60 Forward compatibility with EU draft texts (data governance, AI)

Design choice | Both editions | Search-based only (0 of 2 checked against official text)

WHAT THE SYSTEM DOES

- Data governance:
  - An inventory of record types and fields, each with an owner and a criticality.
  - Documented data flows to and from integrated systems.
  - A periodic data-integrity review report.
- AI:
  - No AI/ML output can change GMP data or status without a human e-signature.
  - Any AI helper is flagged as such. Its output is reviewed and signed, and the model version and confidence are recorded.
  - Critical-use models (e.g. automated visual inspection [I]) must be static, validated and monitored.

REGULATORY BASIS	US: none | EU: none binding | ICH/EMA/draft: EU GMP Ch.4 draft (Jul 2025) 4.10-4.18 (draft); EU GMP Annex 22 draft (Jul 2025) (draft)
INDIA / US / EU DIFFERENCES	EU drafts only. Neither region's current texts reviewed here contain these rules.
HOW IT WILL BE TESTED	In the AI feature inventory, no feature changes GMP data without a human e-signature.

Source requirements: n107, n180

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

(a) Required content: merged US + EU checklists

The MBR designer (PR-03) refuses approval if any item applicable to the product's market set is missing. The completeness gate (PR-16) checks the executed-record lists. "US" means required when US is in the market set, "EU" means required when EU is in the set, and "both" means both texts require it. Conditional items ("where applicable/appropriate/necessary") may be set to N/A only with a signed justification. All wording is paraphrased; see the official texts for the exact requirements.

(a)-1 Processing master (manufacturing formula + processing instructions)

#	Required content	US	EU	Scope
1	Product name, strength, description of dosage form	211.186(b)(1)	4.17(a)-(b)	both
2	Product reference code tied to its specification	-	4.17(a)	EU (kernel: all)
3	Batch size (one master per batch size)	211.186(a)	4.17 lead-in, 4.17(b)	both
4	Each active: name and amount per dosage unit (or per unit weight/measure); total weight or measure of one dosage unit	211.186(b)(2)	-	US
5	Complete component list, with names or codes specific enough to show any special quality characteristic	211.186(b)(3)	4.17(c)	both
6	Accurate quantity of each component in one weight system; any justified variations stated	211.186(b)(4)	4.17(c)	both
7	Flag on any substance that may disappear during processing	-	4.17(c)	EU (kernel: all)
8	Statement of any calculated excess (overage)	211.186(b)(5)	-	US
9	Formula designed to deliver at least 100% of the labeled/established active amount	211.101(a)	-	US
10	Theoretical weight or measure at appropriate phases	211.186(b)(6)	-	US
11	Theoretical/expected yield with min/max limits: final yield (both) and each yield checkpoint (kernel, see D-9)	211.186(b)(7); 211.103	4.17(d)	both
12	Processing location and principal equipment (class/capacity)	211.63; 211.105(b)	4.18(a)	EU explicit; kernel: all
13	Methods, or references to them, for preparing critical equipment (cleaning, assembly, calibration, sterilisation)	211.67(b) (via SOP)	4.18(b)	EU
14	Pre-start check: equipment and workstation clear of previous products, documents and unneeded materials; equipment clean and suitable	211.67(b)(6)	4.18(c); 5.40	both
15	Detailed stepwise instructions: material checks, pre-treatment, sequence of addition, CPPs with ranges	211.186(b)(9)	4.18(d)	both
16	IPC instructions with limits; sampling and test procedures; specifications	211.186(b)(9); 211.110(a)-(b); 211.160(b)(2)	4.18(e); 4.25; 4.26	both
17	Bulk/intermediate storage requirements (container, labelling, storage conditions) where needed	-	4.18(f); 5.41	EU
18	Time limits for production phases where appropriate, with validation reference	211.111	-	US
19	Special notations and precautions	211.186(b)(9)	4.18(g)	both
20	Significant-step flags; performer/checker rule; dispensing and charge-in verification mode (manual or automated)	211.188(b)(11); 211.101(c)-(d); 211.68(c)	4.20(c); 5.38	both (kernel)
21	Container/closure description for steps that fill primary containers [I]	211.186(b)(8)	-	US
22	Preparer and independent checker (each signed and dated), QCU approval, effective date, unique ID and version	211.186(a); 211.100(a); 211.22(c)	4.3	both
23	Registration references per market; registered/EC flags	211.100(a) (via application)	4.2; Ch.5 Principle	both (PR-04)
24	Validation reference for the process and CPP ranges	-	5.24; 5.25; Annex 15 (u)	EU (kernel: all)

(a)-2 Packaging master (packaging instructions)

#	Required content	US	EU	Scope
1	Product name, plus fields for bulk batch number and finished batch number	211.130(c)	4.19(a)	both
2	Dosage form, and strength where applicable	211.186(b)(1)	4.19(b)	both
3	Pack size (number, weight or volume in the final container)	-	4.19(c)	EU (kernel: all)
4	Full list of packaging materials, each with quantity, size, type and specification code; description of containers, closures and packaging materials	211.186(b)(8)	4.19(d)	both
5	A specimen or copy of every label and every other labeling item, bearing the dated signature of whoever approved that labeling; marks where batch number and expiry go	211.186(b)(8) (mandatory)	4.19(e) ("where appropriate")	both; US stricter (D-1)
6	Line-clearance checklist; careful pre-start examination of area and equipment; equipment clean and suitable	211.130(e)	4.19(f)-(g); 5.50	both
7	Special precautions	211.186(b)(9)	4.19(g)	both
8	Description of the packaging operation, including significant subsidiary operations and equipment	211.186(b)(9)	4.19(h)	both
9	IPCs with sampling instructions and acceptance limits; on-line minimum checks; end-of-run label examination	211.134; 211.186(b)(9)	4.19(i); 5.59	both
10	Pre-packaging examination of packaging and labeling materials	211.130(d)	5.52	both
11	Reconciliation limits for labeling, printed materials and bulk (narrow, preset, based on historical data)	211.125(c)	5.61; 4.21(h)	both
12	Theoretical yield with min/max for the packaging phase	211.103; 211.186(b)(7)	4.17(d) (final yield)	both
13	Where cut labeling is used: one of the four 211.122(g) controls	211.122(g)	5.56	US mandatory
14	Expiry-dating rule and storage statement	211.137	4.16(f)-(g)	both
15	Reserve-sample step	211.170(b)	Annex 19 (s)	US (EU: F-8)
16	Market-specific: OTC tamper-evident checks; US product identifier/NDC barcode; EU safety features	211.132; 201.25 (s); DSCSA (s)	Reg. (EU) 2016/161 (s)	per market
17	Approval route as in (a)-1 item 22	211.186(a)	4.3; 4.19 lead-in	both

(a)-3 Executed batch processing record

#	Required content	US	EU
1	Accurate reproduction of the master, checked for accuracy, dated and signed at issue; based on the current approved version	211.188(a)	4.20 lead-in; 4.2

#	Required content	US	EU
2	Product name and batch number	211.130(c)	4.20(a)
3	Dates (US) and times (EU) of start, significant intermediate stages and completion	211.188(b)(1)	4.20(b)
4	ID of each major equipment item and line used	211.188(b)(2); 211.105(b)	4.20(e)
5	ID of each component and in-process material batch used (and analytical control number)	211.188(b)(3)	4.20(d)
6	Quantities actually weighed or measured, including batch number and quantity of any added recovered or reprocessed material	211.188(b)(4)	4.20(d)
7	IPC and lab results, with who performed each IPC	211.188(b)(5)	4.20(f)
8	Sampling performed	211.188(b)(10)	4.25
9	Actual yield and % of theoretical at appropriate phases, with verifier	211.188(b)(7); 211.103	4.20(g)
10	Performer of each significant step and the checker (US: always; EU: where appropriate); for automated steps, who checked the equipment	211.188(b)(11)	4.20(c)
11	Relevant processing operations and events	211.100(b)	4.20(e)
12	Notes on special problems, with signed authorisation for any deviation	211.100(b)	4.20(h); 5.15
13	Any investigation made	211.188(b)(12)	4.29
14	Pre-start clearance confirmation	211.67(b)(6)	4.18(c); 5.40
15	Time-limit outcomes and justified departures	211.111	4.18(f)
16	Cleaning, maintenance and use records for dedicated equipment with no individual log (US-only option; see PR-24)	211.182	-
17	Container/closure description where filling occurs in processing [I]	211.188(b)(9)	-
18	Approval by the person responsible for processing	-	4.20(i)
19	For validated, continuously monitored steps: compliance summary plus exception/OOS reports (a permission, not a requirement)	-	4.20 Note

(a)-4 Executed batch packaging record (per batch or part batch)

#	Required content	US	EU
1	Accurate reproduction of the packaging master, checked, dated and signed at issue	211.188(a)	4.21 lead-in
2	Product name; bulk and finished batch numbers	211.130(c)	4.21(a); 4.19(a)
3	Dates and times of packaging operations	211.188(b)(1)	4.21(b)
4	Performer of each significant step and checker	211.188(b)(11)	4.21(c)
5	Identity and conformity checks against the packaging instructions, including IPC results	211.130(d); 211.188(b)(5)	4.21(d); 5.52
6	Packaging operations carried out, with equipment and line references	211.188(b)(2)	4.21(e)
7	Specimens or copies of all labeling used (US mandatory), and printed samples showing batch coding, expiry and overprint (EU whenever possible)	211.188(b)(8)	4.21(f)
8	Complete labeling control records (issuance, examination, reconciliation)	211.188(b)(8); 211.125	4.21(h)
9	Notes on special problems and unusual events, with signed authorisation for deviations	211.100(b)	4.21(g); 5.60
10	Quantities and references of printed materials and bulk issued, used, destroyed or returned, and product obtained (reconciliation)	211.125(c); 211.184(c)	4.21(h); 5.61
11	Inspection of the packaging/labeling area both prior to and following use (EU: prior only, with checklist)	211.188(b)(6); 211.130(e)	5.50
12	Actual yield and % of theoretical	211.188(b)(7)	-
13	Description of containers and closures	211.188(b)(9)	- (listed in the instructions, 4.19(d))
14	Sampling performed, including reserve samples	211.188(b)(10); 211.170(b)	5.59
15	Results of the in-process and end-of-run label examinations	211.188(b)(13); 211.134	5.59
16	Any investigation made	211.188(b)(12)	5.61
17	Approval by the person responsible for packaging	-	4.21(i)
18	Market-specific: serial commissioning and reconciliation (US Rx, (s)); tamper-evident checks (US OTC)	DSCSA (s); 211.132	Reg. 2016/161 (s)

(b) Release models by market, and one batch destined for several markets

US: quality control unit disposition

- Authority: the QCU approves or rejects drug products, including product made for the firm under contract (211.22(a)). It is an organisational unit; no named individual is required. The disposition is signed by an authorised QCU member using a Part 11 e-signature (11.1(c)).
- Before release:
  - QCU review and approval of all production and control records, packaging and labeling included (211.192);
  - thorough investigation of unexplained discrepancies and specification failures, extended to associated batches and products (211.192);
  - lab conformance to final specifications, including identity and strength of each active (211.165(a)), plus micro testing where required (211.165(b));
  - sterility and pyrogen testing [I] and release-rate testing [T, controlled release] (211.167(a),(c));
  - product held in quarantine until QCU release (211.142(a)).
- Failing product is rejected, and reprocessed material must conform before use (211.165(f)).
- Contract manufacture for a US applicant: the eBMR must be able to hand the complete record to the contract giver's QCU and record its disposition as well.

EU: QP certification

- Authority: each batch must be certified by a Qualified Person of the manufacturing or import authorisation holder before release for sale or supply (Directive 2001/83/EC Art. 51 (s); Annex 16 (s)).
- System rules: every record must be available to the QP, and a system must flag special observations and changes to critical data (4.27). Where the computerised system records certification, only QPs can certify, by e-signature, and the certifier is identified (Annex 11 section 15).
- Certification is entered in a register kept at least 5 years (s).
- Supporting rules:
  - partial-manufacturing confirmations from other sites (Annex 16 App. I (s));
  - the unexpected-deviation route (Annex 16 section 3 (s));
  - batch records kept 5 years after certification if that is longer than expiry + 1 year (4.11).
- Indian export sites: the QP who certifies for the EU market is normally the QP of the EU importing MIA holder, after importation. For third-country batches, Art. 51 also requires testing in the EU unless an agreement provides otherwise (s; F-3). The Indian site's own quality-unit disposition is therefore a manufacturer release, not a QP certification. The eBMR must:
  - give the QP the complete record, plus flags and audit trails of critical-data changes (4.27);
  - accept the external certification reference;
  - keep EU-market status at "awaiting QP certification" until that reference is recorded (PR-55).

One batch destined for several markets (PR-04, PR-56)

1. Declare the market set when the batch is issued. The master version must be effective for every market in the set, which means running the process inside the intersection of the requirements listed ranges and meeting every Established Condition. If a change is approved for one market but still pending for another, a batch for both must use the version effective in both.
2. Testing: every test either market requires is performed. For an attribute tested once, the result must meet the tighter of the two specification limits. EU import testing, if it applies, happens after importation (s).
3. Processing produces one batch processing record. Packaging runs as separate market-specific packaging orders (part batches, 4.21), each with its own packaging record, labeling, identifiers (US DSCSA/NDC; EU safety features), reconciliation and finished-lot identity. Each finished lot must trace to the bulk batch (211.130(c)).
4. The record content is the union of both markets (Section (a)). Examples: times (EU), a checker on every significant step (US), post-use area inspection (US), area logbooks and production-head approval (EU).
5. Disposition is held per lot and per market: US QCU release for US packs, QP certification for EU packs. Releasing a lot in one market implies nothing for the other.
6. Deviations are assessed per market. The Annex 16 section 3 route exists only for the EU. For the US, 211.192 applies and a specification failure means rejection (211.165(f)). A batch rejected for one market can go to the other only if it meets that registration and the recorded investigation concludes there is no impact there. Every investigation link is visible to both release authorities.
7. Retention is the maximum over all markets (PR-59).

(c) Claims made by earlier research that are NOT actually required

- C-1 **Three distinct signers on every master (author, checker, QA).** The rules require a preparer plus an independent second-person checker (211.186(a)) and QCU approval (211.22(c), 211.100(a)). The checker may be a QCU member who also approves. Guardrail 8's three-person rule is a stricter design choice (PR-02).
- C-2 **Wet-ink handwritten signatures on the master record.** 211.186(a) says "full signature, handwritten", but a Part 11-compliant e-signature is its equivalent (11.1(c)), given the firm's 11.100(c) certification to FDA.
- C-3 **A checklist for processing-area clearance.** Ch.5 requires a checklist only for packaging line clearance (5.50). Processing clearance (5.40, 4.18(c)) needs a check, not a checklist (n116).
- C-4 **Second-person verification of line clearance.** Neither 5.50 nor 211.130(e) requires two people. It becomes required only if the firm designates clearance a significant step under 211.188(b)(11) (n123).
- C-5 **Clean-hold and dirty-hold limits required by 211.67 or Ch.5.** The values come from cleaning validation (Annex 15 section 10, (u)). The regulations require cleaning at appropriate intervals and a pre-use inspection (n6, n199).
- C-6 **A personnel-presence list and gowning-qualification gating under 211.28.** These come from Annex 1 and the FDA 2004 guidance. 211.28(c) requires only supervisor-authorised entry to limited-access areas (n4).
- C-7 **Cleaning verification at every product changeover.** 5.21 says cleaning verification after each campaign should be considered for higher-risk products, as a detectability tool (n121).
- C-8 **Colour-coded status labels.** They are optional; 5.13 calls them often helpful (n118).
- C-9 **Potency-adjusted dispensing and automatic filler compensation.** They are one way to meet the at-least-100% intent of 211.101(a), not a mandate (n11).
- C-10 **Excluding the person who entered the quantities from verifying the yield; forbidding manual override.** Both are design choices; 211.103 requires only independent verification (n15).
- C-11 **Zero tolerance for label reconciliation.** US requires narrow preset limits based on historical data (211.125(c)); EU asks that significant or unusual discrepancies be investigated (5.61) (n131).
- C-12 **Review by exception as a requirement, or as a replacement for QCU review.** The 4.20 Note permits shorter reports for validated, continuously monitored processes. It is not an obligation, and US QCU approval of all records still applies (n93, n40).
- C-13 **An e-signature on every release workflow state.** Only QP certification must be an e-signature (Annex 11 section 15). Other approvals must be attributable, and e-signing them is a design choice (n105).
- C-14 **QA approval of every deviation under 5.15.** 5.15 names a competent person, with QC involved when appropriate; routing to QA is a design choice (n119).
- C-15 **Release blocked until every CAPA is complete, or every investigation closed.** Neither is literal text. Annex 16 1.7 (s) requires investigations complete enough to support certification; CAPA completion is not a release criterion (n214; D-8).
- C-16 **Extending batch-record retention automatically when a complaint is linked.** 211.198(b) governs retention of the complaint record only (n42).
- C-17 **Room-level and person-level forward traceability.** These are useful for 211.192 impact assessment but not explicit requirements (n87).
- C-18 **AQL sampling for the 211.134 end-of-run examination.** A representative sample is required; AQL is one way to get one (n26).
- C-19 **Barcode/RFID on every container, and a unique container ID on dispensing labels.** The rules require identification of contents and phase, and the four 211.101(b) label elements (n12, n16).
- C-20 **A binding ban on model-only in-process control.** That is the FDA draft position (Jan 2025), not binding; the 211.110 text is technology-neutral (n18).
- C-21 **The verifier must witness charge-in within a time window, from a terminal in the room.** A policy option, not in 211.101(d) (n14).
- C-22 **A maximum filling-to-labelling interval.** 5.54 says labelling should normally follow as quickly as possible, or else procedures must prevent mix-ups (n125).
- C-23 **A witness signature on destruction of coded or obsolete materials.** 5.48 and 5.62 require destruction to be recorded; 211.125(d) requires destruction (n126, n132).
- C-24 **Blocking the next processing step on an out-of-limit yield.** 5.44 requires recording and investigation, and 211.192 requires investigation. A hard stop is a design choice (n115).
- C-25 **Recording the pre-use cleaning inspection inside the batch record.** 211.182 allows it in the equipment log, except for dedicated equipment without a log (n6).
- C-26 **5.38 requires a second person.** It requires an independent, recorded check. The second person comes from US 211.101(c) (n120).
- C-27 **Supersession of masters is required by 4.5.** The requirement exists, but it comes from 4.3 and 4.20/4.21; 4.5 is about periodic review (n98).
- C-28 **Limits on intermediate yields required by EU 4.17(d).** 4.17(d) requires limits on the final yield and mention of relevant intermediate yields where applicable. The kernel requirement comes from the US text (n89; D-9).
- C-29 **FDA CSA guidance is binding on a drug eBMR.** It is a CDRH/CBER device guidance. It can inform the 211.68 and 11.10(a) validation approach but does not bind Part 211 (n71).
- C-30 **Enhanced testing of reprocessed batches as a US rule.** 211.165(f) requires reprocessed material to meet specifications. EU 5.69 requires QC to consider additional testing (n21).

(d) Conflicts between rows and how they were resolved (with official-text evidence)

- D-1 **Printed-material specimens in the packaging master.** n91 allows justified N/A because 4.19(e) says "where appropriate". But 211.186(b)(8) makes a specimen or copy of every label and every other labeling item mandatory in the master, each bearing the dated signature of the labeling approver. **Resolution:** mandatory whenever US is in the market set; N/A with justification only for EU-only products (PR-03, (a)-2 item 5).
- D-2 **Retest/expiry date block at dispensing.** n13 calls a date-based block good practice under 211.87. n120 and n620 tie it to 5.34, which allows only QC-released starting materials within their retest period. **Resolution:** ENFORCED for all markets, because EU 5.34 is binding and stricter (PR-18).
- D-3 **Independent check of dispensed materials.** n120 says 5.38 does not require a second person, so a validated automated check suffices. n13 and guardrail 7 cite 211.101(c): a second person, or one person when automated. **Resolution:** manual mode needs two distinct people; automated mode needs at least one human verifier; zero-human verification is never allowed (PR-20). The same-user option is open question F-1.
- D-4 **Line clearance after packaging.** 5.50 covers only the pre-start clearance (n123). 211.188(b)(6) requires the record to show that the packaging/labeling area was inspected both prior to and following use (n39). **Resolution:** both (PR-34).
- D-5 **Checklist for clearance.** n116 (5.40, no checklist) vs n123 (5.50, checklist for packaging) vs n90 (4.18(c) checks in processing instructions). **Resolution:** checklist ENFORCED for packaging; processing clearance ENFORCED as a check, with the checklist format CONFIGURABLE (default on) (PR-26, PR-34).
- D-6 **Labeling reconciliation waiver.** 211.125(c) waives labeling reconciliation only for cut or roll labeling with a 100% examination under 211.122(g)(2). 4.21(h) allows the reconciliation quantities to be omitted from the record where robust electronic controls exist. 5.61 still requires discrepancies to be explained before release (n131). **Resolution:** reconciliation always runs by default. The waiver needs the US condition plus, when EU is in the set, documented robust controls. Bulk and batch-coded printed-material reconciliation is never waived when EU is in the set (PR-39).
- D-7 **Retention period.**
  - The candidates are: 211.180(a) at 1 year after expiry (n87, n1); 4.11 at the longer of expiry + 1 year and certification + 5 years (n103); DSCSA 6 years (n82, (s)); complaint-driven extension (n42); guardrail 17.
  - The complaint-driven extension is not required: 211.198(b) governs the complaint record only.
  - **Resolution:** the maximum of the applicable floors (PR-59).
- D-8 **Release while a deviation or investigation is open.**
  - Guardrail 15 blocks disposition while any deviation is open.
  - n214 allows release with a QA-justified exception, and n188/n223 (Annex 16 1.7, (s)) require investigations complete enough to support certification. 211.192 requires review and investigation but does not say "closed before release".
  - **Resolution:** CONFIGURABLE. The default is the guardrail block. The exception needs an e-signed QCU/QP justification, and never applies to reconciliation discrepancies (5.61) or registered-specification failures (PR-53).
- D-9 **Yield limits at intermediate phases.** n89 treats intermediate-yield limits as good practice that 4.17(d) does not require. Guardrail 22 requires min/max at each yield stage. 211.103 requires yields at each appropriate phase, and 211.186(b)(7) ties investigation to master min/max limits. **Resolution:** min/max required at every configured yield checkpoint (US-driven interpretation; see F-12) (PR-31).
- D-10 **Rejected-material storage.** 211.89 and 211.110(d) accept a quarantine system that prevents use, and 211.42(c) accepts "other control systems". 5.66 requires separate storage in restricted areas (n134). **Resolution:** electronic hard status plus a physical restricted location (PR-46).
- D-11 **Superseded-version control.** An earlier claim attributed it to 4.5. n98's correction, checked against the text, shows 4.5 covers only periodic review; blocking comes from 4.3 (effective date) and 4.20/4.21 (currently approved instructions). **Resolution:** enforced, with corrected citations (PR-05).
- D-12 **Returned goods.** 211.204 lets examination or testing prove quality when storage conditions cast doubt. 5.70 requires destruction unless quality is beyond doubt, and forbids re-issue once any doubt arises (n137). **Resolution:** the EU rule for all markets (PR-48).
- D-13 **Checker on significant steps.** 4.20(c) and 4.21(c) require the checker only "where appropriate". 211.188(b)(11) requires the performer and the person supervising or checking each significant step (n39). **Resolution:** the US rule for all markets (PR-14).
- D-14 **Logbook scope.**
  - 211.182 covers major equipment and exempts dedicated equipment whose records are in the batch record.
  - 4.31 requires logbooks for major or critical production and analytical equipment and for areas where product has been processed, and has no exemption.
  - n87 says room-level trace is not required, which is true only for the US.
  - **Resolution:** area logbooks when EU is in the set; the dedicated-equipment exemption is US-only (PR-24).
- D-15 **Scope of automated single-person checking.** n9's correction: 211.68(c) covers only 211.101(c)/(d), 211.103, 211.182 and 211.188(b)(11). The EU has no counterpart, but Annex 11 section 6 accepts validated electronic means as the accuracy check for manually entered critical data. **Resolution:** the automation flag is restricted to those step types and never relaxes other two-person rules (PR-14).
- D-16 **Scope of OTC tamper-evident packaging.** n25 was corrected: the trigger is OTC retail status, not dosage form (211.132(a)); insulin is exempt. **Resolution:** a product-master flag drives it; Tablets edition only in practice (PR-41).
- D-17 **Overage vs "substance that may disappear".** n11 treats overage as a 211.186(b)(5) statement. n89 says an overage is not a lost substance under 4.17(c). **Resolution:** separate fields, calculated excess and lost-in-process flag; both kept (PR-03).
- D-18 **Additional-testing decision.** n135 applies the 5.69 QC decision to reprocessed batches only. n136 notes that 5.69 covers both reprocessed and recovered-material batches. **Resolution:** the decision is mandatory for both (PR-47).
- D-19 **Status of Annex 16 section 3.** n189 is labelled guidance (it merges the EMA Q&A) and n224 is labelled binding, for the same section. **Resolution:** Annex 16 section 3 is part of EU GMP and binding (s). The EMA Q&A details (recurrence as a change; registered in-process specifications; the certified-batch register) are treated as guidance-level configuration (PR-49).
- D-20 **Deviation approver.** 5.15 names a competent person, with QC involved when appropriate; n119 routes to QA; guardrail 12 makes "exception or formal deviation" a QA decision. **Resolution:** the competent-person approval is ENFORCED; QC/QA routing is CONFIGURABLE by class (PR-13, PR-49).
- D-21 **211.186(a) handwritten signature vs an all-electronic master route.** Guardrail 8 is silent. 11.1(c) makes Part 11 e-signatures equivalent to full handwritten signatures, and 11.100(c) requires the certification to FDA. **Resolution:** e-signature, with a tenant-level record of the 11.100(c) certification (PR-02).

(e) "Not an eBMR concern": rows with reasons

- n71 FDA Computer Software Assurance guidance (device-centre guidance, Sept 2025, revised Feb 2026, (s)). It governs how the eBMR is validated, meaning the vendor's and customer's assurance activities. It is not a runtime production control. It belongs in the validation plan and the data-integrity/CSV catalog. It is not binding under Part 211 (C-29).
- n144 EU Ch.5 5.71, which requires telling the MAH about manufacturing constraints that could restrict supply. This is an organisational reporting duty between manufacturer and MAH, and no eBMR function is mandated. Optional support, a supply-impact flag on deviations, is offered in PR-49 as a design choice.
- n611 21 CFR 211.180(f), which requires written notification of responsible officials about complaint, return and salvage investigations, recalls, FDA observations and regulatory actions. This is a QMS and management workflow. The eBMR only supplies the lot links and investigation records (PR-50, PR-57).

(f) Open questions

- F-1 **Automated dispensing under EU 5.38.** Is a validated automated check, with the operator as the only human signer, "independent" for EU purposes? Kernel default: a different human verifier. Also confirm against India Schedule M, which may require a second person regardless of automation.
- F-2 **EU retention for batches never QP-certified** (rejected, destroyed, abandoned). 4.11 is keyed to certification. Provisional default: the later of expiry + 1 year and disposition + 5 years. QA must confirm.
- F-3 **EU import route for Indian sites.** The obligations of Directive 2001/83/EC Art. 51 are not in the uploaded texts: EU QP certification after importation, EU re-testing of imported batches, and whether any mutual-recognition arrangement covers India. The eBMR's role needs confirmation against the official texts: record access for the QP, and recording the external certification reference.
- F-4 **Annex 16 rows are secondary** (n187-n192, n223, n224). Confirm the numbering of sections 1.7 and 3, the Appendix I fields, and the 5-year register retention against the official Annex 16 text.
- F-5 **Annex 15 rows are unverified** (n195-n199). Confirm section numbers (5, 10, 11, 4) and the wording on hold times, OPV and concurrent validation.
- F-6 **Status of drafts:** EU Ch.4 revision (Jul 2025), Annex 22 (Jul 2025), the FDA 211.110 draft guidance (Jan 2025), and the Annex 11 revision. Re-check before each release of the catalog.
- F-7 **Secondary US dates and figures.** Confirm the NDC final rule dates (published March 5, 2026; effective March 7, 2033; transition to March 6, 2036) and the DSCSA figures (6-year retention, 24-hour responses) against official sources.
- F-8 **EU reference and retention samples** (Annex 19) are not in the uploaded texts. Confirm the quantities and retention periods for EU-market batches.
- F-9 **Start of shelf life** (e.g. manufacturing date vs first contact of active with excipients) is not defined in the texts reviewed. It must be a QA-approved product rule.
- F-10 **211.180(b)** keeps component and labeling records for 1 year after "the expiration date", without saying whose. Kernel reading: the last finished lot that used the material. Confirm.
- F-11 **Batch-size ranges under one master.** 211.186(a) and the Ch.4 lead-in above 4.17 require masters for each batch size. Is one master with a validated, registered size range and linear scaling acceptable? Kernel default: discrete approved sizes only.
- F-12 **Intermediate-phase yield limits (US).** The text does not say explicitly that every intermediate yield needs min/max. The kernel requires them at every configured checkpoint (D-9); confirm the customer's regulatory position.
- F-13 **Line clearance as a "significant step"** (dual signature). Recommend the customer's SOPs decide this explicitly. Kernel default: dual.
- F-14 **Part 210.3 definitions** (lot number, yields) are outside the uploaded texts (n1). Confirm wording before quoting them in validation documents.
- F-15 **EU GMP Ch.1** (1.4(xiv) root cause; 1.10 PQR scope) is cited only from secondary rows (n214, n216, n228). Verify against the official Ch.1 text.

PART C

Sterile manufacturing controls

Injectables edition · 79 controls

Conventions

- **EU** = EU GMP Annex 1 (2022). In force since 25 Aug 2023; clause 8.123 (lyophiliser sterilisation frequency) applies from 25 Aug 2024. Annex 1 is mostly written as "should". The Scope section, however, says that any specific limit, frequency or range is a **minimum requirement**. The few declarative or "must/need to" statements are marked **[EU hard]**.
- **US** = 21 CFR 211 ("shall", binding). **FDA-AP** = FDA guidance *Sterile Drug Products Produced by Aseptic Processing* (Sept 2004). It is nonbinding apart from the text boxes that restate Part 211.
- **Market-profile rule (stricter wins).** Each product carries a list of markets (US, EU/PIC/S, others). Product-level rules use the strictest rule among those markets. Line-, room- and person-level rules (APS status, EM limits, requalification, operator qualification) use the strictest rule across every product qualified on that line or room, because one set of APS/EM evidence supports all of them. The default shipped profile is "EU + US". A US-only relaxation can be chosen only for a line that makes no EU/PIC/S product.
- **Type.** ENFORCED = a hard rule in code, not configurable. CONFIGURABLE = a parameter with a shipped default and an allowed range, which can be tightened but never loosened beyond the floor (ST-01). DESIGN-CHOICE = a product design proposal that the text does not demand; it can be switched off per site with a recorded rationale.
- **Gate vocabulary.** *Block* = the step cannot be started or signed. *Deviation-gate* = work continues only after a deviation is raised and QA records a decision. *Release block* = the QA disposition / QP certification e-signature is unavailable.
- Scope values: aseptic, terminal, lyo, all, and the technology profiles isolator, rabs, bfs, ffs, sus.
- "Covers" lists requirement numbers (see Appendix) (nNNN).

Theme 0: Cross-cutting rules (apply to every theme)

ST-01 Regulatory-floor table for every numeric limit

Enforced | Configurable | Injectables | Scope: all | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• The system ships a locked, versioned table of every numeric floor used in this catalog. Examples: Table 5/Table 6 limits; 6-month (Grade A/B) and 12-month (Grade C/D) requalification; about 6-month APS per process, line and shift; 12-month operator APS, gowning and visual-inspection requalification; 4 h maximum settle-plate exposure; 28 L/min minimum Grade A counter flow; weekly steriliser leak test; daily air-removal test; annual worst-case heat-cycle revalidation; 0.22 µm maximum sterilising-filter rating; 100 mL fusion-seal CCI threshold. Each entry stores its source, clause, market profile and whether it is a **hard number** or an Annex 1 **guidance value**. • Saving a site value looser than a hard number is rejected, with no override. Loosening an Annex 1 *guidance value* (10 Pa step between grades, clean-up time under 20 min, 0.36–0.54 m/s UDAF speed) needs a CCS justification reference and QA approval, and is shown on every affected batch. • Every limit is versioned. Each result, check and timer stores the limit version it was evaluated against.
TYPE DETAIL	ENFORCED (floor table locked). Individual site values are CONFIGURABLE only in the tightening direction.
REGULATORY BASIS	EU Annex 1: 1 (Scope, para 2), 4.14, 4.29(iii), 4.30 | US: none equivalent (FDA-AP Table 1 note c lets firms set alternative microbial action levels with justification).
INDIA / US / EU DIFFERENCES	EU is stricter. It turns stated numbers into minimums, while FDA-AP allows justified alternative microbial action levels.
HOW IT WILL BE TESTED	Configure a Grade B active-air action limit of 15 CFU/m³ → rejected. Configure a 7 Pa B/C differential → accepted only with a CCS reference and QA e-signature.

Source requirements: n391

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-02 Product sterility-assurance route, CCS linkage and master-record design checks

Enforced | Configurable | Injectables | Scope: all | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES	• The product master requires a *sterility-assurance route*: terminal / aseptic / aseptic + post-aseptic heat treatment (or another adjunct step). For any route other than terminal, a justification reference is mandatory. The route selects the master-record template and rule set (Themes 7–13). • MBR approval is blocked unless every step marked as a CCS critical control point carries the CCS document ID and version. The executed batch header shows the CCS version in force on the execution date. • Master-record design review: a manual aseptic assembly or connection step where SIP or a pre-assembled sterile kit is feasible raises a mandatory "justify manual manipulation" item (DESIGN-CHOICE). • Optional flag "Annex 1 principles applied to a non-sterile product" with a mandatory list of the principles applied. This is used when the edition is reused for low-bioburden intermediates, and it can never produce a sterility claim. • DESIGN-CHOICE: Grade A/B steps are paperless, using cleanroom-rated devices (see ST-13), because paper sheds fibres.
TYPE DETAIL	CONFIGURABLE (route per product), plus ENFORCED master-record approval checks.
REGULATORY BASIS	EU Annex 1: 1 (Scope, para 3), 2.3, 2.5, 8.7–8.9, 8.15, 8.34, 8.37 | US: 211.113(b); FDA-AP II.B and III (aseptic processing only when terminal sterilisation is not feasible).
INDIA / US / EU DIFFERENCES	Aligned on preferring terminal sterilisation. The CCS is an EU concept; FDA has no CCS requirement, so the CCS fields are EU-driven.
HOW IT WILL BE TESTED	Submit an aseptic product master with no route justification → approval refused. Open an executed batch → every CCP step shows the CCS ID and version effective on that date.

Source requirements: n482, n381, n394, n459, n460, n392

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-03 Change control drives requalification/revalidation blocks

Enforced | Injectables | Scope: all | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every change to an MBR, recipe, limit, line, room, steriliser or filter type needs a CCS-impact assessment before approval, and a post-implementation effectiveness check before closure. Neither can be skipped.

- A change control that sets "requalification/revalidation required" on an asset makes that asset unusable in batch execution until the change is closed or QA records an interim decision. The pre-start check cites the change-control ID.
- Changes tagged "affects aseptic process" feed the APS status engine (ST-29).

REGULATORY BASIS	EU Annex 1: 2.6, 4.32 (requalification after changes) | US: 211.100(a); FDA-AP IX (intro).
INDIA / US / EU DIFFERENCES	Aligned. EU adds the CCS-impact element.
HOW IT WILL BE TESTED	Open a change control on filling line 2 with "requalification required" → batch start on line 2 is blocked. Close it → start allowed, citing the change-control ID.

Source requirements: n396, n318

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-04 Contemporaneous recording and departures from the MBR

Enforced · Injectables · Scope: all · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

- All steps carry server-synchronised timestamps and attributed e-signatures. An entry made more than a configurable window after the recorded activity is flagged "late" and needs a reason. The window is a DESIGN-CHOICE, default 15 min; neither text gives a number.
- A skipped or out-of-sequence step, an out-of-range parameter or substitute equipment forces a deviation with justification before the next dependent step.
- MBRs cannot be executed until QA has approved them.

REGULATORY BASIS	EU Annex 1: 8.7 (defined process) | US: 211.100(a)-(b), 211.186, 211.188; FDA-AP XII (text box).
INDIA / US / EU DIFFERENCES	Aligned. The US wording is binding.
HOW IT WILL BE TESTED	Sign a step 3 h after the activity → flagged "late" and a reason required. Execute step 7 before step 6 → deviation required.

Source requirements: n366

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-05 Instrument calibration status checked at point of use

Enforced · Injectables · Scope: all · All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- Every instrument whose reading enters the batch record is referenced by ID and checked against the calibration system at the moment of use. This covers balances, checkweighers, particle counters and probes, air samplers, integrity testers, steriliser/tunnel/lyo sensors, DP gauges and incubators. An overdue or failed instrument is blocked. Its reading is accepted only through a deviation, with a retrospective impact assessment once as-found data are known.
- An as-found out-of-tolerance result automatically lists every batch and result produced since the last good calibration, for impact assessment.
- Particle-counter records carry counter and probe ID. The ≤1 m tubing guidance is a qualification item and is not re-checked per batch.

REGULATORY BASIS	EU Annex 1: 5.9 (particle counters and tubing qualified) | US: 211.160(b)(4), 211.68; FDA-AP X and XI (text boxes), IX.C.2.
INDIA / US / EU DIFFERENCES	Aligned. The US rule is binding.
HOW IT WILL BE TESTED	Record a fill-weight check on a balance whose calibration expired yesterday → refused or deviation-gated. Enter an as-found failure on an integrity tester → the impacted results list is generated.

Source requirements: n384, n434

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-06 Batch alarm and event register (equipment, utilities, power, interlocks)

Enforced · Configurable · Injectables · Scope: all · All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- For every room, line, isolator, steriliser, lyophiliser and utility used by the batch, the eBMR imports EMS/BMS/SCADA alarms and events for the whole execution window. This includes DP, particle, T/RH, interlock override or simultaneous door opening, leak or backflow alarms, and power events. Each item needs an acknowledgement (user, time) and an assessment of batch impact.
- A *critical* alarm (per master data) pauses the step until an authorised reviewer records an evaluation. Any alarm that breached a limit must be linked to a deviation.
- **Any power interruption that could affect quality, however short, automatically creates a deviation** linked to the batch (US text-box rule).
- A data-feed gap longer than a configurable limit (default 5 min) is treated as an alarm.
- Release is blocked while any alarm in the window is unassessed.

TYPE DETAIL	ENFORCED (register and assessment). Criticality mapping is CONFIGURABLE.
REGULATORY BASIS	EU Annex 1: 4.13, 4.16, 5.2, 6.20–6.22 | US: 211.100, 211.192; FDA-AP IV.C, XII (power interruption).
INDIA / US / EU DIFFERENCES	FDA-AP is more explicit that every power interruption is a deviation. EU adds a requirement for immediate review of critical alarms. Apply both.
HOW IT WILL BE TESTED	Inject a 2 s power dip on the filler feed → deviation auto-created. Leave one of three alarms unassessed → release blocked. Cut the EMS feed for 30 min → a gap item appears.

Source requirements: n428, n365, n444, n409, n281

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 1: Personnel qualification and gowning gate

ST-07 Role and training check at every e-signature (all cleanroom roles)

Enforced | Injectables | Scope: all | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- "Operator" covers every person who records work in a classified area: fillers, set-up, maintenance engineers, cleaners, EM/PM samplers and QA observers. At each e-signature the system checks the LMS for current training on the aseptic curriculum (microbiology basics, hygiene, cleanroom behaviour, gowning, patient-safety consequences) and on the SOPs linked to the step, keyed to the grade and task criticality. Missing or expired training → the signature is refused with a reason.
- Sampling tasks (EM, PM, bioburden, sterility sample pulls) need an "aseptic sampler" qualification. LIMS results carry the analyst ID, and review checks that the sampler and analyst were qualified on the date.
- Investigations of microbiological events need an approver with a microbiology / sterility-assurance role.
- A snapshot of training status at signing is stored in the batch record.

REGULATORY BASIS

EU Annex 1: Glossary (Operator), 7.3, 10.1 | US: 211.25(a)-(c); FDA-AP V.A, V.B.

INDIA / US / EU DIFFERENCES

Aligned. FDA-AP V.B explicitly extends this to lab and sampling staff.

HOW IT WILL BE TESTED

A maintenance technician with expired cleanroom training signs an in-room repair step → refused. An EM task completed by a user without the sampler qualification → refused.

Source requirements: n446, n291, n583, n296, n573

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-08 Unsupervised Grade A/B access gate (gowning qualification + APS participation)

Enforced | Configurable | Injectables | Scope: aseptic | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- Before a person may sign any Grade A/B aseptic step without supervision, the system requires all of the following:
  - gowning qualification passed, with visual and microbial assessment (gloved fingers, forearms, chest, hood/mask/forehead). Results are judged against the Table 6 contact-plate and glove-print limits for the grade concerned (Grade B: 5 CFU; Grade A: no growth), per Table 2 note 3. The qualification must be no more than 12 months old;
  - aseptic training current;
  - participation within the last 12 months in a successful APS on this line or process, **in the role or activity being signed**. Roles include filling, set-up, maintenance intervention, EM sampling and SUS assembly/connection.
- If any check fails, the person is routed to "supervised only" mode, if the site allows it, or blocked.
- The qualification snapshot for each person is stored in the batch roster.
- The site can shorten the gowning requalification interval for manual or intervention-heavy processes. FDA-AP says annual is normally enough only for automated, low-intervention operations.

TYPE DETAIL

CONFIGURABLE. Default: gowning requalification ≤12 months and a successful APS ≤12 months. Allowed range: shorter only. ENFORCED: no unsupervised Grade A/B signature without both.

REGULATORY BASIS

EU Annex 1: 7.4, Table 2 note 3, 8.139, 9.38 (each operator ≥1 successful APS per year) | US: 211.25(a), 211.28(a),(c); FDA-AP V.A, IX.A.2.

INDIA / US / EU DIFFERENCES

Operator APS participation is annual in both. Gowning: EU says at least annually for everyone. FDA says annual is usually adequate for automated operations and implies more frequent requalification for manual work or after adverse findings. Stricter result = ≤12 months for everyone, shortened for manual processes.

HOW IT WILL BE TESTED

Gowning requalified 12 months + 1 day ago → Grade B signature blocked. A technician whose APS covered only EM sampling signs a needle-change intervention → blocked or flagged.

Source requirements: n447, n292, n321, n539

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-09 Disqualification, restriction and requalification triggers

Enforced | Configurable | Injectables | Scope: aseptic | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- Person status can be qualified, supervised-only, retrain, requalification-required or disqualified. The status changes automatically on any of these:
  - a personnel-monitoring action result or an adverse PM trend (rule set in ST-22);
  - implication in a failed APS (ST-33);
  - a gown breach or an observation finding (ST-11, ST-13);
  - manual QA action.
- A disqualified or requalification-required person cannot sign Grade A/B steps on **any** open batch.
- Clearing the status needs QA sign-off referencing linked retraining, a gowning assessment and (default on) a successful APS.

TYPE DETAIL

ENFORCED (the triggers exist). Trend thresholds are CONFIGURABLE.

REGULATORY BASIS

EU Annex 1: 7.6 | US: 211.25(a); FDA-AP V.A, V.C.

INDIA / US / EU DIFFERENCES

Aligned. EU is explicit that requalification should consider a successful APS.

HOW IT WILL BE TESTED

Mark an APS failed → all implicated operators become restricted and their aseptic signatures are refused until the requalification records, including a successful APS, are linked.

Source requirements: n449, n295

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-10 Access authorisation, health declaration and exceptional supervised entry

Enforced | Configurable | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Aseptic-area sign-in checks that the person is on the supervisor-authorised access list and records a fitness-to-enter declaration. This covers illness, open lesions, and recent work with human/animal tissue or microbial cultures. • Declared culture or tissue work blocks entry until a documented decontamination record exists. • A supervisor or medical exclusion flag blocks product-contact steps until a clearance with a medical reference is recorded. The flag state at each signature is kept. • Exceptional entry of an unqualified person (for example a contractor) into Grade B, or into Grade A in operation, needs an "exceptional supervised entry" record: person, reason, times in and out, supervising authorised person (whose qualification is checked), and a cleanliness impact assessment. The record links to every batch in progress in that area and appears in batch review.
TYPE DETAIL	ENFORCED (access list, medical exclusion). The self-declaration form is CONFIGURABLE.
REGULATORY BASIS	EU Annex 1: 7.5, 7.7, 7.8 | US: 211.28(c)-(d); FDA-AP V (text box).
INDIA / US / EU DIFFERENCES	Aligned. The US health-exclusion rule is binding.
HOW IT WILL BE TESTED	Set a medical exclusion → filling signatures refused until clearance. Record a contractor entry during filling → impact assessment mandatory and entry visible in batch review.

Source requirements: n450, n383, n448

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-11 Per-entry gowning record, garment status, wear time and glove events

Enforced | Configurable | Injectables | Scope: all (Grade A/B rules: aseptic) | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each entry is logged with garment and sterile-pack lot, sterilisation expiry, packaging-intact check, visual checks before and after donning, and an integrity check on exit. An expired or unsterilised pack is rejected at scan. • Grade A/B entries need a fresh sterile gown set (suit, hood, mask, eye cover, double sterile gloves, over-boots) on **every** entry. • Garment type is checked against the grade: Grade C (hair covered, high-neck suit, disinfected shoes) and Grade D (hair covered, protective suit). • A per-operator wear timer starts at gowning and warns before, and alerts at, the qualified maximum wear time. • If RFID is used, the garment's laundry/sterilisation cycle count must not exceed the qualified maximum. • Quick-log events cover glove sanitisation, glove change, gown change, gown or glove breach and instrument replacement. A breach during exposed-product work requires an immediate change record and an assessment of units exposed in the breach window.
TYPE DETAIL	ENFORCED (sterile-pack checks, breach handling). Maximum wear time and glove-disinfection interval are CONFIGURABLE (default = the value from garment qualification; no shipped numeric default because Annex 1 sets none).
REGULATORY BASIS	EU Annex 1: 7.10–7.17 | US: 211.28(a); FDA-AP V.A.
INDIA / US / EU DIFFERENCES	EU is more prescriptive: per-grade garment list, fresh sterile garments on every A/B entry, qualified maximum wear time and maximum laundry cycles. Apply EU.
HOW IT WILL BE TESTED	Scan a gown pack past its sterile hold → rejected. Log a torn glove during filling → change record and impact assessment required.

Source requirements: n452, n453, n454, n294

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-12 Headcount limit and personnel presence log

Enforced | Configurable | Injectables | Scope: aseptic | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Records who was present in each aseptic room, with entry and exit times, from access control or sign-in/out. • Computes peak concurrent headcount and compares it with the room maximum. Exceeding it raises a batch event and a deviation. • Counts transfers into the critical zone. • The roster is printable in the batch report.
TYPE DETAIL	CONFIGURABLE (maximum per room is master data, must be ≤ the maximum qualified in the latest successful APS). ENFORCED: deviation when exceeded.
REGULATORY BASIS	EU Annex 1: 7.1, 7.2 | US: 211.42(b) (inferred); FDA-AP IV.E.
INDIA / US / EU DIFFERENCES	Aligned. EU requires the maximum operator number to be documented and challenged in APS.
HOW IT WILL BE TESTED	APS qualified 4 people and 5 are present during filling → deviation, with the peak shown on the timeline.

Source requirements: n445, n285

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-13 Aseptic behaviour observation, QA oversight and in-cleanroom device rules

Configurable | Design choice | Injectables | Scope: aseptic | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• Batch-linked observation record: observer (must hold the aseptic-expert, supervisor or QA role), time window, operator observed, checklist result, findings and coaching. A finding can raise a deviation, and it feeds ST-09. • Line readiness shows when the next observation is due. Batch review shows whether QA oversight took place. • Device rules: only registered, grade-rated tablets and scanners may open sessions in classified areas, with a disinfection record at transfer-in. Station layout keeps data entry out of first-air paths. Entry after the activity is allowed where the MBR permits it.

TYPE DETAIL	CONFIGURABLE (observation frequency per line; default at least one expert observation per line per month, site may tighten). Device rules are DESIGN-CHOICE.
REGULATORY BASIS	EU Annex 1: 7.9, 7.18, 8.19 | US: 211.22(a), 211.25(b); FDA-AP V.A.
INDIA / US / EU DIFFERENCES	Aligned on regular observation. Neither text sets a frequency, so the 1/month default is a design choice.
HOW IT WILL BE TESTED	A user without the expert role creates an observation → refused. Log in from an unregistered tablet in Grade B → refused.

Source requirements: n470, n293, n455, n451

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 2: Cleanroom/area status and environmental and personnel monitoring linked to the batch window

ST-14 Room/zone grade, classification and qualification gate at step start

Enforced | Configurable | Injectables | Scope: all | All 9 source requirements checked against official text

WHAT THE SYSTEM DOES

- Room/zone master data holds the EU grade, ISO class and FDA class, plus qualification status and expiry. These are held separately from routine EM, and EM data never counts as qualification evidence.
- Particle equivalences for mapping: Grade A ≈ ISO 5 (in operation); Grade B in operation ≈ ISO 7; Grade C in operation ≈ ISO 8. Grade D has no predetermined in-operation limit.
- Every MBR step names a minimum grade. Exposed sterile product, open primary containers, aseptic assembly and aseptic connections need Grade A with a Grade B background (isolators: ST-74).
- FDA support rooms: the room next to the aseptic line must be at least ISO 7 in operation; ISO 8 is allowed for less critical work such as equipment washing.
- At step start and at line clearance, the system blocks when any of these holds:
  - the room qualification is expired;
  - requalification is overdue (last requalification more than 6 months ago for A/B, more than 12 months for C/D);
  - a "requalification required" flag is open, for example after an air-movement interruption, an HVAC design or set-point change, a final-filter change, special maintenance or remedial action;
  - the room grade is below the step's requirement.
- The status and due date in effect at execution are stored in the batch record.
- Each step is bound to a room or zone. The scanned location must match. The same room cannot run incompatible activities or two batches at once; component preparation, product preparation and filling are kept separate.
- The 211.42(c)(10) facility systems (T/RH control, HEPA air under positive pressure, EM system, cleaning/disinfection system, maintenance of aseptic-control equipment) appear as room status attributes that the batch reads. It does not re-verify them.
- DESIGN-CHOICE: the scheduler warns if classification sampling is planned inside a commercial batch window, and allows an APS to be tagged as the reclassification run.

TYPE DETAIL	ENFORCED. The maximum requalification intervals are locked: Grade A/B ≤6 months, Grade C/D ≤12 months. Mapping each step to a grade is CONFIGURABLE within the Annex 1 grade rules.
REGULATORY BASIS	EU Annex 1: 4.2, 4.4, 4.23–4.26, 4.32 | US: 211.42(c)(10)(i)-(vi), 211.46(b); FDA-AP IV (intro), IV.B, Table 1.
INDIA / US / EU DIFFERENCES	EU is stricter. It sets fixed maximum requalification intervals (6 and 12 months) and at-rest plus in-operation limits. FDA-AP has no fixed room requalification interval; only HEPA leak testing is about twice a year (ST-17).
HOW IT WILL BE TESTED	A Grade B room last requalified 6 months + 1 day ago → step start blocked. Set a "final filter changed" flag → blocked until requalification is recorded. An aseptic connection step mapped to a Grade C room → MBR validation rejects it.

Source requirements: n404, n421, n424, n275, n279, n402, n5, n388, n422

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-15 Room cleaning and disinfection status

Enforced | Configurable | Injectables | Scope: all | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- The room log records:
  - cleaning **before** disinfection; disinfection without a preceding cleaning entry is flagged;
  - agent, lot, sterility status, preparation time and in-use expiry, contact time, areas and sequence, and operator.
- Agents used in Grade A/B must be sterile (vendor CoA/CoC after qualification, or in-house sterilisation record). Grade C/D agents must be sterile where the CCS says so. An expired dilution or non-sterile lot is rejected at scan.
- Rotation rules apply: at least two agents with different modes of action, and a sporicide at the defined frequency. A spore-former recovery automatically creates a sporicidal-clean task that must be done before the next aseptic batch in that area.
- Fumigation or VHP cycles are recorded with the validated cycle ID. After a failed cycle the room stays unavailable.
- The room shows "ready" at step start only if disinfection and sporicide are not overdue and no expired agent was used.

TYPE DETAIL	ENFORCED (sequence, sterility of agent, expiry). Frequencies and rotation are CONFIGURABLE per validated programme; Annex 1 gives no numbers.
REGULATORY BASIS	EU Annex 1: 4.33–4.36 | US: 211.42(c)(10)(v), 211.56(b)-(c); FDA-AP X.A.3.
INDIA / US / EU DIFFERENCES	EU requires cleaning before disinfection and more than one disinfectant type. FDA-AP requires a sporicide on a written schedule and when spore-formers appear. Apply both.
HOW IT WILL BE TESTED	Use a disinfectant past its in-use expiry → line clearance fails. Record a *Bacillus* recovery → a sporicidal-clean task is required before the next aseptic batch.

Source requirements: n425, n426, n427, n349

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-16 Return to service after maintenance, shutdown or construction, and clean-up period

Enforced | Configurable | Injectables | Scope: all | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- Maintenance inside a cleanroom on a batch line inserts mandatory follow-up steps before restart: extra cleaning and disinfection, extra EM samples, and re-sterilisation of reassembled parts (sterilised after full reassembly where possible).
- *Unplanned* maintenance on equipment flagged sterility-critical needs a sterility-impact assessment **before** work starts. It is automatically linked to the in-process batch and to adjacent batches.
- Sterilisers, HVAC and water systems can be selected only if qualified, not overdue for PM, and formally released after maintenance.
- After an HVAC or utility shutdown or construction work, a return-to-service record is required before aseptic steps can start. It covers cleaning/disinfection, post-shutdown and pre-start viable samples, and requalification where needed. Those results are linked to the next batch.
- After line clearance or any disruption, a clean-up timer set to the room's qualified period must elapse before the next batch or before aseptic work resumes. Optionally, the timer also waits until particle counts are back within at-rest limits.

TYPE DETAIL

ENFORCED (sequence). Clean-up period is CONFIGURABLE: the value from room qualification, where the Annex 1 guidance value is under 20 min. A longer value needs a CCS justification (ST-01).

REGULATORY BASIS

EU Annex 1: 4.29(iii), 5.3, 5.6, 5.7, 9.23, Table 5 note 1 | US: 211.42(c)(10)(vi), 211.100 (inferred); FDA-AP IV.E.

INDIA / US / EU DIFFERENCES

Complementary. EU adds the clean-up period and the pre-maintenance sterility assessment. FDA-AP stresses written return-to-operation procedures after shutdowns and construction.

HOW IT WILL BE TESTED

Log an HVAC shutdown without a return-to-service record → line clearance fails. Finish line clearance and try to start the next batch immediately → blocked until the clean-up period elapses.

Source requirements: n429, n433, n290, n552, n432, n423

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-17 Pressure differentials, airflow velocity/visualisation and HEPA status

Enforced | Configurable | Injectables | Scope: all (continuous Grade B DP: aseptic) | All 7 source requirements checked against official text

WHAT THE SYSTEM DOES

- For every room pair used, the DP trend for the whole batch window is attached, showing minimum values against limits.
  - Critical DPs, including the Grade B background during aseptic preparation and filling, are monitored continuously with instant alarms.
  - Non-critical pairs are captured as recurring manual readings.
- Any time below the limit during open processing needs an impact assessment. Before the next aseptic step can be signed, a restoration and confirmation record is required, for example the clean-up period plus extra EM. Alarms cannot be cleared without an assessment.
- Door-held-open alarms are compared with a configured maximum door-open time.
- Line readiness checks, with block or deviation-gate when overdue:
  - current airflow-visualisation study reference covering the line's interventions;
  - UDAF velocity check current and within the justified range (EU guidance 0.36–0.54 m/s at the working position; FDA 0.45 m/s ±20%, measured about 15 cm below the filter face and at working height);
  - HEPA in-place leak test current, including tunnel and oven HEPAs. Default interval is 6 months. A single probe reading of about 0.01% of the upstream challenge counts as a significant leak.
- Media-fill and sterility-failure investigation templates generate a HEPA retest task.

TYPE DETAIL

ENFORCED (continuous linkage and assessment). Set points are CONFIGURABLE: floor ≥10 Pa between different grades, and ≥12.5 Pa for an aseptic room next to an unclassified room. A lower value between graded rooms needs a CCS justification (EU guidance value).

REGULATORY BASIS

EU Annex 1: 4.4 (Grade B DP continuous), 4.14, 4.15, 4.16, 4.30, 4.32 | US: 211.42(c)(10)(iii)-(iv), 211.46(b); FDA-AP IV.A (footnote 5), IV.C, IV.D.2.

INDIA / US / EU DIFFERENCES

- DP: EU gives a minimum 10 Pa *guidance value*. FDA-AP gives at least 10–15 Pa between classes and at least 12.5 Pa toward an unclassified room at all times, so the 12.5 Pa rule comes from FDA.
- Velocity: the numeric ranges are equivalent.
- HEPA testing: FDA states twice a year; EU covers it through A/B requalification at ≤6 months. Equivalent.

HOW IT WILL BE TESTED

Replay an 8 Pa dip for 5 min during filling → the batch shows the excursion, the next aseptic step needs a restoration record, and release is blocked until assessed. Set the tunnel HEPA test as expired → the depyrogenation step is blocked.

Source requirements: n405, n410, n412, n280, n277, n411, n284

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-18 Batch EM sampling plan and coverage of the processing window

Enforced | Configurable | Injectables | Scope: all (continuous Grade A viable coverage: | All 10 source requirements checked against official text

WHAT THE SYSTEM DOES

- Each batch gets an EM plan generated from the approved EM programme version: locations, method, frequency per activity, timing (during or at end of operation), air volume or surface area, exposure time and limits. Critical locations are flagged.
- The plan also covers the airlocks, change rooms and pass-throughs actually used, taken from gowning and transfer records.
- Tasks are generated for **every shift** the batch spans. A missed or late sample becomes a deviation.
- **Grade A:** viable monitoring (settle plates and/or active air) must be continuous from the start of aseptic set-up to the last unit sealed.
  - The system checks contiguous coverage, and any gap is an exception.
  - Each settle plate may be exposed for no more than **4 h**; a longer plate is invalid or deviation-gated.
  - For Grade C/D, exposure time (still ≤4 h) and frequency come from QRM.
- **Grade B:** a continuous-viable approach is on by default (CONFIGURABLE by risk).
- End-of-operation surface sampling (room and equipment) is mandatory before line clearance. Critical product-contact surfaces such as needles, stopper bowl and feed tracks can be sampled only after the fill-end event; earlier sampling is blocked.
- EM sampling inside Grade A is logged as an inherent intervention.

- Room T/RH trends are attached with limit checks.
- A product flagged hazardous (live organisms, powders, radiopharmaceuticals) switches the plan to classification checks before and after exposure, extra viable sampling and scheduled monitoring during simulated operations.
- Each task records the sampler device ID (calibration current), actual volume, media lot and sampler identity.

TYPE DETAIL	ENFORCED (coverage checks, 4 h plate maximum, end-of-operation sequencing). Plan content is CONFIGURABLE from the approved EM programme.
REGULATORY BASIS	EU Annex 1: 4.1, 9.4–9.8, 9.19–9.24, Table 6 note (a) | US: 211.42(c)(10)(iv), 211.113(b), 211.160(b); FDA-AP IV.A, X.A.1.
INDIA / US / EU DIFFERENCES	EU is stricter. It requires continuous Grade A viable monitoring including set-up, and a maximum 4 h plate exposure. FDA-AP requires sampling each shift, calls settle plates optional, and puts critical-surface sampling at end of operation (same as EU).
HOW IT WILL BE TESTED	Leave a 15-min gap between Grade A plates → coverage exception. Log a plate exposed for 4 h 20 min → flagged invalid. Try a needle-surface sample before fill end → blocked.

Source requirements: n541, n542, n343, n347, n551, n553, n344, n587, n543, n550

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-19 Total particle monitoring (Grade A continuous) and particle excursions

Enforced | Configurable | Injectables | Scope: aseptic | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Grade A monitoring.** The eBMR ingests continuous Grade A counts for **≥0.5 µm and ≥5 µm** at a flow of **≥28 L/min (1 ft³/min)**. Coverage must span all critical processing, including equipment assembly, and the data are aligned on the batch timeline next to interventions. For FDA, the probe is within about 30 cm (1 ft) of the work site and in the airflow, and there is data for every shift the batch spans.
- **Grade B monitoring.** A similar system is used at lower sampling frequency, with alert alarms.
- **Alarms.** An alert-level exceedance raises an alarm, which needs a recorded response. The system prompts for extra microbial sampling.
- **Table 5 monitoring limits** (particles/m³, ≥0.5 µm / ≥5 µm):

Grade	At rest	In operation
A	3,520 / 29	3,520 / 29
B	3,520 / 29	352,000 / 2,930
C	352,000 / 2,930	3,520,000 / 29,300
D	3,520,000 / 29,300	set by the site from risk and routine data

The *classification* table (Table 1) leaves 5 µm "not specified" for Grade A and at-rest Grade B. The 29/m³ Grade A value applies only to *monitoring*.

- **FDA ISO 5 limit.** ≤3,520/m³ ≥0.5 µm during filling and closing.
- **5 µm rule.** An isolated single count is annotated. Consecutive or recurring low-level counts within a configurable window raise an investigation event.
- **Excursions.** Any excursion above the limit becomes a batch event (location, magnitude, duration, severity). It needs an assessment sized to its severity, with a trend look-up, before review can close. A missing counter feed is treated as an alarm (ST-06).

TYPE DETAIL	ENFORCED (continuous Grade A coverage, excursion assessment). Alert levels are CONFIGURABLE from qualification data. Action limits are at most the Table 5 values.
REGULATORY BASIS	EU Annex 1: 9.14–9.18, Table 5 (with notes 1 and 2), Table 1 | US: 211.42(c)(10)(iv); FDA-AP IV.A, X.E.
INDIA / US / EU DIFFERENCES	EU is stricter. It adds 5 µm limits, a 28 L/min continuous requirement and alarm-at-alert. FDA-AP covers only ≥0.5 µm and says continuous systems are *preferred*.
HOW IT WILL BE TESTED	Replay three consecutive samples with 1–2 counts at 5 µm in Grade A → investigation event. Start the Grade A counter 20 min after assembly began → coverage gap.

Source requirements: n548, n549, n547, n276, n353

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-20 EM limits, alert levels and trend rules (EU and FDA profiles)

Enforced | Configurable | Injectables | Scope: all | 7 of 8 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Viable action limits** (EU Table 6, maxima):

Grade	Air (CFU/m³)	Settle plate, 90 mm (CFU/4 h)	Contact plate, 55 mm (CFU/plate)	Glove print, 5 fingers (CFU/glove)
A	no growth	no growth	no growth	no growth
B	10	5	5	5
C	100	50	25	–
D	200	100	50	–

- **FDA Table 1 action levels** (air CFU/m³ / settle plate CFU per 4 h; particles ≥0.5 µm/m³):

ISO class	Air / settle	Particles
5	1 / 1 (normally no growth)	3,520
6	7 / 3	35,200
7	10 / 5	352,000
8	100 / 50	3,520,000

FDA settle plates are optional.

- **Effective limit.** Each location uses the stricter of the EU and FDA values for the markets served. Example: a room classed ISO 8 for US and Grade D for EU gets **100/50** (FDA), not 200/100.
- **Alert levels.** Required for Grade A (total particles only), B, C and D. They are derived from qualification data and periodically re-set from trend data. The system offers no alert field for Grade A viable, because any growth is an action event (ST-21).
- **Trend rules** (configurable; each rule is required):
  - a rising number of alert/action excursions;
  - consecutive alert excursions;
  - recurring isolated action excursions with a common cause, for example after planned maintenance;

- flora change, including a first-seen organism or a spore-former/mould;
- recovery frequency over a rolling window per location.

A triggered rule creates an event linked to every batch in the affected area and period.

- **Result display.** Each result is shown individually against limits; there is no averaged-only view. Records carry location, room, class, shift, operator and batch, so data can be trended along each of these. DESIGN-CHOICE: the review dossier shows 30-day and 12-month snapshots.
- **Rapid methods.** Results from validated rapid or alternative methods may use non-CFU units, with a correlated limit and justification reference. Results from unvalidated methods are refused.

TYPE DETAIL	CONFIGURABLE within floors (ST-01). ENFORCED: no averaged-only views, and no alert tier for Grade A viable.
REGULATORY BASIS	EU Annex 1: 9.9–9.12, 9.28, 9.29, Table 2 note 2, Table 5, Table 6 | US: 211.22(a), 211.42(c)(10)(iv), 211.165(e); FDA-AP IV (Table 1 and notes), X.A.1, X.A.2, X.D.
INDIA / US / EU DIFFERENCES	Grade A is stricter under EU (no growth) than ISO 5 (action 1, normally none). Grade B equals ISO 7 for air and settle plates, and EU adds contact and glove limits. **ISO 8 (FDA 100/50) is stricter than EU Grade D (200/100)**, so it applies for US-market products.
HOW IT WILL BE TESTED	Configure a Grade B air action limit of 15 → rejected. The Grade A viable alert field is absent. Three low-level recoveries at one ISO 7 location in 10 days → the frequency trend rule fires and links to batches.

Source requirements: n544, n545, n346, n348, n74, n274, n556, n352

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-21 Viable excursion handling, identification and batch linkage

Enforced | Injectables | Scope: all (Grade A rules: aseptic) | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Grade A / ISO 5 recovery.** Any recovery above 0 CFU in any sample type (air, settle, contact, glove) automatically opens an investigation. The investigation is linked to every batch whose aseptic window overlaps the sample. Release is blocked until it closes with a documented batch-impact disposition. Rejection is not automatic, because FDA-AP notes a critical-site positive does not by itself require rejection, but a signed rationale is mandatory.
- **Action excursion in any grade.** The result links automatically to **every batch processed in that room/zone from the sampling time until the result was reported**. Those batches go on hold. A root-cause investigation, product-impact assessment and CAPA are required.
- **Alert excursion.** Creates an assessment and follow-up task, and investigation or CAPA is considered.
- **Identification.** Grade A/B isolates must be identified to **species**, and each linked batch needs its own impact statement. A genus-only ID cannot close an A/B record. Grade C/D isolates are identified on excursions, for indicator organisms (spore-formers, moulds) and periodically.
- **Investigation template.** Pre-filled with the context around the sample: interventions, personnel present, PM, particle and DP data, alarms, and the sterilisation records of the sampled item.

REGULATORY BASIS	EU Annex 1: 9.13, 9.30 (Table 6 note c), 9.31 | US: 211.192, 211.42(c)(10)(iv); FDA-AP IV.A (Table 1 note e), X.A.1, X.B.
INDIA / US / EU DIFFERENCES	EU is stricter on identification: species level for A/B, where FDA-AP accepts genus "where appropriate". Both expect investigation of any ISO 5 / Grade A growth.
HOW IT WILL BE TESTED	Enter an action result on day 5 for a sample taken on day 1 → all batches in that area on days 1–5 are linked and held. Enter 1 CFU on a Grade A plate → automatic investigation and release block.

Source requirements: n557, n278, n546, n558, n345

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-22 Personnel monitoring tasks and results

Enforced | Configurable | Injectables | Scope: aseptic | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- For every person on the batch's aseptic roster, the system generates:
  - glove prints (both hands, five fingers), at least daily or in association with each lot (FDA), plus at every exit from Grade B for gloves **and** gown (EU);
  - periodic in-process samples;
  - extra samples for high-intervention or manual-process operators (enhanced gown sampling per CCS).
- **After a critical intervention closes:** each involved operator gives a glove sample. An **outer-glove change** must be recorded before that operator's next action, and the next action is blocked until it is. If a gown sample was taken, a re-gowning record is required before further cleanroom activity.
- **Anti-masking rule:** the glove-sample task records the operator's last glove sanitisation. A sanitisation within a configurable interval before the print (DESIGN-CHOICE default 5 min) needs a justification or a re-sample.
- Where production staff take samples, the frequency of QA oversight is tracked.
- Limits: Grade A gloves no growth; Grade B gloves and gown contact plates 5 CFU.
- Batch review is blocked until every roster member's results are back and assessed.
- An alert or action PM result is linked to every batch the person worked that session. An action result opens an investigation and feeds the person's status (ST-09).

TYPE DETAIL	ENFORCED (sampling after critical interventions and at Grade B exit, outer-glove change). Frequencies are CONFIGURABLE, never below: gloves at least daily or per lot (FDA allows either), and on each Grade B exit.
REGULATORY BASIS	EU Annex 1: 9.25–9.27, Table 6 | US: 211.113(b), 211.42(c)(10)(iv), 211.192 (inferred); FDA-AP V.C, X.A.
INDIA / US / EU DIFFERENCES	Apply both. EU adds sampling on every Grade B exit and after critical interventions, followed by a glove change. FDA-AP adds daily or per-lot glove sampling and forbids sanitising gloves just before sampling.
HOW IT WILL BE TESTED	Close a corrective critical intervention and attempt the next step without a glove-change entry → blocked. Four operators on the roster but only three glove samples → review flags the missing sample.

Source requirements: n554, n555, n297, n298, n299

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-23 EM media qualification and incubation conditions

Enforced | Configurable | Injectables | Scope: all | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• A media lot for EM, PM or APS can be issued only with a passing growth-promotion record. The test uses pharmacopoeial reference organisms and representative local isolates, run by the site, or by the supplier where reliance is justified with transport considered. • Each sample records media lot, GPT status, neutraliser (where disinfectant or product residues could inhibit growth) and incubation conditions. • FDA example scheme for the shipped default: 30–35 °C for 48–72 h (bacteria) and 20–25 °C for 5–7 days (yeasts/moulds). • A result on a lot whose GPT later fails is invalidated, and a resample or deviation is required. • Sterility-sample exterior decontamination (VHP/UV) must be a validated method that does not reduce test sensitivity (ST-65).
TYPE DETAIL	ENFORCED (growth promotion before use). The incubation scheme is CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 10.8, 10.9, 10.11 | US: 211.160(b), 211.165(e); FDA-AP X.B.
INDIA / US / EU DIFFERENCES	Complementary. EU expects GPT with local isolates, normally by the user. FDA-AP gives incubation examples and requires GPT for every prepared lot.
HOW IT WILL BE TESTED	Scan a settle-plate lot without GPT release at the Grade A station → refused.

Source requirements: n579, n350

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-24 EM/PM completeness and review as a release prerequisite

Enforced | Injectables | Scope: all (aseptic **and** terminally sterilis | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• The release package assembles, for every room, zone and line used and the whole processing window: viable and total-particle EM, PM, T/RH, DP, and APS status. • Certification is blocked while any result is outstanding, for example plates still incubating, or while an out-of-trend event is open. • An explicit "EM and trend data reviewed" signature is required. • Exception: the short-shelf-life path (ST-69).
REGULATORY BASIS	EU Annex 1: 9.1–9.3, 10.10 | US: 211.192; FDA-AP XII.
INDIA / US / EU DIFFERENCES	Aligned. EU states explicitly that this applies to terminally sterilised products too.
HOW IT WILL BE TESTED	Attempt certification while two filling-window plates are still incubating → blocked.

Source requirements: n540, n580

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 3: Utilities (WFI, clean steam, gases)

ST-25 High-risk utility points of use linked to steps

Configurable | Injectables | Scope: all | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• The MBR names each high-risk utility point used by each step (WFI outlet, pure-steam header, gas drop). • The batch shows each point's status and trend flag. • An open excursion or adverse trend on a used point becomes a review item.
TYPE DETAIL	CONFIGURABLE. Defaults: WFI, pure steam used as a direct sterilant, and product-contact gases are high-risk; the CCS may add more.
REGULATORY BASIS	EU Annex 1: 6.1–6.4 | US: FDA-AP XII (support-system output is part of the release decision).
INDIA / US / EU DIFFERENCES	Aligned in intent. EU requires critical parameters of high-risk utilities to be trended.
HOW IT WILL BE TESTED	Put an adverse-trend flag on nitrogen drop N-3 → every batch that used N-3 shows a review item.

Source requirements: n435

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-26 Water for injection: draw record, daily sampling, excursions and sanitisation hold

Enforced | Configurable | Injectables | Scope: all | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Draw record.** Each WFI draw records outlet, time and volume. At draw time the system imports loop temperature (for hot storage the validated range, for example circulation above 70 °C), flow or turbulence evidence, and continuous **TOC and conductivity**. Out-of-range values are flagged and need assessment. • **Daily sampling.** At least **one representative sample every day** water is used in manufacture, with all outlets covered over the defined interval. The batch shows the results for the outlets and loop that supplied it. • **Action excursion.** Linked automatically to every batch that drew from the affected outlet or loop in the relevant period. Release is blocked pending investigation of root cause and product impact. • **Alert excursion.** Reviewed to decide whether it is isolated or part of a trend. • **After sanitisation or regeneration:** water use is blocked until chemical results are approved. Batches made with water drawn after the event are tagged, and certification is blocked until post-sanitisation microbial and endotoxin results are approved. • **Tank vent filter.** Integrity is tested before installation and after use. A failed post-use test lists every batch supplied during the filter's service window.

TYPE DETAIL	ENFORCED (post-sanitisation hold, excursion linkage). Loop parameters are CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 6.7–6.15 | US: FDA-AP XII (WFI output reviewed for release); no WFI-specific Part 211 clause in this register.
INDIA / US / EU DIFFERENCES	EU is far more prescriptive (daily sample, continuous TOC/conductivity, post-sanitisation certification hold), so EU governs.
HOW IT WILL BE TESTED	Draw water after chemical release but before microbial release → that batch's certification is blocked until approval. An action TAMC result on outlet 12 → batches that drew from outlet 12 are listed and held.

Source requirements: n436, n437, n438, n439, n440

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-27 Pure (clean) steam quality status for sterilisers and SIP

Enforced | Configurable | Injectables | Scope: all | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• Before an autoclave or SIP cycle, the system checks that the supplying pure-steam system has results within limits and not overdue for: the latest condensate test (meets the WFI monograph; microbial testing not mandatory) and the periodic non-condensable gas, dryness and superheat tests. • If not, the cycle is flagged as "unverified steam" and deviation-gated.
TYPE DETAIL	ENFORCED (check). Test frequencies are CONFIGURABLE per site; Annex 1 says only "regularly" and "periodically".
REGULATORY BASIS	EU Annex 1: 6.16, 6.17 | US: FDA-AP XII (steam generator output reviewed for release).
INDIA / US / EU DIFFERENCES	EU is more specific.
HOW IT WILL BE TESTED	Make the dryness test overdue → the autoclave run is flagged.

Source requirements: n441

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-28 Process gases, point-of-use sterilising gas filters, vent filters and sterile-tank pressure

Enforced | Injectables | Scope: aseptic (gas quality: all) | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• The gas source (bulk tank, cylinder, generator) is recorded with lot/CoA or quality status against a specification covering chemical, particulate, microbial, oil and water content. A source that is not current is blocked. The status of periodic point-of-use microbial monitoring is shown. • Aseptic-use gases pass a **sterilising-grade filter of ≤0.22 µm nominal** at the point of use, and tubing downstream of the final filter is sterilised. • Batch-basis gas filters (for example nitrogen overlay) and product-vessel vent filters are integrity-tested. The results are **release items**, and a missing or failed result blocks certification. • A vent/gas filter register holds, per filter: part/lot, install date, service-life limit and test history. Tests happen at installation, periodically or at end of use, and after any damaging event. Use past the change interval is blocked. A failed test links every batch since the last pass and opens a deviation. Filters must stay dry (hydrophobic media, heated where needed). • A sterilised holding tank must stay under positive pressure, or sealed, from sterilisation to end of use. Loss of pressure is a deviation.
REGULATORY BASIS	EU Annex 1: 6.18, 6.19 | US: 211.46(c), 211.113(b), 211.192; FDA-AP IV.D.1.
INDIA / US / EU DIFFERENCES	EU is stricter per batch: batch-basis filter integrity is reviewed at certification. FDA-AP adds sterile filters on autoclave air lines and lyophiliser vacuum breaks, and back-flow safeguards.
HOW IT WILL BE TESTED	Complete a batch without the overlay filter's post-use test → certification blocked. Record a tank pressure drop to 0 during a hold → deviation required.

Source requirements: n442, n443, n282, n283

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 4: Aseptic process simulation (media fill): status and failure consequences

ST-29 APS status engine and gate at batch start

Enforced | Configurable | Design choice | Injectables | Scope: aseptic | 6 of 7 source requirements checked against official text

WHAT THE SYSTEM DOES	• Status is held per **aseptic process × filling line × shift**, and per container/closure configuration. • **Initial validation.** Status stays "initial validation incomplete" until **≥3 consecutive successful APS** covering every shift the process runs in. • **Periodic revalidation.** A successful APS about **every 6 months** for each process, each line and each shift. • **Back to initial validation (3 runs) or requalification pending** when any of these occurs: ◦ an approved change tagged "affects sterility assurance" (HVAC, equipment, process, number of shifts or personnel, major shutdown); ◦ inactivity longer than a configured period; ◦ a new container or container-closure combination; ◦ FDA triggers flagged "media-fill assessment required": significant staffing change, EM anomaly, extended shutdown, finished-product sterility positive. FDA-AP allows QA to lift the block with a documented risk decision that no new media fill is needed. Where the change affects the aseptic process, EU 9.49 requires the APS. • **Formulation/bulk aseptic stage** (non-filterable products): a separate simulation status, at least **twice a year**. It carries its own envelope: maximum open-operation time, operators, and sterile-bulk hold and transport time. • **Gate at fill start.** Aseptic filling cannot start unless both hold, and the status seen is stored: ◦ the line/shift/format APS status is current and passing; ◦ every sterilisation process used (equipment, components, container-closure, product filtration) is in validated status and not suspended.

	• DESIGN-CHOICE: prompt for an APS after the last batch before a shutdown, long inactivity, decommissioning or relocation.
TYPE DETAIL	ENFORCED (gate, 3-consecutive rule). Periodic interval is CONFIGURABLE with default and maximum = 6 months after the last successful APS. Any grace period is a DESIGN-CHOICE that needs a QA decision on the record (see Open questions).
REGULATORY BASIS	EU Annex 1: 9.38, 9.49 | US: 211.113(b) (validation of aseptic processes is binding); FDA-AP IX.A (intro), IX.A.2, Appendix 3.A.
INDIA / US / EU DIFFERENCES	Both require 3 initial runs and a semi-annual cycle. **EU is stricter:** twice a year for *each process, each line and each shift*. FDA-AP requires semi-annual runs *per line*, with each shift's activities and changeovers represented in the programme. (Row n73 overstates FDA as "per line and shift"; see Conflicts.)
HOW IT WILL BE TESTED	Last successful night-shift APS 6 months + 3 weeks ago → night-shift batch start blocked. Add a new stopper type → "initial APS validation required" for that configuration.

Source requirements: n564, n320, n319, n73, n572, n324, n379

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-30 Manual aseptic operations: per-operator APS qualification per configuration

Enforced | Configurable | Injectables | Scope: aseptic (manual compounding or filling) | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• For processes flagged "manual aseptic", qualification is tracked per **operator × container type × closure × equipment train**: 3 consecutive successful APS initially, then one successful APS about every 6 months per operator. APS batch size mirrors routine batches. • An operator cannot sign a manual aseptic step for a combination not currently qualified.
TYPE DETAIL	ENFORCED for the EU profile. Interval CONFIGURABLE ≤6 months.
REGULATORY BASIS	EU Annex 1: 9.39 | US: FDA-AP IX.A.3 (manual processes simulated at least as long as the real process); no per-operator interval.
INDIA / US / EU DIFFERENCES	EU is stricter: 6-monthly per operator, against FDA's annual participation.
HOW IT WILL BE TESTED	An operator qualified on 10 mL vials is assigned a manual 50 mL fill → refused.

Source requirements: n565

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-31 APS design coverage and the line's "APS-qualified envelope"

Enforced | Injectables | Scope: aseptic (lyo parameters: lyo) | All 8 source requirements checked against official text

WHAT THE SYSTEM DOES	• **APS batch type.** Generated from the commercial MBR with the same step structure. Media or a non-inhibitory surrogate replaces product, with growth-promotion status. The APS never serves as the primary validation of the aseptic process. • **Protocol fields** (worst case must be justified): ◦ container/closure sizes, with bracketing or matrixing justified; ◦ maximum line speed and maximum duration (at least the routine duration; manual or intervention-heavy processes at least as long as the real process); ◦ maximum hold of exposed sterile product and equipment; ◦ fill volume that wets all surfaces and leaves headspace; ◦ **air in place of inert gas**, with occasional anaerobic runs considered; ◦ maximum headcount and maximum operator time in the cleanroom; ◦ shift changes and idle interruptions; ◦ EM as in routine production throughout; ◦ campaign start and end risks (an end-of-campaign APS may only supplement, never replace, routine APS); ◦ all routine inherent and corrective interventions at representative frequency. Rare interventions are simulated periodically, and the catalogue holds each one's simulation interval. • **Lyo APS** covers filling, transport, loading, a representative chamber dwell, unloading and sealing. It records the worst cases challenged: SIP-to-load interval, filtration-to-loading time, number of trays, loading duration and dwell. Media must not boil or freeze, and a partial vacuum is used. • **Unit count.** Justified in the CCS, typically **≥5,000–10,000** units. For batches under 5,000 units, the APS must contain **at least as many units as the production batch**. • **Sterile API.** APS size and surface coverage are justified, and all simulation materials are microbially evaluated. • **Routine checks against the envelope.** A container size not covered, or a batch size above an APS of under 5,000 units, blocks batch creation. The following raise a deviation for QA assessment before release: longer duration, higher peak headcount, more occurrences of an intervention than demonstrated, line speed out of range, longer lyo loading, or longer holds.
TYPE DETAIL	ENFORCED (envelope checks). Envelope values are *derived* from successful APS records and cannot be typed in.
REGULATORY BASIS	EU Annex 1: 9.32–9.37, 9.40 | US: 211.113(b); FDA-AP IX.A.1, IX.A.3–IX.A.6.
INDIA / US / EU DIFFERENCES	Aligned on unit numbers and worst-case design. EU adds explicit lyo worst-case items and campaign start/end. FDA adds the common/rare intervention frequency idea.
HOW IT WILL BE TESTED	APS included 3 needle changes and the routine batch records 5 → flag. APS covered 6 h lyo loading and routine takes 7 h → flag. With a 3,000-unit APS, create a 3,500-unit batch → flagged/blocked.

Source requirements: n559, n560, n561, n562, n322, n389, n563, n566

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-32 APS execution record, unit accountability, incubation and reading

Enforced | Configurable | Injectables | Scope: aseptic | All 12 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Separate batch record per run.** It holds the rationale, a planned-vs-executed intervention matrix, participants and roles, line speed, duration and embedded EM/PM. It cannot be released for distribution and consumes no product inventory. • **Units:** ◦ Every unit is inverted, swirled or agitated before incubation. ◦ **All integral units are incubated**, including cosmetic rejects and units that passed non-destructive IPC. Wrongly rejected units go back to incubation. ◦ Units may be removed without incubation only in the **exact catalogue number** for that intervention type and line location, and **never more than in a production run** [EU hard]. The same catalogue value applies in routine batches. ◦ Set-up and mandatory line-clearance units are incubated as a separate population. ◦ Product-flush equivalents are simulated with media and incubated unless shown irrelevant. • **Reconciliation.** Filled = incubated + justified non-incubated + separately incubated set-up units, with zero unexplained difference. Yield (incubated/filled) is checked against a configured criterion. Excluding an incubated unit needs justification and QA signature. • **Incubation:** ◦ Clear containers are used; amber or opaque containers are replaced by identical-configuration clear ones, or a validated detection method is used. ◦ Incubation starts without unnecessary delay (fill-end to incubation-start limit is a DESIGN-CHOICE, default 24 h). ◦ **20–35 °C at all times, within ±2.5 °C of target, ≥14 days in total. With two temperatures, ≥7 days at each, lower temperature first.** ◦ The incubator record and calibration are linked. • **Reading.** Done by qualified APS readers, with QC oversight if QC does not read. Each suspect unit is referred to the QC microbiologist immediately. Positive controls use reference organisms and local isolates, and isolates are identified to species where practical. • **Media growth promotion.** Must cover Gram-positive, Gram-negative, yeast and mould at **<100 CFU**, plus relevant isolates. The APS cannot close as passed until GPT passes. A GPT failure forces a repeat APS and two investigations. • **Traceability.** Units are time/tray-blocked and mapped to interventions, so a positive shows the surrounding activities. A QC observer signature is mandatory; video reference is optional. • **Abort or invalidation.** Allowed only by selecting an SOP rule that would equally apply to a commercial lot, with justification, QA approval and investigation. Invalidation frequency is reported per line. • **After closure.** The line is set to "post-media-fill clean required" until cleaning, sanitisation and sterilisation records are complete.
TYPE DETAIL	ENFORCED (including the [EU hard] unit-removal rule). Incubation parameters are CONFIGURABLE inside the FDA numbers.
REGULATORY BASIS	EU Annex 1: 9.41–9.45, 9.47, 9.48 | US: 211.113(b), 211.188 (inferred), 211.67(a); FDA-AP IX.A.1, IX.A.7, IX.A.8, IX.A.9.
INDIA / US / EU DIFFERENCES	**EU gives no incubation numbers**; it requires scientifically justified, validated conditions. FDA-AP's 20–35 °C / ±2.5 °C / ≥14 d / ≥7 d per temperature is used as the enforced default. Unit-removal rules are identical in substance.
HOW IT WILL BE TESTED	Record 12 units removed for an intervention whose catalogue value is 8 → refused. A 6-day + 8-day two-temperature scheme → phase rule fails. A GPT inoculum of 150 CFU → invalid.

Source requirements: n323, n567, n568, n569, n571, n327, n328, n329, n330, n331, n332, n390

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-33 APS failure consequences

Enforced | Injectables | Scope: aseptic | All 3 source requirements checked against official text

TYPE DETAIL

ENFORCED. The acceptance profile is selected per line. **The EU rule is mandatory if any product qualified on the line is for an EU/PIC/S market.**

- **Behaviour (EU profile, default):**
- Target is zero growth, and **any contaminated unit fails the APS**, whatever the run size. On recording a positive, the system automatically:
  1. sets the line to "production suspended";
  2. quarantines every product made on the line after the failed APS until the failure is resolved;
  3. builds the list of all batches since the last successful APS, requires a prompt record review with a per-batch sterility-breach risk assessment, and puts unreleased batches in the investigation scope with release blocked pending the outcome;
  4. flags already-released batches for a documented market-impact decision (FDA-AP);
  5. opens a root-cause investigation with CAPA and species identification;
  6. requires **normally ≥3 consecutive successful repeat APS** before the status returns;
  7. restricts implicated operators (ST-09).
- **Behaviour (US-only line profile):** the FDA-AP tiers, where every positive is still investigated:

Units filled	1 contaminated unit	2 contaminated units
<5,000	investigation, then revalidation	–
5,000–10,000	investigation, including whether to repeat	revalidation after investigation
>10,000	investigation	revalidation after investigation

- Intermittent or recurring positives on a line escalate to "revalidation required" even when each run is within the criteria.
- Confirmatory runs: one or more after corrections, and three consecutive when the root cause is not conclusively found.
- Impact on commercial batches made since the last passing media fill is always assessed.

REGULATORY BASIS	EU Annex 1: 9.46 | US: 211.113(b), 211.192 (inferred); FDA-AP IX.A.2, IX.A.9.
INDIA / US / EU DIFFERENCES	**EU is stricter:** any positive fails the APS, product made after the failure is quarantined, and at least 3 repeats are normally required. FDA tolerates one positive in runs of 5,000 or more as investigation-only.
HOW IT WILL BE TESTED	Enter one turbid unit in a 12,000-unit APS on a line with an EU product → failed, line suspended, quarantine and impact list created, 3 repeat runs required. On a US-only line → investigation only.

Source requirements: n570, n325, n326

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 5: Interventions catalogue and log

ST-34 Intervention catalogue (per line)

Enforced | Configurable | Injectables | Scope: aseptic | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- Each catalogue entry holds these fields:
  - ID and description;
  - **type** (inherent / corrective) and **criticality** (enters the critical zone → critical); the two attributes are independent;
  - procedure reference;
  - qualifying APS (ID, date);
  - maximum count and duration demonstrated in APS;
  - frequency class (common / rare) with its simulation interval;
  - airflow-visualisation (smoke) study reference;
  - **number and positions of units to remove**;
  - required clearance (none / local / full) and the duration or exposure beyond which clearance becomes mandatory;
  - post-intervention actions (glove print and glove change, extra sterility sample).
- Only entries qualified in a valid APS for the line can be selected. An entry without a smoke-study reference is flagged unqualified.
- DESIGN-CHOICE: master-record review prefers engineering alternatives, such as in-line weight checks, SIP instead of aseptic connections, and automation.

TYPE DETAIL

ENFORCED structure. Entry values are CONFIGURABLE and QA-approved.

REGULATORY BASIS

EU Annex 1: Glossary (inherent, corrective and critical interventions; critical zone), 4.15, 8.16 | US: 211.63, 211.113(b); FDA-AP IV.E, IX.A.3.

INDIA / US / EU DIFFERENCES

EU formalises the authorised list and classification. FDA-AP stresses minimising interventions and simulating them. Aligned.

HOW IT WILL BE TESTED

Record an intervention: only entries qualified in a valid APS for this line can be selected. Post-intervention glove sampling fires for an inherent *critical* entry and not for a corrective *non-critical* one.

Source requirements: n581, n464, n286

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-35 Intervention and stoppage log, and line clearance

Enforced | Injectables | Scope: aseptic | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Each intervention or stoppage records:**
  - start and end (duration computed, never typed);
  - catalogue ID, inherent/corrective, zone or filling position;
  - operators involved (qualification and APS coverage checked);
  - units removed, and whether line clearance was done.
- **Corrective or major interventions also record:** what failed, what was adjusted or replaced, the tools and parts used with their sterilisation records, and immediate EM/PM where configured.
- **Stops from PLC/SCADA** create entries automatically. A stop with no matching entry is flagged, and operators and reason are required before review.
- **Line clearance.** If the catalogue requires clearance, or the intervention exceeds its maximum duration, restart is blocked until a clearance record exists: scope, units and positions removed, and reconciliation. Removed units are excluded from good output and counted in yield.
- **Removal counts.** Removing a number of units different from the catalogue value in production is a deviation.
- Links: glove sampling (ST-22), extra sterility samples (ST-66), timeline (ST-70).

REGULATORY BASIS

EU Annex 1: 8.17, 9.34 | US: 211.100(a)-(b), 211.188(b); FDA-AP XII.

INDIA / US / EU DIFFERENCES

Aligned. FDA-AP adds line-clearance rules for major interventions.

HOW IT WILL BE TESTED

Replay a 7-min PLC stop with no entry → review flag. Log a needle realignment (local clearance) and restart without a clearance record → blocked.

Source requirements: n466, n363, n364

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-36 Non-qualified interventions and exceedance of APS-demonstrated frequency

Enforced | Injectables | Scope: aseptic | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

- An intervention not in the catalogue can be recorded only as "unqualified". It needs:
  - a real-time QA authorisation e-signature before the operator continues;
  - an automatic deviation and investigation;
  - flagging of the affected units and time window;
  - a mandatory QA disposition item that blocks certification until closed.
- A catalogue intervention done more often than demonstrated in APS, or a rare intervention not simulated within its interval, raises a deviation for QA assessment (see ST-31).

REGULATORY BASIS

EU Annex 1: 8.16, 9.34 | US: 211.113(b), 211.192; FDA-AP IV.E, IX.A.3.

INDIA / US / EU DIFFERENCES

EU is explicit that unqualified interventions are exceptional and are considered at disposition. FDA treats them through the deviation path. Aligned.

HOW IT WILL BE TESTED	Record an "other" intervention → QA sign-off required before continuing, deviation auto-created, certification blocked while open.

Source requirements: n465

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 6: Time limits driven by recorded events

ST-37 Time-limit rule engine (mandatory timer set)

Enforced | Configurable | Injectables | Scope: aseptic (items 1–12); terminal (item 13, | 5 of 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Rule definition.** Each rule has a start event, end event, maximum (minimum where relevant) and validation reference. MBR approval is refused if any rule lacks its reference.
- **Execution.** Timers run automatically from recorded events and warn before the limit. A breach blocks the step and all downstream steps until a deviation with justification and a QA decision exists. 211.111 accepts an overrun only if quality is not compromised.
- **Mandatory set:**
  1. cleaning → drying → sterilisation of equipment, components and containers (dirty/clean hold);
  2. sterilised equipment, components and containers from cycle end to last use, including during filling/assembly;
  3. bio-decontaminated RABS/isolator from end of decontamination to start of aseptic work;
  4. start of product preparation → start of sterilising filtration or sterilisation;
  5. start of preparation → end of aseptic filling (product-specific overall limit reflecting composition and storage);
  6. filtered/sterilised product hold before filling;
  7. total filtration time, pauses included;
  8. aseptic processing time;
  9. filling time;
  10. product exposure on the line (first open container to last unit sealed);
  11. lyo: fill → lyo load, and partial-stopper exposure (ST-59);
  12. stopper → crimp (ST-57);
  13. terminal: preparation → sterilisation (ST-60).
- Each limit is also checked against the longest duration qualified by APS (ST-31). Hold-vessel storage conditions are attached.
- A time-limit compliance summary is part of batch review.

TYPE DETAIL	ENFORCED (the timer set must exist in every aseptic MBR; breach handling). Limit values are CONFIGURABLE, product-specific and validated. Neither text gives numbers, except that the time must be *validated*.
REGULATORY BASIS	EU Annex 1: 8.18(i)–(vii), 8.36, 8.85(ii) (maximum filtration time) | US: 211.111; FDA-AP VIII.
INDIA / US / EU DIFFERENCES	Use the union. EU lists decontaminated-enclosure hold, aseptic processing time and filling time. FDA-AP lists line exposure time and a maximum total filtration time.
HOW IT WILL BE TESTED	Configure a 24 h preparation-to-fill-end limit and run 25 h → next step blocked until a QA-approved deviation. An MBR timer without a validation reference → approval refused.

Source requirements: n315, n316, n72, n467, n468, n469

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-38 Equipment status and hold chain; re-sterilisation between batches

Enforced | Configurable | Injectables | Scope: all (re-sterilisation rule: aseptic) | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Equipment states:** dirty → cleaned → dried → sterilised → in use. The cleaned step records the CIP/wash cycle ID, validated procedure version and final-rinse quality (for example conductivity; endotoxin removal by hot-WFI rinse).
- **Hold enforcement.** Dirty-hold and clean-hold limits are enforced. An item cleaned but not dried or sterilised within the limit cannot be used. A CIP with out-of-spec final rinse leaves the item "not clean".
- **Endotoxin.** Only cleaning or dry heat is accepted as removing endotoxin from equipment. Moist heat and filtration are not treated as depyrogenating.
- **Re-sterilisation.** Product-contact equipment gets a fresh sterilisation per batch by default. A campaign is allowed only with a validated MBR rule (maximum batches or hours plus validation reference); exceeding it is refused.

TYPE DETAIL	ENFORCED. Hold limits are CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 5.4, 8.18(i) | US: 211.63, 211.65(a), 211.67(a), 211.113(b); FDA-AP VII, IX.C (footnote 19).
INDIA / US / EU DIFFERENCES	Aligned. FDA-AP is explicit that equipment is re-sterilised between batches unless a validated interval exists.
HOW IT WILL BE TESTED	Assign a vessel past its clean-hold → blocked. With a validated 3-batch campaign rule, the 4th batch → refused.

Source requirements: n430, n314, n337

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 7: Components, container-closures, depyrogenation, sterilisation cycles and load traceability

ST-39 Component release, including bioburden, endotoxin and microbial tests before use

Enforced | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• Material master flags microbial, bioburden, endotoxin and particulate specifications. This covers APIs, excipients, WFI as an ingredient, containers and closures. • At dispensing or issue, the lot must be QC-released with each flagged result present and within limits. Lots failing endotoxin are rejected. Quarantined, untested or "micro test before use" lots cannot be staged or consumed. • Storage location and condition are checked at issue. • Component lot numbers and result references are written into the batch record.
REGULATORY BASIS	EU Annex 1: 2.1(iv), 10.2 | US: 211.80(a)-(b), 211.84(d)(6) (binding); FDA-AP VI (text box), VI.A.
INDIA / US / EU DIFFERENCES	Aligned. The US rule is binding.
HOW IT WILL BE TESTED	Dispense an excipient with its endotoxin result pending → hard stop.

Source requirements: n393, n300, n301

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-40 Sterile components from suppliers or contract sterilisers, and sterile additions

Enforced | Injectables | Scope: aseptic (supplier checks: all) | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Applies to RTU/RTS items (nested syringes, RTU vials, RTU stoppers, gamma-sterilised assemblies) and to contract-sterilised components. Before issue: ◦ supplier and contract steriliser are in approved status; ◦ the finished-product maker's review of the contractor's validation protocol **and** report is current; ◦ the CoA-reliability confirmation under 211.84(d)(3) is current, otherwise full testing is required; ◦ the lot's CoA, sterilisation certificate and a visual identity check are recorded. • **Aseptically added sterile components.** The method used (filtration, sterile crystallisation, dry heat, irradiation) must match the MBR registration, with evidence linked. Missing evidence blocks the addition. • **Non-filterable products.** Every material addition needs sterility evidence and is made aseptically in Grade A. All contact equipment and bulk/intermediates are sterilised. • **Aseptic upstream stages** (for example aluminium-adjuvanted suspensions): steps that expose product or contact surfaces need ISO 5/Grade A unidirectional air with an ISO 7 / Grade B or better background. EM and PM tasks are generated for these stages. End-of-operation surface EM must be recorded before the cleaning step can be signed. Sterile bulk transfer and transport are logged: vessel ID, integrity or pressure status, route, times, and a hold within the validated vessel-integrity time (ST-29). Where the MBR requires a bulk sterility test, the downstream step is blocked until the result is linked.
REGULATORY BASIS	EU Annex 1: 2.5(vii)-(viii), 8.11 | US: 211.84(d)(3), 211.113(b); FDA-AP VI.A, VI.B.1, Appendix 3.A.
INDIA / US / EU DIFFERENCES	Aligned. FDA-AP is explicit that the product maker must itself evaluate the contractor's validation.
HOW IT WILL BE TESTED	Issue RTU syringe tubs after the CoA-reliability confirmation has lapsed → blocked. Add a sterile API lot without a sterility certificate → blocked.

Source requirements: n395, n309, n302, n461, n378

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-41 Container and closure preparation: wash, rinse, siliconisation, sterilisation/depyrogenation, hold

Enforced | Configurable | Injectables | Scope: all (sterilisation/depyrogenation where | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each container or closure lot needs a complete preparation record before the filler can accept it. The record contains: ◦ wash program ID and water point, with water status in specification (parenterals: final rinse at WFI quality; stoppers: first washes with low-endotoxin purified water or better, final rinse with WFI); ◦ wash-end, drying-end and sterilisation-start timestamps, with a maximum wash-to-sterilise time; ◦ silicone lot (must be released); ◦ validated stopper load configuration (bag count, fill level) and penetration-validated recipe; ◦ depyrogenation or sterilisation cycle ID, equipment and load. • At cycle end a use-by time is computed from the MBR hold limit for that item and pack type. Use after it is blocked or deviation-gated. • The preparation process's revalidation due date is checked at line clearance.
TYPE DETAIL	ENFORCED. Hold limits are CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 8.18(i)-(ii) | US: 211.94(c)-(d) (binding); FDA-AP VI (text box), VI.B.1.
INDIA / US / EU DIFFERENCES	FDA-AP is more specific on stopper washing, siliconisation and water quality. Apply it as written.
HOW IT WILL BE TESTED	Stoppers sterilised at T0 with a 72 h hold are used at T0+73 h → blocked. A stopper load with a wash record but no sterilisation cycle → refused at the filler.

Source requirements: n303, n304, n305, n308, n29

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-42 Product-contact parts sterile and traceable; aseptic assembly and connections

Enforced | Injectables | Scope: aseptic | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• The MBR lists every **direct and indirect** product-contact item: needles, pumps, manifolds, tubing, filter housings, stopper bowls, guides and hoppers.

- At fill start each item must resolve to a passed sterilisation record (steriliser, cycle, validated load pattern), a SIP record or a supplier irradiation certificate. It must also be within sterile hold with packaging intact. Otherwise fill start is blocked and the item is named.
- Each item declares its method: SIP, autoclave, dry heat or irradiation. *Surface decontamination* (for example VHP) is allowed for a product-contact surface only on heat-labile items **and** only with a validation reference showing **≥6-log** reduction. Heat-tolerant items cannot be assigned decontamination.
- Unwrapping, assembly and line set-up are "aseptic" steps. They force Grade A with a Grade B background (isolator background per ST-74), aseptic-qualified operators, and intervention and EM linkage.
- Each aseptic connection records its method:
  - open, in Grade A with Grade B background;
  - post-connection SIP (cycle ID);
  - intrinsic sterile connector (type, lot).

  An open connection outside Grade A is blocked.

REGULATORY BASIS	EU Annex 1: 5.5, 8.12–8.14 | US: 211.67, 211.113(b) (binding); FDA-AP IX.C, Appendix 1.E.
INDIA / US / EU DIFFERENCES	Aligned. The 6-log figure for decontaminating product-contact surfaces is from FDA-AP.
HOW IT WILL BE TESTED	Leave one pump head without an autoclave link → fill start blocked with the item named. Assign VHP to a stainless needle → MBR approval refused.

Source requirements: n431, n336, n462, n463, n371

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-43 Steriliser readiness, validated recipes and load patterns

Enforced | Configurable | Injectables | Scope: all | All 9 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Readiness.** A steriliser, SIP circuit, oven or tunnel can be used only if all of these hold:
  - it is qualified, PM is not overdue and there is no open change control;
  - the controlling and recording sensors, timers and belt-speed sensors are in calibration;
  - where there is a vacuum phase or a return to sub-ambient pressure, a passing vacuum leak test is **≤7 days** old;
  - for porous loads, a passing air-removal (Bowie-Dick type) test was run **that day**, or an air detector works on every cycle.
- **Load building.** Loads are built against a **validated load-pattern ID**: item types, quantities, positions, and minimum/maximum load. Items or quantities outside the pattern are refused.
  - The cycle recipe must be linked to that pattern and to a current validation.
  - Worst-case heat-sterilisation patterns revalidated more than 12 months ago cannot be selected. Other patterns use the CCS interval.
- **Cycle master data.** Stores target SAL $\leq 10^{-6}$ (at most one non-sterile unit per million).
- **BIs.** Where used, the record holds BI lot, verified population and D-value source (qualified CoA or in-house).
- **Manual valves.** Operated through an ordered checklist: valve ID, required and actual position, time, user. Sequence errors are blocked.
- **Performance trending.** Come-up/equilibration time is trended per steriliser. Out-of-range values or drift are flagged before the load is used.

TYPE DETAIL	ENFORCED. Intervals are CONFIGURABLE but never looser than: vacuum leak test **≤7 days**, air-removal test **same day** before porous loads, worst-case heat-cycle revalidation **≤12 months**.
REGULATORY BASIS	EU Annex 1: 8.35, 8.36, 8.38–8.40, 8.60, 8.61 | US: 211.63–211.68, 211.113(b), 211.160(b)(4), 211.188(b) (inferred); FDA-AP IX.C.1, IX.C.2, Glossary (SAL).
INDIA / US / EU DIFFERENCES	EU is stricter numerically (annual worst-case revalidation, weekly leak test, daily air-removal test). FDA-AP says "periodic" and adds manual-valve documentation and SAL 10^{-6}.
HOW IT WILL BE TESTED	Last leak test 8 days ago → autoclave not selectable. Add 2 tubing sets beyond the pattern maximum → load refused.

Source requirements: n483, n484, n338, n339, n496, n497, n340, n341, n342

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-44 Automated cycle acceptance and quarantine of non-conforming cycles

Enforced | Injectables | Scope: all | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Comparison with the validated recipe.** Each imported cycle is checked on:
  - phase durations; a phase **longer or shorter** than validated is a deviation even when exposure passed;
  - temperatures and pressures;
  - porous loads: drain temperature over the whole exposure, equilibration time, exposure time, pressure/temperature agreement with saturated steam, minimum and maximum exposure temperature;
  - fluid loads: time and temperature and/or F0;
  - SIP: condensate-drain temperatures.
- **Exposure timing.** Valid only once the whole load (coldest or load probe) has reached temperature. The cooling medium in contact with sterilised items is recorded with its sterility evidence.
- **Duplex channels.** Both the control channel and the independent monitoring channel are imported with probe IDs and positions. A difference beyond tolerance fails the review.
- **Failure or deviation.** The load is auto-quarantined, its items cannot be consumed, and an investigation opens.
- **BIs.** BI lot verified (population, purity, identity), positions, incubation results and a positive control per cycle. **A passing BI never clears a cycle that failed physical parameters.**
- **Unload.** Load dryness check and per-item inspection for damage, wet packs or breached packaging. Rejects are removed and investigated, and the load cannot be released without the unload record.

REGULATORY BASIS	EU Annex 1: 8.41–8.43, 8.50–8.53, 8.55–8.59 | US: 211.113(b); FDA-AP IX.C.1.
INDIA / US / EU DIFFERENCES	EU is more detailed. Aligned in principle.

HOW IT WILL BE TESTED	Import a cycle with heating phase 20% longer than validated but passing exposure → quarantined. Enter passing BIs on a cycle with a failed temperature → load stays rejected.

Source requirements: n485, n486, n491, n492, n495, n494

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-45 Sterile-state tracking of items and carriers; sterile packaging and hold

Enforced | Configurable | Injectables | Scope: all | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	• Every carrier and item carries ID, product/batch and state: to be sterilised / sterilised, pending review / released / rejected. • Point-of-use scanners refuse anything not "released sterile". Process indicators (tape, dots) are recorded but never serve as release evidence on their own. • Packing time, sterilisation time and sterile expiry are tracked, and the maximum packing-to-sterilisation time and sterile hold are enforced. • At use, the operator confirms sterile-barrier integrity for each item or rejects and replaces it.
TYPE DETAIL	ENFORCED. Packing-to-sterilisation and sterile-hold limits are CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 8.44, 8.46, 8.48 | US: 211.113(b).
INDIA / US / EU DIFFERENCES	EU is explicit. FDA-AP has equivalent expectations in IX.C.
HOW IT WILL BE TESTED	Scan a tray whose cycle is unreviewed at Grade A entry → refused.

Source requirements: n487, n489

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-46 Material transfer into Grade A/B and into the critical zone

Enforced | Injectables | Scope: aseptic (lower-to-higher grade disinfect | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• Each transfer records the item and its method: double-ended steriliser load ID, tunnel run, wipe-down disinfection (agent, lot, expiry, contact time, operator), RTP, or filter. It also records the validated transfer procedure ID. • The item is checked against the **approved transfer list**. An unlisted item is blocked unless a pre-approved QA exception (risk assessment, disinfection and extra monitoring) is linked. The exception shows in batch review. • Moving from a lower to a higher grade needs a disinfection record before acceptance. An expired agent is rejected. • Items for the critical zone need a valid sterilisation record within hold; disinfection alone is refused. • Non-sterilisable items needed in Grade A get validated disinfection and are added to the batch surface-EM plan. • Exits are logged separately with times. Where time-based separation is used, an overlapping entry and exit triggers a warning.
REGULATORY BASIS	EU Annex 1: 4.10–4.12, 8.47, 8.49 | US: 211.42(b), 211.113(b); FDA-AP IV.E.
INDIA / US / EU DIFFERENCES	EU adds the approved-list and exception mechanism. Aligned otherwise.
HOW IT WILL BE TESTED	Use a tool in the filling zone with only a disinfection record → blocked. Scan an unlisted tool at the Grade B hatch → blocked.

Source requirements: n406, n407, n408, n288, n490

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-47 Dry-heat sterilisation/depyrogenation: tunnels and ovens

Enforced | Configurable | Injectables | Scope: all (tunnel: aseptic) | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Per run, attached to the container lot:** equipment, validated recipe and load pattern, container format, belt speed or dwell time, zone minimum and maximum temperatures, time at temperature, DP/airflow profile, alarms, chart, and controller Fh against the validated minimum. • **Readiness.** Belt-speed and temperature instruments must be calibrated, and the tunnel HEPA integrity test must be current. • **Excursions.** Any excursion quarantines the containers processed in that window, identified by time, and a deviation is raised. • **Ovens.** Positive pressure against lower grades is held through the sterilisation and post-sterilisation hold, unless packaging integrity is maintained. Loss of pressure on an unwrapped load rejects it. • **Validation.** Must show **≥3 $\log_{10}$ endotoxin reduction**; once shown, no separate sterilisation demonstration or BI study is needed. Endotoxin-spiked validation containers are fully reconciled, and a run with an unreconciled spiked unit keeps the tunnel unreleased.
TYPE DETAIL	ENFORCED. Tunnel HEPA test interval is CONFIGURABLE, default and maximum 6 months.
REGULATORY BASIS	EU Annex 1: 8.66–8.70 | US: 211.94(c), 211.113(b); FDA-AP VI.B.1, IX.C.2.
INDIA / US / EU DIFFERENCES	Aligned on 3-log. Tunnel HEPA testing: EU text says "at least biannually" (ambiguous); FDA-AP says twice a year, so 6 months is used (see Conflicts).
HOW IT WILL BE TESTED	Import a tunnel trace with a 5-min hot-zone dip → containers in that window quarantined.

Source requirements: n500, n501, n502, n306

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-48 Sterilisation-in-place and post-SIP integrity

Enforced | Configurable | Injectables | Scope: all | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	• The SIP record holds temperature, pressure and time at each monitored location (correlated with the slowest-heating points), plus cycle ID and result.

- After SIP, integrity status is tracked (positive-pressure trend, or sterilising vent-filter ID), along with the maximum hold until product use.
- Loss of pressure invalidates the SIP and blocks product transfer.

TYPE DETAIL	ENFORCED. Maximum post-SIP hold is CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 8.63 | US: 211.113(b); FDA-AP IX.C.1.
INDIA / US / EU DIFFERENCES	Aligned.
HOW IT WILL BE TESTED	Import a post-SIP pressure trend that falls to ambient before use → SIP invalid, transfer blocked.

Source requirements: n499

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-49 Radiation- and ethylene-oxide-sterilised materials; UV excluded

Enforced | Injectables | Scope: all | All 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- **UV** is not offered as a sterilisation method. It can be logged only as a bioburden-reduction aid.
- **Irradiated items** record the certificate, processing lot and a dose within the validated range. A dose below the minimum rejects the lot.
- **EO-sterilised items** cannot be issued until all of the following are recorded:
  - a justification that no other method is practicable;
  - the cycle record: preconditioning time, temperature and RH; gas concentration, pressure and quantity; exposure time;
  - BI results at every validated location, with a positive control per cycle;
  - aeration start, end and conditions, with the **validated minimum aeration elapsed**;
  - residual EO and reaction-product results within limits.

TYPE DETAIL	ENFORCED (UV exclusion is [EU hard])
REGULATORY BASIS	EU Annex 1: 8.71 (UV not acceptable), 8.72–8.78 | US: 211.94(d), 211.113(b); FDA-AP VI.B.1.
INDIA / US / EU DIFFERENCES	EU is stricter: EO only as a last resort, and an explicit UV ban.
HOW IT WILL BE TESTED	Set a component's method to UV → option unavailable. Issue an EO lot with residuals pending → blocked.

Source requirements: n503, n504, n505, n506, n307

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 8: Sterilising filtration, filter integrity (PUPSIT/post-use) and bioburden

ST-50 Sterilising filter selection matches the validated and registered filter

Enforced | Injectables | Scope: aseptic (the 211.72 fibre rule: all inje | 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Filter master data.** Each product holds its validated and registered filter part number(s): polymer, pore rating, and, where used, redundant configuration. It also holds the validation reference: bacterial retention under worst case, challenge **≥10^7 organisms per cm²** of filter area (typically *B. diminuta*, or a justified isolate), preferably in product.
- **Pore rating.** The final filter must be **≤0.22 µm nominal** (EU). FDA-AP accepts 0.2 µm or smaller and treats 0.2 and 0.22 as interchangeable ratings.
- **Installation scan.** The filter must equal the requirements listed and validated part number. The same polymer and rating is not enough if the part differs. Otherwise installation is refused. Lot, serial and sterilisation evidence (in-house load or supplier irradiation certificate) are recorded.
- **211.72 attributes.** The filter master carries "non-fibre-releasing" and "asbestos-free". An MBR that specifies a fibre-releasing filter cannot be approved unless a downstream non-fibre-releasing filter of **≤0.2 µm** follows it. **0.45 µm** is allowed only where manufacturing conditions require it, with a recorded justification. Asbestos-containing filters are never allowed.

REGULATORY BASIS	EU Annex 1: 8.79 (with cross-reference to 8.135), 8.83–8.85 | US: 211.72 (binding), 211.113(b); FDA-AP IX.B.
INDIA / US / EU DIFFERENCES	Numerically equivalent (0.2/0.22 µm). EU also requires the filter to match the marketing authorisation. The US adds the binding fibre-release and asbestos rule.
HOW IT WILL BE TESTED	Scan a 0.45 µm or non-registered part for final filtration → refused.

Source requirements: n507, n333, n29, n76

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-51 Filter train configuration and routine filtration parameters

Enforced | Injectables | Scope: aseptic | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Train master data.** Each position records its role: bioburden-reduction prefilter, primary sterilising filter, or redundant sterilising filter near the point of fill. The redundant filter is a CCS decision; Annex 1 says it should be *considered*. The eBMR generates install, test and sampling steps from the train. It counts aseptic connections downstream of the final filter for review.
- **Validated envelope per product and filter:**
  - integrity-test wetting fluid and specification;
  - pre-filtration hold and its bioburden effect;
  - conditioning;
  - **maximum filtration and contact time**;
  - maximum pressure, flow range, **maximum volume** and temperature;

- time to filter a known volume and the DP across the filter.

- **Execution record.** Start and end (including pauses), volume, flow, upstream and downstream pressure/DP, and temperature. Filtration time per known volume is computed from instrument data where available. The batch record must contain at least the time to filter a known volume and the DP. Values outside the envelope raise a deviation automatically; a significant difference is investigated.
- **Prefilter usage register.** Tracks accumulated use time and lots for prefilters and clarification filters, and blocks use beyond the justified maximum.

TYPE DETAIL	ENFORCED. The validated envelope is master data, not editable at execution.
REGULATORY BASIS	EU Annex 1: 8.80–8.86 | US: 211.111, 211.113(b); FDA-AP VIII, IX.B.
INDIA / US / EU DIFFERENCES	Aligned. Both require a maximum filtration time. EU lists the minimum batch-record content.
HOW IT WILL BE TESTED	Import a DP above the validated maximum → automatic deviation. Record filtration 08:00–16:30 against an 8 h maximum → deviation.

Source requirements: n508, n509, n510, n317

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-52 Filter integrity testing: PUPSIT, post-use test and redundant/serial logic

Enforced | Configurable | Injectables | Scope: aseptic | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **PUPSIT.** A mandatory step between filter sterilisation and product filtration. It can be omitted only where the product/filter configuration carries an approved PUPSIT-waiver risk assessment. The assessment must cover filter-sterilisation process control, the supply chain (contract steriliser, transport, protective packaging) and product knowledge (for example, particle burden that could mask a non-integral filter in the post-use test). The waiver ID is displayed in the batch record and QP review.
- **Post-use test.** A non-destructive test (bubble point, diffusive flow, water intrusion or pressure hold) of **every** sterilising filter used on a fluid. It is run **after use and before removal from the housing**; recording removal first is refused.
- **Test record:** test type, tester ID and calibration, wetting fluid, temperature, test pressure, measured value, the product-specific limit pulled from filter master data (never typed), pass/fail, and the raw printout or e-file. The limit must correlate with the bacterial-retention validation.
- **Failure handling.** A failed or invalid test causes a deviation, batch hold and certification block. Retests are allowed only under the defined procedure, and every attempt is kept.
- **Multi-filter logic:**
  - A train *validated as a serial system*: every filter must pass post-use.
  - A *redundant* train: the primary is tested post-use. If it passes, the backup need not be tested. If it fails, the backup must be tested post-use, and an investigation plus risk assessment into the primary failure opens automatically. Both results appear at QP review.

TYPE DETAIL	ENFORCED. Post-use testing always. **PUPSIT** is ENFORCED in the EU profile with a waiver path. In a US-only profile the pre-use test is CONFIGURABLE, default on.
REGULATORY BASIS	EU Annex 1: 8.82(vi), 8.87, 8.91, 8.92 | US: 211.113(b), 211.192; FDA-AP IX.B, IV.D.1.
INDIA / US / EU DIFFERENCES	**EU is stricter.** EU expects PUPSIT, with a documented risk-based alternative only where it is not feasible. FDA-AP says pre-use testing *can* be done and post-use testing should be routine.
HOW IT WILL BE TESTED	Start filtration without PUPSIT on a product with no waiver → blocked. A redundant train with the primary failing post-use → the backup test step appears, an investigation is created, and the batch is blocked until both are complete.

Source requirements: n511, n512, n514, n335

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-53 Filter single-batch use, maximum duration, and vent filter lifecycle

Enforced | Configurable | Injectables | Scope: aseptic (vent/gas filter rules: all) | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Liquid sterilising filters.** A serial is bound to **one batch**. Continuous use beyond **one working day** is blocked unless validated.
  - Campaign use requires a CCS justification and validation, with a documented maximum duration enforced by recorded times.
  - Under the FDA profile, reuse across lots is allowed only if validation states the maximum number of lots; the system counts lots and blocks beyond.
- **Vent and gas filters:**
  - Critical sterile gas and air vent filters that directly affect product sterility get a post-use integrity test in the housing for each batch, and that test is a release item.
  - Non-critical vent filters are confirmed integral at defined intervals.
  - Long-duration gas filters are tested at installation and before replacement.
  - A maximum use duration (days, uses, SIP cycles) is specified and enforced.
- A "defective/contaminated" flag retires a filter permanently.

TYPE DETAIL	ENFORCED. Maximum durations and cycle counts are CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 8.88–8.90, 8.94, 8.95 | US: 211.113(b); FDA-AP IX.B.
INDIA / US / EU DIFFERENCES	**EU is stricter.** It caps use at one working day unless validated. FDA-AP normally discards after one lot and allows justified multi-lot reuse.
HOW IT WILL BE TESTED	Scan a filter serial used yesterday → refused. Run a tank vent filter beyond its maximum SIP cycles → blocked.

Source requirements: n516, n334, n513

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-54 Pre-sterilisation bioburden per batch

Enforced | Configurable | Injectables | Scope: all (aseptic and terminal) | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Every batch** has a bioburden sample at the worst-case point:
  - aseptic: from the bulk immediately before final sterile filtration; in a redundant set-up, before the *first* filter;
  - terminal: immediately before loading into the steriliser;
  - in both cases, at the end of any hold.
- The sample is timestamped relative to the hold and to the filtration or cycle start. Sampling after filtration started is a sequence violation. Sampling at the start of a long hold raises a warning. The sampling system must not introduce contamination.
- The result is checked against the product limit, which is tied to the filter challenge or cycle lethality. A result over the limit raises a deviation. The result is a mandatory release item and is trended per product.
- For overkill terminal cycles, monitoring at scheduled intervals instead of every batch is allowed with justification. Parametric release: ST-61.

TYPE DETAIL ENFORCED. The overkill-cycle interval option is CONFIGURABLE with justification.

REGULATORY BASIS EU Annex 1: 8.93, 10.3 | US: 211.110(a), 211.113(b); FDA-AP X.C, IX.B, XI.C.5.

INDIA / US / EU DIFFERENCES Aligned. EU is explicit about per-batch sampling and the sampling point.

HOW IT WILL BE TESTED Filtration start without a bioburden sample → blocked. Release with the result pending → blocked.

Source requirements: n515, n574, n351

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 9: Filling, fill weight/volume checks, stoppering and capping

ST-55 Open-container routing under Grade A and validated closure

Enforced | Injectables | Scope: aseptic | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

- The line map is validated so that open primary containers pass only through Grade A zones, with the background suited to the technology, from depyrogenation/sterilisation exit to closure.
- Partially stoppered vials and PFS follow the lyo transfer rules (ST-59).
- The closure step records the validated closing method or equipment and its key settings.

REGULATORY BASIS EU Annex 1: 8.20, 8.21 | US: 211.42(c)(10), 211.113(b).

INDIA / US / EU DIFFERENCES Aligned (ISO 5 critical area).

HOW IT WILL BE TESTED Route open vials through a zone not classified Grade A in the line map → MBR rejected.

Source requirements: n471

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-56 In-process controls: fill weight/volume and device functional tests

Enforced | Configurable | Injectables | Scope: all (device tests: PFS, cartridges, auto | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **IPC definition.** The MBR defines each IPC: trigger (time or unit count), sample size, method, equipment, limits and reaction rules. Examples: fill weight/volume, stopper seating, seal checks. In-line automatic checkweighing is the preferred design because it avoids interventions.
- **Execution.** A missed IPC is flagged. An out-of-limit result raises a deviation and a 211.192 investigation (ST-63).
- **Manual checks in ISO 5.** A manual weight check inside ISO 5/Grade A is also logged as an intervention with its duration.
- **Device presentations.** Scheduled functional IPCs, such as delivery volume and break-loose/glide or activation where relevant. Device-component genealogy is kept (plunger rods, subassemblies).
- The 21 CFR Part 4 cross-references for combination products are **not verified** (see Open questions).

TYPE DETAIL ENFORCED (scheduling, limits, reaction). IPC content is CONFIGURABLE.

REGULATORY BASIS EU Annex 1: 8.101, 8.115 (fill volume as a CPP for FFS/BFS) | US: 211.110(a) (binding); FDA-AP IV.E, VI.B.2; 21 CFR Part 4 (unverified).

INDIA / US / EU DIFFERENCES The US is more explicit through binding 211.110(a). Annex 1 names fill volume only as a CPP for FFS and BFS.

HOW IT WILL BE TESTED Miss a 15-min fill-weight check → flagged. Record a manual weight check in ISO 5 → also logged as an intervention.

Source requirements: n386, n311, n86

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-57 Stoppering and capping: capping mode, Grade A air supply, stopper-height detection, stopper-to-crimp time

Enforced | Configurable | Injectables | Scope: aseptic | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Capping mode** is recorded per line.
  - *Aseptic capping* uses sterilised caps, with sterilisation evidence.
  - *Clean-process capping:* vials stay under Grade A until they leave the aseptic area, then under **Grade A air supply** until crimped. The background of the Grade A air supply must be at least Grade D.

	• **Zone types.** Zone master data separates "Grade A zone" from "Grade A air supply". Air supply means filtered air qualified to Grade A particle quality. For it, the EM engine asks for no continuous particle counting and no Grade A viable limits, but it records air-supply status. A loss of air supply raises a batch event for the vials in that window. • **Stopper-height detection.** Clean-process capping needs **qualified automated stopper-height detection**. A challenge at batch start (and periodically) is mandatory before capping starts. Vials with missing or displaced stoppers are rejected before capping and counted in reconciliation. • **Stopper-to-crimp time.** Checked against the configured maximum. Exceeding it, or losing local protection, raises a deviation. • **Manual capping** happens only in Grade A (isolator, or Grade A with Grade B background). Interventions at the capper must avoid direct vial contact. The crimper sits at a separate station with extraction; this is a facility item.
TYPE DETAIL	ENFORCED. The maximum stopper-to-crimp time is CONFIGURABLE as validated.
REGULATORY BASIS	EU Annex 1: 8.26–8.29, Glossary (Grade A air supply) | US: 211.42(c)(10); FDA-AP IV.E.
INDIA / US / EU DIFFERENCES	EU is stricter. It makes automated stopper-height detection an expectation and defines Grade A air supply. FDA-AP says local protection until crimping, and that on-line detection *adds* assurance.
HOW IT WILL BE TESTED	Run clean-process capping without the stopper-height challenge → capping cannot start. Record a 3 h stopper-to-crimp gap against a 2 h limit → deviation.

Source requirements: n475, n582, n476, n289

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 10: Lyophilisation

ST-58 Lyophiliser readiness: sterilisation frequency, integrity, leak rate, vent filter, trays

Enforced | Configurable | Injectables | Scope: lyo | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Loading mode.** Lyophiliser master data records manual-open (no barrier separation), automated, or closed-barrier. ◦ **Manual-open:** a new successful sterilisation is required **between every unload and the next load**, as a hard block. This has applied since 25 Aug 2024. ◦ Automated or closed-barrier: the CCS-justified frequency is enforced, and the CCS reference is displayed. • **Conditions for loading:** ◦ a passing SIP record exists, no older at loading than the **longest sterilisation-to-use interval challenged in the last successful APS**; ◦ no maintenance or cleaning event has occurred after it; otherwise re-sterilise; ◦ trays and vial rings have their own sterilisation load IDs. • **Every cycle:** ◦ the start-of-cycle leak-rate result is at or below the specified maximum air leakage; a failure blocks the cycle and release; ◦ the integrity/vent filter is sterilised before each use, and its **integrity test result is on the QP release checklist**; ◦ the chamber leak-test frequency is documented. • **Trays.** IDs are tracked with last inspection date. A tray that is overdue or flagged misshapen or damaged is refused.
TYPE DETAIL	ENFORCED. The CCS-justified sterilisation interval for automated or barrier-protected units is CONFIGURABLE.
REGULATORY BASIS	EU Annex 1: 8.122, 8.123 (from 25 Aug 2024), 8.124, 8.125 | US: 211.113(b), 211.68 (inferred); FDA-AP IV.D.1 (sterile filter on the lyophiliser vacuum break).
INDIA / US / EU DIFFERENCES	FDA-AP is silent on lyophiliser sterilisation frequency and leak rate, so EU governs.
HOW IT WILL BE TESTED	On a manual-open lyophiliser, attempt a second load without a new SIP → hard block. Start leak rate above maximum → cycle flagged and release blocked.

Source requirements: n528, n529, n530, n531

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-59 Lyo transfer, loading and unloading as aseptic operations; lyo cycle record

Enforced | Configurable | Injectables | Scope: lyo | 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Aseptic extension.** Transfer, loading, the cycle and unloading are treated as extensions of aseptic processing until containers are fully sealed. They inherit Grade A checks, aseptic-qualified operators, intervention logging, and the EM coverage window, which extends to the end of unloading/sealing. • **Transfer.** Partially closed containers move only under Grade A/ISO 5: conveyor, qualified clean-air cart, or portable UDAF. Alternatively they travel in validated closed trays or boxes, closed in Grade A and not opened in Grade B. The record holds the transfer device ID and qualification, and its Grade A/airflow status in transit. A transit alarm flags the affected trays. • **Loading record:** ◦ shelf-by-shelf pattern (tray IDs, vial counts); ◦ first tray out of the filler, last tray loaded, chamber closure; these drive the **partial-stopper exposure timer** (ST-37); ◦ sterile utensil IDs. • **Unloading.** A stopper-seating check is recorded. If stoppers are not fully seated before the chamber opens, the units stay in Grade A until capped. • **Cycle record** from the lyophiliser SCADA/historian: ◦ recipe ID and version; ◦ per-phase set points and actuals for shelf temperature, chamber pressure and condenser temperature; ◦ product probes, pressure-rise tests and phase durations. Values are compared with the validated (filed) cycle, and excursions raise deviations. This detail is sourced outside Annex 1 and FDA-AP; see "What FDA-AP does not cover".
TYPE DETAIL	ENFORCED (aseptic extension [EU hard], Grade A transfer). Cycle-parameter checks are CONFIGURABLE per validated recipe.

REGULATORY BASIS	EU Annex 1: 8.121, 8.126 | US: 211.42(c)(10), 211.111, 211.113(b), 211.188(b) (inferred); FDA-AP IV.E; FDA *Guide to Inspections of Lyophilization of Parenterals* (1993; not in the uploaded texts, unverified).
INDIA / US / EU DIFFERENCES	Aligned on ISO 5 transfer. EU adds the explicit "extension of aseptic processing" [EU hard] and the closed-tray option.
HOW IT WILL BE TESTED	Load with an unqualified transfer cart → blocked. Exceed the partial-stopper exposure limit → deviation.

Source requirements: n527, n532, n287, n78

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 11: Terminal sterilisation

ST-60 Terminally sterilised products: preparation/filling grades and preparation-to-sterilisation time

Enforced · Configurable · Injectables · Scope: terminal · All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Grade floors:**
  - component and solution preparation: **≥ Grade D**;
  - preparation raised to **≥ Grade C** automatically when the product has an unusual contamination risk (supports microbial growth, long hold before filling, not processed mainly in closed vessels), and for ointments, creams, suspensions and emulsions;
  - filling: **≥ Grade C**;
  - filling moves to **Grade A with ≥ Grade C background** when the CCS shows unusual environmental risk (slow filling, wide-neck containers, exposure of more than a few seconds before closure).

  Product and container risk flags drive these grades.
- Primary containers and components are cleaned by validated processes that control particles, endotoxin and bioburden.
- Where possible, the bulk passes a microorganism-retaining filter before filling; its filter ID is recorded.
- A **preparation-to-sterilisation** timer uses the validated maximum (ST-37 item 13). Pre-sterilisation bioburden: ST-54.
- Cycle master data holds **SAL ≤10^{-6}**, meaning at most one non-sterile unit per million. An "overkill" cycle means a **≥12-log** reduction of organisms with **D ≥1 min**. The FDA-AP glossary's parenthetical states the SAL probability inverted; configuration text must not copy it.

TYPE DETAIL	ENFORCED (grade floors). Risk flags and the time limit are CONFIGURABLE; the time limit must be validated.
REGULATORY BASIS	EU Annex 1: 8.1–8.5, 8.36 | US: 211.113(b), 211.111 (inferred); FDA-AP II.B, Glossary (terminal sterilisation, overkill).
INDIA / US / EU DIFFERENCES	FDA-AP does not regulate terminal sterilisation of product, so EU governs (see the section below).
HOW IT WILL BE TESTED	Flag a terminal product as growth-supporting → a Grade D preparation room is rejected. Set container type "wide-neck" → filling demands Grade A in a Grade C background.

Source requirements: n456, n457, n458, n382

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-61 Terminal sterilisation loads (sub-batches), non-rigid containers, parametric release

Enforced · Injectables · Scope: terminal · 3 of 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Loads as sub-batches.** Each load is a sub-batch. Units are linked by load, tray and pallet to a validated load pattern and cycle (ST-43, ST-44). Routine probe positions are fixed master data correlated with the worst case from load mapping.
- **Superheated-water or spray autoclaves.** A pre-cycle checklist confirms nozzles are not blocked and drains are free of debris; the cycle cannot start otherwise.
- **Non-rigid containers** (BFS, FFS, bags). The cycle is designed to avoid distortion. A post-cycle deformation and leak inspection is recorded.
- **Parametric release** is allowed only where authorised (MA/application reference on the product master). The release gate requires:
  - complete in-limit critical parameters at **every** specified monitoring position (time, temperature, pressure, F0);
  - chemical indicator results (FDA recommends Class 5);
  - per-batch filled-unit bioburden from worst-case positions before the cycle, with organism ID and a resistance assessment for any recovery, plus endotoxin where appropriate.

  Any failure makes parametric release impossible. Falling back to a sterility test is allowed only if the authorisation permits it. Without authorisation, the sterility test under 211.167(a) is required (ST-65).

TYPE DETAIL	ENFORCED. Parametric release is available only with an authorisation reference.
REGULATORY BASIS	EU Annex 1: 8.54 (refers to Annex 17), 8.62, 8.64, 8.65, 10.4 | US: 211.167(a); FDA parametric-release submission guidance (Feb 2010) and CPG 490.200 (secondary sources, not in the uploaded texts).
INDIA / US / EU DIFFERENCES	Both require prior authorisation. Detailed criteria come from EU Annex 17 and the FDA 2010 guidance, neither of which is in the inputs (see Open questions).
HOW IT WILL BE TESTED	One probe below the F0 minimum on a parametric-release batch → parametric release blocked. A cycle start without the nozzle/drain check → blocked.

Source requirements: n498, n493, n575, n77

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 12: 100% visual inspection, AQL and container-closure integrity

ST-62 100% inspection (manual, semi-automated, automated)

Enforced · Configurable · Injectables · Scope: all · 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Mandatory step.** Every filled parenteral unit goes through a 100% inspection step. The step records: ◦ mode, equipment/recipe and inspectors; ◦ units presented, accepted and rejected **by defect code** from a controlled defect library. The library holds criticality (critical/major/minor), particle type (fibre, glass, metal) and integrity defects (cracks, missing or loose stoppers). ◦ reconciliation: filled = accepted + rejected + samples + destroyed, within limits. • **Manual inspection:** ◦ Sign-on requires VI qualification **≤12 months** old, tested with the inspector's usual corrective lenses on defect-library sets under worst-case conditions (inspection time, conveyor speed, container size, fatigue). An eyesight check must also be current. ◦ The station records its illumination/background check. ◦ The inspection rate is controlled. ◦ The station enforces a break after a configurable maximum continuous time. • **Automated inspection (AVI).** Validated to detect at least as well as manual inspection. A challenge-set run is required at start-up and at defined intervals during the batch. A failed interim challenge puts every unit since the last passing challenge on hold for re-inspection. • **Reject rates.** A per-defect reject rate above its alert or action limit opens an investigation and blocks release.
TYPE DETAIL	ENFORCED. Per-defect alert/action limits are CONFIGURABLE and derived from historical data.
REGULATORY BASIS	EU Annex 1: 8.30–8.32 | US: 211.192; FDA-AP VI.B.2; FDA draft guidance *Inspection of Injectable Products for Visible Particulates* (Dec 2021, draft; secondary).
INDIA / US / EU DIFFERENCES	EU is stricter on inspector qualification (annual) and AVI challenges. The FDA draft adds a lifecycle programme with AQL; it is not final.
HOW IT WILL BE TESTED	An inspector with VI qualification 13 months old → sign-on refused. A failed mid-batch AVI challenge → units since the last pass held.

Source requirements: n477, n479, n480, n310, n79

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-63 AQL of accepted units, critical defects, OOS investigations and defect trending

Enforced | Configurable | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	• **AQL sampling.** Accepted units are sampled after inspection. The record holds sample size, acceptance and rejection numbers, and results by class. • **Critical defect in accepted units.** Any **critical** defect found in accepted units, whether in AQL or any later inspection, opens an investigation automatically, blocks release and prompts a re-inspection decision. Re-inspection is allowed only under SOP, with a deviation, and its counts are recorded separately. • **OOS results.** Any out-of-specification IPC or inspection result opens a 211.192 investigation, which includes other potentially affected batches. Release waits for closure. • **Trending.** Defect counts are stored per code per batch, so SPC trending works across batches. An adverse trend opens an investigation with a **mandatory market-impact assessment**.
TYPE DETAIL	ENFORCED. The AQL plan is CONFIGURABLE; see Open questions for values.
REGULATORY BASIS	EU Annex 1: 8.30, 8.33 | US: 211.192 (binding); FDA-AP VI.B.2; FDA draft guidance (2021).
INDIA / US / EU DIFFERENCES	Aligned. Neither final text fixes AQL values.
HOW IT WILL BE TESTED	One cracked vial in the AQL sample → automatic investigation and release block. Ten batches with rising glass rejects → trend rule fires.

Source requirements: n478, n481, n312

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-64 Container-closure integrity

Enforced | Injectables | Scope: all | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES	• **Fusion-sealed containers** (BFS, FFS, SVP/LVP bags, glass or plastic ampoules): ◦ seal-critical parameters (for example heater temperature, dwell, pressure) are recorded and monitored; ◦ **glass ampoules, BFS units and fusion-closed containers of ≤100 mL** get **100% CCI** by a validated method, recording units tested, passes and rejects by reason; ◦ above 100 mL, reduced sampling needs a referenced scientific justification and process-consistency data. • **Visual inspection** is excluded from the CCI method list. A 100% VI record can never satisfy a CCI requirement. • **Non-fusion closures** (stoppered/crimped vials, syringes): CCI on samples by a validated method, under a sampling plan justified by container-closure knowledge, supplier management and component specifications. Any failure opens an investigation and blocks release. • **Vacuum-sealed containers.** A vacuum-retention test is scheduled after the predetermined waiting period. Certification is blocked until it passes. Retention during shelf life is covered by stability. • Integrity-defect units are removed at final inspection (ST-62).
TYPE DETAIL	ENFORCED ("visual inspection is not a CCI method" is [EU hard])
REGULATORY BASIS	EU Annex 1: 8.22–8.25 | US: 211.192; FDA-AP VI.B.2.
INDIA / US / EU DIFFERENCES	**EU is stricter:** 100% CCI for fusion-sealed containers ≤100 mL, and VI not accepted as CCI. FDA-AP only requires removal of defective units and safeguards against shipping units lacking integrity.
HOW IT WILL BE TESTED	Configure a 2 mL ampoule with sampled CCI → MBR rejected. Choose "visual inspection" as the CCI method → option unavailable.

Source requirements: n472, n589, n473, n474

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 13: Sterility, endotoxin and particulate testing, and release

ST-65 Sterility and endotoxin results required per batch; method suitability and test conditions

Enforced | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- The release checklist requires approved LIMS results for **sterility and bacterial endotoxins (or pyrogens)**. The samples must trace to this batch and, where relevant, to the sub-batch or load. Results must be within specification. A result from another batch cannot be linked.
- The only alternative is authorised parametric release (ST-61).
- A sterility result is accepted only if it references a method-suitability (bacteriostasis/fungistasis) validation approved for this product and **formulation version**, current on the test date. A formulation change invalidates the link until it is re-verified.
- Each sterility result must carry:
  - the test environment (isolator or cleanroom ID and status);
  - media lots with sterility and GPT status;
  - an analyst qualified on the test date;
  - incubation start and end.

  A missing or non-compliant element is flagged for QA.
- Sample exterior decontamination (VHP/UV) must be a validated process that does not affect the test.

REGULATORY BASIS

EU Annex 1: 10.5 (test validated per product), 10.8, 10.9 | US: 211.167(a) (binding), 211.165(e), 211.194, 211.160(b), 211.22(b), 211.25(a); FDA-AP VII (text box), XI.A.

INDIA / US / EU DIFFERENCES

Aligned. The US per-batch testing rule is binding.

HOW IT WILL BE TESTED

Release with sterility present and endotoxin pending → refused. A sterility result whose method suitability used the old formulation → refused.

Source requirements: n313, n354, n355

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-66 Sterility sampling plan (start/middle/end, post-intervention, per load and sub-batch)

Enforced | Configurable | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Pull tasks** are generated automatically:
  - at the **start, middle and end** of the aseptic fill, by unit count or elapsed time; EU names start and end, FDA adds middle;
  - after **each recorded critical intervention or environmental excursion**, following the risk rule;
  - for heat-sterilised product in the final container: from the **coolest/slowest-heating positions of each load**. Each load is a sub-batch with its own sample set and test;
  - for lyophilised product: from **each lyophiliser load**.
- Each sample keeps its origin (time, unit counter, tray or position, load) through to LIMS. The release check confirms every required pull was sampled and tested.
- Sampling of sterile materials (sterile bulk, sterile API, post-filtration samples) records the sterile sampling device and its sterilisation record, the aseptic method or SOP, the location or class, and the sampler's qualification. A record missing any of these cannot be completed.
- Design basis: a 20-unit test passes a lot with 0.1% contaminated units about 98% of the time (FDA-AP XI.B). That is why samples are tied to the riskiest moments.

TYPE DETAIL

ENFORCED. Risk rules for extra samples are CONFIGURABLE.

REGULATORY BASIS

EU Annex 1: 10.6 | US: 211.84(c)(3), 211.160(b)(3), 211.165(a); FDA-AP XI.B.

INDIA / US / EU DIFFERENCES

Use the union. EU adds per-load sub-batch sampling for terminal sterilisation and lyo, and risk-based post-intervention samples. FDA adds the middle-of-fill pull and pulls after excursions.

HOW IT WILL BE TESTED

A terminally sterilised batch in 3 autoclave loads → 3 cold-spot sample sets required. A corrective intervention at 40% fill → an extra sample task is generated.

Source requirements: n577, n356, n385

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-67 Sterility-positive handling and investigation

Enforced | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Status on a positive.** A positive sets the batch to "sterility failure: presumed non-sterile" and blocks release.
- **Invalidation and retest.** Invalidation needs a QA-approved record with conclusive evidence that the contamination arose in the laboratory. A retest can be linked only after that approval. A QA conclusion of "inconclusive" allows only "reject". Original results are never replaced. Tests invalidated before incubation remain visible with their deviation reference.
- **Investigation template.** Pre-filled from the eBMR with:
  - species ID of the isolate, compared with EM, PM and bioburden isolates using the same ID method;
  - lab records and lab EM trends;
  - short- and long-term production EM trends, and PM trends;
  - bioburden trend;
  - batch events: interventions, alarms, DP and particle excursions, utility events;
  - maintenance and construction work orders;
  - prior deviations and changes;
  - a HEPA retest task (ST-17).

- **Associated batches** are proposed per 211.192: batches sharing the same line, fill window, sterilisation loads, bulk, component lots or personnel.
- **Trending.** Sterility records carry product, container type, line, sterilisation mode (aseptic/terminal), sampler, analyst and environment. This enables the required trend views, including initial-positive rate for aseptic vs terminal.
- A conclusion of "lab error" with no specific evidence attached is refused, whatever the laboratory's history.

REGULATORY BASIS	EU Annex 1: 3.2 | US: 211.192, 211.165(a), 211.167(a), 211.180(e); FDA-AP XI.C, XI.C.1–7.
INDIA / US / EU DIFFERENCES	FDA-AP is more detailed. EU requires investigation before certification with batch-scope rationale (ST-71). Apply both.
HOW IT WILL BE TESTED	Post a positive, then link a passing retest without an approved invalidation → refused. Mark the outcome "inconclusive" → only "reject" is offered.

Source requirements: n357, n358, n359

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-68 A passing sterility test never overrides a process failure

Enforced | Injectables | Scope: all | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES

- The sterility result is evaluated only after all process, procedural and validation-status checks have passed. A pass never closes or overrides an open nonconformance.
- Certification with only a sterility result attached, while process, EM, utility or sterilisation review items are open, is blocked.

TYPE DETAIL	ENFORCED ([EU hard]: 10.5, second sentence)
REGULATORY BASIS	EU Annex 1: 2.2, 2.7, 10.5 | US: 211.113(b), 211.192.
INDIA / US / EU DIFFERENCES	Aligned. EU states it categorically.
HOW IT WILL BE TESTED	A passing sterility result with an open sterilisation-cycle deviation → status stays "not releasable".

Source requirements: n576, n591, n397

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-69 Short shelf-life products and conditional release before final results

Enforced | Configurable | Injectables | Scope: all (the cell-therapy module is optional | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- For a product flagged "short shelf life", certification may precede the sterility and/or EM results only when all of these hold:
  - a documented risk-mitigation reference exists (process design, extra monitoring, alternative or rapid methods);
  - the **most recent available** EM data are reviewed;
  - late results stay linked to the batch.

  An adverse late result triggers automatic escalation, including a recall assessment.
- Optional cell-therapy module: release requires passing intermediate sterility, stain (for example Gram) and endotoxin results. The final sterility result is tracked afterwards, and a positive triggers escalation and notification.

TYPE DETAIL	CONFIGURABLE (off by default). ENFORCED escalation once enabled.
REGULATORY BASIS	EU Annex 1: 10.7, 10.10 | US: 211.165(a), 211.167(a) (inferred); FDA-AP Appendix 3.B.
INDIA / US / EU DIFFERENCES	Aligned in approach.
HOW IT WILL BE TESTED	Certify a short-shelf-life batch before its sterility result, then enter a positive → escalation linked to the distributed batch.

Source requirements: n578, n380

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 14: Batch record review dossier and release gate

ST-70 Sterile batch review dossier with signed sections and consolidated timeline

Enforced | Design choice | Injectables | Scope: all | 4 of 5 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Dossier sections.** Each section needs a reviewer signature, and QA release is blocked until every section is signed:
  - EM (viable and non-viable) for every room/zone and the whole window;
  - PM for the roster;
  - HVAC/HEPA status and DP;
  - WFI and clean-steam results for the period;
  - equipment/EMS alarm report (ST-06);
  - **all** filter integrity results (product, vent, gas);
  - sterilisation, depyrogenation and SIP cycles (ST-72);
  - interventions and stoppages;
  - time-limit compliance;
  - APS and personnel-qualification status on the fill date;
  - all IPC and lab results.
- **Required results.** The MBR declares a required-results list. Each result is linked as an immutable reference (value, limit, status) and is visible *in the batch record*, not only in LIMS. A later source change shows the revision history and forces re-review.

- **Timeline.** Interventions, particle and EM data, DP and alarms, holds, sterilisation and filter results, and personnel events appear on one exportable time axis. Reviewers judge the batch as a whole.
- **QP/QA access.** QP/QA users reach every linked source from one screen. The certifier role requires a sterile-products training record.

TYPE DETAIL	ENFORCED (sections and signatures). The consolidated timeline is a DESIGN-CHOICE.
REGULATORY BASIS	EU Annex 1: 2.4, 3.1(vii) | US: 211.188, 211.192 (binding); FDA-AP XII.
INDIA / US / EU DIFFERENCES	Aligned. FDA-AP XII lists the essential release-review elements. (Row n75 was unverified; see Conflicts.)
HOW IT WILL BE TESTED	Leave the WFI section unsigned → release refused. Remove the link to the post-use filter test → completeness check fails.

Source requirements: n362, n75, n361, n586, n399

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-71 Release gate: deviations and investigations with impact scope, yield, storage excursions

Enforced | Injectables | Scope: all | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Conditions for the release e-signature:**
  - every section is complete and every review-by-exception item is dispositioned;
  - **every linked deviation and investigation is closed**, or formally assessed as not affecting this batch;
  - yield and reconciliation (bulk, fill count, inspection, labelling) are within MPCR limits or explained.
- **Investigation closure requires:**
  - root-cause category and product-risk statement;
  - CAPA link and residual-risk acceptance rationale;
  - an impact-scope list of batches, with an inclusion/exclusion rationale for each, propagated to every listed batch;
  - a completed "other batches/products assessed" field.

  This applies **even if the batch has already been distributed**.
- **Post-fill handling.** Storage and transport record location and conditions. An excursion from registered storage conditions is linked to the batch and blocks release until assessed.

REGULATORY BASIS	EU Annex 1: 3.1(iii), 3.1(iv), 3.1(vi), 3.2 | US: 211.192 (binding); FDA-AP XII (text box).
INDIA / US / EU DIFFERENCES	Aligned. EU adds a documented inclusion/exclusion rationale for each batch.
HOW IT WILL BE TESTED	An open deviation plus a yield outside limits → release refused, listing both. Close an investigation with the "other batches" field empty → refused.

Source requirements: n400, n360, n401, n398

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-72 Sterilisation-cycle roll-up at release

Enforced | Injectables | Scope: all | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES

- Every item consumed resolves to a unique cycle ID. This covers product, components, equipment, and garments where relevant.
- QA/QP review lists every contributing cycle with its conformity status.
- Certification is blocked while any cycle record is unreviewed or non-conforming.

REGULATORY BASIS	EU Annex 1: 8.45 | US: 211.188(b), 211.192.
INDIA / US / EU DIFFERENCES	Aligned.
HOW IT WILL BE TESTED	Consume stoppers from an unreviewed cycle → certification blocked, with the cycle listed as open.

Source requirements: n488

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-73 Structured execution data for PQR/APR, CCS review and management review

Design choice | Injectables | Scope: all | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES

- The following are stored as structured, queryable data: interventions (type, count, duration), achieved hold times, defects by code, EM/PM excursions, filter test values, cycle results, time-limit exceedances, APS results, sterility and endotoxin results, and deviations.
- Cross-batch exports and dashboards are available per product and line, so the review of at least annual frequency, the PQR, CCS effectiveness review and management review need no re-keying.

TYPE DETAIL	DESIGN-CHOICE for the data model. The annual review itself is a binding US requirement.
REGULATORY BASIS	EU Annex 1: 2.3, 3.1(v), 6.4, 8.33, 9.11 | US: 211.180(e); FDA-AP XI (text box).
INDIA / US / EU DIFFERENCES	Aligned.
HOW IT WILL BE TESTED	A 12-month extract of interventions per line matches the source batch records.

Source requirements: n585, n387

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Theme 15: Technology profiles (isolator, RABS, BFS, FFS, closed/SUS)

ST-74 Barrier-technology profile, background grade, isolator pressure and transfers

Enforced | Configurable | Injectables | Scope: isolator, rabs (and conventional lines) | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Line profile.** Line master data records the technology: isolator (open/closed), RABS (open/closed), conventional cleanroom, BFS, FFS or closed/SUS. The profile selects the mandatory building blocks: glove tests, bio-decontamination, door-opening log. A **conventional (open) aseptic line** needs a CCS justification reference before release for use.
- **Background grades:**
  - open isolator: **≥ Grade C** generally;
  - closed isolator: **≥ Grade D**, decided by CCS risk assessment;
  - RABS for aseptic processing: **≥ Grade B**, with airflow studies covering interventions and door openings;
  - FDA: isolator background is set by interface design, commonly ISO 8, and **never unclassified**.

  Batch start verifies the background room's classification is current.
- **Isolator pressure.** Overpressure against the room is linked continuously for the batch window against qualified limits (FDA typical range **17.5–50 Pa**). Exit-port (mousehole) overpressure is continuously monitored. An excursion becomes an integrity-breach event (ST-76).
- **Transfers** into a running isolator or RABS are logged: technology (RTP alpha/beta, mousehole, transfer isolator), port ID, beta-container sterilisation reference and time. Weaker ports need local HEPA UDAF cover status.
- A negative-pressure isolator is allowed only as a flagged containment configuration.
- RABS door openings are batch events that need assessment.

TYPE DETAIL

ENFORCED (background floors, pressure monitoring). Pressure limits are CONFIGURABLE per qualification.

REGULATORY BASIS

EU Annex 1: 4.3, 4.18–4.20 | US: 211.42(c)(10)(iii)-(iv) (inferred); FDA-AP Appendix 1.B.3-4, 1.C.

INDIA / US / EU DIFFERENCES

For open isolators EU is stricter (Grade C, which is ISO 7 at rest) than FDA's "commonly ISO 8". The 17.5–50 Pa figure is FDA's.

HOW IT WILL BE TESTED

Assign a RABS line to a Grade C room → rejected. Replay exit-port overpressure at 0 Pa for 20 s → integrity-breach event.

Source requirements: n403, n414, n413, n368

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-75 Glove, sleeve and enclosure integrity (isolator and RABS)

Enforced | Configurable | Injectables | Scope: isolator, rabs | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Register per port.** Glove, sleeve or half-suit ID, install date, replacement due date (CCS-defined) and last integrity result. Batch start is blocked if any port on the line is overdue or failed.
- **Isolators:**
  - glove integrity test by a suitable method **at least at the start and end of each batch or campaign**, with more tests if the campaign is long; for small manual batches, at session start and end;
  - a visual check at each use and after any manipulation that could affect integrity;
  - an **enclosure leak test** at defined intervals; overdue blocks batch start.
- **RABS:**
  - gloves are sterilised before fitting, and sterilised or bio-decontaminated by a validated method before **each campaign**;
  - after each exposure to the background during operation, a disinfection entry is required before the next aseptic step;
  - a visual check at each use, and periodic integrity-test status checked at line start.
- **Failed test after use.** The failure links every batch and time window since that glove last passed, for impact assessment. Under FDA-AP, units filled in the suspect window default to rejection unless QA documents that the environment was not compromised.
- **Glove-port manipulations** are logged as interventions (type, port, duration, operator), with tools linked to sterilisation records. Fingertip PM covers each port used.

TYPE DETAIL

ENFORCED. Replacement frequency and campaign test intervals are CONFIGURABLE, with a floor of a glove test at the start and end of each batch or campaign.

REGULATORY BASIS

EU Annex 1: 4.21 | US: 211.113(b); FDA-AP Appendix 1.A, 1.G.

INDIA / US / EU DIFFERENCES

EU is stricter on frequency: a minimum of start and end of each batch/campaign, where FDA says "routine". FDA adds daily attention and inner-glove precautions.

HOW IT WILL BE TESTED

Enter a failed end-of-batch test on port 3 → deviation plus impact window since the last pass. Log a RABS door opening with glove exposure and no disinfection → next aseptic step blocked.

Source requirements: n415, n416, n417, n418, n367, n373

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-76 Barrier cleaning, bio-decontamination, validity interval and integrity breach

Enforced | Injectables | Scope: isolator, rabs | All 6 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Sequence rule.** A bio-decontamination cycle is valid for the batch only if a completed cleaning record for the same enclosure precedes it, with no use in between.
- **Isolator bio-decontamination.** Automated, validated, using a sporicidal agent in vapour or gas form. The record holds:
  - pre-clean confirmation;
  - **gloves extended with fingers separated**;
  - recipe ID and version, and agent lot;
  - key parameters against validated ranges (agent amount or concentration, phase durations);
  - end-of-aeration residual below its limit;

- chemical indicator results;
- pass/fail.

Validation reference: quantified BIs at many locations, normally a **4–6 log** reduction (FDA-AP). The aseptic phase cannot open until the cycle has passed.

- **Validity.** Status reads "decontaminated, valid until" (validated interval; hold per ST-37 item 3). Aseptic steps after that time are blocked. Deteriorating EM trends prompt a review of the interval.
- **Integrity breach.** Overpressure loss, a glove or seam hole, or a power or valve failure sets the status to "decontamination required" and creates a deviation linked to the batch, with the unit-disposition default as in ST-75.
- **Isolator EM.** EM media cannot be introduced until aeration is recorded complete. Per shift: fingertip samples for each port used, air, surfaces (with wipe-down confirmation after contact plates) and exit-port particle monitoring.
- **RABS.** Routine sporicidal disinfection of every interior surface (agent, lot, expiry, SOP version, operator) at the defined frequency. Overdue blocks line start.

TYPE DETAIL	ENFORCED ("cleaning before bio-decontamination" is [EU hard])
REGULATORY BASIS	EU Annex 1: 4.22 | US: 211.113(b) (inferred); FDA-AP Appendix 1.D.1–3, 1.F.
INDIA / US / EU DIFFERENCES	Complementary. EU makes cleaning first essential and requires a sporicide in vapour form. FDA gives the log-reduction expectations and the reject-by-default for compromised units.
HOW IT WILL BE TESTED	A VHP cycle with no prior cleaning record → invalid for batch use. Settle-plate exposure logged before aeration end → blocked.

Source requirements: n588, n419, n369, n370, n420, n372

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-77 Blow-fill-seal (BFS)

Enforced | Injectables | Scope: bfs | All 10 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Environment:**
  - line records shuttle vs rotary machine type;
  - terminal BFS: machine in **≥ Grade D**, with point-of-fill conditions per ST-60;
  - aseptic shuttle: **Grade A** at the parison extrusion, blow-moulding and sealing critical zones, plus a Grade A filling environment;
  - aseptic rotary: **Grade A inside the parison**;
  - machine background **≥ Grade C**, with operators in **Grade A/B garments**, so a Grade C gowning record is refused;
  - FDA: the surrounding room generally ISO 8 or better, and the critical zone meets ISO 5 microbial (normally also particle) levels.
- **EM.** In-operation particle monitoring of the critical zone is not expected; a design-data reference is required instead. Viable monitoring must cover from equipment assembly to the end of filling, with samples spread across the run. Gaps are flagged.
- **Checks at start:**
  - PM is current, and machine-utility integrity (for example mould-cooling leak check) is current;
  - seal-critical tooling has been inspected;
  - a mould change or modification is blocked until its CCI assessment is closed;
  - the polymer lot is released (bioburden/endotoxin where specified) and its supplier is qualified;
  - parison/support air filter IDs and integrity are current;
  - the product-path SIP (and mandrel CIP/SIP) is within the validated recipe.
- **CPPs, imported continuously with limits:** extrusion temperature, speed and throat settings (parison thickness), melt temperature and residence time, mould temperatures and cooling rate. An excursion segregates units made in that window and raises a deviation.
- **IPCs.** Wall thickness, container weight, fill weight/volume, leak checks and process air pressure. **100% CCI** is run with a known-defect challenge verification. Rejects by reason flow into reconciliation. A significant forming or sealing defect creates a deviation with a 211.192 investigation.
- **Interventions.**
  - Entries flagged "stops extrusion/filling" or "requires re-sterilisation" force a passing SIP before resumption.
  - Set-up and troubleshooting are logged as interventions, with media-fill coverage checked.
- **Added components** (for example LVP caps), aseptic case: pre-sterilised and added in Grade A, with validated sealing tests.
- **Campaign.** A campaign-duration timer runs.

REGULATORY BASIS	EU Annex 1: 8.22, 8.105–8.120 | US: 211.100, 211.113(b), 211.192; FDA-AP Appendix 2 (intro), 2.A–2.C.
INDIA / US / EU DIFFERENCES	EU is more prescriptive on zones, the Grade A/B garment rule and 100% CCI. FDA-AP adds extrusion endotoxin/spore challenge validation and polymer supplier control.
HOW IT WILL BE TESTED	A 10-min mould-temperature excursion → units from that window segregated. A modified mould without closed CCI assessment → batch start blocked.

Source requirements: n521, n522, n523, n524, n525, n526, n374, n375, n376, n377

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-78 Form-fill-seal (FFS)

Enforced | Injectables | Scope: ffs | All 4 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Environment** follows the product's route: terminal (ST-60 grades) or aseptic (Grade A/B).
- **Film.** Each roll is scanned against its specification: thickness, strength, microbial and particulate contamination, integrity and artwork. Film bioburden/endotoxin applies where the CCS defines it. A roll with a pending result is blocked.
- **CPPs recorded against validated ranges:**
  - package dimensions and cutting;
  - forming temperatures (including pre-heat and cooling), forming times and pressures;
  - sealing temperature and its uniformity, sealing time and pressure;
  - environmental and product temperature;

- fill volume, speed and uniformity;
- print and emboss settings.

Batch-specific seal strength and uniformity tests are mandatory before the batch can close. CCI is 100% for fusion-sealed containers ≤100 mL (ST-64).

- **Gases.** Product-contact gases (inflation, overlay) are filtered close to the point of use and periodically verified.
- **Machine records.** CIP/SIP records, film decontamination/sterilisation and the campaign-duration limit are held. Machine set-point changes and alarms enter the batch audit trail, and an out-of-range change needs justification.
- **Rejects** by reason (forming, sealing, cutting) go into reconciliation. Sealing issues above a threshold open an investigation. Tooling inspection status is checked at start.

REGULATORY BASIS	EU Annex 1: 8.22, 8.96–8.104 | US: 211.100, 211.192 (general). FDA-AP has no FFS content.
INDIA / US / EU DIFFERENCES	Only EU has FFS-specific requirements.
HOW IT WILL BE TESTED	Omit the batch seal-strength test → batch cannot close. Change a sealing set point outside range on the HMI → recorded with user and time and justification required.

Source requirements: n517, n518, n519, n520

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

ST-79 Closed systems and single-use systems (SUS)

Enforced | Injectables | Scope: sus | All 7 source requirements checked against official text

WHAT THE SYSTEM DOES

- **Assemblies.** Closed-system assemblies are configured items (bags, tubing, connectors, sensors, filters), each with lot and sterilisation evidence. **Every** product-contact element of an aseptic closed system must resolve to a sterilisation record or supplier sterility proof, with no "N/A". Connections to the sterile path after the final filter use intrinsic sterile connectors by design. A pre-use system integrity test (for example pressure decay) is required when the CCS says so.
- **Background.** An aseptic closed system whose integrity could be compromised sits in Grade A. A lower grade needs an integrity result recorded **at every use**. A "system opened" event forces either opening in Grade A (aseptic) or Grade C (terminal), or the extra cleaning, disinfection and, for aseptic use, sterilisation steps.
- **SUS goods receipt.** Per lot or unit: sterilisation evidence (irradiation certificate, dose, indicator status) and approved-supplier status. Unverified lots stay in quarantine.
- **SUS pre-use, per piece:**
  - part number against the approved specification or BOM revision;
  - outer packaging and pouch condition, and label check;
  - CoC and sterilisation proof present;
  - structured visual inspection at unpacking and after installation (damage, holes, kinks, clamp positions), plus the packaging-opening method.

  Each check is documented. Integrated filters follow ST-52.
- **Product compatibility.** Extractables/leachables and adsorption assessment status is held per product × part number; an unassessed part is blocked. A validated maximum contact time is enforced. Freeze/thaw use requires a SUS qualified for that condition.
- Operators for SUS assembly and connections need APS coverage of that manipulation (ST-08).

TYPE DETAIL	ENFORCED ("sterility of product-contact surfaces is critical" is [EU hard])
REGULATORY BASIS	EU Annex 1: 8.127–8.138 | US: 211.113(b) (general). FDA-AP does not address SUS or closed systems.
INDIA / US / EU DIFFERENCES	Only EU has specific requirements.
HOW IT WILL BE TESTED	A closed assembly with one connector lacking sterility evidence → cannot be released for use. A bag part number that differs from the approved BOM revision → refused.

Source requirements: n533, n590, n534, n535, n536, n537, n538

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

What the FDA aseptic guidance (2004) does NOT cover, and where those rules come from

Topic	FDA-AP position	Source used instead	Controls
Terminal sterilisation of the drug product	Out of scope. II.B/III: only aseptic processing and sterilisation of components, containers, closures and equipment are covered; aseptic processing is to be used only when terminal sterilisation is not feasible (row n381).	US binding: 211.113(b) (validated sterilisation), 211.111 (time limits), 211.167(a) (per-batch sterility test), 211.192. EU Annex 1: 8.1–8.5 (preparation and filling grades), 8.34–8.65 (sterilisation, including non-rigid containers and superheated-water loads), 10.3–10.4 (bioburden), 10.6 (per-load sampling). FDA-AP IX.C steriliser principles reused by analogy. Parametric release: EU 8.54 → Annex 17, and FDA's Feb 2010 parametric-release submission guidance + CPG 490.200 (rows n77, n493). **Neither of those two is in the uploaded texts.**	ST-43, ST-44, ST-54, ST-60, ST-61, ST-66
Lyophilisation cycle records (shelf/condenser temperature, chamber pressure, phase times), lyophiliser sterilisation frequency, leak rate	Only ISO 5 protection for transfer and loading (IV.E), sterile filters on vacuum breaks (IV.D.1) and lyo simulation in media fills (IX.A). No cycle-record content.	EU Annex 1 8.121–8.126, including the 8.123 lyophiliser-sterilisation rule in force from 25 Aug 2024. FDA *Guide to Inspections of Lyophilization of Parenterals* (1993) for cycle records (row n78, **not in the uploaded texts; unverified**). US anchors: 211.113(b), 211.68, 211.111, 211.188(b). 211.137(c) (expiry of reconstituted products) is a labelling item, not a batch-execution control.	ST-58, ST-59
RABS	Not mentioned.	EU Annex 1 4.3, 4.18–4.22	ST-74, ST-75, ST-76
Form-fill-seal	Not mentioned. Appendix 2 covers BFS only.	EU Annex 1 8.96–8.104, 8.22	ST-78
Single-use / closed systems	Not mentioned.	EU Annex 1 8.127–8.139	ST-79
100% visual inspection detail, inspector qualification, AQL	VI.B.2 says only that defective units are removed at inspection.	EU Annex 1 8.30–8.33. FDA draft guidance on visible particulates (Dec 2021, **draft**, row n79; AQL figures from secondary sources).	ST-62, ST-63
Container-closure integrity numbers (100% for fusion-sealed ≤100 mL; visual inspection not acceptable)	General statement in VI.B.2.	EU Annex 1 8.22–8.25	ST-64
WFI and pure-steam system detail	WFI and steam-generator output named only as release-review elements (XII).	EU Annex 1 6.7–6.17	ST-26, ST-27
5 µm particle limits; Grade B/C/D viable limits for contact plates and gloves; 28 L/min; 4 h plates	Only ≥0.5 µm ISO classes and air/settle action levels (Table 1).	EU Annex 1 Tables 1, 2, 5 and 6; 9.17; Table 6 note (a)	ST-18, ST-19, ST-20
Contamination Control Strategy	No equivalent concept.	EU Annex 1 2.3–2.6	ST-02, ST-03
Per-grade garment specification, maximum gown wear time, laundry cycles	General gowning expectations (V.A).	EU Annex 1 7.10–7.17	ST-11
Combination-product device controls (PFS, autoinjectors)	Not covered.	21 CFR Part 4 (row n86; post-QMSR cross-references **unverified**)	ST-56

Conflicts between rows and how they were resolved (with evidence)

1. **APS frequency: per shift or per line.**
   - Row n73 (secondary) says FDA requires semi-annual requalification "per line and shift".
   - The primary FDA-AP IX.A.2 text (row n320) says semi-annual qualification **per processing line**, with each shift's activities and changeover *incorporated into the programme*.
   - EU 9.38 (row n564) says twice a year for **each process, each line and each shift**.
   - *Resolution:* EU governs (stricter). n73 is corrected in ST-29.
2. **APS acceptance.**
   - Row n570 (EU 9.46): any contaminated unit fails the APS.
   - Rows n326 and n73 (FDA-AP IX.A.9): tiered by run size. Verified text: under 5,000 units, one positive → revalidation after investigation. 5,000–10,000: one → investigation including a possible repeat; two → revalidation. Over 10,000: one → investigation; two → revalidation.
   - *Resolution:* the rule is chosen per line. The EU rule is mandatory when any EU/PIC/S product is qualified on the line (ST-33).
3. **FDA Table 1 values (row n74, "from recollection, not re-verified").**
   - Checked against the primary FDA-AP Table 1 (row n274): ISO 5 3,520 / 1 / 1; ISO 6 35,200 / 7 / 3; ISO 7 352,000 / 10 / 5; ISO 8 3,520,000 / 100 / 50. All match. Settle plates are optional (note d). ISO 5 samples should normally give no growth (note e).
   - *Resolution:* no conflict. n274 is the citation of record.
4. **FDA-AP XII content (row n75, unverified).**
   - The primary XII text names as essential release-review elements: EM and PM data; support-system output (HEPA/HVAC, WFI, steam generator); equipment functioning (batch alarm report, filter integrity); and documentation of interventions and stoppages (row n362). It does **not** list sterilisation/depyrogenation records or time-limit adherence *in XII*.
   - *Resolution:* ST-70 keeps those dossier sections but bases them on 211.188/211.192, FDA-AP VIII and IX.C, and EU 8.45.
5. **Filter integrity timing (row n76, unverified).**
   - Confirmed by FDA-AP IX.B: a pre-use test *can* be performed, and post-use testing *should routinely* be performed.
   - EU 8.87 (row n511) expects PUPSIT.
   - *Resolution:* EU PUPSIT with a risk-assessed waiver path (ST-52).
6. **Time limits (row n72, secondary).**
   - Confirmed against the FDA-AP VIII text box, which restates 211.111.
   - *Resolution:* merged with EU 8.18 into the union list in ST-37.
7. **Tunnel HEPA test frequency.**
   - Row n500 notes that EU 8.67 says "at least biannually", which is ambiguous (twice a year, or every two years).
   - FDA-AP IV.D.2 (row n284) says twice a year for the aseptic room and names tunnel and oven filters.
   - *Resolution:* default and maximum 6 months (ST-47, ST-17).
8. **Pressure differentials.**
   - Row n410: EU minimum 10 Pa, a *guidance value*.
   - Row n280: FDA at least 10–15 Pa between classes, and at least 12.5 Pa from an aseptic room to an unclassified room at all times.
   - *Resolution:* floor 10 Pa between graded rooms, 12.5 Pa to unclassified rooms (ST-17). Whether to ship a 12.5/15 Pa default is an open question.
9. **Gowning requalification interval.**
   - Row n447 (EU): at least annually for all Grade A/B staff.
   - Row n292 (FDA): annual is normally enough *for automated, low-intervention* operations, with more frequent requalification after adverse findings.
   - *Resolution:* ≤12 months for everyone, shortened for manual processes (ST-08).
10. **Isolate identification level.**

    `- Row n558 (EU 9.31): species level for Grade A/B.`

    `- Row n350 (FDA-AP X.B): species, *or genus where appropriate*.`

    `- *Resolution:* species for A/B (ST-21).`
11. **Glove sampling frequency.**

    `- Row n297 (FDA V.C): daily or with each lot.`

    `- Row n554 (EU 9.25): at every Grade B exit and after critical interventions, followed by an outer-glove change.`

    `- *Resolution:* union (ST-22).`
12. **Filter reuse.**

    `- Row n334 (FDA IX.B): reuse possible if validation states the maximum number of lots.`

    `- Row n516 (EU 8.94): one batch, and no more than one working day of continuous use unless validated.`

    `- *Resolution:* EU cap by default. Multi-lot reuse only on US-only lines with validation (ST-53).`
13. **Filter pore size.** Not a real conflict.

    `- Row n507: ≤0.22 µm nominal.`

    `- Row n333: FDA 0.2 µm; FDA-AP footnote treats 0.2 and 0.22 as interchangeable.`

    `- Row n29 (211.72): the 0.2 µm (0.45 µm if manufacturing conditions require) figure applies to the non-fibre-releasing filter that must follo`

- *Resolution:* a 0.45 µm filter is never accepted as the sterilising filter (ST-50).

14. **Media-fill incubation.**

    - Row n569 (EU 9.44): conditions justified and validated; **no numbers**.

    - Row n327 (FDA IX.A.8): 20–35 °C, ±2.5 °C, ≥14 days, ≥7 days per temperature with the lower first.

    - *Resolution:* FDA numbers enforced as defaults (ST-32). The catalog does not attribute them to Annex 1.

15. **Isolator background.**

    - Row n414 (EU 4.20): open isolator ≥ Grade C, closed ≥ Grade D.

    - Row n368 (FDA App. 1.C): commonly ISO 8, never unclassified.

    - *Resolution:* EU floors (ST-74).

16. **Viable limits: ISO 8 vs Grade D.**

    - Row n274: FDA ISO 8 = 100 CFU/m³ and 50 per 4 h.

    - Row n544 (EU Table 6): Grade D = 200 and 100.

    - *Resolution:* for US-market products the FDA values apply to rooms that FDA would class ISO 8 (ST-20).

17. **Sterility sample points.**

    - Row n577 (EU 10.6): start and end, plus after critical interventions, per load/sub-batch and per lyo load.

    - Row n356 (FDA XI.B): start, middle and end, plus after interventions or excursions.

    - *Resolution:* union (ST-66).

18. **SAL wording.** Row n382 records that the FDA-AP glossary parenthetical states the SAL probability inverted. *Resolution:* configure SAL as $\leq 10^{-6}$ probability of a non-sterile unit (ST-60, ST-43).

19. **Scope of 211.42(c)(10) (row n5, "confirmed").** The regulation requires the facility *systems* to exist. It does not itself require a batch EM plan or a "released for aseptic operation" gate. *Resolution:* those gates cite FDA-AP and 211.192, not 211.42 alone (ST-14, ST-18, ST-24).

20. **Row n29 ("corrected").** 211.94(c)-(d) apply to every dosage form. *Resolution:* only the sterilisation/depyrogenation parts are used in this catalog (ST-41). Filter-integrity rules are *not* based on 211.72 (ST-50, ST-52).

21. **Grade A ≥5 µm limit.** Not a row conflict, but easy to get wrong.

    - Classification Table 1: "not specified".

    - Monitoring Table 5: 29/m³.

    - *Resolution:* ST-19 applies 29/m³ to monitoring only.

"Not an eBMR concern"

Row	Reason
n584	Facility and equipment *design* expectations: impervious surfaces, no recesses, sealed ceilings, no sinks or drains in Grade A/B, trapped drains, windows and cameras, P&IDs, utility installation records, no exposed pipework (EU 4.5–4.9, 4.17, 5.1, 6.5, 6.6). These are verified by facility qualification, not batch execution. The eBMR only reads the resulting qualification status (ST-14).

Facility-only fragments inside otherwise covered rows are also *not* re-checked per batch:

- conveyor partition rule (n434, EU 5.8);
- counter tubing ≤1 m and bends (n434, EU 5.9), which is a qualification item;
- crimper extraction station (n475, EU 8.26);
- hydraulic/heating equipment located outside the filling room (n444, EU 6.21);
- BFS airflow and zone design (n375).

Open questions

1. **APS "approximately every six months".** Should any tolerance be allowed (for example +30 days with a documented QA decision)? The shipped default is none: last success + 6 months.
2. **DP default between graded rooms.** Keep the 10 Pa floor, or ship 12.5 Pa / 15 Pa for customers exporting to the US?
3. **Glove sanitisation before sampling.** The minimum interval (design default 5 min) needs each customer's SOP value.
4. **AQL values for visible particles** (0.65 appears only in secondary sources). USP <790>/<1790> were not verified in this pass, and the FDA 2021 guidance is still draft. Customer QA must set the plan.
5. **Parametric release.** EU Annex 17 and FDA's 2010 guidance and CPG 490.200 are not in the inputs. Verify before building the ST-61 parametric path.
6. **Lyo cycle-record requirements.** The FDA 1993 lyo guide is not in the inputs (row n78). Verify the parameter list for ST-59.
7. **21 CFR Part 4** post-QMSR cross-references for combination products (row n86) are unverified.
8. **"Extended period" of inactivity** that triggers APS revalidation (EU 9.49; FDA "extended shutdowns"): no number exists in either text. Site parameter needed.
9. **Glove/sleeve replacement frequency, maximum gown wear time, maximum laundry cycles, clean-up period, disinfectant rotation.** All come from site qualification. The catalog ships no numbers, and deliberately so.
10. **FDA APS tiers on US-only lines.** Should they be offered at all? Most Indian exporters serve both markets, so an EU-only rule set would be simpler and safer.
11. **Integration boundary.** Water, steam, EMS/BMS, LIMS and LMS data are *read* by the eBMR. Confirm which module owns sampling schedules, such as the daily WFI sample and EM programme versions.
12. **Reconciliation with the India catalog.** Where Revised Schedule M differs (for example grade tables or APS frequency), confirm that the stricter value is loaded into the ST-01 floor table.
13. **Rows marked secondary/unverified** (n72, n73, n74, n75, n76, n77, n78, n79, n86). n72–n76 were checked here against the FDA-AP text. n77, n78, n79 and n86 remain outside the uploaded texts.

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1. **ST-33 APS failure consequences.** The EU rule (any positive fails the APS; quarantine of everything made after the failure; review back to the last successful APS; normally ≥3 repeats) must win on any line that makes EU product. The FDA tiers apply only to US-only lines.
2. **ST-29/ST-30 APS frequency granularity.** EU requires twice a year per *process × line × shift*; FDA is per line. Operators need a successful APS every 12 months. Manual aseptic work needs 3 initial runs, then one about every 6 months *per operator per configuration*.
3. **ST-19/ST-20 EM limits.** Grade A ≥5 µm = 29/m³ for *monitoring* but "not specified" for *classification*. Continuous Grade A monitoring at ≥28 L/min. Grade A viable = no growth, with no alert tier. FDA ISO 8 (100/50) is stricter than EU Grade D (200/100). Settle plates ≤4 h (ST-18).
4. **ST-52 filter integrity.** PUPSIT is expected (EU), with only a documented risk-assessed waiver. Post-use testing happens before removal from the housing. In a redundant train only the primary is tested post-use if it passes. A serially validated train needs *all* filters to pass.
5. **ST-64 CCI.** 100% CCI for glass ampoules, BFS and fusion-sealed containers ≤100 mL. Visual inspection is never a CCI method. Vacuum-retention test before certification.
6. **ST-32 APS incubation and unit accountability.** The 20–35 °C / ±2.5 °C / ≥14 d / ≥7 d per temperature (lower first) rule is **FDA's**; Annex 1 only requires justified, validated conditions. All integral units are incubated, including cosmetic rejects. Removal is capped at the catalogue number and never exceeds production [EU hard].

Runners-up: ST-21 (action-excursion batch linkage from sampling to reporting), ST-58 (sterilise before every load on a manually loaded lyophiliser from 25 Aug 2024), ST-17 (12.5 Pa toward unclassified rooms), ST-14 (6/12-month requalification caps).

PART D

India layer: Revised Schedule M and Indian law

Mandatory for every batch made in India · 52 controls

Scope: the Indian legal layer (Revised Schedule M, G.S.R. 922(E) of 28 Dec 2023; Drugs Rules 1945; IT Act 2000; Bharatiya Sakshya Adhiniyam 2023) and how it combines with US FDA and EU GMP when a batch is also made for those markets. FDA-only and EU-only controls live in the companion catalogs. All text is paraphrased; nothing here is a quotation of a regulation.

Baseline assumption used throughout: every batch is made in India under an Indian manufacturing licence, so the Indian rules always apply. A batch's "market profile" adds US and/or EU rules. Where values differ, the engine applies the stricter one for the markets the batch serves (control IN-52). Label rules are resolved per pack/SKU destination rather than per batch (see section B note).

Status keys used in every control:

- **P** = wording and numbers re-read in the uploaded Gazette text during this review.
- **P~** = present in the Gazette text, but the clause number is inferred (pages are scrambled in the uploaded copy).
- **S** = secondary sources only (not seen in any primary text here); confirm before validation documents cite it.
- **U** = secondary and incomplete or unconfirmed detail; must be confirmed before hard-coding.
- For FDA/EU comparison values: **[chk]** = checked in the supplied primary files (21 CFR 211, 21 CFR 11, EU Ch.4, Ch.5, Annex 1 (2022), Annex 11, FDA 2004 aseptic guidance); **[gk]** = general regulatory knowledge, source not in the supplied set.

Type keys: **ENFORCED** = the engine enforces it and no tenant can switch it off; **CONFIGURABLE** = the tenant sets values within the stated bounds (never looser than the resolved rule); **DESIGN-CHOICE** = our product decision, stricter than or not required by the Indian text, applied for consistency across markets.

Corrections to the requirements list found while re-checking primary texts (applied below):

1. EU Chapter 5 reprocessing/recovery clauses are 5.67-5.69 (not 5.62-5.64); EU per-batch identity testing of starting materials is 5.35 (by reference to Annex 8). EU batch-record retention is Chapter 4 section 4.11.
2. EU Annex 1 (2022) 4.32 caps requalification (which includes the integrity test of final filters) at 6 months for Grades A/B and 12 months for C/D. So for Grade A/B the Indian HEPA interval (recommended 6, maximum 12 months) is weaker than EU, not more prescriptive.
3. Schedule M Part II Table 1 gives 5.0 micrometre limits of 20 (Grade A), 29 (B at rest), 2,900 and 29,000, which are numerically lower than the EU 2022 monitoring/classification values (29, 2,930, 29,300). India is numerically stricter on 5.0 micrometre counts; EU 2022 is stricter on how Grade A is monitored and on microbial limits.
4. Schedule M Part II Table 3 microbial values are stated to be averages and "for information only", not specifications; EU 2022 Table 6 values are maximum action limits with no growth allowed at Grade A.
5. Part VIII 3.8 wording: the 30-minute maximum applies to sampling tablets against the in-process specification; the six named parameters plus lubricant-oil contamination are to be checked "periodically" (frequency left to the firm).
6. EU Annex 1 (2022) 8.94 limits a liquid sterilising filter to a single batch and at most one working day unless validated; India only sets the one-working-day limit.
7. FDA 21 CFR 211.186(a) requires the master record to be signed by one person and independently checked and signed by a second; India only requires approval, signature and date by authorised persons.
8. Release to the EU market: the Indian authorised-person release is still required at the Indian site; EU QP certification takes place at the EU importing/release site. The Indian system supplies the batch package; it does not hold a QP signature slot for the Indian site (correction to requirement n233 implications).

A. Controls

Kernel controls (both editions)

IN-01 Quality-system data feeds: product quality review, management review, CAPA

Enforced · Configurable · Both editions · All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Yields, IPC results, deviations, OOS, rejects, reprocessing, material lots and supplier identity are stored as typed, queryable fields, never only as free text. (b) A PQR extract per product and period (default 12 months) includes every batch, explicitly including export-only SKUs, failed batches, deviations with CAPA status, changes, and flags suppliers that were new in the period; the PQR record carries the authorised person's sign-off. (c) A management-review extract (default 12 months) is available per site. (d) Every CAPA links to its source (deviation, OOS, complaint, PQR, self-inspection) and has an owner, due date and an effectiveness-check task; an ageing/overdue report exists; release screens show open CAPAs for the product/line as information.
TYPE DETAIL	ENFORCED (structured capture and links); CONFIGURABLE (review periods, product grouping with recorded justification).
REGULATORY BASIS	Schedule M Part I 1.6 (management review at least yearly unless justified); section 2.3.1-2.3.3 (quality review normally yearly, export-only products included; CAPA management); section 1 PQS list (CAPA effectiveness monitored); 10.5-10.7 (follow-up of self-inspection actions).
INDIA / US / EU DIFFERENCES	PQR: **same** as FDA 211.180(e) (evaluation at least annually) [chk] and EU Chapter 1 section 1.10 (annual review) [gk]. Management review: **stricter** on frequency (yearly minimum) than FDA Part 211 (no such clause) [chk] and EU Chapter 1 (periodic, no fixed minimum) [gk]. CAPA: **same** as EU Chapter 1 [gk] and FDA 211.192 investigations [chk].
HOW IT WILL BE TESTED	run the PQR extract for a product with domestic and export-only batches; all appear. Create a CAPA with a 90-day effectiveness check; it appears as overdue on day 91.
SOURCE CHECK	P.

Source requirements: n229, n231, n674

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-02 Risk classification recorded on steps, deviations and changes

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	MBR steps, parameters, deviations and change requests carry a risk class from a controlled, versioned scale. Any risk-based reduction (e.g., no second-person verification, "minor" deviation, reduced IPC frequency) cannot be saved without a linked risk assessment and written rationale. Changing a step's criticality in a new MBR version changes its verification requirements.
TYPE DETAIL	CONFIGURABLE (scales, formality levels); ENFORCED (a rationale is mandatory whenever a risk class drives a reduced control).
REGULATORY BASIS	Schedule M Part I section 2 (QRM: systematic, science-based, effort proportionate to risk).
INDIA / US / EU DIFFERENCES	**same** in substance as ICH Q9(R1) as adopted in EU GMP Part III and FDA guidance [gk]; Part 211 has no explicit QRM clause [chk], so India is more explicit than the FDA regulation.
HOW IT WILL BE TESTED	classify a deviation as minor without a rationale; saving is refused.
SOURCE CHECK	P.

Source requirements: n232

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-03 Change control gate and first-batch-after-change evaluation

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) No MBR, packaging-instruction, master-data, limit table, software or configuration version can become effective without a linked, approved change record (approval by the originating unit and by quality). (b) The change record stores the change class (e.g., minor/major) and the list of affected documents; the change cannot be closed until all listed documents are revised. (c) The first batch made or tested under the changed state is automatically flagged "post-change evaluation required"; it cannot be released until QA signs that evaluation. (d) For changes classified critical, the system creates a task to assess retest/expiry impact (optionally a stability commitment).
REGULATORY BASIS	Schedule M Part I 8.1-8.7 (formal change control; quality-unit approval; classification; documents revised on implementation; evaluation of the first batch after the change; expiry/retest impact of critical changes); Part I section 20 (changes to computerised systems follow a change procedure; exact sub-clause number not legible in the uploaded copy).
INDIA / US / EU DIFFERENCES	**stricter** (more explicit) than FDA 211.100(a), which requires changes to procedures to be reviewed and approved by the relevant units and the quality control unit but names no first-batch evaluation [chk]; **same** intent as EU Chapter 1 / ICH Q10 post-implementation evaluation [gk].
HOW IT WILL BE TESTED	approve MBR v2 under change CC-1, execute a batch; release is blocked until the post-change evaluation is signed.
SOURCE CHECK	P (8.6 first-batch evaluation and 8.7 expiry evaluation re-read).

Source requirements: n268

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-04 Role model, segregation of duties, approved technical staff and training gates

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Separate roles for head of production, head(s) of QA/QC, authorised person (AP) and designated production-record reviewer. (b) One user cannot hold both a production-head and a quality-head approval on the same batch. (c) The production-record review signature is distinct from, and must precede, the QA review. (d) An "approved technical staff" register stores, for each person approved for manufacture or testing, the licensing-authority approval reference and date; only these people can sign production-supervisor or QC-approval steps on batches made under an Indian licence. (e) Each MBR step type needs a current training record; the system refuses step sign-on by untrained users and stores the training status at the moment of signing. (f) Visual inspectors need a current eye-check record (see IN-50). (g) Visitor/contractor accounts are view-only or supervised.
TYPE DETAIL	ENFORCED (segregation, training gate); CONFIGURABLE (approved-staff register mandatory for India-licensed sites, on by default).
REGULATORY BASIS	Schedule M Part I 11.3.1 (key personnel), 11.3.4-11.3.5 (heads independent; designated person signs production records), 11.2, 11.4 (training), 11.5 (hygiene; eye checks for visual inspectors); Part I 11.3.2 (qualifications as per the Rules); Drugs Rules 71 and 76 (manufacture under active direction of competent technical staff; testing staff approved by the licensing authority).
INDIA / US / EU DIFFERENCES	training and quality-unit independence **same** as FDA 211.22 and 211.25 [chk] and EU Chapter 2 [gk]. The approved-technical-staff register is **different** (Indian licensing construct; the EU uses the QP under Annex 16 [gk]; FDA has no named-person approval).
HOW IT WILL BE TESTED	give one user production-head and QA-head roles on one batch; the conflict is refused. Remove a user from the approved-staff register; they can no longer sign as production supervisor.
SOURCE CHECK	P for Schedule M (independence of heads and designated reviewer re-read); S for Drugs Rules 71/76.

Source requirements: n234, n662, n271

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-05 Batch disposition by the authorised person

Enforced | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Bulk and finished goods are created in quarantine; only an e-signed disposition changes status (released, rejected, on hold) and warehouse/ERP interfaces must honour that status. (b) The release signature is available only to the AP or to a named delegate whose delegation cites an approved procedure. (c) Hard prerequisites: every step signed; production-record review signed; all deviations closed or formally assessed; head-of-QC approval recorded; a regulatory-notification assessment recorded for any change or deviation; for injectables, sterilisation records, EM data for the processing window and sterility results reviewed (IN-42, IN-47, IN-49). (d) The release checklist (licence and approval conditions met, validated processes, documentation complete and endorsed) is stored with the release record. (e) A rejected batch cannot close without a recorded disposition (return, reprocess, destroy) approved by an authorised role.
REGULATORY BASIS	Schedule M Part I section 1 PQS list (no sale or supply before AP certification; head-of-QC approval), 11.3.6-11.3.10 (AP duties and delegation), 18.2.2 and 18.5.12 (records reviewed before AP; investigation of discrepancies), 14.29-14.31 (quarantine, release, rejected materials).
INDIA / US / EU DIFFERENCES	**different** legal role. FDA: quality control unit approves, after reviewing all production and packaging records, per 211.22 and 211.192 [chk]; the Indian AP plus head-of-QC approval maps onto that and must meet 211.192 for US batches. EU: QP certification (Annex 16 [gk]; Annex 11 section 15 requires the

	system to let only QPs certify, by e-signature [chk]) happens at the EU importing site, not in India; the system supplies the batch package and can store the QP certification reference received back.
HOW IT WILL BE TESTED	attempt release with an open deviation, without head-of-QC approval, with an unsigned step, and as a user without the AP/delegate role; all four are blocked.
SOURCE CHECK	P.

Source requirements: n233, n244, n676

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-06 Document control of masters and corrections to recorded data

Enforced · Design choice · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) MBRs, packaging instructions and SOP references have approval e-signatures, an effective date and a periodic-review date. (b) Once superseded, a version can no longer issue new batches but stays retrievable for a configured period (never shorter than the retention of batches made with it). (c) Every master version needs two signatures from different users: prepared by one, independently checked by another (applied in all profiles; required for US). (d) A correction to a recorded value never overwrites it: original value, new value, user, timestamp and a reason are kept and shown in the rendered record. The reason is mandatory in all profiles.
TYPE DETAIL	ENFORCED; DESIGN-CHOICE for the two-person master approval and the always-mandatory reason.
REGULATORY BASIS	Schedule M Part I 17.2 (design, approval, signature and dating of documents; regular review; preventing use of superseded versions; retention of superseded documents for a set period; clear, indelible entries; alterations signed and dated with the original legible and a reason where appropriate).
INDIA / US / EU DIFFERENCES	corrections **same** as EU Chapter 4 section 4.9 (reason "where appropriate") [chk]; **weaker** than EU Annex 11 section 9 (reason must be documented for any change or deletion of GMP-relevant data) [chk]. Master approval **weaker** than FDA 211.186(a) (one person prepares and signs, a second independently checks and signs) [chk]. The system applies both stricter rules everywhere.
HOW IT WILL BE TESTED	supersede MBR v1; new batches cannot start on it; v1 still opens. Correct a weight; the rendered BMR shows old and new values, who, when, why.
SOURCE CHECK	P~ (text re-read; 17.2 sub-clause numbers inferred).

Source requirements: n236

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-07 Contemporaneous capture, completeness and traceability

Enforced · Both editions · All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Each step is recorded at the time of the action with a server timestamp; entries by hand, by instrument interface, or both, are supported. (b) A batch cannot be completed while any mandatory step is unexecuted; skipped or not-applicable steps need a reasoned, signed justification. (c) Significant deviations are captured at the point of occurrence (IN-20). (d) Full backward genealogy (every input lot) and forward genealogy (packaging batches, distribution records) is queryable per batch. (e) The complete record renders in a human-readable form for the whole retention period (IN-08).
REGULATORY BASIS	Schedule M Part I section 1 PQS list (records during manufacture show each step was done; history of a batch traceable and accessible), 3.1 (basic GMP requirements), 17.2 (records made when each action is taken).
INDIA / US / EU DIFFERENCES	**same** as FDA 211.100(b) (documented at the time of performance) and 211.188(b) [chk], and EU Chapter 4 section 4.8 [chk].
HOW IT WILL BE TESTED	skip a step; completion is refused. From a finished batch, list every input lot and every customer shipment.
SOURCE CHECK	P~.

Source requirements: n230, n235

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-08 Record retention engine

Enforced · Configurable · Both editions · 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	For each batch the system computes a "retain until" date as the maximum of every rule that applies to that batch, stores which rule governed, and blocks deletion or archive-with-loss before that date. Rules: (i) Schedule M: expiry + 1 year (always). (ii) Drugs Rules conditions of licence, pending confirmation: 5 years from manufacture for Schedule U manufacturing and test records (Form 25, Rule 74); for Form 28 licences (Rule 78) test records reportedly 2 years after expiry where an expiry is labelled, otherwise 5 years from manufacture (U). (iii) US market: expiry + 1 year (3 years after distribution for exempt OTC products without expiry). (iv) EU market: the longer of expiry + 1 year or 5 years after QP certification. Audit trails and signatures are retained for the same period as the record they belong to. A legal hold (complaint, recall, litigation) suspends expiry of retention.
TYPE DETAIL	ENFORCED (compute, block purge); CONFIGURABLE (company policy may only lengthen).
REGULATORY BASIS	Schedule M Part I 17.2 (retention at least one year after finished-product expiry); Drugs Rules Rules 74 and 78 with Schedule U (S/U).
INDIA / US / EU DIFFERENCES	Schedule M rule is the **same** as FDA 211.180(a) (expiry + 1 year) [chk] and **weaker** than EU Chapter 4 section 4.11 (longer of expiry + 1 year or 5 years after QP certification) [chk]. If the Rule 74/78 periods are confirmed, the Indian result can exceed FDA for short shelf lives.
HOW IT WILL BE TESTED	24-month tablet made Jan 2026, India only: retention not earlier than Jan 2031 (5 years) because expiry + 1 year is Jan 2029. Same batch for EU with QP certification Mar 2026: not earlier than Mar 2031.
SOURCE CHECK	P~ for Schedule M (retention sentence re-read; numbered 17.2.7 by WHO ordering); U for the Drugs Rules periods.

Source requirements: n235, n661, n678

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-09 Computerised-system controls for electronic batch records

Enforced · Design choice · Both editions · All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Role-based access; only authorised users enter or change data; access by unique credentials. (b) An always-on audit trail records every create, change and delete with previous value, new value, user and timestamp; it cannot be disabled or edited. (c) Completeness controls: sequence and heartbeat checks on instrument feeds so that data not captured while a device or the system was down are detected and flagged in the batch record. (d) Every manually entered critical value needs a second-person verification or a validated system check. (e) System incidents that could affect product quality or record reliability are logged, investigated and linked to affected batches. (f) Configuration and software changes go through change control (IN-03) with test evidence. (g) Validated backup and restore; restore test on a schedule. (h) Optional parallel secondary record (e.g., printout) can be attached. (i) Audit-trail review is a step of batch review (review by exception allowed) and a periodic system evaluation report is available.
TYPE DETAIL	ENFORCED; DESIGN-CHOICE for audit-trail review and periodic evaluation (not in Schedule M).
REGULATORY BASIS	Schedule M Part I 17.2 electronic data-processing clause (electronic or photographic records allowed; authorised persons only; record of changes and deletions; passwords or other means; independent check of critical entries; backup; readily available during retention); Part I 20.2-20.10 (IQ/OQ; qualified commercial software; retrospective validation; access and change controls; preventing omitted data; previous entry, who and when recorded; procedures; second check of critical manual entries; incidents; change procedure; backup; secondary means of recording). The 20.1 sentence is not legible in the uploaded copy.
INDIA / US / EU DIFFERENCES	**same** as EU Annex 11 sections 4, 6, 7, 10, 12, 13 [chk] and FDA 211.68(b) (authorised changes, input/output checks, backup) [chk]. **Weaker** than Annex 11 section 9 (audit trails regularly reviewed; reason for change documented) and section 11 (periodic evaluation) [chk], and than 21 CFR 11.10(e) (secure, computer-generated, time-stamped audit trail kept at least as long as the record) [chk]. The kernel applies the stricter items for every market.
HOW IT WILL BE TESTED	stop an equipment feed mid-step; a gap flag appears. Enter a critical value manually; a verifier is demanded. Restore a backup to a test instance; a closed batch renders completely.
SOURCE CHECK	P (20.2, 20.3, 20.6, 20.7 and the data-processing clause re-read); P~ for the 17.2 sub-clause number.

Source requirements: n237, n238

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-10 Electronic signatures (Indian basis)

Enforced | Configurable | Both editions | 2 of 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Baseline for all signatures: unique user ID plus secret, re-authentication at the moment of signing, signature manifestation (printed name, date/time with time zone, meaning such as performed/verified/approved/released), cryptographic binding of the signature to a hash of the signed content so any later change shows the signature as broken, no shared or reused IDs. (b) Configurable option: for selected meanings (default proposal: MBR approval and AP release) require a Class 3 digital signature certificate from a licensed certifying authority, or an eSign (Second Schedule technique) instead of the password signature. (c) Any record can be printed or exported with signatures and audit trail on demand.
TYPE DETAIL	ENFORCED (baseline); CONFIGURABLE (PKI/eSign option per signature meaning).
REGULATORY BASIS	Schedule M Part I 17.3.9 (packaging-record entries identified by signature or electronic password); Part I 17.3.8 (processing record dated and signed); Part XII 6.1 (e-signatures authenticated and secure; hard copy obtainable where electronic records are used; 6.1.7 originals or true copies); IT Act 2000 sections 3A, 5 and Second Schedule. See section D.
INDIA / US / EU DIFFERENCES	**weaker/less specific** than 21 CFR 11.50 (manifestation), 11.70 (signature/record linking), 11.100, 11.200 (two distinct identification components), 11.300 (identification code/password controls) [chk] and EU Annex 11 section 14 (same effect as handwritten within the company, permanently linked, time and date) [chk]. The baseline in (a) meets all three regimes.
HOW IT WILL BE TESTED	sign a step, then alter the value directly in the test database; signature verification reports the record as altered. Sign a release with a DSC; it validates against the CCA root chain.
SOURCE CHECK	P for Schedule M (17.3.9 wording and Part XII sentence re-read; Part XII sentence number 6.1.9 inferred); S for IT Act.

Source requirements: n663, n667, n242

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-11 Certified evidence export

Design choice | Both editions | Search-based only (0 of 1 checked against official text)

WHAT THE SYSTEM DOES	Any batch record can be exported as a package: rendered human-readable copy, machine-readable data, full audit trail, and a manifest with a SHA-256 hash per file plus system name, version, time source, export operator and time. The hashes re-verify on import. The package supplies the facts needed for the certificate that accompanies electronic evidence in Indian proceedings, and keeps origin and date/time metadata (IT Act s.7).
REGULATORY BASIS	Bharatiya Sakshya Adhiniyam 2023 section 63 (in force 1 Jul 2024, replacing Evidence Act s.65B); IT Act 2000 section 7.
INDIA / US / EU DIFFERENCES	**different**: no equivalent evidence certificate; closest are 21 CFR 11.10(b) (accurate and complete copies in human-readable and electronic form for inspection) [chk] and EU Annex 11 section 8 (clear printed copies; printouts showing changes for batch-release records) [chk].
HOW IT WILL BE TESTED	export a batch; alter one byte; hash verification fails.
SOURCE CHECK	S (certificate form not reviewed; open question G-5).

Source requirements: n668

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-12 Validation package and site master file support

Design choice | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The vendor ships a validation package usable in the customer's validation master plan: URS, risk assessment, IQ/OQ scripts with evidence, traceability matrix, periodic-review template, and a short, accurate system description for the "computerised systems validation" item of the site master file. Supplier-qualified functions are documented so that customer testing can be reduced as allowed for qualified commercial software.
TYPE DETAIL	DESIGN-CHOICE (vendor deliverable).
REGULATORY BASIS	Schedule M Part I section 5 (qualification and validation; validation master plan; attention to automated systems; periodic review); Appendix I site master file item on qualification and calibration including computerised-system validation arrangements; Part I 20.2-20.3.

INDIA / US / EU DIFFERENCES	**same** as EU Annex 11 section 4 [chk] and Annex 15 [gk]; FDA 211.68 and 21 CFR 11.10(a) validation [chk/gk].
HOW IT WILL BE TESTED	the package contains each listed artefact and the SMF paragraph.
SOURCE CHECK	P.

Source requirements: n273

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-13 Master formula (MBR) content and hold times

Enforced | Tablets | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) One approved master per product and per batch size; a batch of a different size needs its own approved master (or a separately approved scaled master). (b) MBR approval fails validation if any of these is empty: product name and product code linked to its specification; dosage form, strength, batch size; each starting material with INN where one exists, quantity, designated name and unique material code; a flag for substances lost in processing (e.g., granulation water, solvents); expected final yield with limits and pertinent intermediate yields; processing location and principal equipment class; references to methods for preparing critical equipment (cleaning, assembly, calibration, sterilisation, use); stepwise instructions; each IPC with its limits; storage requirements for products (container, labelling, special conditions); special precautions; **the permitted hold time for every intermediate and in-process material**. (c) Tablets edition: each mixing, blending and drying step must carry its critical operating parameters (at least time and temperature) with limits, and the batch record captures the observed values (IN-36).
REGULATORY BASIS	Schedule M Part I 17.3.6 (master formula list, including hold time as its final item); Part VIII 2.3 (time and temperature for mixing, blending and drying set in the master formula, monitored and recorded).
INDIA / US / EU DIFFERENCES	per-batch-size masters **same** as FDA 211.186(a) [chk] and EU Chapter 4 (formula and instructions per product and batch size; 4.17-4.18) [chk]. Hold time as a mandatory master element is **stricter** than FDA 211.111 (time limits for each production phase only when appropriate) [chk] and EU 4.17/4.18 (no hold-time item) [chk]. Time/temperature CPPs **same** as EU 4.18(d) [chk].
HOW IT WILL BE TESTED	approve an MBR with one intermediate lacking a hold time; approval is refused. Try to issue a 200 kg batch from a 100 kg master; refused.
SOURCE CHECK	P.

Source requirements: n239, n262, n678

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-14 Master packaging instructions

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	A separate approved packaging master per product x pack size x pack type, linked to the MBR, containing: product name; form, strength and method of application where relevant; pack size; complete packaging bill of materials for the standard batch with quantities, sizes, types and each material's specification code; an image of each approved printed component with the positions of batch number, expiry date and (for India packs) manufacturing date and MRP marked; line-clearance checks before and after packaging; the operation description and equipment; IPCs with sampling instructions and limits. A packaging order cannot start unless every printed component in the bill has an attached approved specimen.
REGULATORY BASIS	Schedule M Part I 17.3.7 (packaging instructions).
INDIA / US / EU DIFFERENCES	**same** as EU Chapter 4 section 4.19 [chk] and FDA 211.186(b) master record content [gk].
HOW IT WILL BE TESTED	create two pack sizes of one product; each needs its own approved packaging master.
SOURCE CHECK	P.

Source requirements: n240

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-15 Batch processing record content

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The executed record is generated from the approved MBR (no re-typing). Before processing, a recorded check that the equipment and workstation are free of previous products, documents and materials and are clean and suitable. Captured at the time of action: product name; batch number; date/time of start of significant intermediate stages and of completion; **named person accountable for each stage** (separate from step performers); initials/IDs of operators of significant steps and, where appropriate, of checkers (e.g., weighing); lot or analytical control number and actual quantity of each starting material, including recovered or reprocessed material; significant operations, events and major equipment IDs; **permitted hold time and actual hold start/stop per intermediate**, with an alarm or deviation when exceeded; IPC results with performer; yields at pertinent stages with a mandatory explanation when outside limits; notes on special problems with signed authorisation for any departure from the master. On completion the processing lead dates and e-signs the record.
REGULATORY BASIS	Schedule M Part I 17.3.8 (batch processing records).
INDIA / US / EU DIFFERENCES	mostly **same** as FDA 211.188(b) (dates, equipment, component lots, weights, IPC results, yields, persons performing and checking each significant step) [chk] and EU Chapter 4 section 4.20 [chk]. **Stricter** on two items absent from both: the person responsible for each stage, and the hold time for intermediates [chk].
HOW IT WILL BE TESTED	exceed a hold time; the system blocks the next step or opens a deviation. Enter an out-of-limit stage yield; a comment and deviation link are forced.
SOURCE CHECK	P (item list re-read, including hold time and responsible person per stage; sub-item letters inferred).

Source requirements: n241

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-16 Batch packaging record content

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	A separate packaging record per packaging batch or part-batch, linked to the bulk batch, built from the approved packaging master. Before packaging: recorded line clearance. Captured at the time of action, with the responsible person identified by signature or electronic password: product name, bulk batch number and quantity to be packed, finished batch number, planned and actual quantity and reconciliation; date and time of operations; responsible person; operator IDs for significant steps; identity and conformity checks against the instructions including IPC results; operations, equipment and lines; handling of unpacked product returned to store; specimens (image capture) of printed materials including coded batch/expiry/manufacturing date/MRP at start and at intervals; notes on problems and deviations with authorisation; quantities and reference numbers of printed materials and bulk issued, used, destroyed or returned.
REGULATORY BASIS	Schedule M Part I 17.3.9 (batch packaging records).
INDIA / US / EU DIFFERENCES	**same** as EU Chapter 4 section 4.21 [chk] and FDA 211.188(b)(6), (8) and 211.134 (examination of packaged products, results in the batch record) [chk].
HOW IT WILL BE TESTED	run two part-batches; two linked packaging records exist, each with start/mid/end coding specimens.
SOURCE CHECK	P.

Source requirements: n242

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-17 Yields and reconciliation

Enforced | Configurable | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Yields are calculated automatically at each stage against MBR limits; printed components and bulk are reconciled automatically (issued = used + destroyed + returned, within limits). Any out-of-limit yield or significant/unusual reconciliation discrepancy opens a deviation that blocks release until explained and closed. No reconciliation waiver is offered for roll labels.
TYPE DETAIL	ENFORCED; CONFIGURABLE (limits per product/component).
REGULATORY BASIS	Schedule M Part I 18.2.3, 18.4.6, 18.5.10 (yield checks and reconciliation; investigation before release).
INDIA / US / EU DIFFERENCES	**same** as EU 5.44 and 5.61 [chk] and FDA 211.103 [gk], 211.125(c) and 211.192 [chk]. FDA 211.125(c) waives label reconciliation for cut or roll labeling when 100% examination is done [chk]; India has no such waiver, so India is **stricter** there and the waiver is not offered.
HOW IT WILL BE TESTED	enter a carton reconciliation with a 2% unexplained loss above the configured limit; a deviation opens and release is blocked.
SOURCE CHECK	P.

Source requirements: n245

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-18 Batch numbering and allocation register

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	A configurable batch-number generator with a site-wide uniqueness constraint: a number is never reissued, including after cancellation or reprocessing; manual entry of an existing number is rejected. Processing and packaging batch numbers are linked by a defined rule. Every allocation writes immediately to an allocation register with at least allocation date, product identity and batch size. A reworked batch always receives a new batch number (IN-19).
TYPE DETAIL	ENFORCED (uniqueness); CONFIGURABLE (number format, bulk-to-pack linkage rule).
REGULATORY BASIS	Schedule M Part I 17.3.10.8-17.3.10.11 (numbering SOP, relation of processing and packaging numbers, no reuse, immediate recording of allocation); 14.32 (reworked batch gets a new number).
INDIA / US / EU DIFFERENCES	new number for rework is **stricter** than FDA 211.115 [chk] and EU 5.67 [chk] (neither requires it). Unique, history-traceable lot numbers **same** as FDA 211.130(c) [chk].
HOW IT WILL BE TESTED	cancel a batch; its number is never offered again. Export the requirements list; the three fields are present.
SOURCE CHECK	P (14.32 re-read); P~ for 17.3.10 sub-numbers.

Source requirements: n243, n249

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-19 Rework, reprocessing and recovery

Enforced | Design choice | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Rework of a rejected batch is modelled as a new batch with a new number and a parent-child link to the rejected batch(es); it requires an approved rework procedure/protocol and a linked risk evaluation; a record of the rework is part of the new batch record. (b) Adding an earlier in-specification batch (recovery) into a later batch of the same product is allowed only at an MBR-defined stage, with prior authorisation, a risk evaluation that addresses shelf-life impact, and recording of source batch numbers and quantities. The expiry of the receiving batch requires QA confirmation; the default proposal is not later than the earliest expiry of any recovered input (design choice). (c) QC receives an automatic flag to consider additional testing of any finished batch that was reworked, reprocessed or contains recovered material. (d) US-market batches: reprocessing also needs quality-unit approval.
TYPE DETAIL	ENFORCED; DESIGN-CHOICE for the default expiry proposal.
REGULATORY BASIS	Schedule M Part I 14.32-14.34.

INDIA / US / EU DIFFERENCES	**same** as EU 5.67-5.69 [chk] and FDA 211.115 (written procedures; quality control unit approval) [chk], except the new-number rule (IN-18), which is stricter.
HOW IT WILL BE TESTED	add recovered material without prior authorisation; the step is blocked.
SOURCE CHECK	P.

Source requirements: n249

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-20 Deviations, OOS and investigation scope

Enforced · Both editions · All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) A deviation is captured at the point of occurrence and linked to the step; a planned or unplanned departure from the MBR cannot continue until a designated role approves it by e-signature (QC co-approval configurable, on by default for product-impacting categories). (b) Closure requires root cause (or most probable cause), CAPA reference and conclusion; the impact assessment can list other batches and products, which are then flagged. (c) IPC results that are specification tests follow the same OOS workflow as QC results: the original result stays in the record, a re-entered passing value never replaces it, and the OOS stays open until the approved investigation closes. (d) Sampling steps print sample labels with material name, batch/lot number, container number, sample number, sampler identity (e-signature) and date. (e) QC results used for release are approved by a checker different from the analyst.
REGULATORY BASIS	Schedule M Part I 18.2.2 (written approval of any departure by a designated person, QC involved where appropriate); 18.5.12 (review before AP; investigation extended to other batches); section 1 PQS list (root cause, CAPA effectiveness); 19.1-19.3 (QC independence; records of sampling and testing; review of production documentation in product assessment); 19.6.6 (sample label items); 19.6.7 (OOS investigated under an approved procedure).
INDIA / US / EU DIFFERENCES	**same** as FDA 211.100(b) (deviations recorded and justified) and 211.192 (investigation extended to other batches) [chk]; sample identification **same** as FDA 211.84(c)(5) [chk]; OOS handling **same** as FDA OOS guidance [gk] and EU Chapter 6 [gk].
HOW IT WILL BE TESTED	enter an out-of-specification IPC assay and then a passing re-test; both remain visible and the OOS stays open.
SOURCE CHECK	P (18.2.2, 19.6.6, 19.6.7 re-read).

Source requirements: n244, n659, n230

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-21 Line clearance, status labelling and identification

Enforced · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Rooms, lines and major equipment carry a status (in use for product/strength/batch/stage; cleaned; to be cleaned; out of service) shown on screen and on printable labels; containers of materials and bulk carry product, strength, batch and stage. An electronic line-clearance checklist, recording the previous product, is a gating step before (and, for packaging, after) each operation; its signer and time appear in the batch record. Each packaging line displays the product and batch in the line status. Required environmental readings for the step are captured into the batch record.
REGULATORY BASIS	Schedule M Part I 18.2.5, 18.4.1-18.4.2 (identification; checks before processing), 18.5.2-18.5.3 (packaging line clearance by checklist; line identification).
INDIA / US / EU DIFFERENCES	**same** as EU 5.40, 5.50, 5.51 [chk] and FDA 211.105(a), 211.130(e) and 211.188(b)(6) [chk].
HOW IT WILL BE TESTED	start compression on a room with status "to be cleaned"; refused.
SOURCE CHECK	P.

Source requirements: n246

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-22 Packaging operation controls (coding, readers, on-line checks, leftovers)

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Coding (batch, expiry, manufacturing date, MRP where applicable) is verified and recorded at start and at intervals; hand-applied printing is re-checked at a shorter interval. (b) Electronic code readers, vision systems, label counters and similar devices are challenged at start, at intervals and at end; a failed challenge places everything since the last good challenge on hold. (c) On-line checks include at least pack appearance, completeness, correct product and materials, correct overprint and working line monitors. (d) Samples taken from the line are not returned. (e) An "unusual event" sub-workflow requires inspection, investigation and authorised approval before product re-enters the line, with a detailed record. (f) Labelling must follow filling/sealing promptly, or the MBR defines controls against mix-ups for the interval. (g) At the end, unused batch-coded materials must be recorded as destroyed; uncoded printed materials return to stock only through a documented check.
TYPE DETAIL	ENFORCED; CONFIGURABLE (check frequencies, never below the MBR minimum; tighter for manual labelling).
REGULATORY BASIS	Schedule M Part I 18.5.4-18.5.11.
INDIA / US / EU DIFFERENCES	**same** as EU 5.54-5.62 [chk] and FDA 211.125(d) (destroy excess coded labeling), 211.130 and 211.134 [chk].
HOW IT WILL BE TESTED	close a packaging order with coded cartons unaccounted for; closure demands a destruction entry.
SOURCE CHECK	P (18.5.4-18.5.9 re-read).

Source requirements: n247

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-23 Dispensing

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Container-level barcode identity check; hard block on any lot not released by QC, expired, or past retest; only users in the designated-dispenser role may dispense; each weighing or measurement is independently checked by a second person or by a validated balance interface configured as the independent check; dispensed materials for a batch are kept and labelled together with a kit label linked to the batch.
REGULATORY BASIS	Schedule M Part I 14.14, 14.16-14.20.
INDIA / US / EU DIFFERENCES	**same** as EU 5.33-5.34 and 5.37-5.39 [chk] and FDA 211.84(a) and 211.101(c)-(d) (second-person or automated check) [chk].
HOW IT WILL BE TESTED	scan a quarantined lot, an expired lot and a lot past retest; all refused.
SOURCE CHECK	P (14.14, 14.16-14.18 re-read).

Source requirements: n248

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-24 Material status, labels, excipient release route, packaging materials and returns

Enforced | Configurable | Both editions | All 3 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) The system is the authority for material status (quarantine, under test, released, rejected, returned, recalled); status changes are e-signed and audit-trailed; with the validated system, labels may carry a barcode plus minimal text. (b) Each lot within a delivery is sampled, tested and released separately; every delivery of primary and printed packaging gets a unique reference. (c) Excipient release: Schedule M 14.15 lets QC release non-API raw materials without batch testing based on vendor approval and statistical analysis of past results. The system supports this route only when QA enables it, stores the statistical justification, and limits such lots to batches whose market profile is India-only. For any batch with a US or EU market, the dispensing check blocks a lot that has no identity test result. Default for new tenants: identity test required for every lot in every profile, because the Schedule M identity-test clause 19.7.2 is truncated in the uploaded copy and may restrict 14.15 (gap F-2). (d) Printed materials are issued only by a designated role against the packaging bill of materials with scan confirmation of artwork version; superseded artwork is blocked and its destruction recorded. (e) Returned goods cannot be made available again without a recorded QC assessment (nature, storage needs, condition, history, time since issue).
TYPE DETAIL	ENFORCED; CONFIGURABLE (vendor-history release route, India-only).
REGULATORY BASIS	Schedule M Part I 14.10-14.13 (receipt, labelling of stored materials, reduced label text with validated systems), 14.15 (non-API raw materials released without batch testing), 14.21-14.25 (packaging materials), 14.36 (returned goods), 19.7.1 (identity, strength, purity before release).
INDIA / US / EU DIFFERENCES	the vendor-history route is **weaker** than FDA 211.84(d)(1)-(2) (at least one identity test on each component; supplier report of analysis accepted only with an identity test and periodic validation of supplier results) [chk] and EU 5.35 (identity testing of each batch per Annex 8) [chk]. Status, labels and packaging-material rules **same** as EU 5.31-5.32, 5.45-5.48 [chk].
HOW IT WILL BE TESTED	release an excipient lot on vendor history and try to dispense it into a batch with markets {India, US}; blocked until an identity result exists.
SOURCE CHECK	P (14.15 and 19.7.1 re-read).

Source requirements: n250, n251, n673

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-25 Equipment status, logbooks, calibration and clean-hold time

Enforced | Configurable | Both editions | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Equipment and room logbooks are written automatically from execution (who, product, batch, start/end, cleaning, maintenance), chronological, correctable only through audited corrections. (b) Before a step can use equipment, the system checks qualified, calibrated (not past due), cleaned and not-out-of-service status; balances need a daily/before-use check record. (c) Each equipment class has a validated clean-hold time (and dirty-hold time where defined); the system computes "clean until" and blocks use after it, forcing re-cleaning. (d) Equipment or utility failure can be signalled from the step and removes the item from use until repaired.
TYPE DETAIL	ENFORCED; CONFIGURABLE (hold-time values from validation data).
REGULATORY BASIS	Schedule M Part I 17.3.10.5 and 17.3.10.16-17.3.10.17 (equipment SOPs near the equipment; logs), 18.4.3 and 18.4.9 (failures, calibration), 18.4.4 (data-based time limits for cleaned equipment and process materials), 18.4.5 and 18.2.8.
INDIA / US / EU DIFFERENCES	**same** as FDA 211.67, 211.68(a) and 211.182 [chk/gk] and EU Chapter 3 and Chapter 4 logbook clauses [gk]. The explicit clean-hold time is **same** in intent as FDA 211.67(b) [gk] and EU Annex 15 cleaning validation [gk].
HOW IT WILL BE TESTED	use a balance with calibration past due; the weighing step is blocked. Wait past a press's clean-hold time; the next batch cannot use it.
SOURCE CHECK	P~ (17.3.10.5 re-read; others from register).

Source requirements: n266, n267

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-26 Cross-contamination and room/product eligibility

Enforced | Configurable | Tablets | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Product-to-room and product-to-equipment restrictions (dedicated, campaign-only, segregated) are master data; campaign length and changeover cleaning-validation references are stored; residue (swab/rinse) results gate the changeover. (b) The system refuses a second open batch of a different product in the same room unless the room is configured as segregated with a recorded risk justification. (c) Tablets: each press sits in its own cubicle unless the same product runs on all presses there or the press is self-enclosed; hoppers, containers and presses carry identity labels verified by scan at charging; highly potent products may only be scheduled in rooms flagged with a negative pressure cascade; corridors are at higher pressure than cubicles and cubicles above atmospheric; pre-use checks include nozzles clean and no lubricant leakage.

TYPE DETAIL	ENFORCED (eligibility rules); CONFIGURABLE (room/equipment master data).
REGULATORY BASIS	Schedule M Part I 18.2.4, 18.2.7, 18.3.1-18.3.5, 18.5.1; Part VIII 1.1, 1.6-1.10, 3.1-3.2.
INDIA / US / EU DIFFERENCES	**same** in intent as EU 5.49 (no packaging in close proximity without segregation) [chk], EU Chapter 3/5 cross-contamination measures [gk] and FDA 211.42(c), 211.46 [gk]; the tablet cubicle and pressure-cascade specifics are **stricter/more prescriptive** than either text.
HOW IT WILL BE TESTED	schedule a highly potent product in a positive-pressure cubicle; blocked. Charge granules from a container labelled for another batch; the scan is refused.
SOURCE CHECK	P (Part VIII 1.6-1.10 and 3.1-3.2 re-read).

Source requirements: n672, n677

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-27 Hazardous products (sex hormones, steroids, cytotoxics)

Enforced · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Product master data carries a hazard class set from a documented risk assessment (which stages fall under Part III). The class restricts rooms, lines and equipment to contained ones and inserts mandatory PPE, containment-check and decontamination/de-gowning steps into the MBR; the executed checks are recorded. Where breathing air is used, the room master data records the 100% back-up with automatic changeover and its status is checked at step start.
TYPE DETAIL	ENFORCED (eligibility and inserted steps once the hazard class is set).
REGULATORY BASIS	Schedule M Part III (general; 2.1-2.3 risk assessment including OELs; 4 PPE and breathing air).
INDIA / US / EU DIFFERENCES	**different/more prescriptive** than EU Chapter 3/5 health-based exposure limits approach [gk] and FDA 211.42/211.46 [gk].
HOW IT WILL BE TESTED	classify a product as cytotoxic; the MBR designer adds PPE and decontamination steps and offers only contained areas.
SOURCE CHECK	P~ (Part III 2.1-2.3 re-read; the rest is interleaved with Part IV text in the uploaded copy).

Source requirements: n666

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-28 Self-inspection support and supplier approval

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	A read-only auditor role with batch sampling and exports of deviation/CAPA status; findings link to batches and CAPA both ways; the self-inspection schedule reminds QA when 12 months since the last self-inspection are about to pass (frequency may be set shorter, never longer, for Indian sites). The approved-supplier list is master data used at receipt and dispensing.
TYPE DETAIL	CONFIGURABLE (audit mode, schedule); ENFORCED (read-only auditor role).
REGULATORY BASIS	Schedule M Part I 10.1-10.8 (10.4: frequency set in the procedure, at least once a year).
INDIA / US / EU DIFFERENCES	**stricter** than FDA Part 211 (no self-inspection clause) [chk] and EU Chapter 9 (programme-based, no fixed minimum) [gk].
HOW IT WILL BE TESTED	an auditor opens a closed batch record; edit functions are unavailable.
SOURCE CHECK	P (10.4 re-read).

Source requirements: n269

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-29 Complaints, recalls and genealogy queries

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Complaint and recall records link to batch IDs; genealogy answers within seconds "which batches used material lot X" and "which batches contain reprocessed or recovered material from batch Y"; per-batch distribution data (customer, quantity, date) supports a recall reconciliation of delivered versus recovered quantities; the AP role owns recall execution; a flag records that the licensing authority was informed of an intended recall.
TYPE DETAIL	ENFORCED (links); CONFIGURABLE (notification templates).
REGULATORY BASIS	Schedule M Part I section 6 (complaints; pharmacovigilance) and section 7 (7.1-7.12 recalls; AP responsible; final report with reconciliation).
INDIA / US / EU DIFFERENCES	**same** in substance as FDA 211.198 and 211.204 [gk] and EU Chapter 8 [gk]; licensing-authority notification is **different** (Indian authority).
HOW IT WILL BE TESTED	from a defective API lot, list every finished batch including those with recovered material from affected batches.
SOURCE CHECK	P (7.1-7.4 and 7.12 re-read).

Source requirements: n270

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-30 Loan-licence and contract manufacture

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	A batch can belong to a loan licensee or contract giver while made on the host site; the record header carries both the host manufacturing licence number and the loan licence number (IN-31, section C); completed records can be shared read-only with the loan licensee/contract giver; the release workflow shows which party's authorised person releases, per the written contract.
TYPE DETAIL	CONFIGURABLE (party model); ENFORCED (read-only sharing).
REGULATORY BASIS	Schedule M Part I 9.1-9.5 (loan licence and contract production; 9.3.x duties of the loan licensee/contract giver).
INDIA / US / EU DIFFERENCES	**different**: the loan licence is an Indian legal construct (Drugs Rules Forms 25-A/28-A, S); EU Chapter 7 outsourced activities [gk] and FDA contract-manufacturing expectations [gk] cover the contract part.
HOW IT WILL BE TESTED	create a loan-licence batch; both licence numbers appear and a contract-giver reviewer cannot edit.
SOURCE CHECK	P.

Source requirements: n272

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-31 Finished-pack label content and coding verification

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Artwork templates are typed by destination class: India domestic (Schedule M 17.3.1 plus Drugs Rules Rules 96/97, section C), India export (Rule 94), US, EU. (b) The India-domestic template cannot be approved without fields for: drug name; active ingredients with INN and amount each, and net contents; batch number; **date of manufacture and expiry date in plain (uncoded) form**; special storage/handling; directions, warnings and precautions; name and address of the manufacturer and of the party placing the product on the market. (c) The packaging-record coding check compares every printed variable (batch, manufacturing date, expiry, MRP where applicable) against batch data at start, at intervals and at end. (d) Status labels on containers, equipment and premises follow the company format, optionally colour-coded.
TYPE DETAIL	ENFORCED (per destination template); CONFIGURABLE (template contents per market).
REGULATORY BASIS	Schedule M Part I 17.3.1 (items (a)-(g), including expiry and manufacture date uncoded); Drugs Rules Rule 96 (S).
INDIA / US / EU DIFFERENCES	**stricter** than FDA (lot number and expiration date required under 21 CFR 201.17 and 211.137; no manufacturing date) [gk] and EU Directive 2001/83/EC Article 54 (expiry and batch number; no manufacturing date) [gk]. Applies to India-domestic packs; export packs follow IN-52 destination rules and Rule 94.
HOW IT WILL BE TESTED	load an India-domestic template without a manufacturing-date field; approval fails. Print an expiry one month off; the coding check fails.
SOURCE CHECK	P for Schedule M (item list re-read); S for Rule 96.

Source requirements: n252

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-32 Reference and retention samples

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The packaging record creates a mandatory reference-sample pull for each finished batch, in the final pack (or an equivalent pack for very large packs), with quantity not less than twice that needed for the full test plan. The FDA allowance to leave sterility and pyrogen tests out of the doubling is not used, because the Indian condition (if confirmed) covers all tests on the batch. Starting-material samples are also retained. Retention date = maximum over the batch's markets: India expiry + 3 months (3 years from manufacture where no expiry is labelled) pending confirmation; US expiry + 1 year; EU expiry + 1 year. For US batches, an annual visual examination task is scheduled for representative reserve samples.
TYPE DETAIL	ENFORCED (quantity and retention date); CONFIGURABLE (company policy may lengthen).
REGULATORY BASIS	Schedule M Part I 19.3(g) (sufficient samples retained in final pack for an appropriate time); Drugs Rules Rules 74 and 78 conditions of licence (quantity and period, S).
INDIA / US / EU DIFFERENCES	period **weaker** than FDA 211.170(b)(1) (1 year after expiry) [chk] and EU Annex 19 (at least 1 year after expiry) [gk]. Quantity **same** (twice the full testing need; FDA excludes sterility and pyrogen tests) [chk]. FDA's yearly visual examination of reserve samples [chk] has no Indian counterpart (**stricter** FDA).
HOW IT WILL BE TESTED	batch with markets {India, US}: sample retention = expiry + 1 year; {India}: expiry + 3 months unless company policy is longer.
SOURCE CHECK	P for 19.3(g) (re-read); S for the Drugs Rules quantity and period.

Source requirements: n660

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-33 Pharmacopoeia and edition references in specifications and limits

Enforced · Configurable · Both editions · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Each specification and each IPC limit in an MBR stores its source (IP, BP, Ph. Eur., USP, in-house) and edition as data. Marking a new edition effective (e.g., IP 2026 from 1 Jul 2026) lists every MBR whose limits cite the previous edition and opens a change-control task. For multi-market products, each limit shows its source and the narrower range is enforced (IN-34, IN-52). Reference standards record their source (IP reference standards from the Indian Pharmacopoeia Commission). The latest IP edition, the 1940 Act and the 1945 Rules are listed as controlled reference documents.
TYPE DETAIL	ENFORCED (reference field mandatory); CONFIGURABLE (edition effective dates).
REGULATORY BASIS	Schedule M Part I PQS resources list (Act, Rules, latest IP, official guidance), 15.2 (IP reference standards from IPC), 17.3.2 (specifications revised for new pharmacopoeia editions); packaging materials to meet IP requirements.
INDIA / US / EU DIFFERENCES	**different**: US products are held to USP-NF [gk], EU products to Ph. Eur. [gk]; the engine keeps the IP limit for Indian compliance and adds the importing market's compendial limit.
HOW IT WILL BE TESTED	mark IP 2026 effective; all MBRs citing IP 2022 appear on the review list.

SOURCE CHECK	P (17.3.2 revision sentence re-read); S for the IP 2026 dates.

Source requirements: n665

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Tablets edition

IN-34 Weight-variation limits by pharmacopoeia

Enforced · Configurable · Tablets · Search-based only (0 of 1 checked against official text)

WHAT THE SYSTEM DOES	Individual-weight IPC limits are computed from the target tablet weight using a pharmacopoeia band table stored as versioned data. IP/BP/Ph. Eur. uniformity-of-mass bands as reported by secondary sources (to be confirmed against IP 2026): 80 mg or less 10%; more than 80 mg and less than 250 mg 7.5%; 250 mg or more 5%. Legacy USP in-process bands often used by US-market firms: 130 mg or less 10%; 130-324 mg 7.5%; more than 324 mg 5%. The MBR shows which table applies; for a product sold in India and the US the tighter band applies (IP is equal or tighter at every weight: e.g., 100 mg is 7.5% vs 10%; 300 mg is 5% vs 7.5%). Release testing of uniformity of dosage units (USP chapter 905 / Ph. Eur. 2.9.40 acceptance value) is a separate calculation.
TYPE DETAIL	CONFIGURABLE (band tables as versioned reference data); ENFORCED (tighter band for multi-market products).
REGULATORY BASIS	Schedule M Part VIII 3.3-3.4, 3.8 (weight variation against pharmacopoeial parameters); Indian Pharmacopoeia uniformity-of-weight test (not seen).
INDIA / US / EU DIFFERENCES	**same** as Ph. Eur. 2.9.5 bands [gk]; **stricter** than the legacy USP bands for 80-130 mg and 250-324 mg tablets [gk]; **different** from current USP chapter 905, which judges content/weight uniformity by acceptance value [gk].
HOW IT WILL BE TESTED	300 mg tablet: India or India+US gives plus/minus 5%; US-only legacy table gives 7.5%.
SOURCE CHECK	U (band values from secondary sources; IP 2026 wording and any adoption of the harmonised UDU chapter not confirmed).

Source requirements: n669

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-35 Compression in-process controls

Enforced · Configurable · Tablets · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Calibrated balance status is checked at step start. (b) Start of each compression run: a sample from every station/compression point must be tested for the pharmacopoeial parameters (appearance, weight variation, disintegration, hardness, friability, thickness); the start check cannot be signed until each station has a result; on multi-point machines this station-wise check repeats at fixed intervals set in the MBR. (c) During compression an IPC timer forces a sample check against the in-process specification at intervals of at most 30 minutes; the interval may be configured shorter but never longer; a missed check raises an alarm and a deviation. (d) The same parameters plus lubricating-oil contamination are checked periodically at an MBR-defined frequency. (e) De-dusting and foreign-matter/defect checks are recorded; tablets are collected in clean, labelled containers. (f) Rejected or discarded tablets are held in identified containers and their quantity is recorded and fed into yield reconciliation.
TYPE DETAIL	ENFORCED (30-minute ceiling for India-made batches); CONFIGURABLE (shorter intervals; frequency of the periodic parameters).
REGULATORY BASIS	Schedule M Part VIII 3.3-3.8.
INDIA / US / EU DIFFERENCES	**stricter** than FDA 211.110(a) (lists weight variation and disintegration as IPCs, no frequency) [chk] and EU 5.43 (necessary IPCs carried out and recorded, no frequency) [chk].
HOW IT WILL BE TESTED	run 45 minutes without an IPC entry; the alarm fires at 30 minutes. On a 16-station press the start check cannot be signed with 15 results.
SOURCE CHECK	P (3.3-3.8 re-read; see correction 5 on what the 30-minute rule covers).

Source requirements: n261

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-36 Sifting, mixing, granulation and drying controls; sieve, punch and die checks

Enforced · Tablets · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Step types for sifting, dry mixing, granulation, drying (fluid-bed/tray), blending and lubrication carry typed CPP fields with limits (time and temperature at minimum; others such as impeller speed, inlet/outlet air temperature, LOD as the MBR defines); observed values are captured. (b) Sieves/screens record ID and mesh, with an integrity check both before and after each use; punches and dies record an inspection for wear or breakage before and after each use; a failed after-use check opens a deviation covering the material processed. (c) Sieve residues are examined for foreign material at the MBR-defined frequency. (d) Fluid-bed filter bags are tracked by ID: a bag registered to another product is blocked unless a washing record exists; for products flagged highly potent or sensitising, product-dedicated bags only. (e) Metal-detector challenge records where fitted. (f) Granulation/coating solutions record preparation time and use-by limits. (g) Dry ingredients are sifted before use unless the MBR records that input quality is assured.
REGULATORY BASIS	Schedule M Part VIII 1.3-1.4 (metal detector recommended; wooden equipment avoided; screens, sieves, punches and dies checked before and after each use; sifting), 2.1-2.5 (dust extraction; sieve residues; CPPs in master formula and batch record; FBD bags; solutions).
INDIA / US / EU DIFFERENCES	**stricter/more prescriptive** than FDA 211.63-211.67 [gk] and EU Chapter 3/5 [gk], which do not name before-and-after checks of sieves, punches and dies or FBD bag rules.
HOW IT WILL BE TESTED	omit the after-use sieve check; the step cannot be closed. Use an FBD bag from another product without a wash record; blocked.
SOURCE CHECK	P (1.3-1.4 and 2.1-2.5 re-read).

Source requirements: n262

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-37 Coating, capsule shells and printing of tablets/capsules

Enforced | Tablets | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	Coating steps record solution/suspension preparation time and a use-by limit (freshly prepared; use after the limit is blocked), spray and air parameters, and room temperature/RH. Empty capsule shells are handled as a component lot with storage-condition checks. After printing of tablets or capsules, a QC approval of the printed bulk is a hard gate before packaging or sale; simultaneous printing of different products/batches requires a segregation record.
REGULATORY BASIS	Schedule M Part VIII 4.1-4.2, 5, 6.1-6.2; Part I 14.29-14.31.
INDIA / US / EU DIFFERENCES	the post-printing QC approval gate is **stricter** (no FDA/EU counterpart) [chk: not in 211 or EU Ch.5]; coating controls **same** in intent as FDA 211.113(a) [chk].
HOW IT WILL BE TESTED	open a packaging order for printed tablets without the QC approval; refused.
SOURCE CHECK	P (4.1-4.2, 5, 6.1-6.2 re-read).

Source requirements: n263, n676

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-38 Strip and blister packaging

Enforced | Configurable | Tablets | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Before a new batch, the counting/strip/blister equipment clearance (removal of rogue tablets, capsules, foils) is performed by one user and independently verified by a different user; both signatures are recorded. (b) Strip/blister IPC records defect categories (misprint, foil cut, missing tablet, poor sealing). (c) A periodic vacuum leak test of pockets is a scheduled IPC with frequency, sample size and pass/fail; a missed test flags the record. (d) Products flagged uncoated, potent or beta-lactam are only schedulable on eligible lines; potent and beta-lactam packing only in isolated areas.
TYPE DETAIL	ENFORCED; CONFIGURABLE (leak-test frequency and sample size).
REGULATORY BASIS	Schedule M Part VIII 7.1-7.4.
INDIA / US / EU DIFFERENCES	**stricter** than FDA 211.130(e) and 211.188(b)(6) (inspection of the area before and after use documented; no explicit independent-check requirement for the clearance) [chk] and EU 5.50 (line clearance by checklist) [chk]; periodic pocket leak testing is not named in either [chk].
HOW IT WILL BE TESTED	one user signs both "performed" and "verified" for blister clearance; refused. Skip the scheduled leak test; the record flags a missed IPC.
SOURCE CHECK	P (7.1-7.4 re-read).

Source requirements: n264

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-39 OSD room environment: pressure differentials, temperature and humidity

Enforced | Configurable | Tablets | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	Each step records room ID and captures differential pressure, temperature and RH at start and during long steps (manual entry or BMS/EMS import) against room-specific alert and action limits. The room master data validator refuses pressure ranges of adjacent rooms that overlap (e.g., 5-15 Pa next to 15-30 Pa is rejected). An excursion notifies Production and QA immediately and links to the batch as a deviation. Moisture-sensitive products may carry tighter product-specific RH limits.
TYPE DETAIL	ENFORCED (limit checks); CONFIGURABLE (room limits).
REGULATORY BASIS	Schedule M Part VIII 1.5, 1.14 (non-overlapping limits, 5-15 Pa vs 15-30 Pa example), 1.17, 1.25, 1.27-1.28.
INDIA / US / EU DIFFERENCES	**stricter/more prescriptive** than FDA 211.46 [gk] and EU Chapter 3 [gk] (no numeric cascade rules for non-sterile products).
HOW IT WILL BE TESTED	enter a differential pressure below the action limit at compression start; the step is blocked or a deviation opens and both departments are notified.
SOURCE CHECK	P for 1.5 and 1.14 (re-read); P~ for the others.

Source requirements: n265, n677

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Injectables edition

IN-40 Clean-area grades, particle limits and airflow

Enforced | Configurable | Injectables | All 2 source requirements checked against official text

WHAT THE SYSTEM DOES	(a) Room/zone master data carry grade A-D and a market-keyed limit table. India table (maximum particles per m3, 0.5 / 5.0 micrometre): Grade A 3,520 / 20 at rest and in operation; Grade B 3,520 / 29 at rest, 352,000 / 2,900 in operation; Grade C 352,000 / 2,900 at rest, 3,520,000 / 29,000 in operation; Grade D 3,520,000 / 29,000 at rest, in operation not defined. (b) Imported EM counts for the batch's processing window are checked against the resolved limits (section B row B-13). (c) Grade A unidirectional airflow records: velocity at 15-30 cm below the terminal filter within 0.36-0.54 m/s (guidance) and a hard minimum of 0.36 m/s at working level; readings below 0.36 m/s at working level are flagged. (d) Clean-up (recovery) time after operations: about 15-20 minutes (guidance) with a 100-fold reduction demonstrated; stored per room from qualification. (e) Classification sampling data: at least 1 m3 per location in Grade A; for the lower grades (C while operating, D while at rest) at least 2 litres sampled for no less than 1 minute per location. Classification results are kept separate from routine monitoring.
TYPE DETAIL	ENFORCED (market-resolved limit table); CONFIGURABLE (alert limits, locations).
REGULATORY BASIS	Schedule M Part II 4.1-4.6 and Table 1; 4.6.2-4.6.3 (sampling volumes, portable counters, isokinetic heads); 4.7.5 (clean-up period).

INDIA / US / EU DIFFERENCES	0.5 micrometre values **same** as EU Annex 1 (2022) Tables 1 and 5 [chk] and FDA 2004 Table 1 (ISO 5: 3,520; ISO 7: 352,000; ISO 8: 3,520,000, measured during activity) [chk]. 5.0 micrometre values **stricter (lower)** than EU 2022 (Table 5 monitoring: Grade A 29; B 29 at rest / 2,930 in operation; C 2,930 / 29,300; D 29,300; Table 1 classification: Grade A not specified) [chk]; FDA sets no 5.0 micrometre limit [chk]. EU 2022 is **stricter on monitoring method**: Grade A monitored continuously for both sizes at no less than 28 L/min (9.17) and consecutive low-level 5 micrometre counts investigated (Table 5 note 2) [chk]; EU Grade D in-operation limits must be set by the site from risk assessment [chk]. Airflow: EU 0.36-0.54 m/s at the working position (guidance) [chk]; India adds the hard 0.36 m/s minimum at working level (**stricter**). Clean-up: EU under 20 minutes guidance [chk] (**same**).
HOW IT WILL BE TESTED	import a Grade B in-operation 5.0 micrometre count of 2,910/m3: it exceeds the India limit (2,900) and therefore fails for every profile, although it would pass the EU-only value (2,930).
SOURCE CHECK	P (Table 1 cells reassembled from scattered lines; values consistent with WHO TRS 961 Annex 6 and the stated ISO 4.8 classification for Grade A).

Source requirements: n253, n671

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-41 Minimum grade per sterile operation

Enforced | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Each injectables MBR step type carries a required minimum grade and background grade derived from process type and product risk flags; the MBR validator rejects a step mapped to a lower-grade room; execution checks the room's current status and EM state. Terminally sterilised: filling at least Grade C; Grade A with at least Grade C background when the product is unusually at risk (slow filling, wide-necked containers, exposure for more than a few seconds); solutions and components prepared at least in Grade D (Grade C when at unusual microbial risk); ointments, creams, suspensions and emulsions prepared and filled in Grade C before terminal sterilisation. Aseptic: components after washing at least Grade D; sterile materials not later sterilised or filtered handled in Grade A with Grade B background; solutions to be sterile-filtered prepared in Grade C (D if closed system); aseptic handling/filling in A with B background; partially stoppered lyophilisation containers moved in A/B or in sealed trays in B; exposed sterile semi-solids/suspensions not later filtered made in A/B.
REGULATORY BASIS	Schedule M Part II 1.1-1.3, 4.11, 4.11.1-4.11.2 (and Table 2 examples).
INDIA / US / EU DIFFERENCES	**same** in structure as EU Annex 1 (2022) Tables 3 and 4 [chk] and FDA 2004 (ISO 5 critical area with ISO 7 supporting area) [chk]; EU 2022 adds a contamination control strategy overlay [chk].
HOW IT WILL BE TESTED	map an aseptic filling step to a Grade C room; MBR validation fails.
SOURCE CHECK	P (4.11.1 and 4.11.2 fragments re-read).

Source requirements: n675

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-42 Environmental monitoring data in the batch record and microbial limits

Enforced | Configurable | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) The batch record links the EM data for its processing window: continuous Grade A particle counts for the full critical processing including equipment assembly (capturing interventions and transients, alarming at alert limits), Grade B monitoring, and viable results (settle plates, active air, contact plates, glove prints, swabs) per location and session. (b) Release is blocked until all EM results for the filling window are complete (no plate still incubating) and reviewed, and every excursion is investigated. (c) Each viable result is evaluated individually against the resolved action limit (design choice: the Indian table gives averages, but averaging must not hide a single Grade A recovery). India values (CFU; air per m3 / settle plate 90 mm per 4 h / contact 55 mm / 5-finger glove): Grade A under 1 for all four; Grade B 10 / 5 / 5 / 5; Grade C 100 / 50 / 25 / not set; Grade D 200 / 100 / 50 / not set. (d) Alert and action limits must be configured for every location; exceeding an action limit or an adverse alert trend opens an investigation with CAPA.
TYPE DETAIL	ENFORCED (release gate, limit check); CONFIGURABLE (alert limits, locations).
REGULATORY BASIS	Schedule M Part II 4.7-4.10 and Table 3 (values stated to be averages and for information, not specifications); 4.8 (EM results considered in batch release review).
INDIA / US / EU DIFFERENCES	Grades B-D numbers **same** as EU Annex 1 (2022) Table 6 [chk]; EU is **stricter** because its values are maximum action limits and Grade A is "no growth", with any Grade A growth investigated [chk]. FDA 2004: ISO 5 action level 1 CFU/m3 active air and 1 CFU settle plate (settle plates optional); ISO 5 samples should normally yield no contaminants; averaging can mask localized excursions [chk]; broadly **same** as India at Grade A.
HOW IT WILL BE TESTED	record a Grade A settle plate of 1 CFU during a fill; an excursion is raised for every profile and release is blocked until the investigation closes.
SOURCE CHECK	P (Table 3 cells reassembled from scattered lines; the "average values" and "for information only" notes re-read).

Source requirements: n254

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-43 HEPA filter leak-test interval

Enforced | Configurable | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Room/HEPA master data hold last and next installed-filter leak-test dates; the test aerosol must be non-nutritive. A batch cannot start in a room whose HEPA test is overdue under the resolved interval (section B row B-16); a warning appears at the recommended interval.
TYPE DETAIL	ENFORCED (block when overdue); CONFIGURABLE (warning threshold).
REGULATORY BASIS	Schedule M Part II 4.5 (ISO leak test, recommended every 6 months, not exceeding 12 months).
INDIA / US / EU DIFFERENCES	**weaker** for Grade A/B than EU Annex 1 (2022) 4.32 (requalification including integrity test of final filters at most every 6 months for A/B; 12 months for C/D) [chk] and FDA 2004 (leak tests twice yearly in the aseptic processing room) [chk]; **same** as EU for C/D (12 months).
HOW IT WILL BE TESTED	India-only profile, Grade B room tested 13 months ago: batch start blocked; tested 7 months ago: warning only. India+EU profile, same 7-month case: blocked.
SOURCE CHECK	P (4.5 re-read).

Source requirements: n671

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-44 Clean-area sanitation, disinfectants and water monitoring

Enforced | Configurable | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Cleaning/disinfection records for Grade A-D rooms record agent, lot, dilution preparation time and use-by, sterility status for Grade A/B use, and the rotation (at least two agent types, including a sporicidal agent). A dilution past its defined storage period, or a non-sterile agent in Grade A/B, is rejected. The batch record links the cleaning events covering its window. Water results (WFI endotoxin, microbial, chemical) for the points used by the batch are visible at release relative to batch dates.
TYPE DETAIL	ENFORCED; CONFIGURABLE (agents, rotation).
REGULATORY BASIS	Schedule M Part II 3.1-3.4 (more than one type of disinfectant; sporicidal agent; residue removal validated; sterile agents for A/B), 5.9 (water monitoring and records).
INDIA / US / EU DIFFERENCES	**same** as EU Annex 1 (2022) 4.33 (more than one agent type; periodic sporicidal use) [chk] and FDA 211.67/211.113(b) expectations [gk].
HOW IT WILL BE TESTED	record use of a dilution past its storage period; refused.
SOURCE CHECK	P (3.1-3.3 and 5.9 re-read).

Source requirements: n670

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-45 Aseptic process simulation (media fill) envelope and acceptance

Enforced | Configurable | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Each aseptic line holds its qualified APS envelope: maximum fill duration, processing hold times, intervention types simulated, shifts, operators, container sizes. A batch whose duration, interventions, shift or operators fall outside the last successful APS is flagged and needs a deviation. (b) Initial qualification requires three consecutive successful runs; requalification at the resolved interval and after significant HVAC, equipment or process changes; each shift and shift changeover represented; for small batches the units filled are at least the batch size. (c) Acceptance resolved per profile (section B row B-10). India/FDA rule: fewer than 5,000 units, no contaminated unit; 5,000-10,000 units, one unit means investigation including a possible repeat, two mean revalidation after investigation; more than 10,000 units, one means investigation, two mean revalidation after investigation; intermittent contamination at any size is investigated. (d) A failed APS lists every batch filled on that line since the last successful APS for impact assessment.
TYPE DETAIL	ENFORCED (envelope checks; resolved acceptance); CONFIGURABLE (requalification interval where the resolved rule allows).
REGULATORY BASIS	Schedule M Part II 5.3-5.7; 4.11 (media fills set hold times and maximum fill duration).
INDIA / US / EU DIFFERENCES	acceptance **same** as FDA 2004 (plus FDA: under 5,000 units, one contaminated unit is cause for revalidation after investigation) [chk]; **weaker** than EU Annex 1 (2022) 9.46 (any contaminated unit is a failed APS, normally three successful repeats) [chk]. Frequency: India only "defined intervals"; FDA 2004 semi-annual per line and each authorised person at least yearly [chk]; EU 9.38 twice a year per process, line and shift, and each operator at least one successful APS per year [chk] (both **stricter**).
HOW IT WILL BE TESTED	India+EU profile with one contaminated unit in 12,000: recorded as a failed APS. Mark an APS failed; all batches since the previous pass are listed.
SOURCE CHECK	P (5.3-5.7 re-read).

Source requirements: n255

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-46 Processing time limits, bioburden and endotoxin

Enforced | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The injectables MBR carries timers with product-specific maxima and hard alarms: washing/drying to sterilisation and sterilisation to use of components, containers and equipment; start of solution preparation to sterilisation or sterile filtration; (EU profile adds) start of preparation to end of aseptic filling, sterilised product hold before filling, aseptic processing time and filling time. Processing stage of components and equipment is identified. A pre-sterilisation bioburden sample is mandatory for every batch (aseptic and terminally sterilised), with result entry against working limits; scheduled intervals are allowed only for products flagged overkill-terminal; for parametric release it is an IPC on every batch. Purge/blanket gas steps require a sterilising-filter reference. Endotoxin steps on WFI, intermediates and finished product; for LVPs the water/intermediate endotoxin step cannot be removed from the MBR.
REGULATORY BASIS	Schedule M Part II 5.12-5.17, 2.4.
INDIA / US / EU DIFFERENCES	**same** core as EU Annex 1 (2022) 8.18 and 8.93 (bioburden immediately before final sterile filtration; before the first filter where redundant) [chk] and FDA 211.110(a)(6) and 211.113(b) [chk]; EU 8.18 is **stricter**: it limits more intervals, including preparation start through to the end of aseptic filling, and the filling time itself [chk].
HOW IT WILL BE TESTED	let the preparation-to-filtration timer exceed its maximum; a deviation is forced. Close a batch without a bioburden result; blocked.
SOURCE CHECK	P (5.12-5.16 and 2.4 re-read).

Source requirements: n256

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-47 Sterilisation cycle records and revalidation

Enforced | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Steriliser/autoclave/tunnel/EO/irradiation interfaces import each cycle into the batch record (cycle ID, validated load pattern selected from a list, trace, coolest-point temperature with independent second probe where fitted, pressure, F0 where used, alarms; for EO also time, pressure, temperature, humidity, gas concentration and biological indicators; for irradiation dosimeter results). A reviewer signs each cycle before batch release, and cycle faults must be dispositioned. A steriliser whose revalidation (at least yearly and after significant change) or chamber leak test is overdue cannot be selected. Carrier labels (basket, tray) print material name, batch number and sterilised/not-sterilised status. Cooling fluid in contact with product must be recorded as sterile.
REGULATORY BASIS	Schedule M Part II 6.3-6.11 (6.5 at least annual verification; 6.8 carrier labelling; 6.10 records approved in batch release; 6.11.1 heat sterilisation recording and faults; chamber leak tests).
INDIA / US / EU DIFFERENCES	yearly revalidation of every process is **stricter** than EU Annex 1 (2022) 8.39 (at least yearly for worst-case load patterns; others per CCS) [chk] and FDA (periodic) [gk]; chamber leak testing **weaker** than EU 8.60 (normally weekly when a vacuum phase is used) [chk], where India only says "regular".
HOW IT WILL BE TESTED	select a steriliser whose annual revalidation lapsed; blocked. Import a cycle with an alarm; release is blocked until the alarm is dispositioned.
SOURCE CHECK	P (6.3-6.11.1.4 re-read).

Source requirements: n257

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-48 Sterile filtration

Enforced · Injectables · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The filtration step records filter ID/lot and pore rating (0.22 micrometre or equivalent), pre-use integrity result (after sterilisation for EU-profile batches), post-use integrity result (instrument import where possible), filtration start/stop time and differential pressure against validated ranges (outside range forces an investigation note), post-use integrity of critical vent filters, and the product hold time from end of compounding to filling. A failed integrity test puts the batch on hold with a deviation. A filter ID cannot be used beyond one working day unless flagged as validated for extended use; for EU-profile batches a liquid sterilising filter is also limited to a single batch. Filtration is refused as the sole method for a product flagged as terminally sterilisable.
REGULATORY BASIS	Schedule M Part II 6.11.2 (6.11.2.4-6.11.2.9).
INDIA / US / EU DIFFERENCES	pre-use testing **stricter** than FDA 2004 (pre-use test may be performed; post-use testing routine) [chk]; **weaker** than EU Annex 1 (2022) 8.87 (pre-use post-sterilisation integrity test unless justified by risk; post-use before removal from housing) [chk] and 8.94 (single batch and at most one working day) [chk].
HOW IT WILL BE TESTED	enter a failing post-use result; the batch goes on hold. Reuse a filter on day 2 without validation; blocked. EU profile, same filter for a second batch on day 1: blocked.
SOURCE CHECK	P (6.11.2.5-6.11.2.8 re-read).

Source requirements: n258

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-49 Sterility-test sampling and release review

Enforced · Injectables · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The record marks mandatory sterility-sample pulls: first and last units of an aseptic fill and after every significant interruption (logging an interruption automatically adds a pull after restart); for heat-sterilised loads, units from the potentially coolest position. The release review screen presents processing record, EM data (aseptic) and sterility result together and requires each to be opened and acknowledged. Parametric release mode is available only when the product licence permits it.
REGULATORY BASIS	Schedule M Part II 2.1-2.3; 4.8.
INDIA / US / EU DIFFERENCES	**same** as EU Annex 1 (2022) sterility sampling expectations [gk] and FDA 211.167(a) / 2004 guidance [gk].
HOW IT WILL BE TESTED	log a significant stoppage; a sterility pull after restart becomes mandatory.
SOURCE CHECK	P (2.1-2.3 re-read).

Source requirements: n259

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-50 Capping, vacuum-sealed containers and 100% inspection

Enforced · Configurable · Injectables · All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	(a) Capping records the station (separate, with extraction), confirms Grade A air supply to vials until crimped when capping is outside the aseptic core, and counts vials rejected for missing/displaced stoppers before capping. (b) Vacuum-sealed containers get a vacuum-hold test after a predetermined period. (c) 100% inspection step records inspector identity, booth lighting/background check, start/stop times with enforced breaks, quantities inspected and rejects by defect category. (d) Sign-on is refused if the inspector's eyesight check (with their own corrective lenses) or inspection qualification has lapsed; EU-profile batches require inspection qualification renewed at least yearly. (e) Automated inspection machines: validation reference and challenge-set results at start and at intervals; a missed challenge flags the lot.
TYPE DETAIL	ENFORCED; CONFIGURABLE (break interval, challenge frequency).
REGULATORY BASIS	Schedule M Part II 10.3-10.8; Part I 11.5.1 (periodic eye checks for visual inspectors).
INDIA / US / EU DIFFERENCES	**weaker** than EU Annex 1 (2022) 8.31-8.33 (inspector qualification at least yearly using defect libraries; automated systems challenged before start-up and at intervals; defect trending) [chk]; EU 8.22 (100% integrity testing of fusion-sealed containers of 100 ml or less) [chk] has no verified Indian counterpart. FDA: 21 CFR 211 has no inspector-qualification clause [chk]; FDA expects 100% visual inspection of injectables under USP chapters 790/1790 [gk].

HOW IT WILL BE TESTED	assign an inspector with an expired eye check; sign-on refused.
SOURCE CHECK	P (10.3-10.5, 10.7, 10.8 re-read). Part II 10.1-10.2 (closure integrity) not found in the uploaded copy (gap F-7).

Source requirements: n260

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

IN-51 Part XIII area and equipment classes for parenterals

Configurable | Injectables | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	The Injectables edition ships area types (water management, container/closure preparation, solution preparation, filling/capping/sealing, sterilisation, quarantine, visual inspection, packaging; separate FFS/BFS set) and equipment classes (RO/EDI plus multi-column distillation, SS 316L tank and circulating loop, washers, double-ended dryer/steriliser, filtration, filling under laminar flow, steam and hot-air sterilisers, pressure leak tester, visual inspection units with white and black backgrounds, batch coding and labelling). Leak-test steps may only reference the pressure-leak-tester class; each water draw records the WFI loop/point of use. Recommended minimum areas (SVI 150 m2 basic plus 100 m2 ancillary; LVP 150 m2 each; LVP packaging materials 100 m2; FFS LVP 250 m2 plus 150 m2) are stored as site-profile information, not enforced.
TYPE DETAIL	CONFIGURABLE (master data shipped as defaults).
REGULATORY BASIS	Schedule M Part XIII (parenteral preparations section; FFS section).
INDIA / US / EU DIFFERENCES	**different**: no FDA or EU counterpart to prescriptive minimum areas and equipment lists [chk: not in 211 or Annex 1].
HOW IT WILL BE TESTED	build an SVI MBR using only shipped classes; every needed area type exists.
SOURCE CHECK	P (area figures re-read; equipment list fragmentary in the uploaded copy).

Source requirements: n664

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

Kernel rule engine

IN-52 Market-profile rule resolution

Enforced | Both editions | All 1 source requirement checked against official text

WHAT THE SYSTEM DOES	Each batch carries its target markets (India domestic; export country list), and each pack/SKU carries its destination class. A versioned, market-keyed rule table is consulted at MBR approval, batch start, execution, release, sample retention and archiving. For each parameter the engine resolves the strictest value across India (always) and the batch's markets, stores the resolved value and the market that drove it in the batch record, and re-resolves if the market list changes before release (which forces QA review). Section B lists the resolved values. Resolution never loosens an Indian requirement for an export-only batch, because Indian GMP applies to everything made under the licence.
REGULATORY BASIS	Schedule M (all Parts cited above); Drugs Rules; 21 CFR 211; EU GMP Chapters 4-5, Annexes 1 and 11.
INDIA / US / EU DIFFERENCES	**different** (cross-cutting mechanism); implements "stricter rule wins".
HOW IT WILL BE TESTED	set up one tablet and one injectable product, each with a batch for {India} and a batch for {India, US, EU}. The {India, US, EU} batches resolve to: retention max(expiry + 1 y, mfg + 5 y, QP certification + 5 y); reference samples expiry + 1 y; annual self-inspection; new number on rework; identity test per excipient lot; plus, for the tablet, the 30-minute compression IPC and IP weight bands; and, for the injectable, any contaminated APS unit fails, Grade A viable no growth, PUPSIT and a single-batch sterilising filter. The resolution log names the governing market for each value.
SOURCE CHECK	P for Indian values marked P elsewhere; U where the underlying control is S/U (IN-08, IN-32, IN-34).

Source requirements: n678

Reviewer decision ☐ Follow ☐ Don't follow ☐ Modify Comments

B. Market-profile resolution table

How to read: "India" is always present because every batch is made under an Indian licence; columns show the value the engine (IN-52) applies. "India + US" means the batch serves the US market (with or without Indian domestic sale); an export-only batch for the US resolves like "India + US" for GMP rows. Label rows (B-15 and section C) resolve per pack destination: an India-domestic pack follows Rules 96/97, an export pack follows Rule 94 plus the importing market's rules. Status keys as in the header.

#	Topic	India only	India + US	India + EU	India + US + EU	Basis and status
B-01	Batch record retention (IN-08)	Longest of expiry + 1 y and mfg + 5 y (Rule 74); Form 28 test records per Rule 78 variant	Same as India only (FDA expiry + 1 y is not longer)	Longest of India value and QP certification + 5 y	Longest of India value and QP certification + 5 y	Schedule M 17.2 (P~); Rules 74/78 (U); 211.180(a) [chk]; EU 4.11 [chk]. If counsel does not confirm the Rule 74/78 periods, India only falls back to expiry + 1 y
B-02	Reference/retention samples (IN-32)	Expiry + 3 months (3 y from mfg if no expiry); 2x all tests; final pack	Expiry + 1 y; 2x all tests; yearly visual check of representative samples	Expiry + 1 y; 2x all tests	Expiry + 1 y; 2x all tests; yearly visual check	Schedule M 19.3(g) (P); Rules 74/78 (S); 211.170(b) [chk]; EU Annex 19 [gk]
B-03	Compression IPC frequency (IN-35)	Sample vs IPC spec at most every 30 min; station-wise check at run start (and fixed intervals on multi-point presses); other parameters periodically per MBR	Same	Same	Same	Part VIII 3.4, 3.8 (P); FDA 211.110(a) and EU 5.43 set no frequency [chk]
B-04	Sieve, screen, punch and die checks (IN-36)	Before and after each use	Same	Same	Same	Part VIII 1.3 (P)
B-05	Strip/blister line (IN-38)	Independent second-person recorded equipment check before each batch; periodic pocket vacuum leak test	Same	Same	Same	Part VIII 7.1, 7.4 (P)
B-06	Hold times in master and batch record (IN-13, IN-15)	Mandatory for every intermediate and in-process material; actual hold recorded	Same	Same, plus EU aseptic timers (preparation start to end of filling; filling time) for injectables	Same as India + EU	Part I 17.3.6, 17.3.8 (P); 211.111 [chk]; EU 4.17-4.20 and Annex 1 8.18 [chk]
B-07	Reworked batch number (IN-18)	New batch number	Same	Same	Same	Part I 14.32 (P)
B-08	Self-inspection frequency (IN-28)	At least once a year	Same	Same	Same	Part I 10.4 (P)
B-09	Excipient (non-API) release (IN-24)	Identity test per lot by default; vendor-history route of 14.15 allowed if QA enables it with statistical justification	Identity test on each lot (supplier CoA only with own identity test and periodic verification)	Identity test on each batch; identity sampling per container unless a validated reduced procedure applies	As India + US and India + EU combined	Part I 14.15, 19.7.1 (P; 19.7.2 truncated); 211.84(d) [chk]; EU 5.35 [chk], Annex 8 [gk]
B-10	Media-fill (APS) acceptance (IN-45)	Under 5,000 units: none; 5,000-10,000: 1 = investigate (consider repeat), 2 = revalidate; over 10,000: 1 = investigate, 2 = revalidate	As India, and under 5,000 units one positive = revalidation after investigation	Any contaminated unit = failed APS; root cause, CAPA, normally 3 repeat APS, review of batches since last pass	As India + EU	Part II 5.6-5.7 (P); FDA 2004 [chk]; EU Annex 1 9.46 [chk]
B-11	Media-fill frequency (IN-45)	Site-defined interval must be configured (recommended default 6 months) and after significant change	Every 6 months per line incl. all shifts; each authorised person yearly	Every 6 months per process, line and shift; each operator one successful APS yearly	As India + EU	Part II 5.5 (P); FDA 2004 [chk]; EU 9.38 [chk]
B-12	Grade A viable limits; status of B-D values (IN-42)	Grade A under 1 CFU (air per m3, settle, contact, glove), evaluated per sample by design; B-D per Table 3; site sets alert/action limits no looser	Same; any Grade A recovery is an excursion (FDA ISO 5 action level 1 CFU)	Grade A no growth, any growth investigated; B-D values are maximum action limits; A/B settle plates exposed for the whole operation, changed at most every 4 h	As India + EU	Part II Table 3 (P, stated as averages and informative); FDA 2004 Table 1 [chk]; EU Table 6 [chk]
B-13	Non-viable particle limits (IN-40)	India Table 1 (0.5 / 5.0 micrometre: A 3,520/20; B 3,520/29 rest, 352,000/2,900 op; C 352,000/2,900 rest, 3,520,000/29,000 op; D 3,520,000/29,000 rest)	India table (FDA 0.5 micrometre values identical; no FDA 5 micrometre limit)	India numeric limits (lower 5 micrometre values) plus EU monitoring rules: continuous Grade A for both sizes at 28 L/min or more, investigation of repeated low 5 micrometre counts, site-set Grade D in-operation limits	As India + EU	Part II Table 1 (P); FDA 2004 [chk]; EU Tables 1, 5 and 9.17 [chk]
B-14	Tablet weight-variation IPC bands (IN-34)	IP bands (80 mg or less 10%; over 80 to under 250 mg 7.5%; 250 mg or more 5%)	IP bands (tighter than legacy USP bands at every weight); release UDU per USP chapter 905 as registered	IP bands (= Ph. Eur. 2.9.5); release per Ph. Eur. 2.9.40 as registered	IP bands; both release tests as registered	IP values (U); USP/Ph. Eur. [gk]
B-15	Date of manufacture on label (IN-31)	India-domestic pack: mandatory, uncoded, verified in coding check	Domestic packs as India; US packs: not required by FDA	Domestic packs as India; EU packs: not required by EU	Per pack destination as in the other columns	Schedule M 17.3.1(d) (P); Rule 96 (S); 21 CFR 201.17 and Dir. 2001/83 Art. 54 [gk]; Rule 94 export particulars (U)
B-16	HEPA leak-test interval (IN-43)	Block beyond 12 months; warn beyond 6 months	Aseptic processing rooms (A/B): 6 months; others 12 months	A/B: 6 months; C/D: 12 months	A/B: 6 months; C/D: 12 months	Part II 4.5 (P); FDA 2004 [chk]; EU 4.32 [chk]
B-17	Sterilising-filter integrity and use (IN-48)	Integrity before use and right after use; one working day maximum unless validated	Same as India (stricter than FDA)	Pre-use test after sterilisation (PUPSIT) unless risk-justified; post-use test before removal; single batch and one working day maximum unless validated	As India + EU	Part II 6.11.2.7-6.11.2.8 (P); FDA 2004 [chk]; EU 8.87, 8.94 [chk]
B-18	Sterilisation revalidation; chamber leak test (IN-47)	Every process at least yearly and after significant change; chamber leak test regularly (SOP interval)	Same	Same yearly rule; chamber leak test normally weekly where a vacuum phase is used	As India + EU	Part II 6.5, 6.11.1.4 (P); EU 8.39, 8.60 [chk]
B-19	Visual-inspector qualification (IN-50)	Regular eyesight checks (interval configured; recommended yearly); frequent breaks	Same	Inspection qualification at least yearly with defect library, eyesight considered; breaks; automated inspection challenged at start and intervals	As India + EU	Part II 10.8, Part I 11.5.1 (P); EU 8.31-8.33 [chk]
B-20	Master record approval (IN-06)	Two-person (prepare + independent check) by design	Two-person mandatory	Two-person by design	Two-person	Part I 17.2 (P~); 211.186(a) [chk]
B-21	Reason for any data change (IN-06)	Mandatory by design ("where appropriate" in Schedule M)	Mandatory	Mandatory	Mandatory	Part I 17.2 (P~); EU 4.9, Annex 11 section 9 [chk]

#	Topic	India only	India + US	India + EU	India + US + EU	Basis and status
B-22	Audit-trail review (IN-09)	Part of batch review by design (not in Schedule M)	Part of batch record review before release	Regular review; part of batch review	Part of batch review	Annex 11 section 9 [chk]; FDA DI Q&A 2018 Q7-Q8 [chk]
B-23	Batch release authority (IN-05)	Authorised person, with head-of-QC approval	AP release that also meets 211.192 quality-unit review of all production and packaging records	Indian AP release; EU QP certification at the EU site (system supplies the batch package)	Both	Schedule M section 1, 11.3.6-11.3.10 (P); 211.22, 211.192 [chk]; Annex 11 section 15 [chk], Annex 16 [gk]
B-24	Management review	At least yearly	Same	Same	Same	Part I 1.6 (P)
B-25	Grade A airflow (IN-40)	0.36-0.54 m/s at 15-30 cm below the terminal filter; never below 0.36 m/s at working level	Same	India positions plus 0.36-0.54 m/s at the working position (EU)	As India + EU	Part II 4.3 Grade A text (P); EU Annex 1 airflow guidance [chk]
B-26	Sample-retention quantity basis	Twice all tests, including sterility	Same (the Indian rule, if confirmed, leaves no room for the FDA sterility/pyrogen exclusion)	Same	Same	Rules 74/78 (S); 211.170(b) [chk]

C. India-specific fields

Fields the Indian layer adds to the kernel data model. "Record" = where the field lives; "Check" = what the system verifies. Status keys as in the header.

C.1 Batch manufacturing record (header and body)

Field	Record / check	Legal basis	Status
Manufacturing licence number and licence form (Form 25 for drugs outside Schedules C/C(1), which covers most tablets; Form 28 for Schedule C/C(1) drugs, where parenterals fall, and a tablet too if its drug is listed in C(1); Form 28-D reported for LVPs, sera and vaccines)	BMR/BPR header, from site master data; drives which conditions of licence (Rule 74 or Rule 78) feed retention (IN-08)	Drugs Rules licence rules and Schedule A forms; Rules 74 and 78	S (form mapping U, see G-10)
Loan licence number (Form 25-A / 28-A) plus the host facility's licence number	BMR/BPR header for loan-licence batches; both numbers printed (IN-30)	Schedule M Part I section 9; Drugs Rules loan-licence forms	P for section 9; S for forms
Approved technical staff (manufacture and testing) with licensing-authority approval reference and date	Register; gates supervisory and QC approval signatures (IN-04)	Drugs Rules Rules 71 and 76; Schedule M Part I 11.3.2	S / P
Schedule U particulars: batch size; dates of start and completion; each raw material with quantity and analytical report (control) number; finished-product analytical report number; actual production and packing quantities; date of release for distribution or sale	BMR fields (ARN per dispensed lot; release date written by the release workflow)	Drugs Rules Rule 74 (and 78) with Schedule U	S
Permitted hold time per intermediate and actual hold start/stop	MBR and BMR (IN-13, IN-15)	Schedule M Part I 17.3.6, 17.3.8	P
Person responsible for each production stage; dated completion signature of the person responsible for processing	BMR (IN-15)	Schedule M Part I 17.3.8	P
Batch-number allocation entry (allocation date, product, batch size); link between processing and packaging numbers	Allocation register (IN-18)	Schedule M Part I 17.3.10.8-17.3.10.11	P~
Pharmacopoeia and edition behind every specification/IPC limit (e.g., IP 2026)	MBR limit metadata (IN-33)	Schedule M Part I 17.3.2	P
Target markets and export destinations	BMR header; input to rule resolution (IN-52)	Design (needed to apply stricter-rule-wins)	n/a
Marketer name and address and marketer-manufacturer agreement reference, where a marketer sells the product	Product master and BMR header; also on label	Drugs Rules Rules 84D and 84E (2020 amendment); Rule 96	S
Tablets: compression IPC entries at no more than 30-minute intervals; station-wise start-of-run results; sieve/screen/punch/die checks before and after use; FBD bag ID; rejected-tablet quantity; QC approval of printed tablets/capsules	BMR (IN-35 to IN-37)	Schedule M Part VIII 1.3, 2.4, 3.4, 3.7-3.8, 6.2	P
Injectables: inspector eye-check validity at sign-on; APS envelope reference; filter integrity results; approved sterilisation cycles; EM summary for the processing window; bioburden result	BMR (IN-42 to IN-50)	Schedule M Part II 2.3, 4.8, 5.16, 6.10, 6.11.2.7, 10.8; Part I 11.5.1	P

C.2 Batch packaging record

Field	Record / check	Legal basis	Status
Coded variables: batch number with an allowed prefix (e.g., "Batch No.", "B. No.", "Lot No."), manufacturing date and expiry date in plain form	Coding verification at start, intervals and end against batch data (IN-22, IN-31)	Schedule M Part I 17.3.1; Drugs Rules Rule 96	P / S
MRP value used for the run (inclusive of all taxes)	Overprint verified, coded packs reconciled; value traceable to the approved price for the period	Drugs (Prices Control) Order 2013; Legal Metrology (Packaged Commodities) Rules 2011 exemption for DPCO formulations	S
Schedule H2 QR/barcode: payload generated from batch master data (unique product identification code, proper and generic name, brand name, manufacturer name and address, batch number, manufacturing date, expiry date, manufacturing licence number; from the 2026 amendment reportedly excipient details where applicable) and reader verification results	Code generation step plus reader challenge and per-pack or sampled verification; on primary pack label (secondary if space is insufficient)	Drugs Rules Rule 96 and Schedule H2 (G.S.R. 823(E) of 17 Nov 2022; G.S.R. 506(E) of 22 Jun 2026)	S (2026 data elements U, see G-6)
Export pack particulars (destination-specific artwork; Indian particulars kept under Rule 94; approved code number instead of manufacturer name when the consignee asks)	Artwork template by destination; coding check	Drugs Rules Rule 94	U (see G-9)
Reference-sample pull (quantity, storage location, retain-until date)	Packaging record step (IN-32)	Schedule M 19.3(g); Rules 74/78	P / S
Blister/strip: independent second-person clearance check; periodic pocket vacuum-test results	Packaging record (IN-38)	Schedule M Part VIII 7.1, 7.4	P

C.3 Label (India-domestic artwork template)

Field	Check in the packaging master	Legal basis	Status
Drug name; actives with INN and amount; net contents; batch number; manufacturing and expiry dates uncoded; storage/handling; directions, warnings, precautions; name and address of manufacturer and of the party placing the product on the market	Template cannot be approved without each field (IN-31)	Schedule M Part I 17.3.1(a)-(g)	P
Proper (generic) name with "I.P." where the drug is in the Indian Pharmacopoeia, printed at least two font sizes larger than the brand name (combinations of three or more actives exempt)	Artwork review checklist item; font-size comparison recorded by the approver	Drugs Rules Rule 96(1)(i) as amended in 2018, mandatory from 1 Apr 2019	S
Manufacturing licence number ("Mfg. Lic. No.") and, for loan-licence products, the loan licence number	Template field bound to master data	Drugs Rules Rule 96	S
Schedule H, H1 or X caution text in a box with red border and the Rx / NRx / XRx symbol at top left, per the product's classification	Product master carries the Indian schedule; template selects the matching panel	Drugs Rules Rule 97; Schedules H, H1, X	S
MRP	Template field; overprint (C.2)	DPCO 2013	S
Marketer name and address where different from the manufacturer	Template field	Drugs Rules Rule 96 and Rules 84D/84E	S
QR/barcode for Schedule H2 products (from 1 Aug 2023 for the 300 listed brands; vaccines, NDPS narcotic/psychotropic and anticancer drugs from 1 Jul 2027; all antimicrobials from 1 Jul 2028)	Template reserves the code area; C.2 generates and verifies content	Rule 96 and Schedule H2	S

C.4 Incoming material data

Field	Use	Legal basis	Status
API QR code at every packaging level (unique product identification code, API name, brand if any, manufacturer name and address, batch number, batch size, manufacturing date, expiry or retest date, serial shipping container code, licence number, special storage conditions)	Parsed at receipt/dispensing scan to pre-fill and cross-check lot data	G.S.R. 20(E) of 18 Jan 2022, from 1 Jan 2023	S
Export track-and-trace (DGFT) data	Not required: DGFT withdrew the export barcoding/iVEDA upload regime on 31 Jan 2025. Destination serialisation (US DSCSA, EU FMD) is handled by the FDA/EU layers	DGFT Public Notice 44/2024-25	S

D. Electronic records and signatures in India

D.1 What Schedule M allows and requires (P unless marked)

- **Electronic records are expressly allowed.** Part I 17.2 (data-processing clause): data and records may be kept by electronic data-processing, photographic or other reliable means, provided the system's master formulae and SOPs are available and record accuracy is checked. With electronic handling: only authorised persons enter or change data; changes and deletions are recorded; access is limited by passwords or other means; critical data entry is independently checked; electronically stored batch records are protected by back-up (tape, microfilm, discs, printouts or other means) and stay readily available throughout retention. Retention is at least one year after the finished product's expiry (17.2).
- **Computerised systems section.** Part I 20.2-20.10: suitability shown by IQ/OQ; qualified commercial software needs less testing; retrospective validation possible with documentation; controls against unauthorised access and changes; controls against omitted data (e.g., system switched off); each change records the previous entry, who changed it and when; written operating/maintenance procedures; an extra accuracy check (second person or system) for critical manual entries; incidents recorded and investigated; system changes authorised, documented, tested and recorded to show the validated state is maintained; back-up against permanent loss; data protection for all systems; a second means of recording is permitted. The opening 20.1 sentence is not legible in the uploaded copy (F-3). The API Part repeats this set in Part XII 5.4.x and adds that validation effort scales with how diverse, complex and critical the application is (5.4.1).
- **Signatures.** Batch processing records are dated and signed by the person responsible (17.3.8); batch packaging entries identify the responsible person by signature or electronic password (17.3.9); corrections are signed and dated (17.2). The only explicit e-signature sentence sits in Part XII (APIs, 6.1): e-signatures on documents must be authenticated and secure; where electronic or reduction techniques are used, retrieval equipment and a way to produce a hard copy must be readily available (6.1.7). Part XII binds API manufacture only, but it is the clearest Indian statement and the product applies it to both editions.
- **Validation and site master file.** Part I section 5 requires a validation master plan with attention to automated systems; Appendix I requires the site master file to describe computerised-system validation arrangements.
- **Not in Schedule M:** audit-trail review, e-signature manifestation (name, date/time, meaning), signature-to-record linking, password rules, periodic system evaluation, data-integrity (ALCOA+) terminology. No CDSCO guidance, circular or FAQ on electronic batch records, e-signatures or audit-trail review was found by the research (S). The product therefore applies 21 CFR Part 11 and EU Annex 11 detail in all editions (IN-09, IN-10).

D.2 IT Act 2000 position (S; primary Act text not read)

- **s.4:** where a law requires information in writing, an electronic record satisfies it if accessible for later reference.
- **s.5:** where a law requires a signature, the requirement is met by an electronic signature affixed in the manner the Central Government prescribes.
- **s.3A:** an electronic signature technique counts as reliable if the signature creation data are linked to and controlled by the signatory alone, and any later change to the signature or to the signed information is detectable. Recognised techniques are listed in the Second Schedule: digital signatures under certificates from licensed certifying authorities (PKI, CCA root), and Aadhaar/e-KYC-based eSign.
- **s.7:** electronic retention satisfies a retention duty if the information stays accessible, is kept in its original format or one shown to represent it accurately, and origin, destination and date/time details are kept.
- **Bharatiya Sakshya Adhiniyam 2023 s.63** (in force 1 Jul 2024, replacing Evidence Act s.65B): computer output is admissible subject to conditions; a certificate in the prescribed form, reportedly signed by the person in charge of the device and an expert and including a hash of the record, accompanies it (format not reviewed).

D.3 Password-based signatures versus DSC/eSign

- Schedule M itself accepts "signature or electronic password" for packaging records and allows electronic records with password-restricted access, which supports username/password e-signatures for GMP purposes under the Drugs Rules.
- The IT Act s.5 route is narrower: it deems a legal signature requirement met by a Second-Schedule technique. A password signature is not a listed technique. Whether a signature required by the Drugs Rules (e.g., the AP's release, MBR approval, the processing-lead's completion signature) must satisfy s.5, or is satisfied by Schedule M's own acceptance of electronic passwords and validated systems, is **legally uncertain**.

D.4 Recommendation for the system

1. **Baseline (all signatures, all editions, ENFORCED):** meet 21 CFR Part 11 subpart C and EU Annex 11 section 14 in full: unique ID plus secret, re-authentication at each signing (both components at the first signing of a session), signature manifestation (name, date/time with zone, meaning) shown on screen and in print, cryptographic binding to a hash of the signed content so that any change shows the signature as invalid, account controls (no reuse, lockout, periodic credential review). This also meets the s.3A(2) reliability tests in substance (sole control, change detection), and Schedule M's "authenticated and secure".
2. **Option (CONFIGURABLE per signature meaning; recommended default ON for Indian licensees for MBR/packaging-master approval and AP release):** a Class 3 DSC from a licensed CA (e.g., USB token or HSM-backed signing) or an eSign service, producing a PKI signature over the record hash that validates to the CCA root. This places the most consequential signatures squarely within IT Act s.5. The password signature is still captured for Part 11/Annex 11 purposes. eSign involves Aadhaar/e-KYC through a licensed eSign service provider; privacy and contract terms need review.
3. **Evidence export (IN-11):** a certified export package with SHA-256 manifest and system details to support a BSA s.63 certificate.
4. **Printed/PDF rendering** of any record with signatures, meanings and audit trail on demand (Part XII 6.1.7; Annex 11 section 8; 21 CFR 11.10(b)).

D.5 Questions for Indian legal counsel (flagged; do not treat as settled)

- Q1. Does a signature requirement in the Drugs Rules/Schedule M (release by the authorised person, approval of master documents, completion signature of processing records) count as a "law requiring a signature" under IT Act s.5, so that only a Second-Schedule technique (DSC/eSign) satisfies it? Or is Schedule M's acceptance of electronic passwords and validated systems a sufficient, specific authority?
- Q2. If a password signature is used, what is its evidential weight in a prosecution for a not-of-standard-quality or spurious drug under the Drugs and Cosmetics Act, and what must the BSA s.63 certificate contain (signatories, hash algorithm)?
- Q3. Would an Indian drug inspector accept an electronic batch record with password signatures and no wet-ink or DSC signature for the AP release? (No CDSCO guidance found.)
- Q4. For loan-licence batches, which party's authorised person must sign, and may the loan licensee sign remotely in the host's system?
- Q5. Are the Drugs Rules retention periods (Rules 74/78, Schedule U) as summarised in IN-08 current, and do they apply to electronic records identically?

E. Status, in-force dates and deadlines (as of 27 Sep 2026)

Item	Date / value	Effect on the product	Status
Revised Schedule M notified	G.S.R. 922(E), New Delhi, 28 Dec 2023 (draft G.S.R. 999(E) of 5 Oct 2018); made under sections 12 and 33 of the 1940 Act after DTAB consultation; substitutes Schedule M in the Drugs Rules and retitles it (GMP plus premises, plant and equipment); Rules 74, 76 and 78 cross-references updated	Legal basis for every IN control	P
Commencement	Amending rules in force on publication; paragraph 6 sets implementation by turnover: large manufacturers (over Rs 250 crore) within 6 months of publication (about 28 Jun 2024); small and medium (Rs 250 crore or less) within 12 months (about 28 Dec 2024)	Site profile stores category and dates	P
MSME extension	G.S.R. 127(E) of 11 Feb 2025 (draft G.S.R. 10(E), Jan 2025): MSMEs applying in Form A with an upgradation plan within 3 months got until 31 Dec 2025; non-applicants kept the original date	None today (expired)	S
Current position	Binding on all licensed manufacturers from 1 Jan 2026 at the latest; CDSCO asked State/UT regulators in Nov 2025 to begin risk-based Schedule M inspections with monthly reporting; no further extension notice found in 2026 searches (egazette not reachable)	All Indian customers need full compliance now	S (see G-7)
Draft licence-form change	Draft G.S.R. 299(E) of 21 Apr 2026 reportedly widens the licence forms listed in Rule 89 (25A, 25F, 28A, 28B, 28D, 28DA, 28F); 30-day consultation draft	May change licence-form labels in C.1; not Schedule M content	S, single source
Indian Pharmacopoeia 2026	10th edition released 2 Jan 2026; effective 1 Jul 2026, replacing IP 2022 as the "latest edition" Schedule M requires; Amendment List 01 issued; draft harmonised Uniformity of Dosage Units chapter circulated Oct 2024 (adoption in IP 2026 not confirmed)	Change-control sweep of MBR limits citing IP 2022 (IN-33); weight-band table to be confirmed (IN-34)	S
Schedule H2 QR codes (formulations)	G.S.R. 823(E) of 17 Nov 2022: 300 listed brands from 1 Aug 2023. G.S.R. 506(E) of 22 Jun 2026 (Drugs (Seventh Amendment) Rules 2026; draft 16 Oct 2025): all vaccines, NDPS narcotic/psychotropic drugs and anticancer drugs from 1 Jul 2027; all antimicrobials from 1 Jul 2028; reportedly adds excipient details where applicable	Packaging QR generation/verification must be ready for domestic packs of these categories by 1 Jul 2027 / 1 Jul 2028	S (2026 details U)
API QR codes	G.S.R. 20(E) of 18 Jan 2022: every API made in or imported into India, at each packaging level, from 1 Jan 2023	Receipt/dispensing scan can use it (C.4)	S
DGFT export track-and-trace	Withdrawn by DGFT Public Notice 44/2024-25 of 31 Jan 2025 (removed Handbook of Procedures para 2.76); tertiary (2011) and secondary (2013) barcoding and iVEDA uploads no longer required; primary-level barcoding had been deferred repeatedly; no replacement Indian export rule found	Drop DGFT/iVEDA features from the Indian layer; destination serialisation belongs to FDA/EU layers	S
Bharatiya Sakshya Adhiniyam 2023	In force 1 Jul 2024; s.63 replaces Evidence Act s.65B for electronic evidence	IN-11 export	S
Items to re-check before each release	Any 2026-27 corrigendum or amendment to Schedule M; final text of G.S.R. 506(E); outcome of draft G.S.R. 299(E); IP 2026 amendment lists	Rule-table versions (IN-52)	open

F. Gaps: text that could not be verified in the uploaded Gazette copy

The uploaded copy is a compressed extraction with scrambled paragraph order and missing pages. The following could not be checked and must be verified against a clean copy of G.S.R. 922(E) (e.g., the CDSCO PDF) before validation documents cite them.

- **F-1** Part I section 13 (equipment) and 14.1-14.4 (general materials clauses): not found. Equipment controls in IN-25 rest on 17.3.10 and 18.4 text instead.
- **F-2** Part I section 19 after 19.7.2: the identity-test sentence is cut off mid-word, and later QC clauses (possibly identity testing per container, retained-sample periods, stability, finished-product testing and release) are missing. Affects IN-24 (scope of the 14.15 excipient route) and IN-32.
- **F-3** Part I 20.1 opening sentence of the computerised-systems section: not legible.
- **F-4** Sub-clause numbering inside Part I 17.2 (general documentation: retention sentence, data-processing clause), and item letters within 17.3.8 and 17.3.9 and numbering of 17.3.10: inferred from WHO TRS 986 ordering; the wording itself was found.
- **F-5** Part II sections 7-9 (personnel, premises, equipment for sterile areas: gowning per grade, training, premises finishes, equipment design): only fragments found. The FDA/EU layers' gowning and APS-participation controls are therefore not yet matched to Indian text.
- **F-6** Part II Tables 1 and 3: values reassembled from scattered cells; consistent with WHO TRS 961 Annex 6 and with the stated ISO classifications, but the table layout (including footnotes) could not be read as a whole.
- **F-7** Part II 10.1-10.2 (closure integrity of fusion-sealed and other containers): not found. No Indian container-closure-integrity value is catalogued; EU Annex 1 (2022) 8.22 applies for EU-profile batches.
- **F-8** Part III (hazardous substances): text interleaved with Part IV (biologicals); IN-27 detail needs re-checking.
- **F-9** Part X and Part XI headings: not found (not needed for either edition).
- **F-10** Part XIII tablet section (areas and equipment for tablets, referred to as item 3 for tablets): not found; the Tablets edition has no Part XIII master-data list yet.
- **F-11** Part XIII parenteral equipment list: fragmentary; area figures found (IN-51).
- **F-12** Outside Schedule M, no primary text was available for: Drugs Rules 71, 74, 76, 78, 84D/84E, 94, 96, 97 and Schedules C/C(1), H, H1, H2, U, X; the IT Act; the BSA; IP 2026; G.S.R. 127(E), 823(E), 506(E), 20(E); DGFT Public Notice 44/2024-25. Everything relying on them is marked S or U.

G. Rows not treated as eBMR concerns, and open questions

G.1 "Not an eBMR concern"

No requirement is wholly out of scope: every row maps to at least one control above. The following parts of rows are outside the eBMR and are handled by facility engineering, HR/EHS or the wider QMS; the eBMR only consumes their status or supplies data:

- n664: recommended minimum floor areas and room partitioning (facility design; stored as site-profile information only).
- n677 and n265: building structure, HVAC design, airlock layouts and pressure-cascade diagrams (engineering documents; the eBMR only reads room status and readings).
- n666: breathing-air plant design, effluent treatment, facility containment engineering.
- n271: organisation charts, job descriptions, pre-employment and periodic health checks, hygiene rules (HR/EHS; the eBMR only reads training and eye-check status).
- n270: pharmacovigilance collection and ADR reporting to the authority (PV system; the eBMR only supplies batch genealogy).
- n269: supplier audits and external quality audits (QMS/supplier module; the eBMR only uses the approved-supplier list).
- n229 and n273: quality manual, senior-management responsibilities and the validation master plan document itself (customer QMS documents).
- n668: the signatories and legal execution of the BSA s.63 certificate (legal process; the eBMR supplies the data package).

G.2 Open questions

- **G-1** Drugs Rules retention and sample conditions: confirm Rule 74 (Form 25) five-year retention of Schedule U records and its start point; Rule 78 (Form 28) test-record period (reported two years after expiry, or five years from manufacture without expiry); reference-sample quantity (twice all tests) and period (three months after expiry; three years from manufacture without expiry); whether Schedule U applies unchanged alongside revised Schedule M. Drives B-01, B-02, B-26.
- **G-2** Legal weight of password e-signatures under IT Act s.5 versus Schedule M's acceptance of electronic passwords (section D.5, Q1-Q3). Obtain a legal opinion and, if possible, informal CDSCO or State Licensing Authority confirmation.
- **G-3** Scope of Part I 14.15 (release of non-API raw materials without batch testing) versus the truncated 19.7.2 identity-test clause; until clarified the default is an identity test on every lot (IN-24).
- **G-4** Part VIII 3.8: the text ties the 30-minute ceiling to sampling against the in-process specification and makes the six named parameters periodic; confirm with Indian QA practice/inspectors which parameters inspectors expect at the 30-minute interval (usually at least weight/appearance), as this sets the default IPC schedule.
- **G-5** Bharatiya Sakshya Adhiniyam s.63 certificate: prescribed form, signatories, hash expectations (IN-11).
- **G-6** Schedule H2 after G.S.R. 506(E): exact data elements (excipient details), whether the new elements also apply to the original 300 brands, whether a GS1 GTIN is required as the unique product identification code, and whether any aggregation or portal reporting starts on 1 Jul 2027.
- **G-7** MSME status: confirm no gazette notification after G.S.R. 127(E) extended or relaxed revised Schedule M, whether Form A applicants that missed upgrades have any transition, and whether any corrigendum to G.S.R. 922(E) exists.
- **G-8** IP 2026: wording of uniformity of weight / uniformity of dosage units, disintegration and friability tests, and whether State regulators expect IP in-process bands for export-only products registered to USP or Ph. Eur. specifications (IN-34).
- **G-9** Rule 94 export labelling: whether "for export only / not for domestic consumption" comes from Rule 94 or from export NOC conditions, and which Indian particulars (licence number, manufacturer name or approved code, batch, expiry, manufacturing date) remain mandatory when the importing country's rules differ.
- **G-10** Licence-form mapping (Form 25 vs 28 vs 28-D; Schedule X forms) and the outcome of draft G.S.R. 299(E).
- **G-11** Part II Table 3 describes microbial values as averages and informative; confirm that Indian inspectors accept the product's stricter per-sample evaluation and that site-set action limits no looser than Table 3 satisfy 4.10.
- **G-12** Whether CDSCO expects audit-trail review or has issued any data-integrity guidance for electronic records (none found); until then the Annex 11/FDA expectation is applied (B-22).
- **G-13** Whether Indian inspectors expect the EU/FDA requalification detail for media fills (frequency, operator participation) at Indian sites making India-only product, given Part II only says "defined intervals" (B-11).

APPENDIX

Clause-level requirements register

678 requirements · 587 checked against official texts

Each row is one requirement as paraphrased from its source. "Used in" shows the controls built from it. Rows are grouped by market and document. "Checked" means compared with the official text you supplied.

Other Indian law

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n661	Rule 74 (records of manufacture per Schedule U; records of tests) and Rule 78 equivalents	Separately from Schedule M, the conditions of a manufacturing licence require batch-wise manufacturing records with the particulars listed in Schedule U (e.g., product name, batch size and number, dates of start and completion of manufacture, ingredients and quantities, raw-material control numbers, analytical report references, actual production and packing details, date of release for sale) and records of tests, kept for five years (reported as from the date of manufacture). Summaries of Rule 78 (Schedule C/C(1) products on Form 28) describe test records kept for two years after expiry for products with an expiry date and five years from manufacture for others.	binding	Unverified	IN-08
n667	s.4 legal recognition of electronic records; s.5 legal recognition of electronic signatures; s.3A reliability criteria; s.7 retention of electronic records	Where a law requires information in writing, an electronic record satisfies it if it stays accessible for later reference (s.4). Where a law requires a signature, the requirement is met by an electronic signature affixed in the manner prescribed by the Central Government (s.5). An electronic signature technique is treated as reliable if the signing data is linked to and controlled by the signer alone and any later change to the signature or to the signed information is detectable (s.3A(2)); recognised techniques are listed in the Second Schedule (digital signatures under licensed certifying authorities, and Aadhaar/e-KYC based eSign). Where a law requires records to be retained, electronic retention qualifies if the information stays accessible, is kept in its original format or one shown to represent it accurately, and origin, destination and date/time details are kept (s.7).	binding	Search-based	IN-10
n668	s.63 admissibility of electronic records; certificate in the prescribed Schedule form	Computer output (printouts, copies on media) of an electronic record is treated as a document and admissible without the original if section 63 conditions are met; for such secondary electronic evidence a certificate in the prescribed form is needed, reportedly signed by the person in charge of the device and an expert and including a hash value of the record.	binding	Search-based	IN-11

Revised Schedule M (India)

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n229	Part I, section 1 (PQS), 1.1-1.7 and the PQS 'shall ensure' list	The manufacturer must run a documented pharmaceutical quality system covering GMP and QRM, with a quality manual, senior-management responsibility and management reviews of the PQS at least once a year unless otherwise justified. Product and process monitoring results must feed into batch release and deviation investigations, and planned changes must be evaluated and approved before implementation and reviewed afterwards.	binding	Checked	IN-01
n230	Part I, section 1 (PQS), 'shall ensure' item (s); section 3.1 (basic GMP requirements, items 6 and 7)	Records must be made during manufacture, by hand, by recording instruments or both, showing that every required step was actually done and that quantity and quality are as expected. Significant deviations must be fully recorded and investigated to root cause, with CAPA whose effectiveness is monitored. Manufacturing and distribution records must let the whole history of a batch be traced and must be kept in an understandable, accessible form.	binding	Checked	IN-07, IN-20
n231	Part I, section 2.3 (product quality review), 2.3.1-2.3.3	Regular, periodic or rolling quality reviews are required for every product, expressly including products made for export only, normally each year. Minimum inputs include starting and packaging materials (especially new sources and API supply-chain traceability), critical in-process and finished-product results, failed batches, significant deviations and CAPA effectiveness, process or method changes, dossier variations, stability results, returns, complaints and recalls, earlier CAPA adequacy, post-marketing commitments, equipment and utility qualification status, and technical agreements. Results must be assessed for CAPA or revalidation; grouping by product type is allowed if justified; the authorised person must ensure the review is timely and accurate.	binding	Checked	IN-01
n232	Part I, section 2 (QRM)	There must be a QRM system: a systematic process to assess, control, communicate and review risks to product quality, usable both before and after the fact. Risk evaluation must rest on scientific knowledge and process experience and link to patient protection, and the effort, formality and documentation of QRM must match the level of risk.	binding	Checked	IN-02
n233	Part I, 11.3.6-11.3.10 (authorised person and batch release) and section 1 PQS 'shall ensure' list item (i)	No batch may be sold or supplied until the authorised person has certified it. The authorised person answers for technical and regulatory compliance and approves release after assessing production conditions, in-process results, the manufacturing and packaging documentation, specification compliance and the finished pack. Before release they must confirm, among other things: licence and product approval conditions met; GMP followed; main processes validated; all checks done with production conditions and records considered; planned changes or deviations reported through a defined system (some may need licensing-authority notification or approval) and covered by extra testing; all production and QC documentation completed and endorsed by trained supervisors; the head of QC has approved; and all relevant factors considered. The release function may be delegated to a qualified designated person under an approved procedure, normally QA through batch review.	binding	Checked	IN-05
n234	Part I, 11.3.1, 11.3.4(c), 11.3.5(b)	Key personnel are the head(s) of production, the head(s) of the quality units (QA and QC, which may be combined) and the authorised person, normally full-time. Heads of production and quality must be independent of each other; tasks may be delegated but not responsibility. The production head must ensure a designated person reviews and signs the production records; the quality heads evaluate batch records and approve or reject materials and products.	binding	Checked	IN-04
n235	Part I, 17.2 (general documentation clauses; retention sentence probably 17.2.7)	Records must be made or completed at the time each action is taken so that all significant manufacturing activities are traceable, and they must be kept until at least one year has passed since the finished product expired.	binding	Checked	IN-07, IN-08
n236	Part I, 17.2 (general documentation clauses: approval, review, entries, alterations)	Documents need careful design, drafting, review and distribution, must match the licence, be approved, signed and dated by responsible persons, and not be changed without authorisation. They must be clear, orderly and easy to check; copies made from master documents must not introduce errors. Documents are reviewed regularly; a system must stop use of superseded versions, which are kept for a defined period. Data entries must be clear, legible and indelible with enough space. Any alteration must be signed and dated, leave the original readable and, where appropriate, give the reason.	binding	Checked	IN-06

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n237	Part I, 17.2 (electronic data-processing clause; probably 17.2.8)	Data and records may be held electronically, photographically or by another dependable method, provided the master formulae and detailed SOPs for the system are available and record accuracy is checked. With electronic handling, only authorised persons may enter or change data, changes and deletions must be recorded, access must be limited by passwords or similar, and critical data entry must be independently checked. Electronically stored batch records must be protected by backup (tape, microfilm, discs, paper printouts or other means) and stay readily available throughout retention.	binding	Checked	IN-09
n238	Part I, section 20 (computerised systems), 20.2-20.10 (20.1 heading text lost in extraction; the Part XII twin 5.4.1 requires validation scaled to diversity, co	Computerised systems must be shown fit for their tasks by installation and operational qualification; qualified commercial software needs less testing, and an unvalidated legacy system may be validated retrospectively if documentation exists. Systems must stop unauthorised access or data changes, prevent missing data (for example data not captured while the system was off) and record each data change with the previous value, who changed it and when. Written procedures must cover operation and maintenance. Manual entry of critical data needs an extra accuracy check by a second person or by the system. Incidents that could affect product quality or record reliability must be recorded and investigated. System changes go through change control, formally authorised, documented and tested, with records showing a validated state. A backup must prevent permanent record loss, with data protection for all systems, and a secondary (non-computer) record of the data is permitted as well.	binding	Checked	IN-09
n239	Part I, 17.3.6 (master formula records), 17.3.6.1-17.3.6.2	A formally authorised master formula must exist for each product and each batch size. It must cover: product name with a product reference code linked to its specification; dosage form, strength and batch size; all starting materials (with INNs where applicable) with amounts, designated name and unique material reference, noting any substance lost during processing; expected final yield with limits and relevant intermediate yields; processing location and principal equipment; methods (or references) for preparing and operating critical equipment such as cleaning, assembly, calibration, sterilisation and use; detailed stepwise instructions (material checks, pre-treatments, order of addition, mixing times, temperatures); in-process controls with limits; storage requirements including container, labelling and special conditions; special precautions; and the permitted hold time for intermediate and in-process material.	binding	Checked	IN-13
n240	Part I, 17.3.7 (packaging instructions)	Formally authorised packaging instructions are needed for each product, pack size and pack type. They normally cover: product name; dosage form, strength and (where relevant) method of application; pack size by count, weight or volume; a complete list of packaging materials for a standard batch with quantities, sizes, types and each material's specification code; where appropriate a specimen of the printed packaging material showing where batch number and expiry date go; special precautions including line-clearance checks of the area and equipment before and after packaging; a description of the operation, subsidiary operations and equipment; and in-process controls with sampling instructions and acceptance limits.	binding	Checked	IN-14
n241	Part I, 17.3.8 (batch processing records), 17.3.8.1-17.3.8.3	A batch processing record is kept for every batch, built from the currently approved documents, prepared so as to avoid errors (copying or validated computer programs are recommended; transcription from approved documents is to be avoided). Before processing, a recorded check confirms that equipment and workstation are free of previous products, documents and materials and that equipment is clean and suitable. During processing each item is recorded when the action is taken, and on completion the record is dated and signed by the person responsible for processing. Items: product name; batch number; dates and times of starting significant intermediate stages and of completing production; name of the person responsible for each stage; initials of operators of significant steps and, where appropriate, of checkers (e.g., weighing); batch or analytical control number and actual quantity of each starting material weighed, including any recovered or reprocessed material added; relevant operations, events and major equipment; permitted hold time for intermediate and in-process material; IPCs with performer initials and results; yields at pertinent stages with explanations of significant deviations from expected yield; notes on special problems with signed authorisation for any departure from the master formula.	binding	Checked	IN-15
n242	Part I, 17.3.9 (batch packaging records), 17.3.9.1-17.3.9.3	A batch packaging record is kept for each batch or part-batch packed, based on the approved packaging instructions and prepared to avoid errors (copying or validated computer programs recommended). Before packaging, a recorded check confirms equipment and workstation are clear of previous products, documents and materials and are clean and suitable. Each item is recorded when the action happens, with date and responsible person identified by signature or electronic password: product name, batch number and bulk quantity to be packed, finished-product batch number, planned and actual quantity and the reconciliation; date and time of packaging operations; name of the responsible person; initials of operators of significant steps; identity and conformity checks against the instructions including IPC results; details of operations, equipment and lines, plus instructions or records for unpacked product returned to store; where possible, samples of printed packaging materials used, including specimens showing print approval and periodic checks of batch number, expiry date and overprinting; notes on special problems and deviations with written authorisation; and quantities and reference numbers of all printed materials and bulk issued, used, destroyed or returned, and product obtained, allowing a proper reconciliation.	binding	Checked	IN-10, IN-16
n243	Part I, 17.3.10.8-17.3.10.11 (batch numbering)	An SOP must define the batch (lot) numbering system so each intermediate, bulk and finished batch has its own number; processing and packaging numbering must be related; numbers must never be reused, including for reprocessing; each allocation must be recorded at once (e.g., in a logbook) with at least the allocation date, product identity and batch size.	binding	Checked	IN-18
n244	Part I, 18.2.2 (deviations) and 18.5.12 (batch record review and failure investigation)	Departures from instructions or procedures should be avoided; where they happen they must follow an approved procedure and be authorised in writing by a designated person, with QC involvement where appropriate. Production records must be reviewed as part of batch-release approval before going to the authorised person. Any divergence or specification failure must be investigated thoroughly, extended where needed to other batches and products, and written up with conclusion and follow-up.	binding	Checked	IN-05, IN-20
n245	Part I, 18.2.3, 18.4.6 and 18.5.10 (yields and reconciliation)	Yield checks and reconciliation of quantities must be done as needed to show no discrepancy outside acceptable limits. Any significant deviation from expected yield must be recorded and investigated. Any significant or unusual discrepancy found when reconciling bulk, printed packaging materials and units produced must be investigated, explained and recorded before release.	binding	Checked	IN-17
n246	Part I, 18.2.5, 18.4.1-18.4.2, 18.5.2-18.5.3 (identification and line clearance)	Throughout processing, materials, bulk containers, major equipment and, where appropriate, rooms and packaging lines must show the product, strength and batch number, and the stage where applicable; recording the previous product can help. Before any processing or packaging, the area, lines, printers and equipment must be checked clean and free of earlier products, materials, labels and documents; packaging line clearance follows a procedure and checklist and is recorded. Needed in-process and environmental controls must be done and recorded. Each packaging station or line must display the product name and batch number.	binding	Checked	IN-21
n247	Part I, 18.5.4-18.5.11 (packaging operations)	Labelling should follow filling and sealing quickly, or controls must prevent mix-ups. Any printing of codes or expiry dates, on-line or separate, must be checked and recorded, and hand printing re-checked at intervals. Extra care is needed with cut labels, off-line overprinting and hand packing; electronic code readers, label counters and similar devices must be checked for correct operation, and manual labelling needs more frequent checks. Printed or embossed information must be distinct and fade-resistant. On-line checks must at least cover pack appearance, completeness, correct product and materials, correct overprint and working line monitors; samples taken off the line are not returned. Product from an unusual event goes back only after inspection, investigation and approval, with a detailed record. After packaging, unused batch-coded materials are destroyed with a record; uncoded printed materials go back to stock only after documented checks.	binding	Checked	IN-22
n248	Part I, 14.14, 14.16-14.20 (dispensing and use of starting materials)	Only starting materials released by QC and within shelf life may be used. There must be means of confirming the identity of the contents of every container, and sampled bulk containers must be identified. Dispensing is done only by designated persons under a written procedure, into clean, properly labelled containers, with accurate weighing or measuring. Each dispensed material and its weight or volume must be independently checked and recorded. Materials dispensed for a batch are kept together and clearly labelled as such.	binding	Checked	IN-23
n249	Part I, 14.32-14.34 (reworking, recovery, reprocessing)	Reworking or recovering rejected product must be exceptional, allowed only if final quality is unaffected and specifications are met, and done under a defined, authorised procedure after risk evaluation; a record is kept and the reworked batch gets a new batch number. Blending some or all of an earlier in-specification batch into a later batch of that product, at a set process stage, needs prior authorisation, follow a defined procedure after risk evaluation	binding	Checked	IN-18, IN-19

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		(including shelf-life impact) and be recorded. QC must consider extra testing of any finished product that was reprocessed, reworked or contains recovered product.			
n250	Part I, 14.15 (release of non-API raw materials without batch testing)	QC may release raw materials other than APIs for manufacturing without specific batch testing, but only on the basis of vendor approval and statistical analysis of earlier test results for that material.	binding	Checked	IN-24
n251	Part I, 14.13 (labelling of starting materials) and 14.10-14.12, 14.23-14.25 (receipt and packaging materials)	Stored starting-material labels must show at least the designated name and internal code, the supplier's batch number and the manufacturer's own control or batch number, the current status (for example quarantined, under test, approved, rejected, returned, recalled) and, where appropriate, an expiry or retest date; with fully validated computerised storage systems, not all of this needs to be human-readable on the label. Each batch within a delivery is treated separately for sampling, testing and release. Every lot or delivery of primary and printed packaging gets a unique reference, and packaging materials are checked on delivery to the packaging area for quantity, identity and conformity with the instructions.	binding	Checked	IN-24
n252	Part I, 17.3.1 (labels of finished drugs, reference standards and containers)	Every finished drug must be labelled with at least: drug name; active ingredients (with INNs) and amount of each, plus net contents; the manufacturer's batch number; expiry date and date of manufacture in uncoded form; any special storage or handling conditions; directions for use, warnings and precautions; and the name and address of whoever made it or is responsible for marketing it. Labels on containers, equipment or premises must be clear, unambiguous and in the company's format, optionally colour-coded for status.	binding	Checked	IN-31
n253	Part II (sterile products), 4.1-4.6 and Table 1 (clean-area grades and particle limits)	Sterile manufacture uses four grades (A-D) classified to ISO standards. Table 1 sets maximum particles per cubic metre, at rest and in operation, for sizes of 0.5 and 5.0 micrometres. Grade A: 3,520 and 20 in both states. Grade B: 3,520 and 29 at rest; 352,000 and 2,900 in operation. Grade C: 352,000 and 2,900 at rest; 3,520,000 and 29,000 in operation. Grade D: 3,520,000 and 29,000 at rest; not defined in operation. Grade A unidirectional air must run at about 0.36-0.54 m/s (guidance) measured 15-30 cm below the terminal filter, never below 0.36 m/s at working level, shown by airflow visualisation. Classification is kept separate from routine monitoring.	binding · Injectables	Checked	IN-40
n254	Part II, 4.7-4.10 and Table 3 (environmental monitoring; microbial limits)	Clean rooms must be monitored in operation at risk-based locations. Grade A needs particle monitoring for the whole of critical processing, including equipment assembly (with stated exceptions), capturing interventions and transient events and alarming on alert-limit excursions; Grade B uses a similar system at lower frequency. Recovery ('clean-up') after operations of about 15-20 minutes to the at-rest limits. Microbial monitoring (settle plates, active air, contact plates, swabs, glove prints) must be frequent during aseptic work, and monitoring results must be considered when reviewing batch documentation for release. Recommended in-operation microbial limits (average values): Grade A under 1 CFU for air per m3, 4-hour settle plate, contact plate and 5-finger glove print; Grade B 10, 5, 5, 5; Grade C 100, 50, 25; Grade D 200, 100, 50. Alert and action limits must be set; exceeding action limits or an adverse alert trend triggers investigation and CAPA per SOP.	binding · Injectables	Checked	IN-42
n255	Part II, 5.3-5.7 (process simulation / media fills) and 4.11 (hold times and fill duration)	Aseptic processing must be validated by media fills: three consecutive successful runs initially, repeated at defined intervals and after significant HVAC, equipment or process changes, simulating routine steps, normal and worst-case interventions and each shift and shift change. Container numbers must allow valid evaluation and, for small batches, at least equal the batch size. Target is zero growth; acceptance: under 5,000 units no contaminated unit; 5,000-10,000 units one contaminated unit means investigation and possible repeat, two means revalidation after investigation; over 10,000 units one means investigation, two means revalidation. Intermittent contamination at any size must be investigated, and gross failures must consider impact on batches made since the last good media fill. Media fills are also used to set processing hold times and the maximum fill duration.	binding · Injectables	Checked	IN-45
n256	Part II, 5.12-5.17 (processing time limits, bioburden, endotoxins) and 2.4	The processing stage of cleaned components, bulk containers and equipment must be identified. The intervals from washing/drying to sterilisation and from sterilisation to use must be as short as possible and within validated time limits. The time from starting solution preparation to sterilisation or sterile filtration must be as short as possible, with a maximum permitted time set for each product. Purge or blanket gases must pass a sterilising filter. Bioburden must be tested before sterilisation on every batch for both aseptic and terminally sterilised products (scheduled intervals allowed only for overkill cycles; each batch as an IPC for parametric release), against working limits; endotoxins monitored where appropriate. WFI, relevant intermediates and finished injectables must be monitored for endotoxins by a validated pharmacopoeial method; for LVPs, water or intermediate monitoring is always required.	binding · Injectables	Checked	IN-46
n257	Part II, 6.5-6.11 (sterilisation: validation, loading, records)	All sterilisation processes must be validated, with effectiveness in every part of each load type shown by physical measurements and, where appropriate, biological indicators, and re-verified at least once a year and after significant equipment changes. Validated loading patterns are required. Every carrier (basket, tray) must be labelled with material name, batch number and sterilised/not-sterilised status. A sterilisation record must exist for every run and be approved as part of batch release. Heat cycles are recorded on suitably accurate equipment with the probe at the validated coolest point, preferably checked against an independent second probe; temperature and pressure both monitored; control instruments independent of monitoring; automated control/monitoring validated; system and cycle faults registered by the system and seen by the operator; vacuum-phase chambers leak-tested regularly. For EO, each cycle records time, pressure, temperature, humidity and gas concentration with a continuous pressure/temperature chart and biological indicators; radiation dose is measured by dosimeters; these records form part of the batch record. Cooling fluids contacting product must be sterile.	binding · Injectables	Checked	IN-47
n258	Part II, 6.11.2 (aseptic processing and sterile filtration)	Aseptic processing must control environment, personnel, critical surfaces, container/closure sterilisation and transfer, the maximum holding time of product before filling, and the sterilising filter. Solutions that cannot be terminally sterilised may be filtered through a 0.22 micrometre (or equivalent) sterile filter into sterilised containers; filtration alone is not enough when terminal sterilisation is possible. A second (redundant) filtration just before filling may be advisable; the final filter should be as close to the fill point as possible; filters must not shed fibres (asbestos filters banned). Filter integrity must be checked before use and confirmed immediately after use (bubble point, diffusion or pressure hold); filtration time for a known volume and the pressure difference are set during validation, with significant routine differences noted and investigated, and the results go into the batch record. Critical vent filters are confirmed after use. A filter may be used for at most one working day unless longer use has been validated.	binding · Injectables	Checked	IN-48
n259	Part II, 2.1-2.3 (sterility testing and release) and 4.8	The sterility test is only the last of a series of controls and must be validated for the product using pharmacopoeial methods. Sterility samples must represent the whole batch and include the riskiest parts: for aseptic fills, units filled first, last and right after any major stoppage; for heat-sterilised products, the potentially coolest part of the load. Batch processing records, and for aseptic processing the environmental records, must be examined together with sterility results. Parametric release, where authorised, needs special attention to validation and monitoring of the whole process.	binding · Injectables	Checked	IN-49
n260	Part II, 10.3-10.8 (finishing: capping, vacuum-sealed containers, inspection) and Part I 11.5.1	Cap crimping generates particles and needs a separate station with extraction; if capping is done outside the aseptic core, vials must stay under Grade A air until crimped; vials with missing or displaced stoppers are rejected before capping, and human intervention at capping must avoid direct contact. Vacuum-sealed containers must be tested for maintained vacuum after a set period. Every filled parenteral container must be inspected individually for extraneous contamination or defects; visual inspection needs controlled lighting and background, inspectors must pass regular eyesight checks (with their own corrective lenses) and get frequent breaks; automated inspection must be validated and its performance checked at intervals; results must be recorded. Staff doing visual inspection must also have periodic eye checks under Part I.	binding · Injectables	Checked	IN-50
n261	Part VIII (oral solid dosage forms), 3.3-3.8 (compression in-process controls)	Calibrated weighing equipment must be at hand for in-process tablet weight checks, with procedures able to catch out-of-limit tablets. At the start of each compression run, and for multi-station or multi-point machines at fixed intervals, enough individual tablets must be examined so that each station/point is checked for pharmacopoeial parameters (appearance, weight variation, disintegration, hardness, friability, thickness), with results recorded in the batch documentation. During compression, samples must be taken at intervals of no more than thirty minutes against the in-process specification, with periodic checks of the same parameters plus contamination by lubricating oil. Tablets must be de-dusted (preferably automatically) and checked for foreign matter and defects, collected in clean labelled containers, and rejected or discarded tablets must be kept in identified containers with their quantity recorded in the BMR.	binding · Tablets	Checked	IN-35

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n262	Part VIII, 2.1-2.5 (sifting, mixing, granulation) and 1.3-1.4	Critical operating parameters such as time and temperature for every mixing, blending and drying step must be defined in the master formula, watched while the step runs and written into the batch record. Unless closed, mixing, sifting and blending equipment needs dust extraction or a dedicated area. Sieve residues must be examined periodically for foreign material. Fluid-bed drier filter bags must be washed between products, and product-dedicated for certain highly potent or sensitising products; inlet air must be filtered. Granulation and coating solutions must be made, stored and used so as to limit contamination and microbial growth. All dry ingredients are sifted before use (normally in dedicated areas) unless input quality is assured. A metal detector is recommended, wooden equipment is to be avoided, and screens, sieves, punches and dies must be checked for wear or breakage before and after each use.	binding · Tablets	Checked	IN-13, IN-36
n263	Part VIII, 4 (coating), 5 (capsule filling), 6 (printing of tablets and capsules)	Air supplied to coating pans for drying must be filtered and of suitable quality, with exhaust and temperature/humidity control in the area. Coating solutions and suspensions are made fresh and used so as to minimise microbial growth, and their preparation and use are documented. Empty capsule shells count as a drug component and must be stored away from excess heat and moisture. Printing on tablets or capsules needs special care against mix-ups, with segregation if different products or batches are printed at the same time, edible-grade colours and suitable inks; printed tablets or capsules must be approved by QC before release for packaging or sale.	binding · Tablets	Checked	IN-37
n264	Part VIII, 7.1-7.4 (strip and blister packaging)	Automatic counting, strip and blister machines must be cleared of every 'rogue' tablet, capsule or foil before a new packaging operation, and an independent, recorded check of the equipment is required before a new batch is handled. Uncoated tablets are packed on equipment designed to limit cross-contamination, in an isolated area for potent or beta-lactam tablets. Strips leaving the machine are inspected for misprints, cut foil, missing tablets and poor sealing. Strip and blister pocket integrity must be checked periodically by a vacuum (leak) test, with records.	binding · Tablets	Checked	IN-38
n265	Part VIII, 1.5, 1.14, 1.17, 1.25, 1.27-1.28 (pressure cascades, temperature and humidity)	Where rooms are designed with pressure differentials, these must be monitored regularly and any deviation reported at once to Production and QA. Pressure-differential limits of adjacent rooms must not overlap. Pressure control and monitoring devices must be calibrated and qualified, compliance verified regularly and recorded, and alarms set from risk analysis. Room pressure indicators must show normal range and alert/action limits at the point of reading. Where relevant, temperature and relative humidity must be controlled, monitored and recorded, with alert and action limits.	binding · Tablets	Checked	IN-39
n266	Part I, 17.3.10.5, 17.3.10.16-17.3.10.17 (equipment SOPs and logs) and 18.4.3, 18.4.9 (equipment failure, calibration)	Each instrument and piece of equipment needs SOPs (use, calibration, cleaning, maintenance) kept close to it. Records must be kept for major and critical equipment of validations, calibrations, maintenance, cleaning and repairs with dates and who did them, and use of major/critical equipment and of processing areas must be logged in chronological order. There must be a means of signalling equipment or utility failures; defective equipment is withdrawn until fixed; equipment is cleaned promptly after use and stored clean and dry. Measuring, weighing, recording and control instruments are serviced and calibrated at set intervals with records, checked daily or before use for analytical tests, and labelled with the calibration date and next due date.	binding	Checked	IN-25
n267	Part I, 18.4.4 (time limits after cleaning) and 18.4.5, 18.2.8	Time limits for storing process materials and cleaned equipment before use must be stated and based on data. Containers for filling must be cleaned before filling, with attention to glass fragments and metal particles. In-process controls done in production areas must not harm the product or another product.	binding	Checked	IN-25
n268	Part I, section 8 (change control), 8.1-8.7	A formal change-control system must assess all changes that could affect production and control. Written procedures must govern how changes are identified, documented, reviewed and approved, whether they concern materials and their specifications, test methods, buildings and utilities, equipment, process steps, labels and packs, or computer hardware and software. Proposals are drafted, reviewed and approved by the relevant units and by the quality units. Impact on product quality is evaluated, and changes may be classified (e.g., minor/major) to decide testing and validation. All affected documents must be revised when the change is implemented. The first batch made or tested after the change must be evaluated, and effects of critical changes on retest or expiry dates assessed (possibly with accelerated or ongoing stability).	binding	Checked	IN-03
n269	Part I, section 10 (self-inspection, quality audits, suppliers), 10.1-10.8	Self-inspections must be routine and at least once a year (frequency stated in the procedure), done by an objective team of GMP-knowledgeable staff (internal or external), and may also be triggered by recalls, repeated rejections or announced inspections. A report must cover results, evaluation and conclusions and recommended corrective actions, with management follow-up; the self-inspection checklist covers documentation, production and IPCs, QC, sanitation, validation, complaints, recalls, labels control and results of previous self-inspections. Suppliers are approved by QC together with other departments after evaluation, audited where needed, and listed as approved.	binding	Checked	IN-28
n270	Part I, sections 6 (complaints and adverse reactions) and 7 (product recalls)	Complaints about possible defects are recorded in full, investigated (normally with the QC head), and all decisions and measures recorded and cross-referenced to the batch records; other batches, including any containing reprocessed material from the defective batch, must be considered; complaint records are reviewed for trends; the licensing authority is told when action is contemplated for serious quality problems. A pharmacovigilance system must collect and forward ADR reports to the licensing authorities. Recalls: written SOP able to reach every distribution level down to retail quickly (print and electronic media may be used); the authorised person runs recalls; distribution records with wholesaler and direct-customer details must be readily available; recalled stock is segregated securely; the licensing authority is informed of any intended recall; progress is recorded and a final report reconciles delivered and recovered quantities; recall arrangements are tested periodically.	binding	Checked	IN-29
n271	Part I, 11.2, 11.4 (training), 11.5 (personal hygiene)	Staff must be qualified and trained, with written job descriptions, an organisation chart, no gaps or unexplained overlaps in GMP responsibilities, and deputies only if suitably qualified. Training follows a written programme for everyone entering production or QC areas, including maintenance and cleaning staff; there is job-specific training for new recruits, continuing training with periodic assessment of effectiveness, specific training for clean areas and hazardous materials, and training records (including evidence of qualification for consultants and contract staff). Health checks are required before and during employment, with periodic eye checks for visual inspectors; sick staff with lesions are excluded from handling products; untrained visitors are kept out or supervised.	binding	Checked	IN-04
n272	Part I, section 9 (production under loan licence or contract, contract analysis), 9.1-9.5	Production under a loan licence or contract must follow the product licence and a written contract that sets each party's responsibilities (including release, sampling and testing) and states how the authorised person releases each batch. Only a holder of a valid manufacturing licence may manufacture under contract; subcontracting needs the contract giver's prior approval. The loan licensee or contract giver must review records and results, confirm GMP and licence compliance and release by the authorised person, and must hold or have access to manufacturing, analytical and distribution records and reference samples, including any needed for complaints or suspected fraud.	binding	Checked	IN-30
n273	Part I, section 5 (qualification and validation), 5.1-5.12; Appendix I (site master file) item on computerised-system validation	Each company must identify the qualification and validation needed to show critical aspects are controlled, in a validation master plan covering DQ, IQ, OQ, PQ and process validation, per approved protocols, with written reports; any change that may affect quality must be qualified or validated; validation is ongoing with periodic review, and processes are set on validation results. Particular attention goes to analytical methods, automated systems and cleaning procedures. The site master file must describe qualification and calibration, including recording systems and arrangements for computerised-system validation.	binding	Checked	IN-12
n659	Part I, 19.1-19.7 (quality control: independence, basic requirements, sampling, OOS, material release)	QC must be independent of production and other departments and headed by a suitably qualified person. Samples are taken by QC-approved methods and people; records (manual or instrument) must show all sampling and testing was done and deviations investigated; product assessment must include review of production documentation and deviations; finished products must match the licensed composition. Tests follow written procedures and results are checked by a supervisor before release or rejection. Sample containers are labelled with material name, batch, container number, sample number, sampler's signature and date. Out-of-specification results must be investigated under an approved procedure and recorded. Before releasing starting or packaging materials the QC in-charge must ensure testing for identity, strength, purity and other quality parameters.	binding	Checked	IN-20
n660	Part I, 19.3(g) (retained samples); Drugs Rules Rule 74 (Form 25 licences) and Rule 78 (Form 28 licences)	Schedule M requires enough samples of starting materials and products to be retained for future examination, kept for an appropriate time in the final pack (or an equivalent pack if the marketed one is very large). The conditions of licence in the Drugs Rules set the amount and period: at least twice the quantity needed for all tests on the batch, kept for three months beyond the expiry date, or three years from manufacture where no expiry is labelled.	binding	Checked	IN-32

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
	reference-sample conditions				
n662	Rule 71(1)/Rule 76 (competent technical staff; testing staff); Schedule M Part I 11.3.2	Manufacture must be carried out under the active direction and personal supervision of competent technical staff (at least one whole-time person with prescribed pharmacy, science, chemical engineering or medical qualifications and experience), and testing under qualified staff approved by the licensing authority. Schedule M says key personnel must hold the qualifications and experience specified under the rules.	binding	Checked	IN-04
n663	Part XII (APIs, ICH Q7-based), 6.1 documentation clauses (e-signature sentence, probably 6.1.9), 6.1.5-6.1.7	The only explicit electronic-signature rule in Schedule M sits in the API Part: where electronic signatures are used on documents they must be authenticated and secure. The same Part requires entries made indelibly right after the activity, identifying the person, with corrections dated and signed leaving the original readable; records may be kept as originals or accurate copies, and where electronic records are used there must be retrieval equipment and a way to produce a hard copy; records must be readily available at the site during retention, or promptly retrievable electronically.	binding	Checked	IN-10
n664	Part XIII, item 11 (parenteral preparations: areas, equipment, minimum areas) and FFS/BFS subsection	For parenterals, Part XIII divides operations into separate areas (water management, container and closure preparation, solution preparation, filling/capping/sealing, sterilisation, quarantine, visual inspection, packaging; FFS/BFS lines have their own list) and recommends equipment for each, e.g., RO or EDI plus multi-column distillation, SS 316L storage tank and continuously circulating loop, automatic washers, double-ended dryer/steriliser, cartridge/membrane filtration, filling under laminar air flow, steam steriliser preferably with computer-controlled cycles, hot-air steriliser, pressure leak tester, visual inspection units with white and black backgrounds and lighting, batch coding machine and labelling unit. Recommended minimum areas are about 150 m2 basic plus 100 m2 ancillary for SVI, 150 m2 each for LVP, 100 m2 for LVP packaging materials, and 250 m2 plus 150 m2 for FFS LVP lines, partitioned with airlocks.	binding · Injectables	Checked	IN-51
n665	Part I (PQS resources list; 15.2 reference standards; 17.3.2 specifications; packaging-material clause) - Indian Pharmacopoeia references	The PQS must make the key reference texts available, expressly the 1940 Act and 1945 Rules, the latest IP edition and official MoHFW guidance documents. Specifications may need periodic revision to follow new editions of the IP or other official pharmacopoeias. IP reference standards must be obtained from the Indian Pharmacopoeia Commission. Packaging materials must meet IP requirements, and purified water must meet pharmacopoeial specifications (potable water per BIS or municipal standards as feed).	binding	Checked	IN-33
n666	Part III (hazardous substances: sex hormones, steroids, cytotoxics) general, 2 (risk assessment), 4 (PPE and breathing air)	Where a tablet or injectable contains sex hormones, steroids or cytotoxics, Part III applies in addition to Part I: a risk assessment must decide the hazard to operators, public and environment and which stages fall under Part III; facility and equipment should give containment, otherwise operator protection (suits, double gloves where contact is unavoidable, respirators with breathing air) is required; central breathing-air systems need a 100% back-up with automatic changeover; de-gowning and decontamination procedures apply.	binding	Checked	IN-27
n669	Part VIII 3.4 and 3.8 (weight variation checks against pharmacopoeial parameters); IP general test for uniformity of weight	Part VIII requires in-process checks of weight variation and other pharmacopoeial parameters, which for Indian products means IP limits. IP (like BP/Ph. Eur.) sets average-weight bands of 80 mg or less (10% deviation), above 80 mg and under 250 mg (7.5%) and 250 mg or more (5%), whereas the older USP weight-variation bands commonly used as in-process limits are 130 mg or less (10%), 130-324 mg (7.5%) and above 324 mg (5%); current USP <905> instead judges uniformity of dosage units by an acceptance value. The same tablet can therefore fall in different bands (e.g., a 200 mg tablet: 7.5% under both; a 300 mg tablet: 5% under IP but 7.5% under old-USP bands).	binding · Tablets	Unverified	IN-34
n670	Part II, 3.1-3.4 (sanitation of clean areas, disinfectants) and 5.9 (water monitoring)	Clean areas must be cleaned often and thoroughly under an approved programme; disinfectants, where used, must be of at least two kinds, and the programme must include a sporicidal agent; effectiveness must be demonstrated, interactions between cleaning agents validated and disinfectant residues shown to be removed. Disinfectants and detergents are monitored for microbial contamination, dilutions kept in cleaned containers and stored only for defined periods unless sterilised, and those for Grade A/B areas must be sterile before use. Fumigation may help for inaccessible places. Water sources, treatment equipment and treated water are monitored regularly for chemical, microbial and endotoxin contamination, with records of results and actions.	binding · Injectables	Checked	IN-44
n671	Part II, 4.5 (HEPA filter leak testing), 4.6 (classification sampling)	Installed HEPA filters must be leak-tested to ISO standards, recommended every six months and never more than twelve months apart, using an aerosol that does not support microbial growth; patching is allowed within ISO limits. For classification, Grade A needs at least 1 m3 per location; lower grades at least 2 litres and one minute per location; portable counters with short tubing and isokinetic heads in unidirectional flow.	binding · Injectables	Checked	IN-40, IN-43
n672	Part I, 18.3.1-18.3.5 (prevention of cross-contamination) and 18.2.4, 18.2.7, 18.5.1	Cross-contamination must be prevented by technical or organisational measures: dedicated self-contained areas (which may be needed for penicillins, cytotoxics, sex hormones, spore-formers, live vaccines and some biologicals), or campaign production followed by validated cleaning, airlocks and pressure differentials, limiting recirculation of untreated air, protective clothing, cleaning of known effectiveness, closed systems, residue testing and cleanliness status labels on equipment; effectiveness is checked periodically. Two products may share a room at the same time or back-to-back only when mix-up and cross-contamination risks are excluded; non-medicinal products may not be made in drug areas or equipment; different products are not packed in close proximity without physical segregation. Dust generation from dry materials must be controlled.	binding	Checked	IN-26
n673	Part I, 14.21-14.25 (packaging materials) and 14.36 (returned goods)	Primary and printed packaging materials are bought, handled and controlled like starting materials. Printed materials are stored securely against unauthorised access, roll-feed labels preferred, cut labels and loose printed items kept and moved in separate closed containers, and issued only by designated staff under an approved, documented procedure. Every packaging lot or delivery is given a unique identifier; outdated or obsolete material is destroyed with a record. Returned products are destroyed unless their quality is certain, and are reissued, relabelled or otherwise dealt with only after a documented QC assessment considering product nature, storage needs, condition, history and time since issue.	binding	Checked	IN-24
n674	Part I, 2.3.3 (CAPA management) with PQS item (s) and 10.5-10.7 (follow-up)	Corrective and preventive actions must be completed promptly and effectively under documented procedures; there must be procedures for ongoing management and review of these actions, and their effectiveness must be verified during self-inspection. Investigations must go to an appropriate depth of root-cause analysis, identify the most likely root causes and monitor CAPA effectiveness; management evaluates self-inspection reports and corrective actions.	binding	Checked	IN-01
n675	Part II, 4.11 and 4.11.1-4.11.2 (grade required per operation) and 1.1-1.3	Sterile operations are split into terminally sterilised and aseptic categories, and component preparation, product preparation, filling and sterilisation must be done in separate areas of appropriate grade. Terminally sterilised: filling generally at least Grade C; Grade A with at least C background where the product is at unusual environmental risk (slow filling, wide-necked or long-open containers); solutions and components prepared at least in D, or C if at unusual microbial risk; ointments, creams, suspensions and emulsions prepared and filled in C before terminal sterilisation. Aseptic: washed components handled in at least D; sterile materials not later sterilised or filtered handled in A with B background; solutions to be sterile-filtered prepared in C (D if closed); aseptic handling and filling, and exposed sterile equipment, in A with B background; partially stoppered lyophilisation containers moved in A/B or in sealed trays in B; exposed sterile semi-solids and suspensions not later filtered made in A/B.	binding · Injectables	Checked	IN-41
n676	Part I, 14.29-14.31 (finished-product quarantine, release, rejected materials) and Part VIII 6.2	Finished products stay in quarantine until final release and are then stored as usable stock under the manufacturer's conditions; the evaluation and documents needed for release are those of section 19. Rejected materials and products are clearly marked, stored separately in restricted areas and returned, reprocessed or destroyed promptly, with the action approved by authorised personnel and recorded. Printed tablets and capsules need QC approval before release for packaging or sale.	binding	Checked	IN-05, IN-37
n677	Part VIII, 1.1, 1.6-1.10, 3.1-3.2 (dust control, cubicles, pressure cascade, mix-up prevention at compression)	Dry-product processing needs special dust and cross-contamination control, with enclosed dust-control systems where required. Each tablet press needs effective dust control and must sit in its own cubicle unless the same product runs on each machine or the press provides its own enclosed controlled environment; physical, procedural and labelling measures must prevent mix-ups of materials, granules and tablets at compression. Equipment must be checked so that no material lodges undetected, nozzles stay clean and lubricants do not leak into product. In a multi-	binding · Tablets	Checked	IN-26, IN-39

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		product OSD site, dust must not move between cubicles; the corridor is kept at a higher pressure than cubicles and cubicles above atmospheric, while highly potent products run under a negative pressure cascade.			
n678	Cross-cutting: per-market rule matrix (derived from SM-07, SM-11, SM-13, SM-21, SM-22, SM-24, SM-25-SM-27, SM-33, SM-41, SM-47, SM-48, SM-56, SM-58)	Where Indian rules and FDA/EU rules differ, a batch made for several markets must satisfy the strictest applicable rule. Known differences to encode: batch-record retention (India: longest of expiry+1y under Schedule M and 5 years under Rule 74, or expiry+2y for Rule 78 test records; FDA expiry+1y; EU longest of expiry+1y and certification+5y); retained/reference samples (India expiry+3 months under the Drugs Rules; FDA and EU expiry+1y); media-fill acceptance (India and FDA allow limited contaminated units with investigation; EU 2022 treats any growth as failure); Grade A viable limit (India '<1' average; EU 'no growth'); clean-room particle tables (India/WHO 2011 values vs EU 2022 values); HEPA leak-test interval (India 6-12 months); compression IPC interval (India at most 30 minutes; FDA/EU firm-defined); annual self-inspection (India only); new batch number for reworked batches (India/WHO only); hold time in MFR/BPR, date of manufacture on label, MRP and Schedule H/H1/X panels (India only); non-API excipient release without testing (allowed in India, not for FDA/EU identity testing); pharmacopoeial limits (IP vs USP vs Ph. Eur.).	binding	Checked	IN-08, IN-13, IN-52

21 CFR Part 11

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n3	211.25(a)-(c)	Everyone who manufactures, processes, packs or holds a drug product needs education, training and experience (any combination is acceptable) sufficient for the assigned functions. Training must cover the specific operations the person performs and CGMP as it applies to their job, including the CGMP regulations and the SOPs they require; CGMP training is delivered by qualified people, continually and often enough that staff stay familiar with the requirements that apply to them. Supervisors need qualifications that let them supervise in a way that assures the product's safety, identity, strength, quality and purity, and there must be enough qualified staff to perform and supervise each product's operations.	binding	Checked	DI-21
n43	11.1(a), (b), (e); 11.2(a); 11.3(b)	Part 11 sets the criteria for FDA to regard electronic records, electronic signatures, and handwritten signatures applied to electronic records as reliable, trustworthy and broadly equivalent to paper records with handwritten signatures. It covers any record kept in electronic form (created, modified, maintained, archived, retrieved or transmitted) under an FDA records requirement, but not paper records that have merely been sent electronically. Computer systems (hardware and software), controls and associated documentation kept under Part 11 must be open to FDA inspection and readily available to inspectors. For records kept but not submitted to FDA, electronic records and signatures may replace paper and handwritten signatures in whole or in part if Part 11 is met (11.2(a)). Definitions: in a closed system, access is controlled by the people responsible for the records' content; in an open system it is not. Other defined terms include electronic record, electronic signature (symbols an individual executes, adopts or authorizes as the legally binding equivalent of a handwritten signature), digital signature (cryptographic, allowing signer identity and data integrity to be verified), biometrics, and handwritten signature (which also covers a scripted name or legal mark applied with a stylus to a capturing device).	binding	Checked	DI-01, DI-08
n44	11.10(a)	Closed systems must be validated to show that they are accurate and reliable, perform consistently as intended, and can detect records that are invalid or have been altered.	binding	Checked	DI-03, DI-23, DI-44
n45	11.10(b)	The system must be able to output full, accurate copies of its records, both as human-readable renderings and in electronic form, suitable for FDA to inspect, review and copy; firms should contact FDA with any questions about the agency's ability to review and copy the records.	binding	Checked	DI-40
n46	11.10(c)	Records must be protected so they can be retrieved accurately and readily for the whole retention period.	binding	Checked	DI-44
n47	11.10(d)	Access to the system must be limited to authorized individuals.	binding	Checked	DI-14, DI-16, DI-38
n48	11.10(e)	The system must use secure, computer-generated, time-stamped audit trails that capture on their own the date and time of each operator entry or action that creates, changes or deletes an electronic record. A change must leave the previously recorded information visible. Audit trail documentation is kept at least as long as the underlying electronic records and is available to FDA for review and copying.	binding	Checked	DI-22, DI-24
n49	11.10(f)	Where appropriate, operational system checks must enforce the permitted order of steps and events.	binding	Checked	DI-34
n50	11.10(g)	Authority checks must confirm that a person is authorized before letting them use the system, sign a record electronically, reach the operation or the system's input/output devices, change a record, or carry out the operation in question.	binding	Checked	DI-17
n51	11.10(h)	Where appropriate, device (e.g., terminal) checks must confirm that the source of data input or operational instructions is valid.	binding	Checked	DI-20
n52	11.10(i), (j), (k)(1)-(2)	(i) It must be established that the people who develop, maintain or use the electronic record/signature system have the education, training and experience for their assigned tasks. (j) Written policies, established and followed, must make individuals accountable and responsible for actions taken under their electronic signatures, so that record and signature falsification is deterred. (k) Systems documentation must be appropriately controlled: its distribution, access and use for system operation and maintenance must be controlled, and revision and change control procedures must keep an audit trail showing, in time order, how that documentation was developed and modified.	binding	Checked	DI-09, DI-21
n53	11.30	Anyone using an open system to create, modify, maintain or transmit electronic records must apply procedures and controls that protect the records' authenticity, integrity and, where appropriate, confidentiality from creation until receipt. These include the 11.10 controls as appropriate plus additional measures, such as document encryption and suitable digital signature standards, as needed in the circumstances.	binding	Checked	DI-08
n54	11.50(a)-(b)	Signed electronic records must show, with each signing, the signer's printed name, the date and time of signing, and what the signature means (e.g., review, approval, responsibility or authorship). These three items are controlled in the same way as the electronic records themselves and must appear in every human-readable form of the record, such as a screen display or printout.	binding	Checked	DI-36
n55	11.70	Electronic signatures, and handwritten signatures applied to electronic records, must be linked to their records so that nobody can, by ordinary means, remove, copy or transfer a signature to falsify an electronic record.	binding	Checked	DI-37
n56	11.100(a)-(c) (as amended 2 March 2023)	(a) Each electronic signature belongs to exactly one individual and is never reused by or reassigned to anyone else. (b) The organization verifies a person's identity before it establishes, assigns, certifies or otherwise sanctions that person's electronic signature or any element of it. (c) Before or when they first use them, persons using electronic signatures certify to FDA that the signatures in their system (used on or after 20 August 1997) are meant to be legally binding like traditional handwritten signatures; the certification carries a traditional handwritten signature and is sent electronically or on paper, with submission details on FDA's Letters of Non-Repudiation Agreement web page. On request, they give FDA additional certification or testimony that a particular e-signature legally binds the signer as a handwritten one would.	binding	Checked	DI-15
n57	11.200(a)(1)-(3), (b)	Electronic signatures not based on biometrics must use at least two distinct identification components (such as an ID code and a password). Within one continuous period of controlled system access, the first signing uses all components and later signings use at least one component that only the individual can execute and that is meant for that individual's use alone; signings made outside such a continuous period each use all components. Non-biometric signatures may be used only by their genuine owners and must be administered and executed so that any attempt by someone else to use them would need two or more people to collude. Biometric signatures must be designed so that nobody other than their genuine owner can use them.	binding	Checked	DI-38
n58	11.300(a)-(e)	Where e-signatures rely on ID codes combined with passwords, controls must protect their security and integrity, including: keeping every ID-code/password combination unique to one person; periodically checking, recalling or	binding	Checked	DI-19

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		revising issued codes and passwords (e.g., password aging); procedures for managing losses, so that tokens, cards or other devices that hold or generate codes/passwords and are lost, stolen, missing or possibly compromised are electronically deauthorized, with temporary or permanent replacements issued under rigorous controls; transaction safeguards that prevent unauthorized use of passwords or ID codes and that detect any attempted misuse and report it immediately and urgently to the system security unit and, where appropriate, to management; and initial and periodic testing of such tokens or cards to confirm they work and have not been altered without authorization.			
n59	Sections III.A-III.C (narrow interpretation of scope; enforcement discretion; predicate rules)	FDA interprets Part 11's scope narrowly. Part 11 applies to predicate-rule records kept electronically in place of paper; to records kept electronically in addition to paper and relied on for regulated activities (business practice decides); to records submitted electronically; and to e-signatures intended as handwritten equivalents. FDA exercises enforcement discretion on the validation, audit trail, record retention and record copying requirements, but predicate rules (e.g., 211.68, 211.180) still apply. Audit trails are particularly appropriate where users are expected to create, modify or delete regulated records during normal operation. FDA intends to enforce all other provisions, including access limits, operational, authority and device checks, training, accountability policies, documentation controls, open-system controls, and all e-signature requirements (11.50, 11.70, 11.100, 11.200, 11.300). Enforcement discretion also applies to legacy systems operational before August 20, 1997, under certain circumstances.	guidance	Search-based	DI-01
n614	11.1(c)-(d); 11.2(a); 11.3(b)(7)-(8)	Electronic signatures, with their records, that meet Part 11 are treated as equivalent to full handwritten signatures, initials and other signings required by FDA regulations, unless a regulation effective on or after 20 August 1997 specifically excludes them; compliant electronic records may be used instead of paper unless paper is specifically required. A handwritten signature applied to an electronic record (e.g., a name or mark written with a stylus on a device that captures it) is a handwritten signature, not an electronic signature, and must still satisfy 11.50 and 11.70.	binding	Checked	DI-35
n615	11.10 (introductory text)	Anyone using a closed system to create, modify, maintain or transmit electronic records must apply procedures and controls designed to protect the records' authenticity and integrity, their confidentiality where appropriate, and to ensure a signer cannot readily repudiate a signed record as not genuine; paragraphs (a) to (k) list the minimum controls.	binding	Checked	DI-08, DI-13

21 CFR Part 211

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n2	211.22(a), (c), (d)	A quality control unit (QCU) must exist with the responsibility and authority to approve or reject every component, drug product container, closure, in-process material, packaging material, labeling item and drug product, and to review production records to make sure no errors occurred or that any errors were fully investigated. The QCU also approves or rejects products that another company makes, processes, packs or holds for the firm under contract. The QCU approves or rejects every procedure or specification that affects identity, strength, quality or purity. QCU responsibilities and procedures must be written down and followed. (211.22(b), laboratory facilities available to the QCU, is not an eBMR function.)	binding	Checked	DI-17, DI-39
n4	211.28(a)-(d)	Staff wear clean clothing suited to their duties, plus protective apparel (head, face, hand and arm coverings) as needed to protect product from contamination, and practise good sanitation and health habits. Only people authorized by supervisors may enter areas designated as limited-access. Anyone who appears ill or has open lesions that could harm product safety or quality (whether a medical exam or a supervisor noticed it) is kept away from direct contact with components, containers, closures, in-process materials and product until the condition is corrected or competent medical staff clear them; all staff are told to report such health conditions to supervisors.	binding	Checked	PR-15
n5	211.42(c)(10)(i)-(vi)	Aseptic processing areas must include, as appropriate: smooth, hard, easily cleaned floors, walls and ceilings; temperature and humidity control; HEPA-filtered supply air at positive pressure, whether or not the airflow is laminar; a system to monitor environmental conditions; a system to clean and disinfect the room and equipment so aseptic conditions are achieved; and a system to maintain the equipment that controls aseptic conditions. These are facility requirements; the eBMR's role is to check and record their status for the batch.	binding · Injectables	Checked	ST-14
n6	211.63; 211.67(a)-(c)	Equipment must be suitably designed, large enough and placed so that it supports its intended operations and its cleaning and maintenance. Equipment and utensils are cleaned and maintained, and sanitized and/or sterilized where the drug requires it, at intervals that prevent malfunction or contamination that would push product quality outside established requirements. Written, followed procedures must cover: who is responsible; cleaning, maintenance and (where needed) sanitizing schedules; enough detail on methods, equipment and materials, including dismantling and reassembly; removing or obliterating the previous batch's identification; protecting clean equipment until use; and checking equipment is clean immediately before use. Cleaning, maintenance, sanitizing and inspection are recorded as 211.180 and 211.182 require.	binding	Checked	PR-23, PR-24
n7	211.68(a)	Automatic, mechanical or electronic equipment, including computers and related systems, may be used in manufacturing, processing, packing and holding if it performs its function satisfactorily. When it is used, it must be routinely calibrated, inspected or checked under a written program designed to assure proper performance, and written records of those calibration checks and inspections must be kept.	binding	Checked	DI-11, DI-20, DI-32
n8	211.68(b)	Computer or related systems must be controlled so that only authorized personnel can change master production and control records or other records. Formulas and other records or data entered into and produced by the system must be checked for accuracy, with the depth and frequency of checking based on the system's complexity and reliability. A backup file of entered data must be kept, except where computerization eliminates certain data (for example laboratory calculations), in which case a written record of the program is kept with validation data. Hard-copy or alternative backups must keep the backup data exact, complete and protected against alteration, accidental erasure or loss.	binding	Checked	DI-09, DI-31, DI-45
n9	211.68(c)	For the operations covered by 211.101(c) and (d) (dispensing checks and addition of components), 211.103 (yield calculation), 211.182 (equipment cleaning and maintenance log) and 211.188(b)(11) (people performing and checking significant steps), automated equipment can satisfy the requirement that one person performs and another checks, provided the equipment is used in conformity with 211.68 and one person checks that it performed the operation properly. (The statement that the 2008 rule codified a long-standing FDA view is preamble commentary and cannot be verified from the uploaded text; the eCFR source notes do confirm 211.68 was amended in September 2008.)	binding	Checked	PR-14
n10	211.100(a)-(b)	There must be written production and process control procedures designed to assure the product has the identity, strength, quality and purity it claims, covering every requirement of Subpart F. These procedures, and any changes, are written, reviewed and approved by the relevant organizational units and then reviewed and approved by the QCU. They must be followed when production and process control functions are carried out, and documented at the time each is performed; any departure from them must be recorded and justified.	binding	Checked	DI-27, DI-28
n11	211.101(a)	Production and control procedures must make sure each batch is formulated with the intent of delivering at least 100% of the labeled or established amount of active ingredient.	binding	Checked	PR-19
n12	211.101(b)	Procedures must require components to be weighed, measured or subdivided as appropriate. When a component is transferred out of its original container, the new container must be labeled with the component name or item code, the receiving or control number, the weight or measure it holds, and the batch it was dispensed for (product name, strength and lot number).	binding	Checked	PR-19
n13	211.101(c)	Weighing, measuring or subdividing of components must be adequately supervised. A second person must examine each container dispensed to manufacturing to confirm that the QCU released the component, that the weight or measure matches the batch production record, and that the containers are properly identified. If automated equipment under 211.68 does the weighing, measuring or subdividing, one person is enough to confirm these three points.	binding	Checked	PR-18, PR-20

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n14	211.101(d)	One person charges each component into the batch and a different person confirms the addition; if automated equipment under 211.68 performs the charge, a single person's verification is enough.	binding	Checked	PR-21
n15	211.103	At the close of every relevant stage of manufacturing, processing, packaging or holding, the actual yield and the percentage of theoretical yield must be worked out. One person does the calculation and a second person verifies it independently; if automated equipment under 211.68 calculates the yield, one person verifies it independently.	binding	Checked	PR-31
n16	211.105(a)-(b)	Every compounding or storage container, processing line and item of major equipment used for a batch must be properly marked at all times to show what it contains and, where needed, the processing phase. Major equipment carries a distinctive identification number or code, which is recorded in the batch production record so the exact equipment used for each batch is known; where a facility has only one item of a given type, its name can serve instead.	binding	Checked	PR-22
n17	211.110(a)-(d)	To assure batch uniformity and integrity, written procedures, established and followed, must describe the in-process controls, tests or examinations performed on suitable samples of in-process material from each batch. These controls monitor output and validate the performance of processes that could cause variability in the in-process material or the finished product. Where appropriate they include, among others: tablet or capsule weight variation, disintegration time, adequacy of mixing (uniformity and homogeneity), dissolution time and rate, clarity, completeness or pH of solutions, and bioburden testing. In-process specifications must be consistent with final product specifications and, where possible, derived from historical acceptable process averages and variability using suitable statistics; sample testing must assure conformance. The QCU approves or rejects in-process materials during production based on identity, strength, quality and purity testing as appropriate (for example when a significant phase starts or ends, or after long storage). Rejected in-process materials are identified and held under a quarantine system that prevents their use.	binding	Checked	DI-29, DI-32
n18	FDA draft guidance 'Considerations for Complying With 21 CFR 211.110' (January 2025) - not in the uploaded texts	FDA's nonbinding draft thinking: in-process sampling and testing may rely on in-line, at-line and on-line measurements and on process models, including for advanced manufacturing such as continuous manufacturing or 3D printing; because a process model cannot guarantee that its assumptions stay valid at all times (for example during unplanned disturbances), a control strategy that relies on a model alone is not considered enough to satisfy 211.110. Per the original search the draft was not yet final (status not re-checked here).	draft	Search-based	PR-29
n19	211.111	Where appropriate, time limits for completing each production phase must be set to assure product quality. Exceeding a limit can be acceptable if quality is not compromised, but the departure must be justified and documented.	binding	Checked	PR-30
n20	211.113(a)-(b)	(a) For drug products that need not be sterile, appropriate written procedures designed to prevent objectionable microorganisms must be established and followed. (b) For products that purport to be sterile, appropriate written procedures designed to prevent microbiological contamination must be established and followed, and they must include validation of every aseptic and sterilization process.	binding	Checked	PR-28
n21	211.115(a)-(b)	Written, followed procedures must set out a system for reprocessing batches that fail standards or specifications, including the steps that make sure the reprocessed batch will meet all established standards, specifications and characteristics. Reprocessing may take place only after QCU review and approval.	binding	Checked	PR-47
n22	211.122(a)-(h)	Written, followed procedures must describe in enough detail how labeling and packaging materials are received, identified, stored, handled, sampled, and examined and/or tested; these materials are sampled representatively and examined or tested on receipt and before use. Materials that meet their written specifications may be approved and released; those that do not are rejected so they cannot be used. For each shipment of each different labeling and packaging material, records show receipt, examination or testing, and acceptance or rejection. Labels and labeling for each different product, strength, dosage form or fill quantity are kept apart with suitable identification, in a store open only to authorized staff. Obsolete or outdated labels, labeling and other packaging materials are destroyed. Gang-printed labeling for different products, strengths or net contents is banned unless the sheets are clearly distinguished by size, shape or color. When cut labels are used on immediate containers, on individual unit cartons, or on multi-unit cartons whose immediate containers are not individually cartoned, one special control is required: (1) lines dedicated to each strength of each product; (2) electronic or electromechanical equipment that checks 100% of output for correct labeling during or after finishing; (3) for hand-applied labeling, a 100% visual check by one person, independently verified by a second; or (4) any automated method (label size or shape differentiation included) that physically stops wrong labeling from passing through the labeling and packaging equipment. Printing devices on or linked to the lines that imprint the unit label or case must be monitored so that every imprint matches what the BPR specifies.	binding	Checked	PR-32
n23	211.125(a)-(f)	Labeling issued for labeling operations is strictly controlled. Labeling issued to a batch is carefully checked for identity and for conformity with the labeling specified in the master or batch records. Procedures balance the labeling quantities issued, used and returned, and call for evaluation whenever the difference between finished product quantity and labeling issued falls outside tight preset limits derived from historical operating data; such discrepancies are investigated under 211.192. Reconciliation is waived for cut or roll labeling when a 100% electronic or electromechanical examination under 211.122(g)(2) is performed. All surplus labeling printed with lot or control numbers is destroyed. Returned labeling is kept and stored so that mixups are prevented and it stays properly identified. Written procedures describing labeling issuance controls in sufficient detail are established and followed. (Documenting the issuance examination in the BPR is required by 211.130(d), 211.184(d) and 211.188(b)(8).)	binding	Checked	PR-33, PR-39
n24	211.130(a)-(e)	Written, followed procedures must make sure the correct labels, labeling and packaging materials are used, and must include: physical or spatial separation from operations on other products to prevent mixups and cross-contamination; identification and handling of filled containers held unlabeled for later labeling so that containers, lots or parts of lots are not mislabeled (identification need not be on each container but must allow name, strength, quantity and lot/control number to be determined); a lot or control number on the product that allows the batch's manufacturing and control history to be traced; checking before packaging starts that packaging and labeling materials are suitable and correct, recorded in the BPR; and inspecting the packaging and labeling facilities right before use to confirm that product from earlier operations and unsuitable materials have been removed, with results recorded in the BPR.	binding	Checked	PR-33, PR-34, PR-35, PR-38
n25	211.132(a)-(c) (with (d)-(f) for exemptions, NDA notification and PPPA)	Manufacturers and packers of OTC human drug products for retail sale (except dermatological, dentifrice, insulin and lozenge products) must use a tamper-evident package if the public can reach the product while it is held for sale; otherwise the product is adulterated and/or misbranded. The package needs one or more indicators or barriers to entry that, if breached or missing, would give consumers visible evidence of tampering; it must be distinctive by design or use an identifying characteristic (e.g., a pattern, name, trademark, logo or picture), and the feature must stay intact under reasonable handling through manufacture, distribution and retail display. Two-piece hard gelatin capsules must also be sealed with an acceptable tamper-evident technology. Each retail package (except ammonia inhalant in crushable glass ampules and gas-propelled aerosols) must carry a prominently placed statement, unaffected if the feature is breached or missing, that names all tamper-evident features and any capsule-sealing technology and refers to any identifying characteristic used. Exemptions require a citizen petition; NDA holders report related packaging and labeling changes under 314.70, and special packaging under the Poison Prevention Packaging Act still applies.	binding · Tablets	Checked	PR-41
n26	211.134(a)-(c)	During finishing operations, packaged and labeled product is examined to give assurance that the containers and packages in the lot bear the correct label. When finishing is complete, a representative sample of units is collected and visually checked for correct labeling. The results of both examinations are recorded in the batch production or control records.	binding	Checked	PR-37
n27	211.137(a)-(c), with (d)-(h)	Each drug product must carry an expiration date, set by the stability testing of 211.166, so that it still meets its identity, strength, quality and purity standards at the time of use; the date must relate to any storage conditions stated on the labeling. Products reconstituted at dispensing need expiration information on the labeling for both the reconstituted and the unreconstituted product. Expiration dates appear on labeling as 21 CFR 201.17 requires (outside the uploaded	binding	Checked	PR-43

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		texts). Exempt: homeopathic products; allergenic extracts labeled 'No U.S. Standard of Potency'; investigational new drugs that meet standards shown by stability studies during clinical use (reconstituted investigational products still need reconstituted-product expiry information). FDA does not enforce the section, pending a proposed exemption, for human OTC products whose labeling has no dosage limitations and that are shown stable for at least 3 years.			
n28	211.84(a); 211.87; 211.89	Every lot of components, drug product containers and closures is kept from use until the QCU has released it after appropriate sampling, testing or examination. Approved materials are retested or re-examined for identity, strength, quality and purity as needed (for instance after long storage or exposure to air, heat or other harmful conditions) and approved or rejected by the QCU under 211.84. Rejected materials are identified and controlled in a quarantine system designed to stop their use in operations for which they are unsuitable.	binding	Checked	PR-18, PR-45
n29	211.94(c)-(d); 211.72	Drug product containers and closures must be clean and, where the drug's nature requires, sterilized and treated to remove pyrogenic properties so they are fit for use; depyrogenation processes must be validated. Written, followed standards or specifications and test methods, and where indicated cleaning, sterilizing and depyrogenation methods, are required for containers and closures. These container/closure provisions apply to all dosage forms (sterilization and depyrogenation only where the product requires them). For liquid filtration when making injectable drugs for human use, filters must not shed fibers; a fiber-releasing filter may be used only when the product cannot be made without it, and must then be followed by a non-fiber-releasing filter rated at most 0.2 micron nominal pore size (0.45 micron if manufacturing conditions require). Filters containing asbestos are prohibited.	binding · Injectables	Checked	ST-41, ST-50
n30	211.150(a)-(b); 211.196	Written, followed distribution procedures must provide for shipping the oldest approved stock of a drug product first (temporary, appropriate departures allowed) and a system that makes each lot's distribution readily determinable so it can be recalled if necessary. Distribution records must show the product name and strength, a description of the dosage form, the consignee's name and address, the date and quantity shipped, and the lot or control number. (211.196 as currently worded after the June 2024 amendment.)	binding	Checked	PR-57
n31	211.160(a); 211.165(a)-(f); 211.167(a)	Specifications, standards, sampling plans, test procedures and other laboratory control mechanisms, and changes to them, are drafted by the appropriate unit and reviewed and approved by the QCU; laboratory controls are followed and documented when performed, and departures are recorded and justified. Before release, each batch needs an appropriate laboratory determination that it meets final specifications, including identity and strength of each active ingredient (short-lived radiopharmaceuticals may be released before sterility/pyrogen tests finish, provided those tests are completed as soon as possible). Batches that must be free of objectionable microorganisms get appropriate microbial testing. Sampling and testing plans are written, state the sampling method and number of units tested per batch, and are followed. QCU acceptance criteria must be adequate to assure each batch meets every specification and appropriate statistical quality control criteria, including acceptance and/or rejection levels, as a condition of approval and release. Test methods have their accuracy, sensitivity, specificity and reproducibility established and documented. Products failing standards, specifications or other quality criteria are rejected; reprocessed material must meet appropriate standards before acceptance and use. Every batch purporting to be sterile and/or pyrogen-free is tested for those attributes under written, followed procedures.	binding	Checked	DI-39
n32	211.180(a)-(d)	(a) Production, control and distribution records required by Part 211 and specifically tied to a batch are kept for at least 1 year after that batch's expiration date or, for certain OTC products exempt from expiry dating under 211.137, 3 years after the batch is distributed. (b) Records for all components, containers, closures and labeling are kept at least 1 year after the expiration date (commonly read as the expiry of the last product lot that used them) or, for exempt OTC products, 3 years after distribution of the last lot that used them. (c) All Part 211 records, or copies, must be readily available for authorized inspection during retention at the establishment where the activities took place and must be reproducible (e.g., photocopied) during inspection; records that the site can pull up instantly from elsewhere by computer or other electronic means count as available. (d) Records may be kept as originals or as true copies (e.g., photocopies, microfilm, microfiche or other accurate reproductions); if reduction techniques are used, suitable reader and copying equipment must be on hand.	binding	Checked	DI-40, DI-47
n33	211.180(e)(1)-(2)	Written records must be kept so their data can support an evaluation, at least once a year, of the quality standards of each drug product, to decide whether its specifications or its manufacturing or control procedures need to change. Written, followed procedures for this review must cover a representative number of batches (approved or rejected) and, where applicable, their associated records, plus complaints, recalls, returned or salvaged products, and 211.192 investigations for each product.	binding	Checked	PR-58
n34	211.182	Cleaning, maintenance (other than routine lubrication and adjustments) and use of major equipment are recorded in individual equipment logs that record, for every batch processed, its lot number and product plus the date and time. Individual logs are not needed for equipment dedicated to one product if its lots follow, and are made in, numerical sequence, but then the cleaning, maintenance and use records belong in the batch record. The person doing the cleaning or maintenance and the person double-checking it date and sign or initial the log (for cleaning or maintenance done by automated equipment under 211.68, only the verifying person). Log entries are kept in chronological order.	binding	Checked	PR-24
n35	211.184(a)-(e)	Component, container, closure and labeling records must include: (a) for each shipment of each lot, its identity and quantity, the supplier name, the supplier lot number(s) if known, the 211.80 receiving code, the receipt date and, if known and different from the supplier, the name and location of the prime manufacturer; (b) the results of every test or examination (including those under 211.82(a), 211.84(d) and 211.122(a)) and the conclusions drawn; (c) an individual inventory record for each component, container and closure and, for components, a reconciliation of the use of each lot, with enough information to identify every drug product batch or lot that used each component, container or closure; (d) documentation that labels and labeling were examined and reviewed for conformity with specifications (see 211.122(c) and 211.130(c)); (e) a record of what was done with rejected components, containers, closures and labeling.	binding	Checked	PR-17
n37	211.186(b)(1)-(9)	The master record must contain: the product name, strength and dosage-form description; each active ingredient's name and quantity, expressed per dosage unit or per weight/volume unit of product, and the total weight or measure of one dosage unit; a complete component list using names or codes specific enough to show any special quality characteristic; an accurate amount for each component, all expressed in one weight system (metric, avoirdupois or apothecary), with reasonable variations allowed only if justified in the record; a statement of any calculated excess of a component; theoretical weight or measure at appropriate processing phases; theoretical yield together with the upper and lower percentage limits that trigger a 211.192 investigation; a description of containers, closures and packaging materials, plus a sample or copy of every label and other labeling piece bearing the approver's signature and date; and full manufacturing and control instructions together with the sampling and test procedures, specifications, and any special notes and precautions.	binding	Checked	PR-03
n39	211.188(b)(1)-(13)	The batch record must document that each significant step of manufacture, processing, packing or holding was accomplished, covering: the dates; which individual major equipment and lines were used; specific identification of each component or in-process material batch used; component weights and measures used during processing; in-process and laboratory control results; checks of the packaging/labeling area both before and after it is used; actual yield and percentage of theoretical yield at appropriate phases; complete labeling control records with specimens or copies of all labeling used; a description of containers and closures; any sampling performed; who performed each significant step and who directly supervised or checked it (or, for a step done by automated equipment under 211.68, who checked the equipment's work); any 211.192 investigation; and the results of the 211.134 label examinations.	binding	Checked	PR-14, PR-16, PR-34
n40	211.192	Before a batch is released or distributed, the QCU must review and approve all its production and control records, packaging and labeling records included, to establish compliance with all established, approved written procedures. Any unexplained discrepancy (including a percentage of theoretical yield outside the master-record limits), or any failure of a batch or its components to meet specifications, requires a thorough investigation, even if the batch has already shipped, and that investigation must extend to other batches of the same product and to other products that may be associated with the failure or discrepancy. A written investigation record with conclusions and follow-up is required.	binding	Checked	PR-50, PR-52

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n41	211.194(a)(1)-(8)	Laboratory records must contain complete data from every test needed to assure compliance with specifications and standards, including: a description of each sample with its source (where it was taken), quantity, lot number or other distinctive code, date sampled and date received for testing; a statement of each test method used, with the location of data showing the method is accurate and reliable for the product (a citation suffices for unmodified compendial, AOAC, other recognized or approved-application methods), plus verification that every method is suitable under actual conditions of use; the sample weight or measure used for each test where appropriate; a complete record of all raw data, including instrument graphs, charts and spectra, each identified to the specific material and lot tested; all calculations with units, conversion and equivalency factors; the results and how they compare with the standards; the tester's initials or signature and the test dates; and a second person's initials or signature confirming the original records were checked for accuracy, completeness and compliance.	binding	Checked	DI-29, DI-32
n42	211.198(a)-(b)	Written, followed procedures must govern handling of all written and oral complaints about a drug product, including QCU review of any complaint about a possible failure to meet specifications, a decision on whether a 211.192 investigation is needed, and a review of whether the complaint is a serious and unexpected adverse drug experience that must be reported under 21 CFR 310.305 or 514.80. Each complaint is recorded in a designated complaint file kept at the establishment that made, processed or packed the product, or elsewhere if the records are readily available for inspection there. Where known, the record gives product name and strength, lot number, complainant name, nature of the complaint and the reply. If investigated, the record includes findings and follow-up (the investigation record itself stays where the investigation was done, per 211.180(c)); if not, it gives the reason and names the person who decided no investigation was needed. Complaint records are kept for the later of 1 year after the product's expiry or 1 year after receipt of the complaint; for OTC products exempt from expiry dating, 3 years after distribution.	binding	Checked	PR-57, PR-59
n78	FDA Guide to Inspections of Lyophilization of Parenterals (1993) - not in the uploaded texts; the 211.137(c) citation was verified	FDA's 1993 investigator guide on lyophilization of parenterals covers the freeze-drying cycle and its controls, cycle validation, and lyophilizer sterilization and design; cycle records are reviewed against the validated/filed cycle (details from recollection, not verified here). The one Part 211 citation, 211.137(c), is confirmed: products reconstituted at dispensing need labeled expiry information for both the reconstituted and the unreconstituted forms.	guidance · Injectables	Unverified	ST-59
n592	211.42(b), (c)(1)-(9)	Operations must take place in specifically defined areas of adequate size, using separate or defined areas, or other control systems, that prevent contamination and mixups during: receipt, identification, storage and withholding of components, containers, closures and labeling until the QCU has sampled, tested or examined and released them; holding of rejected materials pending disposition; storage of released materials; storage of in-process materials; manufacturing and processing; packaging and labeling; quarantine of drug products before release; storage of released drug products; and control and laboratory work. Buildings need space for orderly placement of equipment and materials, and material and product flows must be designed to prevent contamination and mixups.	binding	Checked	PR-25, PR-45
n593	211.46(a)-(c)	Ventilation must be adequate, and where appropriate for the operation there must be equipment giving adequate control of air pressure, microorganisms, dust, humidity and temperature. Air filtration (prefilters and particulate filters) is used where appropriate on air supplied to production areas; if air is recirculated, measures must control recirculation of production dust; and areas where air is contaminated during production need adequate exhaust or equivalent control systems.	binding	Checked	PR-25
n594	211.42(d); 211.46(d); 211.176	Penicillin must be manufactured, processed and packed in facilities, and with air-handling systems, completely separate from those used for other human drug products. If there is a reasonable possibility that a non-penicillin product was exposed to penicillin cross-contamination, that product must be tested for penicillin by the method incorporated by reference, and it may not be marketed if detectable levels are found.	binding	Checked	PR-27
n595	211.56(a)-(d); 211.58	Buildings used to make, process, pack or hold drug products must be clean, sanitary, free of vermin (laboratory animals excepted) and in good repair, with trash and organic waste removed promptly and hygienically. Written, followed procedures must assign responsibility for sanitation and describe cleaning schedules, methods, equipment and materials in enough detail. Further written, followed procedures govern the use of suitable rodenticides, insecticides, fungicides, fumigants and cleaning or sanitizing agents so that equipment, materials and product are not contaminated; pesticides may be used only if registered and used under FIFRA. These procedures also apply to contractors and temporary staff.	binding	Checked	PR-25
n596	211.65(a)-(b)	Equipment surfaces that touch components, in-process materials or product must not react with, add to or absorb from them to a degree that moves the product's safety, identity, strength, quality or purity outside official or established requirements. Substances needed to run equipment, such as lubricants or coolants, must not contact components, containers, closures, in-process materials or product in a way that does so.	binding	Checked	discussed in notes of Part A
n597	211.80(a), (d)	Detailed written, followed procedures must cover how components, containers and closures are received, identified, stored, handled, sampled, tested, and approved or rejected. Each container or group of containers of these materials must carry a distinctive code for every lot of every shipment received; the code is then used when recording the lot's disposition, and each lot's status (quarantined, approved or rejected) must be identified.	binding	Checked	PR-17
n598	211.82(a)-(b)	On arrival and before acceptance, each container or group of containers of components, containers and closures is visually examined for correct content labeling, container damage or broken seals, and contamination. The materials then stay in quarantine until tested or examined, as appropriate, and released, with storage meeting 211.80.	binding	Checked	PR-17
n599	211.84(b)-(c)	Representative samples must be taken from each shipment of each lot of components, containers and closures; the number of containers sampled and the amount taken per container are based on sound criteria such as statistics on variability, confidence and precision, the supplier's quality history, and the quantity needed for analysis and for 211.170 reserve samples. Sampling must: clean container exteriors where needed so the material is not contaminated; open, sample and reseal containers without contaminating their contents or other materials; use sterile equipment and aseptic technique when needed; not composite top, middle and bottom sub-samples when such stratified sampling is required; label sample containers so the material, lot, source container, sampling date and sampler can be determined; and mark sampled containers to show samples were removed.	binding	Checked	PR-17
n600	211.84(d)-(e)	Each component must get at least one identity test (a specific one where it exists) and be tested for conformity with its written purity, strength and quality specifications; instead of full in-house testing, a supplier's report of analysis may be accepted if the manufacturer runs at least one specific identity test and establishes, and re-checks at appropriate intervals, the reliability of the supplier's results. Containers and closures are tested against their written specifications, or a supplier's certificate of testing is accepted provided the manufacturer does at least a visual identification and periodically validates the supplier's results. Components are examined microscopically when appropriate; lots liable to filth, insect infestation or other extraneous adulterants are examined against specifications for them; and lots with potential for objectionable microbial contamination, given their use, are micro-tested before use. Lots meeting the specifications may be approved and released; others are rejected.	binding	Checked	discussed in notes of Part A
n601	211.86	Approved components, drug product containers and closures must be used in rotation so that the oldest approved stock goes first; temporary and appropriate departures from this are allowed.	binding	Checked	PR-18
n602	211.94(a)-(b)	Drug product containers and closures must not react with, add to or absorb from the drug to a degree that changes its safety, identity, strength, quality or purity beyond official or established requirements, and the container-closure system must adequately protect the product against foreseeable external factors during storage and use that could cause deterioration or contamination.	binding	Checked	discussed in notes of Part A
n603	211.142(a)-(b)	Written, followed warehousing procedures must provide for holding drug products in quarantine until the QCU releases them, and for storing drug products under temperature, humidity and light conditions that keep their identity, strength, quality and purity unaffected.	binding	Checked	discussed in notes of Part A

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n604	211.160(b)(1)-(3); 211.165(c)	Laboratory controls must include scientifically sound specifications, standards, sampling plans and test procedures, and must determine conformance: for each lot within each shipment of components, containers, closures and labeling (with written descriptions of sampling and testing, and retesting of materials that can deteriorate); for in-process materials; and for drug products. In every case the samples must be representative and properly identified. Sampling and testing plans for drug products must state the sampling method and how many units per batch are tested, and must be followed.	binding	Checked	PR-29
n605	211.160(b)(4); 211.194(d)	Instruments, apparatus, gauges and recording devices must be calibrated at suitable intervals under an established written program that gives specific directions, schedules, limits for accuracy and precision, and remedial action when those limits are not met; devices that do not meet their specifications must not be used. Complete records of this periodic calibration must be kept.	binding	Checked	PR-23
n606	211.194(b)-(c)	Complete records must be kept of any modification of an established test method, giving the reason and data showing that the modified method is at least as accurate and reliable for the material tested; and complete records must show how laboratory reference standards, reagents and standard solutions were tested and standardized.	binding	Checked	PR-06, PR-29
n607	211.166(a)-(b); 211.194(e)	A written, followed stability program must assess product stability, and its results are used to set storage conditions and expiration dates. It includes statistically based sample sizes and test intervals, storage conditions for retained samples, reliable and specific methods, testing in the marketed container-closure system, and, for products reconstituted at dispensing, testing both as dispensed and after reconstitution. Enough batches must be tested to set an appropriate expiration date, with the data kept; accelerated data may support a tentative date only while full shelf-life studies continue until that date is confirmed or corrected. Complete records of all stability testing are kept.	binding	Checked	PR-43
n608	211.167(c) (211.167(b), ophthalmic ointments, is outside the tablets/injectables scope)	Each batch of a controlled-release dosage form must undergo appropriate laboratory testing showing it meets the specification for the release rate of each active ingredient, using written test procedures that are followed.	binding · Tablets	Checked	PR-54
n609	211.170(a)	For each lot in each shipment of each active ingredient, an appropriately identified, representative reserve sample must be kept, of at least twice the amount needed for all tests that determine whether it meets its specifications, excluding sterility and pyrogen tests. It is kept until 1 year past the expiry of the last drug product lot that contains that active ingredient; for radioactive products (other than nonradioactive reagent kits), 3 or 6 months after that expiry depending on whether the dating period is 30 days or less or longer; and for OTC products exempt from expiry dating, 3 years after distribution of the last lot containing it.	binding	Checked	PR-44
n610	211.170(b)	For each lot or batch of drug product, an appropriately identified, representative reserve sample must be kept under conditions consistent with the labeling, in the same immediate container-closure system as the marketed product (or one with essentially the same characteristics), of at least twice the amount needed for all required tests except sterility and pyrogen tests. Except for radioactive products, reserve samples from representative lots chosen by acceptable statistical procedures are visually examined at least once a year for signs of deterioration, unless examination would compromise the sample; any deterioration is investigated under 211.192, and results are recorded with the product's other stability data. Retention is 1 year after the product's expiration date, with shorter periods for radioactive products and 3 years after distribution for OTC products exempt from expiry dating.	binding	Checked	PR-44
n611	211.180(f)	Procedures must make sure the firm's responsible officials, if they are not personally involved or already aware, are told in writing about investigations under 211.198 (complaints), 211.204 (returns) or 211.208 (salvage), about recalls, about FDA inspection-observation reports, and about any FDA regulatory action relating to good manufacturing practice.	binding	Checked	discussed in notes of Part B
n612	211.204	Returned drug products must be identified as returns and held. If the conditions under which they were held, stored or shipped before or during return, or the state of the product, container, carton or labeling, cast doubt on their safety, identity, strength, quality or purity, they must be destroyed unless examination, testing or other investigation shows they meet appropriate standards; reprocessing is allowed if the result meets appropriate standards, specifications and characteristics. Return records must show the product name and label potency of the dosage form, the lot (or control or batch) number, the reason for return, the quantity returned, the date of disposition and the final disposition. If the reason for return implicates associated batches, a 211.192 investigation is required. Holding, testing and reprocessing of returns follow written procedures.	binding	Checked	discussed in notes of Part A
n613	211.208	Drug products exposed to improper storage conditions (extreme temperature, humidity, smoke, fumes, pressure, age or radiation) because of natural disasters, fires, accidents or equipment failures must not be salvaged and put back on the market. Where it is unclear whether products were so exposed, salvaging is allowed only with laboratory test and assay evidence (including animal feeding studies where applicable) that the products meet all applicable identity, strength, quality and purity standards, and with evidence from inspection of the premises that the products and their packaging were not exposed to improper conditions; organoleptic examination counts only as supplementary evidence. Records including name, lot number and disposition are kept.	binding	Checked	discussed in notes of Part A

FDA Aseptic Processing Guidance

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n72	Time limits for aseptic processing phases (Section on time limitations/validation)	Time limits should be established for each phase of aseptic processing. These include the period between the start of bulk product compounding and its sterilization, filtration processes, product exposure on the processing line, and storage of sterilized equipment, containers and closures. Total filtration time should be limited to an established maximum to prevent microorganisms from penetrating the filter. Time limits should be supported by data, and bioburden and endotoxin load should be assessed when setting them (e.g., for the formulation stage).	guidance · Injectables	Search-based	ST-37
n73	Section IX.A - Process simulations (media fills): frequency, interventions, acceptance criteria	Aseptic processes are validated by process simulations. The guidance sets initial qualification with multiple (three) successful runs and semi-annual requalification per line and shift. Simulations should include activities and interventions representative of each shift and shift changeover. Acceptance: when filling fewer than 5,000 units, no contaminated units should be detected, and one contaminated unit is cause for revalidation after investigation. For 5,000-10,000 units, one contaminated unit should result in an investigation including consideration of a repeat media fill, and two are cause for revalidation after investigation. For more than 10,000 units, one contaminated unit should result in an investigation, and two are cause for revalidation after investigation.	guidance · Injectables	Search-based	ST-29
n74	Section IV / Table 1 (air classifications) and Section X.A (environmental monitoring program: alert/action levels, trending, investigations)	The critical zone should meet ISO 5 (3,520 particles ≥ 0.5 micron per cubic meter, in operation). The area adjacent to the aseptic line should be at least ISO 7, with ISO 8 for less critical activities. Table 1 gives microbiological action levels. From recollection of the guidance, not re-verified: ISO 5 active air 1 cfu/m3 and settling plates (90 mm, 4 h) 1 cfu; ISO 6 7 and 3; ISO 7 10 and 5; ISO 8 100 and 50. The EM program should include alert/action levels, trending, investigation of excursions, and evaluation of their impact on batches.	guidance · Injectables	Unverified	ST-20
n75	Section XII - Batch Record Review: Process Control Documentation	The guidance has a dedicated section on batch record review for aseptic processing (confirmed section title). From recollection of the text, not re-verified because the document could not be fetched: before batch release, the QCU review should cover all in-process and laboratory control results together with process-control documentation for the aseptic operation. That documentation includes environmental and personnel monitoring data, interventions and deviations during filling, sterilization and depyrogenation records for equipment, containers and closures, filter integrity results, and adherence to time limits. Discrepancies and excursions should be investigated and their impact on the batch assessed.	guidance · Injectables	Unverified	ST-70

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n76	Sections on filtration efficacy and sterilization of equipment, containers and closures (integrity testing; qualification/validation of cycles)	From recollection of the guidance, not re-verified: sterilizing-grade filter integrity testing can be done before processing and should be done routinely after use to detect filter leaks or perforations during processing. Sterilization and depyrogenation cycles for equipment, containers and closures should be validated and each run's physical parameters recorded and reviewed.	guidance · Injectables	Unverified	ST-50
n274	IV (intro) and Table 1	Classified areas are to be qualified and routinely monitored with most weight on in-operation (dynamic) data taken near exposed materials. Recommended limits per class: ISO 5 / Class 100 = 3,520 particles ≥0.5 um per m3, active air action 1 cfu/m3, settle plate action 1 cfu per 4 h; ISO 6 = 35,200, 7, 3; ISO 7 = 352,000, 10, 5; ISO 8 = 3,520,000, 100, 50. Firms may set alternative microbial action levels with justification; settle plates are optional; ISO 5 is roughly EU Grade A.	guidance · Injectables	Checked	ST-20
n275	IV (intro)	Clean-area control parameters rest on qualification data, and the routine monitoring program should confirm on an ongoing basis that each area still meets its class while operations are running.	guidance · Injectables	Checked	ST-14
n276	IV.A	In the ISO 5 critical zone, non-viable particles (≥0.5 um) should stay at or below 3,520 per m3. Counts are taken during filling/closing at representative high-risk points, generally within about one foot of the work site and inside the airflow, with a justified probe orientation. Monitoring should happen in every production shift, preferably with a remote (continuous) counting system. For powder operations, sampling may be adapted to separate product particles from extrinsic contamination.	guidance · Injectables	Checked	ST-19
n277	IV.A; IV.D.2	HEPA air in the critical zone should flow fast enough to sweep particles away and keep airflow unidirectional. Each line needs a justified velocity range (commonly 0.45 m/s +/-20%). In-situ smoke/airflow studies under dynamic conditions, including the effect of interventions, should be documented with conclusions. Periodic velocity readings should be taken close to the filter (roughly 15 cm / 6 in below its face) and at a fixed height over the work surface, then compared with the range established in the airflow studies. Filters showing non-uniform velocity should be replaced.	guidance · Injectables	Checked	ST-17
n278	IV.A; Table 1 note e	Viable air samples from the critical ISO 5 zone should normally show no growth. Any contamination found there should get proper investigative attention.	guidance · Injectables	Checked	ST-21
n279	IV.B	Support areas are classed by what happens in them. The room right next to the aseptic line should be at least ISO 7 when in operation (ISO 6, or the whole room at ISO 5, is also acceptable). ISO 8 suits less critical work such as washing equipment. Support areas should keep bioburden low on items that will be sterilised later.	guidance · Injectables	Checked	ST-14
n280	IV.C	Cleaner rooms should hold a clear positive pressure over adjacent less clean rooms: about 10-15 Pa between different classes with doors closed, and at least about 12.5 Pa at all times between the aseptic room and any unclassified neighbour. Door-open time should be tightly controlled. If the differential falls below its minimum, the aseptic room's environmental quality should be restored and confirmed before work continues.	guidance · Injectables	Checked	ST-17
n281	IV.C	Differential pressures between cleanrooms should be monitored continuously through each shift and recorded often. Every alarm should be documented and every out-of-limit condition investigated. The monitoring system should use alarms to catch a developing low-pressure trend early, so conditions can be restored before action levels are reached.	guidance · Injectables	Checked	ST-06
n282	IV.D.1	Compressed gases used in cleanrooms (air, nitrogen, CO2 for purging or overlay) should be suitably pure, for example oil-free. After filtration they should be at least as clean, microbiologically and in particles, as the room air they enter. Sterile membrane filters are advised on autoclave air lines, lyophiliser vacuum breaks, and tanks holding sterilised material. Sterilised holding tanks and their contents should be kept under positive pressure or sealed, with safeguards against back-flow of non-sterile air or liquid.	guidance · Injectables	Checked	ST-28
n283	IV.D.1	Gas and vent filters should stay dry (hydrophobic media, heated where appropriate). Filters that form a sterile boundary or supply sterile gas that can affect product should be integrity tested at installation, periodically afterwards (for example at end of use), and after any event that could damage them. Integrity-test failures should be investigated, and filters replaced at defined intervals.	guidance · Injectables	Checked	ST-28
n284	IV.D.2	HEPA filters should be leak tested in place at installation and then at set intervals (about twice a year for the aseptic processing room). This includes filters in depyrogenation tunnels and ovens. Extra tests are appropriate after poor air results or renovation, or during a media fill or sterility failure investigation. A single probe reading of about 0.01% of the upstream challenge means a significant leak: repair or replace, then retest. Scans should be fully documented, and the manufacturer stays responsible for the specifications and acceptance criteria used by contractors.	guidance · Injectables	Checked	ST-17
n285	IV.E	Personnel and material flows should avoid unnecessary activity. The number of people in the aseptic room should be kept to a minimum. Entries to and exits from the room, and especially the critical zone, should be limited, as should transfers into the critical zone of a cleanroom or isolator. Movement next to the critical zone should be restricted.	guidance · Injectables	Checked	ST-12
n286	IV.E	Every aseptic intervention or line stoppage adds contamination risk, so equipment and processes should keep interventions few and simple. Examples: in-line weight checks instead of manual ones, sterilise-in-place of pre-assembled connections instead of aseptic connections, automation or robotics.	guidance · Injectables	Checked	ST-34
n287	IV.E	In lyophilisation, partially stoppered sterile product should only move through critical (ISO 5) conditions. The path from filler to lyophiliser, and the transport and loading steps, should give ISO 5 protection, for example with qualified transfer equipment.	guidance · Injectables	Checked	ST-59
n288	IV.E	Materials, supplies, equipment and utensils moving from lower to higher classified areas should follow written procedures that protect room conditions. Items are disinfected by a defined method or, if used in the critical zone, sterilised by a suitable method. Double-door or integrated sterilisers, airlocks and interlocked doors support controlled transfer.	guidance · Injectables	Checked	ST-46
n289	IV.E	If stoppered but uncapped vials leave the aseptic zone or room, the product should be protected (for example with local air protection) until crimping is done. On-line detection of stoppers that are not properly seated adds assurance.	guidance · Injectables	Checked	ST-57
n290	IV.E	Deviation or change control should cover abnormal states caused by shutdowns of air handling or other utilities, and the effect of construction work on facility control. Written procedures should say how a facility is brought back to operating conditions after a shutdown.	guidance · Injectables	Checked	ST-16
n291	V.A	Before anyone may enter the aseptic manufacturing area they should be trained in aseptic technique, cleanroom behaviour, microbiology, hygiene, gowning, the patient-safety risk of non-sterile product, and the area's specific SOPs. They should then take part in ongoing periodic training.	guidance · Injectables	Checked	ST-07
n292	V.A	Only people who are qualified and correctly gowned should be allowed into the aseptic area. Gowns should be sterile and non-shedding and cover skin and hair. Gowning qualification should include microbial surface sampling at several justified gown sites (for example glove fingers, mask, forearm, chest). Requalification should be periodic: yearly is usually enough for automated operations with little personnel involvement and good environmental data, and more often after adverse findings.	guidance · Injectables	Checked	ST-08
n293	V.A	Supervisors should routinely watch each operator during real operations to check they follow procedures. The quality unit should regularly oversee adherence to SOPs and aseptic technique on the floor.	guidance · Injectables	Checked	ST-13
n294	V.A	Sterilised materials should only be handled with sterile instruments. Between uses, instruments are kept under ISO 5 in sterilised containers and replaced as needed. Gloves should be sanitised or changed often. Any torn or defective	guidance · Injectables	Checked	ST-11

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		gown part should be changed at once. Operators should never touch sterile product, containers, closures or critical surfaces with their gown or gloves.			
n295	V.A	Written procedures should define when personnel must be retrained, requalified, or moved to other areas.	guidance · Injectables	Checked	ST-09
n296	V.B	People who take aseptic samples or run microbiology tests should meet the same training, aseptic-technique and qualification principles as manufacturing staff. Otherwise the laboratory data, and so the state of control, cannot be trusted.	guidance · Injectables	Checked	ST-07
n297	V.C	There should be an active personnel monitoring program. Each operator's gloves are surface-sampled every day or with each lot, plus other chosen gown sites at a set frequency. The quality unit should apply more thorough monitoring to people doing labour-intensive work that involves repeated or complex aseptic manipulations.	guidance · Injectables	Checked	ST-22
n298	V.C	Gloves should not be sanitised just before sampling, because that can hide organisms that were present during the aseptic manipulation.	guidance · Injectables	Checked	ST-22
n299	V.C; X.A	If an operator exceeds set levels or shows an adverse trend, an investigation should start promptly. Follow-up can include more sampling, closer observation, retraining, gowning requalification, or reassignment outside the aseptic area.	guidance · Injectables	Checked	ST-22
n300	VI.A	For every component that could carry contamination (API, WFI, excipients), the microbial content (bioburden) and endotoxin level should be characterised and acceptance limits set. Written procedures and specifications should govern accepting or rejecting each lot of any component that may carry endotoxin, and lots that fail endotoxin limits should be rejected.	guidance · Injectables	Checked	ST-39
n301	VI (text box)	Any lot of a component, container or closure that could carry microbial contamination objectionable for its use must pass microbiological testing before use. Components, containers and closures must always be handled and stored so as to prevent contamination.	guidance · Injectables	Checked	ST-39
n302	VI.A	Each component is sterilised on its own, or components are combined and the mixture sterilised, and knowing the bioburden helps judge whether the sterilisation is adequate. Options include filtering a solution (for heat-labile soluble materials), sterile crystallisation or precipitation or freeze-drying after filtration (more handling, so higher risk), dry heat for heat-stable insoluble powders (with heat distribution and penetration studies), and irradiation (with supporting studies).	guidance · Injectables	Checked	ST-40
n303	VI (text box); VI.B.1	Containers and closures must be clean and, where the drug requires it, sterilised and processed to remove pyrogens. There must be written and followed standards, test methods and, where indicated, cleaning, sterilising and depyrogenation methods for them.	guidance · Injectables	Checked	ST-41
n304	VI.B.1	The container and closure preparation process should be validated to show it makes them sterile and, for parenterals, non-pyrogenic. Written procedures should state how often the process is revalidated and how long sterile, depyrogenated containers and closures may be held before use.	guidance · Injectables	Checked	ST-41
n305	VI.B.1	Glass containers are washed and rinsed before sterilisation using high-purity water, with the final rinse at WFI quality for parenterals. Rubber closures are washed with at least Purified Water of low endotoxin at first, then WFI for the final rinse, and hot WFI rinses usually depyrogenate them. The time between washing, drying and sterilising stoppers should be kept short, because residual moisture supports microbial growth and endotoxin formation. Validation should show stopper washing removes endotoxin.	guidance · Injectables	Checked	ST-41
n306	VI.B.1; IX.C.2	Dry heat on glass usually both sterilises and depyrogenates. It should be validated with heat distribution and penetration studies, worst-case cycles, container characteristics such as mass, and defined loading patterns. The endotoxin challenge should show at least a 3-log (99.9%) reduction; once that is shown, a separate biological indicator study is not needed. In depyrogenation tunnels, belt-speed sensors and transmitters should be calibrated routinely.	guidance · Injectables	Checked	ST-47
n307	VI.B.1	Plastic containers may be sterilised by gas (for example ethylene oxide), irradiation or other suitable methods. For EtO, the cycle's parameters and limits should be specified and closely monitored: temperature, pressure, humidity, gas concentration, exposure time, degassing and aeration, and residual testing. Biological indicators are especially important to show the cycle works, and EtO and its by-products should be held within specified residual limits.	guidance · Injectables	Checked	ST-49
n308	VI.B.1	Silicone used to treat rubber stoppers is a possible contamination source. It should meet defined quality criteria and should not harm the drug's safety, quality or purity. Because rubber conducts heat poorly, heat-based stopper cycles need extra attention to penetration into the stopper load.	guidance · Injectables	Checked	ST-41
n309	VI.B.1	Contractors who sterilise or depyrogenate containers and closures are held to the same CGMP expectations as in-house processing. The finished-product maker should itself evaluate both the protocol and the final report of the contractor's validation. Under 211.84(d)(3), once the supplier's test results have been shown reliable at appropriate intervals, containers or closures may be accepted on visual identification plus review of the certificate of analysis.	guidance · Injectables	Checked	ST-40
n310	VI.B.2	Any container-closure system that lets microorganisms in is unfit for sterile product. Damaged or defective units should be found and removed when the final sealed product is inspected, and safeguards should rule out shipping units that may have lost integrity. If hard-to-detect damage causes integrity loss, improved procedures to prevent and detect it should be put in place quickly.	guidance · Injectables	Checked	ST-62
n311	VI.B.2	Functional defects in delivery devices, for example syringe defects or incorrect delivery volume, can cause quality problems and should be watched through suitable in-process testing.	guidance · Injectables	Checked	ST-56
n312	VI.B.2	Defects or results outside the specifications for in-process and final inspection have to be investigated under the 211.192 requirements.	guidance · Injectables	Checked	ST-63
n313	VII (text box)	Every batch that claims to be sterile and/or pyrogen-free must get suitable laboratory testing showing it meets those claims, using written test procedures that are followed.	guidance · Injectables	Checked	ST-65
n314	VII	Endotoxin control should cover components, containers, closures, storage time limits and equipment. Bioburden and endotoxin build-up are controlled by cleaning, drying and storing equipment properly. Sterilising filters and moist heat have not been shown to remove endotoxin, so endotoxin on equipment has to be removed by cleaning (for example CIP ending in hot WFI rinses) or destroyed by high-temperature dry heat. Equipment should be dried after cleaning unless it goes straight to sterilisation.	guidance · Injectables	Checked	ST-38
n315	VIII (text box and first sentence)	Where appropriate, time limits must be set for each phase of aseptic processing. Going past a limit can be accepted only if product quality is not compromised, and any such deviation must be justified and documented.	guidance · Injectables	Checked	ST-37
n316	VIII	Time limits should cover, for example: from start of bulk compounding to its sterilisation; the filtration process; how long product is exposed on the processing line; and storage of sterilised equipment, containers and closures. Each limit should be backed by data, and bioburden and endotoxin load should be considered when setting limits for stages such as formulation.	guidance · Injectables	Checked	ST-37
n317	VIII	Total product filtration time should have a set maximum, both to stop organisms growing through the filter and to prevent a significant rise in upstream bioburden and endotoxin. Maximum use times should also be set and justified for upstream clarification or particle-removal filters, because they can harbour attached microbes.	guidance · Injectables	Checked	ST-51
n318	IX (intro)	Any change to facility, equipment, process or test method should go through the written change-control program, and that review should decide whether revalidation or requalification is needed.	guidance · Injectables	Checked	ST-03

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n319	IX.A	Sterilisation, aseptic filling and closing must be adequately validated so that products claiming to be sterile are sterile. Aseptic operations are validated by process simulation (media fill): growth medium replaces product and is exposed to the same product-contact surfaces, container-closures, environments and manipulations; the filled units are then incubated. EM data gathered during the simulation also helps evaluate the line.	guidance · Injectables	Checked	ST-29
n320	IX.A.2	Initial line qualification should include at least three consecutive, separate, successful media fills. After that, each line is requalified about every six months. The semi-annual program should include activities and interventions typical of each shift and of shift changeover.	guidance · Injectables	Checked	ST-29
n321	IX.A.2	Every person permitted into the aseptic room while production is running, maintenance staff and technicians included, should join a media fill no less often than annually, in a role that matches the work they normally do in production.	guidance · Injectables	Checked	ST-08
n322	IX.A.1, IX.A.3-6	Media fills should reproduce the line's real contamination risks and worst-case conditions. These include the longest permitted run and its time effects, the normal and non-routine interventions (with their types and frequency), aseptic set-up and connections, additions and sampling, how many people are present, shift, break and gown changes, weight checks, line speed and configuration, and container-closure formats. Lyophilisation should be simulated (partial vacuum, no freezing, medium kept aerobic), and manual or intervention-heavy processes should be simulated for at least as long as the real process. Run size and speed should be justified, usually 5,000-10,000 units, and at least the maximum batch size when batches are under 5,000. Media fills must not be run under unusually clean conditions or used to justify risky practices.	guidance · Injectables	Checked	ST-31
n323	IX.A.1	Every media-fill run should have its own written batch record showing the production conditions and the activities simulated. It should be run with the same vigilance as routine production. The firm's reasons for what was simulated should be clearly recorded.	guidance · Injectables	Checked	ST-32
n324	IX.A.2	Every product or line change should go through written change control. Any change or event that could weaken the aseptic process's ability to keep out contamination should be assessed with additional media fills. Examples include facility or equipment modifications, a new line configuration, major staffing changes, unusual EM results, container-closure changes, long shutdowns, and sterility-test failures in finished product.	guidance · Injectables	Checked	ST-29
n325	IX.A.2; IX.A.9	Any contaminated media-fill unit counts as objectionable and should be investigated, with organisms identified to species and possible causes surveyed. A failure investigation should also assess the impact on commercial product made on that line since the last media fill. Once corrections are made, repeat run(s) should confirm the line is back in control. If the root cause stays unclear, three consecutive successful runs together with closer scrutiny of production may be needed.	guidance · Injectables	Checked	ST-33
n326	IX.A.9	The target is zero contaminated units whatever the run size. Suggested criteria: under 5,000 units, one positive calls for investigation and then revalidation. From 5,000 to 10,000 units, one positive calls for investigation (including whether to repeat), and two call for revalidation after investigation. Above 10,000 units, one positive calls for investigation, and two call for revalidation after investigation. Intermittent or recurring positives on a line, even within the criteria, point to an adverse trend that should lead to finding the cause, correcting it, and revalidating. The criteria never make it acceptable to ship a non-sterile unit.	guidance · Injectables	Checked	ST-33
n327	IX.A.8	Media-filled units should be incubated at a temperature suited to recovering bioburden and environmental isolates. The temperature should always be between 20 and 35 C and within +/-2.5 C of target, for a total of at least 14 days. If two temperatures are used, each should be held at least 7 days, lower one first. Units should be read by people trained to inspect media fills, with QC oversight if QC is not doing the reading, and any suspect unit should go immediately to the QC microbiologist. Amber or opaque containers should be swapped for clear ones that are otherwise identical, or another method used so growth can be seen.	guidance · Injectables	Checked	ST-32
n328	IX.A.8	If units are inspected right after a media fill, every integral unit should be incubated, including units with cosmetic defects. Only units that have lost integrity are rejected, and units rejected by mistake should promptly go back for incubation. Units damaged after incubation starts stay in the run's results; excluding any must be fully justified in the report. If hard-to-see damage turns out to correlate with growth, that should be investigated. Yield and reconciliation criteria (units incubated versus units filled) should be set, with a full accounting and description of every rejected unit.	guidance · Injectables	Checked	ST-32
n329	IX.A.8	Written procedures for aseptic interventions should be clear and specific, stating for example the intervention type and how many units must be removed afterwards. Batch records for both production and media fills should show the procedure was followed. A media-fill intervention should never remove more units than the same intervention would in production. Units removed at a well-documented intervention need not be incubated, but vague procedures do not justify leaving them out. Large line clearances should not undermine the media fill's ability to detect contamination from a simulated activity.	guidance · Injectables	Checked	ST-32
n330	IX.A.9	The QC unit should watch process-simulation runs. It should be possible to trace any contaminated unit back to roughly when it was filled and what activity was being simulated. Video recording can help spot poor personnel practices.	guidance · Injectables	Checked	ST-32
n331	IX.A.9	Invalidating a media-fill run should be rare. A run should only be aborted where written procedures would require a commercial lot to be handled the same way, and the abort should be supported by documented justification.	guidance · Injectables	Checked	ST-32
n332	IX.A.7	Media fills generally use a general growth medium such as soybean casein digest, and anaerobic media in special cases. The medium should be shown to grow gram-positive and gram-negative bacteria, yeast and mould, using fewer than 100 CFU, and relevant EM or sterility-test isolates should be added where appropriate. If growth promotion fails, any contamination found should still be investigated and the media fill repeated promptly, and the reason for the growth-promotion failure should be investigated too. Each unit should contain enough medium to wet all inner surfaces when inverted or swirled.	guidance · Injectables	Checked	ST-32
n333	IX.B	A sterilising-grade filter (usually rated 0.2 um or finer, with redundant filters considered in many cases) should be validated to consistently produce sterile effluent. Validation uses a microbial challenge under worst-case conditions for the product, at least 10^7 organisms per cm2 of filter area, typically B. diminuta or a justified bioburden isolate, with direct inoculation into product preferred. It also accounts for factors such as viscosity, pH, compatibility, pressure, flow rate, maximum use time, temperature, osmolality and hydraulic shock. The filter user remains responsible for reviewing vendor or lab validation data, which should fit their product and conditions. Production should use filters identical to those validated (same polymer and pore rating).	guidance · Injectables	Checked	ST-50
n334	IX.B	Sterilising filters should normally be discarded after one lot. Where reuse can be justified, the filter validation should include the maximum number of lots the filter will process.	guidance · Injectables	Checked	ST-53
n335	IX.B; IV.D.1	Filter integrity may be tested before use, and should routinely be tested after use to catch leaks or perforations that happened during filtration. Suitable tests include forward flow and bubble point. The production integrity-test specification should be consistent with the bacterial-retention validation data.	guidance · Injectables	Checked	ST-52
n336	IX.C	Equipment surfaces that touch sterilised product or its sterilised containers and closures must be sterile so the drug's purity is not altered. Where contamination is reasonably possible, surfaces near the sterile product should also be free of viable organisms. Sterilisation of such critical equipment needs the same validation rigour as sterilisation of product, containers and closures.	guidance · Injectables	Checked	ST-42
n337	IX.C; footnote 19	Aseptic processing equipment should normally be re-sterilised between batches. Other intervals are possible only if defined, justified and supported by validation. After sterilisation, equipment, containers and closures should be moved and assembled with strict aseptic technique so they stay sterile.	guidance · Injectables	Checked	ST-38

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n338	IX.C.1	Sterilisation cycles should be validated, and periodically requalified, with the load configurations and the positions of biological indicators and temperature sensors documented. Batch production records should then show that each load followed the validated load pattern.	guidance · Injectables	Checked	ST-43
n339	IX.C.1	Cycle specifications should deliver enough lethality to the slowest-heating spots and demonstrate an SAL of at least 10^-6. Air removal is essential in steam cycles. Hard-to-sterilise items (filters, filling manifolds, pumps, tightly wrapped or dense packs, long tubing, hydrophobic filters, stopper loads) and SIP points downstream of filters need particular attention. A steriliser's continued suitability is maintained through qualification, maintenance, change control (including load-configuration changes or steriliser modifications) and periodic verification with biological challenge.	guidance · Injectables	Checked	ST-43
n340	IX.C.2	The data from sterilisation cycle monitoring devices must be reliable. Temperature, pressure and dwell-time devices should be calibrated at suitable intervals, validation sensors checked before and after each study, and steam-purity instruments and tunnel belt-speed sensors calibrated. Biological indicators should have their counts confirmed and be stored properly. Vendor D-values can be accepted only where the CoA's reliability has been qualified, otherwise the D-value should be determined. Instruments that fail their specifications may not be used.	guidance · Injectables	Checked	ST-43
n341	IX.C.2	When a steriliser or SIP operation needs valves operated by hand, those steps should be written into the manufacturing procedures and recorded in the batch records. Measuring devices should sit at the control points most likely to show process variation quickly.	guidance · Injectables	Checked	ST-43
n342	IX.C.2	Steriliser performance indicators such as equilibration (come-up) time should be reviewed routinely to confirm the unit still runs under its validated conditions, and sterilising equipment should be kept well maintained.	guidance · Injectables	Checked	ST-43
n343	X.A.1	Environmental monitoring (EM) is one of the key laboratory controls. It shows the state of the aseptic environment while a particular batch is being made, and shows trends in the surrounding clean areas, so contamination routes can be found and fixed before product is affected. There should be a written program with sound methods. It should cover every production shift and sample air as well as floors, walls, equipment and the critical product-contact surfaces, and extend across all classified areas such as aseptic corridors and gowning rooms. Locations should be listed, and their timing and frequency chosen according to the operation. Sample sizes should be large enough to detect the contamination levels expected.	guidance · Injectables	Checked	ST-18
n344	X.A.1	The critical zone's microbiological quality should be monitored during filling and closing, with air and surface samples where activity or product exposure is significant. Critical sites should be chosen by considering how hard set-up is, how long processing lasts, and the effect of interventions. Product-contact critical surfaces should remain sterile throughout. They are sampled at the end of the aseptic operation, so sampling does not touch sterile surfaces during processing.	guidance · Injectables	Checked	ST-18
n345	X.A.1	Finding contamination at a critical site does not automatically mean the batch must be rejected. It should, however, trigger an investigation of the operational information and data, while bearing in mind that a small number of false positives can occur.	guidance · Injectables	Checked	ST-21
n346	X.A.1; X.A.2	EM methods can miss organisms, so adverse trends include a rising frequency of recoveries over time, not only consecutive hits. Each individual result should be compared with the alert and action levels without averaging, because averaging can hide local problems. An alert result calls for attention, and an action result calls for a deeper investigation. Even without a trend, a single action-level result should prompt an evaluation and a decision on whether remediation is needed, and unfavourable trends in any room class should lead to remediation. Written procedures should set how often data are reviewed and what actions follow.	guidance · Injectables	Checked	ST-20
n347	X.A.1; X.A.4	SOPs should describe each EM location in enough detail that it can be sampled the same way every time. They should also define frequency, timing (during or at the end of operations), duration, sample size (air volume or surface area), equipment and technique, alert and action levels, and the response to excursions. Surfaces (product-contact, floors, walls, equipment) should be sampled regularly with contact plates or swabs. Active air samplers should be used every production shift at chosen points, and should be suitable (collection efficiency, cleanability, sterilisability, minimal airflow disruption), calibrated, and sample a meaningful volume. Settle plates are qualitative or semi-quantitative, should sit at the highest-risk points, and should be exposed under conditions that avoid drying out.	guidance · Injectables	Checked	ST-18
n348	X.A.2	Monitoring levels should reflect each location's relationship to the operation and draw on historical data, media fills, qualification and sanitisation studies. The quality unit should oversee short-term (daily to quarterly) and long-term trends in EM and personnel monitoring data. Trend reports should break data down by location, shift, room and operator, and should show significant changes in the microbial flora. Senior management should be kept informed of trends and investigations under a written system.	guidance · Injectables	Checked	ST-20
n349	X.A.3	The suitability, efficacy and limits of disinfectants and disinfection procedures should be assessed. Disinfectants should be sterile, kept in suitable (for example sterile) containers, and used only within the period set by written procedures. Routine agents should work against the facility's normal vegetative flora, and a sporicide should be used on a written schedule and whenever EM suggests spore-formers. Procedures should specify preparation, sequence and contact time, and EM should confirm they work.	guidance · Injectables	Checked	ST-15
n350	X.B	Isolates from the critical and immediately surrounding areas, and from personnel, should routinely be identified to species (or, where appropriate, genus). Isolates from ancillary areas such as ISO 8 should be identified at frequent intervals, so the facility has a current database of its flora. Genotypic methods are especially useful when investigating sterility or media-fill failures. EM media should be validated to recover both bacteria and fungi and incubated under suitable conditions, for example 30-35 C for 48-72 h for bacteria and 20-25 C for 5-7 days for yeasts and moulds. Every prepared media lot should pass growth promotion, with neutralisers used where disinfectant or product residues could inhibit growth.	guidance · Injectables	Checked	ST-23
n351	X.C; IX.B; XI.C.5	Process controls should keep bioburden in the unfiltered product low, because bioburden raises the challenge to the sterilising filter and can add impurities such as endotoxin or cause degradation. A pre-filtration bioburden limit should be set. Bioburden of unsterilised bulk should be measured so the types of organisms can be trended, and bioburden trends should be reviewed in any sterility-failure investigation.	guidance · Injectables	Checked	ST-54
n352	X.D	Other microbiological methods, such as rapid methods, can be used for EM, in-process testing and release testing once they have been shown to be equivalent to or better than traditional (for example USP) methods.	guidance · Injectables	Checked	ST-20
n353	X.E; IV.A	Routine particle monitoring quickly detects departures from qualified air cleanliness. Any result above the classification level at a location should be investigated for cause. The depth of the investigation should match how severe the excursion was and include a review of trend data, with corrective action taken as needed to stop it happening again.	guidance · Injectables	Checked	ST-19
n354	XI.A	Sterility test methods are required to be accurate and reproducible, with validation and documentation per 211.165 and 211.194. USP <71> is the main reference. Method validation should show that representative organisms are reliably recovered in the presence of the product, and the method should be modified (dilution, extra rinses, neutralisers) if growth is inhibited, so there is no opportunity for false negatives.	guidance · Injectables	Checked	ST-65
n355	XI (intro); XI.A	The sterility testing environment should have facilities and controls comparable to aseptic filling; isolators reduce false positives. If the lab is weaker than production, a real product failure risks being blamed on the lab. Test media should be sterile and proven to support growth. Analysts should be qualified and trained, with a written program that keeps training current and confirms good test practice.	guidance · Injectables	Checked	ST-65
n356	XI.B	Because the sterility test samples so few units, it has limited power. For example, a lot with 0.1% contaminated units tested with 20 units passes about 98% of the time. Samples should therefore represent the whole batch and its processing conditions. They should be taken at the start, middle and end of the aseptic operation, and also in connection with interventions or excursions. Any positive is a serious CGMP issue that needs thorough investigation.	guidance · Injectables	Checked	ST-66

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n357	XI.C	If growth appears in a sterility test, the lot should be treated as non-sterile and investigated. A first positive can be invalidated only when the growth is unequivocally shown to come from laboratory error, and a retest is justified only by conclusive, documented evidence that contamination happened during testing. If the evidence is inconclusive, the batch should be rejected as failing sterility. Any testing deviation should be documented, investigated and fixed, and a test whose integrity was compromised should be invalidated immediately, before incubation.	guidance · Injectables	Checked	ST-67
n358	XI.C.1-7	A sterility-positive investigation should end in written specific conclusions and corrective actions, supported by evidence. That evidence should include: species ID of the isolate, compared with EM, personnel and bioburden isolates using the same ID method; the lab's deviation and test records and lab EM trends; short- and long-term production EM trends, not only the lot's own day or shift; personnel monitoring trends and practices; pre-sterilisation bioburden trends; a full review of batch and production records for events affecting the critical zone, utility or support-system problems (for example air imbalance or high particles), and construction or maintenance; and the manufacturing history of the product and similar products, meaning past deviations and changes. Because the cause may extend beyond one batch, the whole process should be investigated. Under 211.192 the investigation must extend to other associated batches.	guidance · Injectables	Checked	ST-67
n359	XI.C.2	Keep separate sterility-failure trends by product, container type, filling line, and sampling and testing personnel. Where sample handling in the lab is similar, a higher initial failure rate for aseptically filled product than for terminally sterilised product suggests problems in aseptic production. Upward trends in the lab's aseptic-area counts should be investigated promptly and fixed. A lab's poor track record must not be used to automatically blame the lab and miss a real production problem.	guidance · Injectables	Checked	ST-67
n360	XII (text box)	Before a batch is released or distributed, the quality unit must review and approve all production and control records, including packaging and labelling, to confirm the approved procedures were followed. Any unexplained discrepancy, including yield outside the MPCR limits, and any failure of the batch or a component to meet specifications must be thoroughly investigated, even if the batch has already shipped. The investigation must extend to other batches and products that may be affected, and must be written up with conclusions and follow-up.	guidance · Injectables	Checked	ST-71
n361	XII	For an aseptically processed product, every IPC and lab control result has to form part of the documentation for the batch record.	guidance · Injectables	Checked	ST-70
n362	XII	Process and environmental controls should be built into daily aseptic operations and strictly maintained. Before the release decision, the batch records and data are reviewed against procedures, operating parameters and specifications, which amounts to a holistic assessment of how the process and its supporting systems performed during that production run. The guidance treats these as essential elements of the release decision: review of EM and personnel monitoring data; data showing support systems gave acceptable output (HEPA/HVAC, WFI, steam generator); and evidence that equipment worked properly (for example the batch alarm report and integrity of the various filters).	guidance · Injectables	Checked	ST-70
n363	XII	Interventions and stoppages are normally recorded in the batch record, but in inconsistent ways. Line stoppages and any unplanned interventions should be documented well enough, with the time and duration of each event. Extensive interventions both lengthen the time sterile items spend in the critical zone and add contamination risk, and sterility failures have often been traced to atypical or extensive interventions made in response to problems during filling. Major interventions such as machine adjustments or repairs should be documented in more detail than minor events.	guidance · Injectables	Checked	ST-35
n364	XII	Written procedures should say when interventions such as machine adjustments or repairs require line clearance. An intervention that involves substantial activity near exposed product or container-closures, or that goes beyond a reasonable exposure time, should lead to a local or full line clearance where appropriate.	guidance · Injectables	Checked	ST-35
n365	XII	Any interruption of the power supply that could affect product quality, however brief, is a manufacturing deviation and must be recorded in the batch record.	guidance · Injectables	Checked	ST-06
n366	XII (text box)	Written production and process control procedures, approved by the quality unit, must be followed and documented at the time each activity is performed. Any departure from them must be recorded and justified. Master records and batch records must meet 211.186 and 211.188.	guidance · Injectables	Checked	ST-04
n367	Appendix 1.A	A leak in an isolator's gloves, sleeves, half-suits or seams is a serious integrity breach. These parts need daily attention and a preventive-maintenance program, with written replacement frequencies that ensure parts are changed before they degrade. Transfer systems, gaskets and seals should be covered too. Gloves should be visually checked for obvious defects every time they are used, and physically integrity-tested routinely. Any glove lacking integrity should be removed from service. Because microscopic holes may go undetected, attention to the inner-glove surface and wearing a thin second pair of gloves are recommended.	guidance · Injectables	Checked	ST-75
n368	Appendix 1.B.3-4; 1.C	Isolators should keep a qualified positive pressure against the room (typically about 17.5-50 Pa, depending on design and exit port). Air balance with directly connected equipment such as a depyrogenation tunnel should also be qualified. Openings should be designed so room air cannot be drawn in, for example with local ISO 5 protection. The interior should be ISO 5. The background room is set by interface design and transfer frequency (commonly ISO 8) and should never be unclassified. Rapid transfer ports work well, but weaker ports need local HEPA unidirectional cover. Overpressure at the product exit (mousehole) should be supplied and monitored continuously.	guidance · Injectables	Checked	ST-74
n369	Appendix 1.D.1-2	Decontamination should reach every isolator surface: gloves fully extended with fingers apart, and the interior cleaned first. The cycle should leave inner surfaces free of viable organisms. It should be validated with quantified biological indicators on a range of materials and in many locations, including hard-to-reach spots, with a margin of extra kill (usually a 4-6 log reduction, and 4 log can be enough for low-bioburden wrapped items brought into a transfer isolator). Validation should also show uniform distribution of the agent, and chemical indicators can confirm the agent reached a location.	guidance · Injectables	Checked	ST-76
n370	Appendix 1.D.3	Isolators should be decontaminated often. If one is used for several days between cycles, the interval should be justified by validation and shortened if data show the isolator's microbial quality getting worse. A loss of integrity, from causes such as power or valve failure, insufficient overpressure, glove or seam holes or other leaks, should normally trigger decontamination and an investigation. If the environment may have been compromised, any product that could be affected should be rejected.	guidance · Injectables	Checked	ST-76
n371	Appendix 1.E	The whole sterile processing stream should be sterilised so product-contact surfaces are sterile from the start of every operation. Equipment and supplies for use inside the isolator should be chosen so they can be steam-sterilised or equivalent, and materials that tolerate heat (for example by SIP) are expected to be heat-sterilised. Where surface decontamination is used to make a product-contact surface free of viable organisms, it should show at least a 6-log reduction with a suitable BI.	guidance · Injectables	Checked	ST-42
n372	Appendix 1.F	Isolator EM should routinely cover air, surfaces, gloves or half-suits, and particles inside the isolator. Medium left on surfaces after contact plating should be wiped off. Air quality should be checked periodically in every shift, with particle monitoring at the exit port recommended. EM media should never be exposed to decontamination residues, which would suppress recovery.	guidance · Injectables	Checked	ST-76
n373	Appendix 1.G	Even though gowning demands are lower with isolators, manual work through gloves and half-suits still carries risk: those are hard to decontaminate and glove defects may go unnoticed. Contaminated isolator gloves can make product non-sterile, so rigorous aseptic technique must be observed, including using sterile tools for manipulations.	guidance · Injectables	Checked	ST-75
n374	Appendix 2 (intro); 2.A	The guidance's aseptic standards apply to blow-fill-seal (BFS) unless stated otherwise. The steps most exposed to contamination are cutting the parison, moving it under the fill mandrel, and withdrawing the mandrel before sealing. The product path should be sterilised by a validated SIP or equivalent, and any other surface that could contaminate product should also be sterile. The room around the machine should generally be ISO 8 or better. HEPA-filtered or sterile membrane-filtered air should be used whenever product or materials are exposed (parison forming, moulding,	guidance · Injectables	Checked	ST-77

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		filling). The critical zone should meet ISO 5 microbial limits during operation and normally ISO 5 particle levels. Only qualified, gowned staff should enter the surrounding classified room.			
n375	Appendix 2.A	Particles from extrusion, cutting and sealing should be controlled through airflow design, separation of the filling zone, and similar measures, with smoke studies and particle data from multiple locations used in qualification. A sound preventive-maintenance program is needed. The cooling, heating and other utility systems attached to the BFS machine can contaminate product, so their integrity should be maintained and routinely monitored.	guidance · Injectables	Checked	ST-77
n376	Appendix 2.B	BFS set-up, equipment troubleshooting and the associated aseptic procedures deserve special attention. Validation should cover equipment sterilisation, media fills, polymer extrusion and its sterilising effect, compatibility between product and plastic, integrity of forming and sealing, and variation in unit weight. The extrusion time and temperature should be shown to deal with endotoxin or spore challenges, so containers are sterile and, for parenterals, non-pyrogenic. The polymer should be pharmaceutical grade and pass the relevant plastics criteria, and polymer suppliers should be qualified and monitored.	guidance · Injectables	Checked	ST-77
n377	Appendix 2.C	In-process parameters such as container weight variation, fill weight, leakers and air pressure support ongoing process control. Microbial air monitoring is essential and should follow a comprehensive plan that represents the whole fill, and continuous particle monitoring is valuable. The line should be set up to make integral units consistently, and every unit should get a reliable, sensitive final examination able to catch defects such as leakers. Significant defects from heat or mechanical causes (wall thickness, container or closure interface problems, badly formed closures) should be investigated in line with 211.100 and 211.192.	guidance · Injectables	Checked	ST-77
n378	Appendix 3.A	Some products must be handled aseptically from early steps onwards: after a point beyond which they cannot be filtered, or when the final product cannot be sterile-filtered, so each component is sterilised and mixed aseptically (for example aluminium-adjuvanted vaccines). Every transfer, transport and storage stage then needs tight control, and bulk may in some cases need sterility testing. Steps that expose product or contact surfaces, such as aseptic connections, should happen under ISO 5 unidirectional air with an ISO 7 or better background. Microbial and particle monitoring should run during operations, surface sampling should be done at the end but before cleaning, and personnel monitoring should accompany the operations.	guidance · Injectables	Checked	ST-40
n379	Appendix 3.A	Process simulations of the steps before filling should include every manipulation, addition and exposure of contact surfaces that could affect sterility, under worst-case conditions such as the longest open operations and the most operators. They need not match total manufacturing time if the manipulations are fully represented. They should also cover storage and transport of sterile bulk, with assurance that bulk vessels stay intact for the specified hold times. The formulation-stage simulation should be performed at least twice a year.	guidance · Injectables	Checked	ST-29
n380	Appendix 3.B	Cell therapy and some cell-derived products cannot be filter-sterilised and are handled aseptically throughout, ideally in closed systems. They are often given to patients before final sterility results are available. In that case additional controls should be considered, such as sterility tests at intermediate stages (for example after the last manipulation before harvest). Rapid indicators such as microscopy, Gram or other stains, and endotoxin should pass before release.	guidance · Injectables	Checked	ST-69
n381	III; II.B	The guidance does not cover terminal sterilisation of drug products, only aseptic processing and the sterilisation of components, containers, closures and equipment. It holds that aseptic processing should be used only where terminal sterilisation is not feasible, although a packaging advantage (for example some dual-chamber syringes) may justify aseptic processing with adjunct steps that raise sterility assurance.	guidance · Injectables	Checked	ST-02
n382	II.B; Glossary (terminal sterilization, overkill); IX.C.1	For terminal sterilisation, product is filled and sealed in a high-quality environment so bioburden stays low and the later sterilisation in the final container succeeds. The glossary defines the target sterility assurance level as usually 10^-6, but its parenthetical phrases the probability of a non-sterile unit as greater than one in a million, an apparent inversion: the intended meaning is at most one in a million. An overkill process is defined as giving at least a 12-log reduction of organisms with a D-value of at least 1 minute.	guidance · Injectables	Checked	ST-60
n383	V (text box)	Only people authorised by supervisors may enter limited-access areas. A person whom a medical check or a supervisor identifies as ill, or as having open sores, in a way that could affect product quality must not have direct contact with components, containers, closures, in-process materials and product until the condition is resolved or cleared by competent medical staff. Staff must report such health conditions to their supervisor.	guidance · Injectables	Checked	ST-10
n384	X and XI (text boxes); IX.C.2	Instruments, apparatus, gauges and recorders must be calibrated at suitable intervals under a written program with directions, schedules, accuracy and precision limits, and remedial actions. Any that do not meet specifications must not be used.	guidance · Injectables	Checked	ST-05
n385	IX (text box)	When needed, samples must be collected with sterile equipment and aseptic technique.	guidance · Injectables	Checked	ST-66
n386	X and XI (text boxes); IV.E; IX.A.1	To ensure batch uniformity and integrity, written procedures must be established and followed for the in-process controls, tests or examinations performed on samples of in-process material from each batch. These controls must monitor output and validate the performance of process steps that can cause variability. In aseptic filling this includes fill weight or volume checks, which the guidance prefers automated (in-line) to reduce interventions.	guidance · Injectables	Checked	ST-56
n387	XI (text box)	Records must be kept so their data can be used, at least once a year, to evaluate each product's quality standards and decide whether specifications or manufacturing or control procedures need to change.	guidance · Injectables	Checked	ST-73
n388	IV (text box)	Aseptic processing areas must have, as appropriate: smooth, hard, cleanable floors, walls and ceilings; temperature and humidity control; HEPA-filtered air supplied under positive pressure; a system for monitoring environmental conditions; a cleaning and disinfection system for rooms and equipment; and a system for maintaining the equipment that controls aseptic conditions. Equipment to control air pressure, micro-organisms, dust, humidity and temperature must be provided where appropriate.	guidance · Injectables	Checked	ST-14
n389	IX.A.3	How long a media fill runs should depend on the time needed to include the manipulations and interventions, and on how long the real aseptic operation lasts. Common interventions should be simulated in routine media fills, and rare ones can be simulated periodically.	guidance · Injectables	Checked	ST-31
n390	IX.A.7	Concern about nutrient medium contaminating the facility is addressed by handling the medium properly and promptly following each media fill with cleaning, sanitising and, where needed, sterilisation of the equipment. Done this way, later product runs are unlikely to be compromised.	guidance · Injectables	Checked	ST-32

FDA Data Integrity Q&A

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n36	FDA DI Q&A Q11 (e-signature satisfies the 'full signature, handwritten' wording); related Q3 (creating an electronic MPCR is a workflow to validat	Binding (211.186(a), wording checked against the local CFR text): for each drug product and each batch size, one person prepares, dates and signs the MPCR and a second person independently checks, dates and signs it; MPCR preparation follows a written procedure. DI Q&A Q11: an electronic signature whose controls securely tie it to its record meets the handwritten full-signature wording (via 11.2(a)); Part 11 sets when e-signatures are legally equivalent, and firms should document the controls that let them identify the individual signer.	guidance	Checked	DI-04, DI-35

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n38	FDA DI Q&A Q6 (blank-form control, reconciliation, keeping spoiled forms); binding basis 211.188(a); document controls per 211.100, 211.160(a), 211.186; ret	Binding (211.188(a), checked against the local CFR text): every batch has a production and control record with complete information on its production and control, starting with an accurate reproduction of the applicable master record that has been checked for accuracy, dated and signed. DI Q&A Q6: document controls must exist; blank forms, if used (Q6 names electronic worksheets, lab notebooks and MPCRs), should fall under quality-unit control or some other document-control method; where appropriate, numbered sets may be issued and should be reconciled once all issued forms are completed; incomplete or erroneous forms should remain in the permanent record with a written reason for their replacement.	guidance	Checked	DI-42
n60	FDA DI Q&A Q1a-Q1f (definitions) and footnote 5; binding basis: ALCOA letters mapped to 211.101(d), 211.122, 211.186, 211.188(b)(11) (attributable), 211.180	Q1a: data integrity means data are complete, consistent and accurate, and such data should show the ALCOA traits (Attributable; Legible; Contemporaneously recorded; Original or true copy; Accurate); the letters are listed, not individually defined. Integrity matters from first capture through use, storage and transfer to disposal after retention, and system design and controls should make errors, omissions and aberrant results easy to spot. Q1b: metadata is the context needed to make sense of a value (e.g., units, timestamps, who acquired it, which instrument, material identity/status, and audit trails); data should be retained for the full retention period with the metadata needed to reconstruct the activity, and the data-metadata link kept secure and traceable. Q1c: an audit trail is a secured, time-stamped electronic record generated by the computer itself, from which the creation, change or deletion of an electronic record can be reconstructed; it covers data-level changes and record/system-level actions such as access attempts or renaming/deleting files, and reprocessing should be documented with its justification. Q1d: static = fixed (paper, image); dynamic = the user can interact with the content (e.g., re-integrate a chromatogram, alter spreadsheet formulas that compute results or a calculated yield). Q1e: the 211.68(b) backup is a true copy of the original kept securely for the retention period, holding data plus metadata in the original or a compatible format; FDA equates it with an archive; temporary copies kept against crashes or interruptions do not meet 211.68(b). Q1f: 'computer or related systems' reach beyond software and servers to attached devices, networks, cloud hosting, the people who run them and their documentation (manuals, SOPs).	guidance	Checked	DI-02, DI-45
n61	FDA DI Q&A Q3; binding basis 211.63, 211.68(b), 211.110(a); MPCR-review analogy 211.100, 211.186	Yes: each CGMP workflow run on a computer system (FDA's example is building an electronic MPCR) is an intended use that should be checked through validation, with validation effort proportionate to the risk of the automated system. Qualifying a platform such as an MES shows it meets its own requirements but not that a particular MPCR it generates calculates correctly; validating the workflow shows the MPCR's steps, requirements and calculations work as intended - comparable to reviewing a paper MPCR and its supporting procedures before production use. If one system serves both CGMP and non-CGMP purposes, the firm should assess, and mitigate as appropriate, the chance that the non-CGMP features influence CGMP operations. Controls should cover software, hardware, personnel and documentation.	guidance	Checked	DI-04, DI-05, DI-26
n62	FDA DI Q&A Q4; binding basis 211.68(b) (changes only by authorized personnel)	Must (211.68(b)): controls ensuring only authorized personnel can change computerized MPCRs or other CGMP records or enter laboratory data into computerized records; Q4 extends the restricted-control expectation to records from automated visual inspection, electronic materials management and automated dispensing/weighing systems. Should: technically limit, where feasible, who can change specifications, process parameters, data, or manufacturing and testing methods (e.g., through permissions over settings and data); give the system administrator role, including rights over files and settings, to people independent of those responsible for record content; establish a documented record of each CGMP system's authorized users and their privileges (e.g., a maintained list).	guidance	Checked	DI-09, DI-16, DI-18
n63	FDA DI Q&A Q5; binding basis 211.100, 211.68(b), 211.188(b)(11), 211.194(a)(7)-(8) (actions attributable to a specific individual)	With shared login credentials the login cannot identify a unique individual, so the system would not meet the part 211/212 CGMP requirements; system and documentation controls must be designed to assure product quality (211.100), and actions such as changing records or entering data must be traceable to one specific person (211.68(b), 211.188(b)(11), 211.194(a)(7)-(8)). Read-only accounts shared among users, which cannot alter data or settings, may be used just to look at data, but not for actions such as second-person review, which must be attributable to one individual.	guidance	Checked	DI-14
n64	FDA DI Q&A Q6; binding basis 211.100, 211.160(a), 211.186 (document controls required); 211.192, 211.194 (retaining replaced forms)	Must: document controls that assure product quality are in place. Should: blank forms, if used - Q6 explicitly names electronic worksheets, laboratory notebooks and MPCRs - be controlled by the quality unit or another document-control method (bound, paginated, officially stamped notebooks are cited as good practice because unofficial books and missing pages stand out); where appropriate, issue numbered sets and reconcile them once all issued forms are completed; retain spoiled or incomplete forms in the permanent record together with a written reason they were replaced; keep all data needed to recreate the CGMP activity in the complete record.	guidance	Checked	DI-42
n65	FDA DI Q&A Q7 (who) and Q8 (how often), footnote 13; binding basis 211.192 (QU reviews and approves all production and control records, audit trails include	Q7: audit trail review is comparable to examining cross-outs on paper; whoever is responsible for reviewing a CGMP record should review the audit trails of changes to that record's data during the same review; the quality unit must review and approve every production and control record, audit trails being part of those records (211.192); some review may be done by the person directly supervising or checking a step (211.188); FDA advises overseeing and reviewing CGMP records through a quality-system approach. Q8: where CGMP regulations fix a review frequency for the data, use the same frequency for its audit trail - for batch records, review after each significant step (211.188(b)) and before batch release (211.22); otherwise set the frequency from process knowledge and a risk assessment of data criticality, control mechanisms and product-quality impact; the approach must satisfy CGMP and its reliability should be proven. Footnote 13 names risks such as unauthorized or undetected deletion, amendment or exclusion, and offers logs of instrument status and communications, and alerts, as candidates for risk-based review.	guidance	Checked	DI-25
n66	FDA DI Q&A Q12 and footnote 14; binding basis 211.100(b), 211.160(a) (documented at the time of performance), 211.180(d)	Any data created to meet a CGMP requirement is a CGMP record and must be documented or saved when the activity is performed (211.100(b), 211.160(a)). FDA expects processes designed so that required data cannot be changed unless the change itself is recorded (its lab example: write to durable media as each step or injection finishes, including incomplete or aborted ones, rather than at the end of a set; audit-trail changes and investigate aborted runs). Jotting data on scrap paper for later transcription is not acceptable, and neither is any electronic storage that lets records be manipulated before a permanent record exists. Technical and procedural controls may be combined - e.g., an EBR or LIMS that commits automatically after every entry (the electronic counterpart of indelible, contemporaneous entries on a paper batch record) plus a procedure requiring data to be entered as soon as generated. Footnote 14: FDA also reviews records not intended for CGMP that contain CGMP information.	guidance	Checked	DI-28
n67	FDA DI Q&A Q9 (true copies) and Q10 (printouts vs dynamic originals); binding basis 211.180(d) (original records or true copies; suitable reader available),	Q9: yes, electronic copies may serve as true copies of records originally on paper or in electronic form if they keep the original's content and meaning - all metadata needed to reconstruct the activity and the original's static or dynamic nature; true copies of dynamic records may be kept in the original format or another format that preserves content and meaning, provided suitable reader and copying equipment (software, hardware, media readers) remains readily available. Q10: a printout or static record can meet retention requirements when it is itself the original or a true copy - e.g., balances and pH meters that print at acquisition, in which case the printout or a true copy must be kept (211.180); for instruments whose records are dynamic (e.g., a reprocessable FT-IR spectral file), a printout or static record does not satisfy 211.180(d), and a partial spectrum display could hide contaminants. Original lab records, paper or electronic, must receive second-person review (211.194(a)(8)).	guidance	Checked	DI-32, DI-41
n68	FDA DI Q&A Q11; binding basis 211.186(a); 21 CFR 11.2(a), with Part 11 signature controls 11.50, 11.70, 11.100, 11.200, 11.300	Yes: electronic signatures with appropriate controls may stand in for handwritten signatures or initials in any record CGMP requires; for the handwritten full-signature wording of 211.186(a), an electronic signature with controls that securely link it to its record meets the requirement (11.2(a)); Part 11 sets the criteria for legal equivalence. Firms should document the controls that let them identify the specific person who signed. PET MPCRs (part 212) have no handwritten-signature requirement.	guidance	Checked	DI-35
n69	FDA DI Q&A Q2 (with Q13 on retaining all data and testing into compliance); binding basis	The quality unit must assess all data generated within a CGMP record when deciding on release, and those data are retained for CGMP purposes; electronic data include the metadata needed to reconstruct the activity. Excluding test results from QU decisions on conformance to a specification needs a valid, documented,	guidance	Checked	DI-29

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
	211.22, 211.180 (QU evaluation, retention), 211.160(b), 211.188, 2	scientifically sound justification (see the OOS guidance). Even after legitimate invalidation based on a scientifically sound investigation, the complete batch record handed to the QU still carries the invalidated original data plus the investigation report. Retention and review requirements are the same for paper and electronic records. Q13 adds that every result - obvious errors, failing, passing and suspect - must be retained and reviewed, and invalidation needs an investigation.			
n70	FDA DI Q&A Q17 (and footnote 14 to Q12); binding basis 211.180(c) (records available for authorized inspection and copying), FD&C Act 704	Every record required under CGMP is open to FDA inspection, including records created and kept on computerized systems and electronic communications that support CGMP activities (FDA's example: an email authorizing batch release is a CGMP record). Firms must permit authorized inspection and review, and copying of records including electronic data (211.180(c)). Footnote 14: FDA routinely asks for records not intended to meet a CGMP requirement that still contain CGMP information (e.g., shipping records used to reconstruct an activity). Metadata and audit trails are covered because Q1b and Q7 treat them as part of the record.	guidance	Checked	DI-40
n88	Composite. DI Q&A support: Q1a (ALCOA), Q1b (audit trails are metadata kept for the retention period), Q1c (audit trail definition; data not lost or obscure	The composite is consistent with the DI Q&A and the cited rules: records attributable and contemporaneous (211.100(b); Q1a, Q5, Q12); changes must not obscure earlier entries (11.10(e); Q1c, Q12); audit trails secure, computer-generated, time-stamped and retained at least as long as the records (11.10(e); Q1b); signatures linked so they cannot be removed, copied or moved to falsify a record (11.70; Q11); backups exact, complete and protected from alteration, erasure or loss (211.68(b); Q1e).	guidance	Checked	DI-02, DI-22, DI-27, DI-31, DI-45
n639	Q1a (whole life cycle; design for detection); binding basis 211.68(b) (input/output checked for accuracy), 211.188, 211.192	Data integrity should hold from first capture through use, storage and transfer to disposal after the retention period, and system design and controls should make errors, omissions and aberrant results easy to spot throughout (recommendation; the binding hooks are the 211.68(b) accuracy checks and the record-review duties in 211.188/211.192).	guidance	Checked	DI-26
n640	Q1b (audit trails count as metadata; data-metadata link secure and traceable), Q1c (audit trail scope; justification for reprocessing); binding basis 11.10(e);	Audit trails should let reviewers reconstruct how each electronic record was created, changed or deleted, covering data-level events (e.g., parameters, results) and record/system-level events (e.g., access attempts, renaming or deleting files); FDA's HPLC example expects who ran it, when, the processing settings, and any reprocessing with its reason. Audit trails are themselves metadata, to be retained with the data for the retention period with the relationship kept secure. Binding: 11.10(e) requires secure, computer-generated, time-stamped audit trails of operator entries and actions, kept as long as the records.	guidance	Checked	DI-02, DI-22
n641	Q1d (spreadsheet formulas computing results or a calculated yield are dynamic content), Q3 (validated MPCR calculations), Q6 (electronic worksheets are controll	FDA gives spreadsheet formulas that compute results or a calculated yield as an example of dynamic record content (Q1d); workflow validation should show MPCR calculations are correct (Q3); electronic worksheets are forms to control (Q6). Binding: yields are determined at appropriate phases and, when an automated system under 211.68 computes them, one person independently verifies the result (211.103); the batch record states actual and percent-of-theoretical yield (211.188(b)(7)); computer input and output are checked for accuracy (211.68(b)).	guidance	Checked	DI-33
n642	Q1f; binding basis 211.68(a)-(b)	For 211.68, 'computer or related systems' means more than software and servers: FDA includes attached devices, networks, cloud hosting, the people who operate the system and its documentation such as manuals and SOPs (drawing on an ANSI description of a system as people, machines and methods organized for set functions).	guidance	Checked	DI-03
n643	Q3; binding basis 211.63, 211.68(b), 211.110(a)	When one system handles both CGMP and non-CGMP work, the firm should assess, and mitigate as appropriate, the chance that its non-CGMP features influence CGMP operations; the extent of validation should match the risk the automated system poses, and controls should address software, hardware, personnel and documentation.	guidance	Checked	DI-05
n644	Q4; binding basis 211.68(b)	Manufacturers should set up and use a way of documenting which personnel are authorized on each CGMP computer system and with what privileges (e.g., a maintained list of authorized individuals), supporting the binding 211.68(b) duty that only authorized personnel make changes.	guidance	Checked	DI-16
n645	Inference from Q4 (technical restriction of settings changes; administrators independent of record content), Q12 (no change to required data without a record),	The DI Q&A does not explicitly say who may switch audit trails off. It does recommend technically restricting who can change settings and reserving administrator rights over files and settings for people independent of record content (Q4), and it expects required data to be unchangeable without a record of the change (Q12); Part 11 requires secure, computer-generated audit trails of record creation, modification and deletion (11.10(e), binding). Together these mean audit trailing of GMP records should never be something that users responsible for record content can disable.	guidance	Checked	DI-18, DI-23
n646	Q8 and footnote 13; binding basis 211.68(b), 211.192	Where the regulations fix no review frequency for a kind of data, derive the audit-trail review frequency from process knowledge and a risk assessment of data criticality, control mechanisms and product-quality impact, considering risks such as deletion, amendment or exclusion without authorization or detection; FDA's examples for risk-based review are logs of instrument status and communications, and alerts. The approach should meet CGMP and its reliability should be demonstrated.	guidance	Checked	DI-26
n647	Q9, Q1e; binding basis 211.180(a), 211.180(d), 211.68(b)	Copies of dynamic electronic records can count as true copies when held in the original format or in one that keeps content and meaning, as long as a suitable reader and copying equipment (software, hardware, media readers) stays readily available (211.180(d)); backups/archives are true copies with metadata kept for the whole retention period (Q1e).	guidance	Checked	DI-41, DI-46, DI-48
n648	Q12 (data are a CGMP record when generated), Q10 (printouts as originals vs dynamic records), Q2 (same rules for paper and electronic); binding basis 211.180, 2	Data generated to meet a CGMP requirement are a CGMP record from the moment of generation (Q12). A printout or static record can be the original for devices such as balances and pH meters, and then it or a true copy must be kept (Q10, 211.180); dynamic electronic records cannot be replaced by printouts (Q10). Retention and review requirements do not differ between paper and electronic records (Q2).	guidance	Checked	DI-01, DI-32
n649	Q10 (second-person review of original lab records; microbiology plate-count example); binding basis 211.194(a)(8); for production steps 211.188(b)(11)	Original laboratory records, paper or electronic, must get second-person review so that all results and related information are properly reported (211.194(a)(8)); FDA's example is a contemporaneous written record of petri-dish colony counts that a second person then reviews.	guidance	Checked	DI-30
n650	Q12 (no scrap-paper transcription), Q6 (controlled blank forms); binding basis 211.100(b), 211.160(a), 211.180(d), 211.186	Writing data on loose scraps that are thrown away once copied into the permanent record is not acceptable (Q12); blank forms that are used should be controlled and reconciled, with spoiled forms retained with justification (Q6).	guidance	Checked	DI-43
n651	Q12 (lab example: save each step or injection including aborted ones; aborted/incomplete injections audit-trailed, investigated, justified) - applied by analogy	FDA expects processes designed so that required data cannot be changed without a record; in its chromatography example, data are written to durable media as each step or injection completes (incomplete or aborted ones included), and aborted or incomplete injections are captured in audit trails, investigated and justified.	guidance	Checked	DI-29
n652	Q13 (applied by analogy to in-process testing); binding basis 211.160, 211.165, 211.188, 211.192, 211.194, 211.68(b)	Sampling or testing aimed at obtaining a particular result or overcoming an unacceptable one (so-called testing into compliance) conflicts with CGMP, and running actual samples in test, prep or equilibration runs to disguise it is violative. If an actual sample is used for system suitability, it should be a characterized secondary standard from a different batch, under written procedures. Every result, including plain mistakes and failing, suspect or	guidance	Checked	DI-29

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		passing values, has to stay in the retained CGMP records and be reviewed; invalidation needs an investigation with a documented, scientifically sound justification.			
n653	Q14 (applied by analogy to production calculations and in-line data); binding basis 211.160, 211.165(c), 211.194(a)(4), 211.194(a); for production 211.188	Saving only the final result of reprocessed lab chromatography is not acceptable; reprocessing should rarely be needed, must follow written procedures, and each result must be kept for review; lab records must contain complete data, e.g., notebooks, worksheets and instrument outputs such as spectra.	guidance	Checked	DI-33
n654	Q15; binding basis 211.22(a), 211.125(c), 211.192, 211.198, 211.204	Suspected or known falsification or alteration of CGMP records must be fully investigated within the CGMP quality system, whatever the source or intent, to establish its impact on patients, on product quality and on the reliability of data, find the root cause and ensure corrective action; it cannot be handled informally. FDA also invites reports of suspected DI issues by email to its drug information address.	guidance	Checked	DI-50
n655	Q16; binding basis 211.25	Staff should be trained, as part of routine CGMP training, to prevent and to detect data integrity problems; FDA links this to 211.25 (education, training and experience for assigned duties).	guidance	Checked	DI-21
n656	Q17 and footnote 14; binding basis 211.180(c)	Electronic communications that support CGMP activities are CGMP records open to FDA review (an email approving batch release is FDA's example), and FDA also reviews non-CGMP records that contain CGMP information (footnote 14). Firms must permit inspection and review, and copying of records including electronic data (211.180(c)).	guidance	Checked	DI-40
n657	Q18 and Introduction; binding basis 211.22(a), 211.192	FDA encourages remediation that establishes the scope and root causes of DI problems, assesses their potential effects (including on submission data) with a scientifically sound risk assessment, and puts in place a management strategy with a global corrective action plan; this may involve a third-party auditor, moving responsible individuals out of positions of influence over CGMP data, stronger quality oversight, enhanced computer systems and mechanisms such as anonymous reporting and data governance. The Introduction expects risk-based DI strategies covering system design, operation and monitoring, backed by management and a quality culture that encourages reporting.	guidance	Checked	DI-50
n658	Q1a (life cycle through disposition), Background (data secure until disposition), Q8 footnote 13 (deletion/amendment/exclusion risks), Q12, Q1c, Q3; binding bas	Data should be protected from creation until disposition after the retention period (Q1a, Background); FDA names unauthorized or undetected deletion, amendment or exclusion as risks to data (footnote 13); required data should not be changeable without a record (Q12); audit trails cover deletions (Q1c); each workflow, including any that can remove data, is an intended use to validate (Q3).	guidance	Checked	DI-24, DI-47

Other US FDA law / guidance

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n1	210.3(b) definitions (Part 210; adopted into Part 211 by 211.3 but not in the uploaded files)	Part 210 definitions, which Part 211 adopts through 211.3: a lot, control or batch number is any unique code of letters, digits or symbols from which the full manufacturing, processing, packing, holding and distribution history of a batch or lot can be reconstructed. Theoretical yield is what a phase would produce from the components used if nothing were lost and no error occurred; actual yield is what the phase really produces; percentage of theoretical yield is actual divided by theoretical for the same phase, expressed as a percent. (Could not be checked here: Part 210 is not among the uploaded texts.)	binding	Search-based	PR-10, PR-31, PR-57
n71	Risk-based assurance framework: intended use; process-risk determination (high vs not high); assurance activities (scripted, unscripted: ad hoc, error-guessing,	FDA recommends a risk-based approach to establish confidence that automation used in production or the quality system performs as intended. Identify intended use; determine whether a feature is high process risk (failure could compromise product quality/safety) or not; choose proportionate assurance activities (scripted testing for high-risk features; unscripted or exploratory testing, leveraging vendor activities, etc., for lower risk); and keep a record of intended use, risk determination, assurance activities, issues and their disposition, conclusion, date and responsible person. The guidance was issued by CDRH and CBER for medical device production/quality system software. It supersedes Section 6 of 'General Principles of Software Validation'. The Feb 2026 revision aligns it with the QMSR (effective Feb 2, 2026) and reportedly adds explicit consideration of revalidation when software is updated (from a secondary source).	guidance	Search-based	discussed in notes of Part B
n77	Parametric release data and monitoring positions; 211.167(a) sterility testing otherwise required	FDA has accepted parametric release for moist-heat terminally sterilized products since 1985, where approved in the application. Positions in the load should be identified so the load's exposure to the sterilization process is measured and recorded against defined criteria for parametric release, with the location and number of monitors described and justified in the application. A Class 5 chemical indicator is recommended for most applications. Without an approved parametric release, 211.167(a) requires sterility testing of each batch.	guidance · Injectables	Search-based	ST-61
n79	100% visual inspection plus AQL sampling; lifecycle inspection program	FDA's draft recommends a holistic program in which every unit of an injectable is inspected (100% inspection, manual, semi-automated or automated), followed by statistical sampling (AQL) of accepted units to verify batch quality. Meeting a USP compendial standard alone is generally not sufficient for CGMP. USP <790> frames the 'essentially free' expectation with a sampling plan (AQL 0.65 cited in secondary sources). Defects should be classified, trended and investigated.	draft · Injectables	Search-based	ST-62
n80	Stage 2 (PPQ) and Stage 3 (Continued Process Verification) - data collection and trending	Stage 3 provides ongoing assurance in routine production that the process stays in a state of control. FDA recommends continued monitoring and sampling of process parameters and quality attributes at the level established during process qualification until enough data exist for significant variability estimates. It recommends that a statistician or person with adequate statistical training develop the data collection plan and analysis methods, that intra-batch and inter-batch variation be evaluated, and that variation be detected through timely assessment of defect complaints, OOS findings, process deviation reports, process yield variations, batch records, incoming raw material records and adverse event reports.	guidance	Search-based	PR-09, PR-58
n81	Product identifier definition; data carriers; manufacturer obligation to affix to each package and homogeneous case	The product identifier is a standardized graphic containing the standardized numerical identifier (NDC plus a unique alphanumeric serial number of up to 20 characters), lot number and expiration date, in both human-readable and machine-readable form. On packages it must be a 2-dimensional data matrix barcode; on homogeneous cases, a linear or 2D data matrix barcode (case carrier rule from the statute's text). Since Nov 27, 2017, manufacturers must affix or imprint a product identifier on each package and homogeneous case of product intended to be introduced in a transaction into commerce. The guidance allows human-readable elements to appear in different locations on the label if space requires.	guidance	Search-based	PR-39, PR-42
n82	Transaction information/statement; verification of suspect/illegitimate product; response to verification requests; record retention	Manufacturers provide transaction information and statements to subsequent owners and keep them for at least 6 years. They must verify product at package level (including the standardized numerical identifier) when they determine product is suspect or when FDA requests it. They must respond to verification requests from authorized trading partners within 24 hours (secondary-source confirmation). Illegitimate product must be quarantined, investigated and notified (Form 3911 to FDA and trading partners within 24 hours per statute), and investigation records are kept at least 6 years. The 24-hour notification and 6-year retention for investigations are partly from statutory text knowledge.	binding	Search-based	PR-42, PR-59
n83	Package-level interoperable electronic tracing; EPCIS; exemptions timeline	Since Nov 27, 2023 (with a one-year stabilization period to Nov 27, 2024), transaction information must include the product identifier at package level and be exchanged securely, interoperably and electronically. FDA recommends the GS1 EPCIS standard. FDA's exemption for eligible manufacturers and repackagers ended May 27, 2025 (wholesale distributors Aug 27, 2025; dispensers with 26+ FTE Nov 27, 2025; small dispensers exempt to Nov 27, 2026). Manufacturers are therefore fully subject to the enhanced requirements as of Sept 2026. The package-level guidance discusses use of aggregation/inference by trading partners; statute does not explicitly mandate aggregation (interpretation; verify with regulatory counsel).	guidance	Search-based	discussed in notes of Part A

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n84	201.18 - Drugs; significance of control numbers	The lot number on a drug's label should be capable of yielding the complete manufacturing history of the package. An incorrect lot number may be regarded as causing the article to be misbranded.	binding	Search-based	PR-36, PR-57
n85	201.25(c) barcode content and symbology; NDC format (12-digit, 6-4-2) and transition	Drug products subject to the bar code rule must carry a barcode containing at least the NDC. Before the 2026 rule, only a linear barcode meeting EAN/UCC or HIBCC standards (or another approved standard) was allowed. The March 5, 2026 final rule standardizes all FDA-assigned NDCs to 12 digits (6-digit labeler, 4-digit product, 2-digit package), effective March 7, 2033. It adds a further transition to March 6, 2036 for labeling updates and depletion of old labels, after which drugs with 10-digit NDCs introduced into commerce may face action. It revises 201.25 to allow linear or nonlinear (e.g., 2D Data Matrix) barcodes that meet recognized standards.	binding	Search-based	PR-36, PR-42
n86	4.4(b) streamlined approach for drug-led combination products (e.g., prefilled syringes, autoinjectors)	Drug-device combination products manufactured under Part 211 must also meet specified device QS/QMSR provisions (historically 820.20 management responsibility, 820.30 design controls, 820.50 purchasing controls, 820.100 CAPA, 820.170 installation, 820.200 servicing). The exact post-QMSR cross-references were not verified in this pass.	binding · Injectables	Unverified	ST-56
n87	Composite. In uploaded texts: 211.130(c); 211.188(b)(2), (b)(3), (b)(11); 211.105(b); 211.182; 211.184(c); 211.150(b); 211.196; 211.192. Outside: 210.3(b), 201.	Within Part 211, traceability is required as follows. Backward: the lot or control number must allow the batch's manufacturing and control history to be determined (211.130(c)), and the batch record must identify the major equipment, the component and in-process lots used, and the people who performed and checked each significant step (211.188(b)(2), (b)(3), (b)(11); 211.105(b)). Forward: component, container and closure inventory records must identify every product lot that used them (211.184(c)); equipment logs must show each batch processed (211.182); and the distribution of each lot must be readily determinable for recall (211.150(b), 211.196). Investigations must reach other batches and products possibly associated with a failure or discrepancy (211.192). The wording 'complete history including distribution' belongs to the Part 210 lot-number definition (outside the uploaded texts), and serial-number/aggregation traceability comes from DSCSA (outside). Room-level and person-level forward tracing are not explicit requirements; they are useful for 211.192 impact assessments.	binding	Checked	PR-57

EMA GMP Q&A

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n183	Q&A on assessing data risk / data lifecycle	Data risk should be judged by how exposed the data are to accidental or intentional amendment, deletion or re-creation; controls that stop unauthorised actions or make them more visible/detectable reduce risk. The assessment should focus on the business process (production, QC) and factors such as complexity, consistency, degree of automation vs human input, and subjectivity of outcomes, at every lifecycle stage. Holding data in temporary memory before it is saved is flagged as a risk because changes there are often not audit-trailed; removing or shortening such temporary storage lowers the risk.	guidance	Search-based	DI-07, DI-28
n184	Q&A on review of electronic data / review by exception	Quality risk management may be applied to reviewing electronic data, and review by exception is acceptable when scientifically justified; exception reports are named as a common way to focus review of electronic batch records, and how much of the full eBR is reviewed can depend on the exceptions and on experience with the process. The automated system and its batch exception report must be properly tested and validated. Reviewing printouts alone risks missing data (e.g., un-investigated OOS or anomalies); reviewing the electronic raw data helps detect deletion, amendment, duplication, re-use and fabrication.	guidance	Search-based	DI-26
n185	Q&A on control of blank/template forms	Paper templates for GMP records must be handed out in a controlled, traceable way so that a record cannot be quietly re-created: the site should be able to show who received how many uniquely identified copies, when and for which area, and the copies should be made hard to duplicate without detection.	guidance	Search-based	DI-43
n186	Q&A on outsourced activities / chain of contracts	Through the chain of contracts it must be ensured that the integrity of records about manufacturing activities is secured throughout their retention period at a site holding a manufacturing/import authorisation; a recipient of supplier documents (e.g., certificates of analysis) should understand the supplier's systems and procedures for producing them and be told of significant changes.	guidance	Search-based	DI-08
n201	Q&A on risk-based review of electronic data (review by exception)	QRM principles may be used when reviewing electronic data. Review by exception is acceptable where it is scientifically justified. Exception reporting is the usual way to focus review of records such as electronic batch records, because it brings the exceptions (the most critical part of batch review) straight to the reviewer. How deeply the rest of the eBR is reviewed can depend on the exceptions found and on experience and confidence with the process. The automated system and its batch exception report output must be tested and validated so they meet GMP and business needs.	guidance	Search-based	DI-26
n202	Q&A on data criticality and the data lifecycle	Data integrity controls must match how critical the data is and how risky it is, at every stage of the lifecycle (creation, processing, checking, use in decisions, storage, disposal). Criticality depends on how the data drives decisions. Data showing compliance with critical quality attributes, such as assay results for a tablet, matter more to a release decision than peripheral records such as warehouse cleaning logs.	guidance	Search-based	DI-07
n203	Q&A on assessing data risk	Data risk assessment should look at how vulnerable data is to accidental or deliberate change, deletion or recreation. Controls that prevent unauthorised actions, or that make them more visible and detectable, count as mitigations. The assessment should cover the whole business process, including manual steps at system interfaces, not just the IT system's features.	guidance	Search-based	DI-07, DI-24
n221	Q&A on data lifecycle risks when data are used for decisions (temporary vs permanent memory)	If someone can judge whether data are acceptable before the record is committed to permanent storage, they can adjust the data until they get a passing result without the audit trail or the reviewer seeing it. EMA flags as a specific risk systems that warn users about an out-of-specification entry before the entry is saved, or that hold entries in temporary storage with little or no audit trail.	guidance	Search-based	DI-28
n222	Q&A on designing and controlling paper documentation to prevent unauthorised re-creation of GMP data	Where paper records are still used, blank templates must be controlled so no one can quietly create a fresh copy and rewrite GMP data. Templates should be uniquely identified and version-controlled, their issue should be traceable (for example through a numbered issue log), and issued copies should be reconciled. How strict the controls are depends on data risk and criticality.	guidance	Search-based	DI-43

EU GMP Annex 1

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n391	1 (Scope, para 2)	Any numeric limit, frequency or range stated in the Annex is a floor, not a target; site values may be tighter but never looser.	binding · Injectables	Checked	ST-01
n392	1 (Scope, para 3)	If a site applies Annex 1 principles to non-sterile or low-bioburden products it should record which principles are applied and show compliance with them.	binding · Injectables	Checked	ST-02
n393	2.1(iv)	Starting and packaging materials need adequate control and testing so their bioburden and endotoxin/pyrogen levels suit use.	binding · Injectables	Checked	ST-39
n394	2.3; 2.5	A site-wide Contamination Control Strategy (CCS) defines the critical control points and the controls and monitoring that manage contamination risk; it is reviewed and updated actively.	binding · Injectables	Checked	ST-02

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n395	2.5(vii)-(viii)	The CCS covers vendor approval (component sterilisation, single-use systems, critical service providers) and the transfer of critical information with outsourced parties such as contract sterilisers.	binding · Injectables	Checked	ST-40
n396	2.6	Changes to systems are assessed for their effect on the CCS both before and after they are implemented.	binding · Injectables	Checked	ST-03
n397	2.2; 2.7	Monitoring or testing alone does not prove sterility; sterility should not rest solely on a terminal process or a finished-product test.	binding · Injectables	Checked	ST-68
n398	3.1(vi)	Finishing, storage and transport should not compromise the sterile product; container integrity, contamination and degradation risks and the registered storage conditions are to be considered.	binding · Injectables	Checked	ST-71
n399	3.1(vii)	Persons certifying or releasing sterile product need appropriate access to manufacturing and quality information and knowledge of sterile manufacture and its critical quality attributes.	binding · Injectables	Checked	ST-70
n400	3.2	Every non-conformity (e.g. sterility failure, EM excursion, procedural deviation) is investigated before the batch is certified; the investigation assesses process/product impact and other potentially affected batches, and records why batches were included or excluded.	binding · Injectables	Checked	ST-71
n401	3.1(iii); 3.1(iv)	Root cause analysis of procedural, process or equipment failures should correctly identify product risk so CAPA is appropriate; risk decisions and residual-risk acceptance are documented with rationale.	binding · Injectables	Checked	ST-71
n402	4.2	Component preparation, product preparation and filling are kept technically and operationally separated to prevent mix-ups and contamination.	binding · Injectables	Checked	ST-14
n403	4.3	RABS or isolators should be considered in the CCS for aseptic operations; any alternative (open) approach needs justification.	binding · Injectables	Checked	ST-74
n404	4.4	Four cleanroom grades apply: A for high-risk critical-zone operations (filling, stopper bowl, open primary packs, aseptic connections under first air); B as background to A for aseptic work (unless isolator); C and D for less critical stages, isolator background, and preparation/filling of terminally sterilised products. Direct operator interventions in Grade A should be minimised.	binding · Injectables	Checked	ST-14
n405	4.4 (Grade B)	For aseptic preparation and filling, pressure differences of the Grade B background room are monitored continuously.	binding · Injectables	Checked	ST-17
n406	4.10; 4.11	Transfers into Grade A/B follow a one-way route, preferably sterilising items through a double-ended autoclave or depyrogenation oven/tunnel; otherwise a validated equivalent (transfer disinfection, RTP for isolators, bacteria-retentive filter for gases/liquids). Removal of items (materials, waste, EM samples) uses a separate one-way route or procedural time separation.	binding · Injectables	Checked	ST-46
n407	4.12(ii)	Only items on an approved list, assessed in transfer validation, may enter Grade A/B through airlocks or hatches; items destined for Grade A are protected while crossing Grade B; any unlisted item needs prior exceptional approval with a recorded risk assessment and a QA-approved disinfection and monitoring plan.	binding · Injectables	Checked	ST-46
n408	4.12(ii) (3rd bullet)	Items moving from lower-grade or unclassified areas to higher-grade areas are cleaned and disinfected in proportion to risk and per the CCS.	binding · Injectables	Checked	ST-46
n409	4.13	Airlock and pass-through doors should not open simultaneously; A/B airlocks use interlocks, C/D airlocks at least visual/audible warnings; a time delay between interlocked doors may be set.	binding · Injectables	Checked	ST-06
n410	4.14	Cleanrooms keep positive pressure/airflow relative to lower grades in all operating states; adjacent rooms of different grades should differ by at least 10 Pa (guidance value); containment designs may differ, but air flowing into a critical zone should come from an equal or higher grade.	binding · Injectables	Checked	ST-17
n411	4.15	Airflow visualisation (smoke) studies at rest and in operation, including simulated interventions, demonstrate no ingress from lower to higher grades; videos are retained and results feed the EM programme.	binding · Injectables	Checked	ST-17
n412	4.16	Critical pressure differentials are continuously monitored and recorded with an instant alarm for air-supply failure or DP below limit; alarms may not be overridden without assessment; a response procedure exists; alarm delays are justified in the CCS; non-critical DPs are recorded at regular intervals.	binding · Injectables	Checked	ST-17
n413	4.18; 4.19	Isolators and RABS separate Grade A from the surrounding room; item entry/removal risks during processing are minimised using high-capability transfer technologies; open isolators and RABS give first-air protection with unidirectional flow over exposed product; closed isolators may be turbulent only for simple operations; negative-pressure isolators only where containment is essential.	binding · Injectables	Checked	ST-74
n414	4.20	Minimum background grade: open isolator generally Grade C, closed isolator Grade D (decided by CCS risk assessment covering bio-decontamination, automation, glove manipulations, glove/barrier breach, transfers, door-open set-up); RABS for aseptic processing at least Grade B with airflow studies during interventions and door openings.	binding · Injectables	Checked	ST-74
n415	4.21	Glove material suitability is demonstrated and the glove replacement frequency is defined in the CCS.	binding · Injectables	Checked	ST-75
n416	4.21(i)(a)	Isolator glove leak testing uses a suitable method at defined intervals, generally at minimum when each batch or campaign starts and when it ends (more often if campaign length requires); gloves are visually inspected at each use and after any manipulation that may affect integrity; for small manual batches the session start/end may be used.	binding · Injectables	Checked	ST-75
n417	4.21(i)(b)	Integrity/leak testing of the isolator enclosure is performed at defined intervals.	binding · Injectables	Checked	ST-75
n418	4.21(ii)	RABS Grade A gloves are sterilised before fitting and sterilised or bio-decontaminated by a validated method before each campaign; if exposed to the background during operation they are disinfected by an approved method after each exposure; they are visually checked at each use and integrity-tested periodically.	binding · Injectables	Checked	ST-75
n419	4.22; 4.22(i)	Cleaning and bio-decontamination of barrier systems are defined and controlled; cleaning comes before bio-decontamination (declared essential, see AN1-198); agents should be shown not to harm the product. Isolator interior bio-decontamination is automated, validated, run within defined cycle parameters with a sporicidal agent in gaseous/vapour form, with gloves extended and fingers separated.	binding · Injectables	Checked	ST-76
n420	4.22(ii)	RABS interiors receive routine sporicidal disinfection by a validated method covering all interior surfaces.	binding · Injectables	Checked	ST-76
n421	4.23; 4.24; 4.25	Cleanrooms and clean-air devices (UDAF, RABS, isolators) are qualified per Annex 15 for at-rest and operational states; qualification/classification is kept distinct from routine EM.	binding · Injectables	Checked	ST-14
n422	4.26	Classification is scheduled so it does not affect product; initial classification in simulated operations; reclassification in simulated operations or during an APS.	binding · Injectables	Checked	ST-14
n423	4.28; 4.29	Grade A/B classification should include extra locations at critical points (point of fill, closure feeder bowls) chosen by documented risk assessment; at-rest limits should be regained after operations and line clearance within a clean-up period set in qualification (guidance under 20 minutes), which procedures should respect when cleanliness is disrupted.	binding · Injectables	Checked	ST-16

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n424	4.32	Cleanrooms and clean-air equipment are requalified periodically (Grade A/B at most every 6 months, C/D at most every 12 months) and after remedial actions or significant changes (air interruption, HVAC or design changes, final filter replacement).	binding · Injectables	Checked	ST-14
n425	4.33	Cleanrooms are cleaned then disinfected per a written programme; residues are removed; more than one disinfectant type with different modes of action is used; a sporicide is applied periodically; effectiveness is monitored including shifts in flora or resistant organisms.	binding · Injectables	Checked	ST-15
n426	4.34; 4.35	Disinfection is validated, including in-use expiry of prepared solutions. Disinfectants/detergents for Grade A/B are sterile before use (C/D where CCS requires); in-house dilutions are prepared to avoid contamination, monitored microbiologically, kept in cleaned (sterilised where applicable) containers and stored only for the defined period; ready-made agents may rely on CoA/CoC after vendor qualification.	binding · Injectables	Checked	ST-15
n427	4.36	Fumigation or vapour disinfection (e.g. VHP) of rooms requires a validated, understood agent and dispersion system.	binding · Injectables	Checked	ST-15
n428	5.2	Process and equipment alarms are acknowledged and trended; review frequency follows criticality, with critical alarms reviewed immediately.	binding · Injectables	Checked	ST-06
n429	5.3	Maintenance inside a cleanroom that cannot keep cleanliness/asepsis needs restricted access, defined work protocols, and consideration of extra cleaning, disinfection and EM; equipment sterilisation is done after full reassembly where possible.	binding · Injectables	Checked	ST-16
n430	5.4	Equipment cleaning is validated to remove residues that impair disinfectants and to minimise chemical, microbial and particulate contamination before disinfection.	binding · Injectables	Checked	ST-38
n431	5.5	In aseptic processing both direct product-contact parts (e.g. filling needles, pumps) and indirect contact parts whose sterility is critical (e.g. stopper bowls, guides, sterilised components) are sterilised.	binding · Injectables	Checked	ST-42
n432	5.6	Sterilisers, HVAC (including filtration) and water systems are qualified, monitored and maintained on plan; return to use after maintenance is formally approved.	binding · Injectables	Checked	ST-16
n433	5.7	Before unplanned maintenance on sterility-critical equipment, the potential impact on product sterility is assessed and recorded.	binding · Injectables	Checked	ST-16
n434	5.8; 5.9	Conveyors should not cross a Grade A/B partition to a lower grade unless continuously sterilised (e.g. tunnel). Particle counters and tubing are qualified (tubing typically no longer than 1 m, minimal bends, isokinetic heads in UDAF, close to critical point).	binding · Injectables	Checked	ST-05
n435	6.1; 6.2; 6.3; 6.4	Utility controls are proportionate to risk, documented in the CCS; high-risk utilities include those contacting product, materials becoming product, or product-contact surfaces (e.g. rinse water, process gases, sterilising steam); critical parameters of high-risk utilities are trended.	binding · Injectables	Checked	ST-25
n436	6.7; 6.8; 6.9; 6.10	Water systems are designed and qualified (including seasonal variation) to prevent microbial, particulate and endotoxin contamination; water meets the pharmacopoeial monograph; flow stays turbulent and is monitored; WFI is made by distillation or an equivalent process and stored/circulated to minimise growth (e.g. continuous circulation above 70 C).	binding · Injectables	Checked	ST-26
n437	6.11	Hydrophobic bacteria-retentive vent filters on WFI tanks should not contaminate; their integrity is tested before installation and after use, and condensation on them is prevented (e.g. heating).	binding · Injectables	Checked	ST-26
n438	6.12	Water systems are sanitised/regenerated on a schedule and after out-of-limit results; chemical sanitisation is followed by validated rinsing; water is tested afterwards; chemical results are approved before return to use, and microbiological/endotoxin results should be within specification and approved before batches made with that water are considered for certification.	binding · Injectables	Checked	ST-26
n439	6.13; 6.14	Water is monitored chemically and microbiologically per a CCS-based plan covering all outlets/points of use, with at least one representative sample every day of water used in manufacture; alert levels derive from qualification data and are reassessed; alert excursions are reviewed for isolated vs trend; every action-limit excursion is investigated for root cause and impact on products and processes.	binding · Injectables	Checked	ST-26
n440	6.15	WFI systems include continuous monitoring such as TOC and conductivity, with risk-based sensor locations.	binding · Injectables	Checked	ST-26
n441	6.16; 6.17	Pure steam used as a direct sterilant comes from purified feed water, has no contaminating additives, its condensate meets the WFI monograph (microbial testing not mandatory), is sampled regularly, and is periodically checked for non-condensable gases, dryness and superheat.	binding · Injectables	Checked	ST-27
n442	6.18	Gases directly contacting product or primary container surfaces have specified chemical, particulate, microbial, oil and water quality meeting pharmacopoeia or product requirements.	binding · Injectables	Checked	ST-28
n443	6.19	Process gases for aseptic use pass a sterilising-grade filter (max nominal 0.22 um) at point of use; filters used on a batch basis (e.g. overlay gas) or as product vessel vents are integrity tested and the results reviewed during batch certification; tubing downstream of the final filter is sterilised; gas is periodically monitored microbially at point of use.	binding · Injectables	Checked	ST-28
n444	6.20; 6.21; 6.22	Backflow from vacuum/pressure systems into product is prevented; hydraulic, heating and cooling equipment is outside the filling room where possible, spillage is contained, and leaks posing product risk are detectable.	binding · Injectables	Checked	ST-06
n445	7.1; 7.2	Enough qualified, trained and experienced staff are available for sterile manufacture and the specific technologies used; only the minimum staff are present in cleanrooms, and the maximum operator number per cleanroom is set, documented and considered in qualification and APS.	binding · Injectables	Checked	ST-12
n446	7.3	Everyone who enters cleanrooms (including cleaners, maintenance and monitoring staff) receives regular training, gowning qualification and assessment covering microbiology basics, hygiene, cleanroom practice, contamination control and (for Grade B entry/Grade A intervention) aseptic technique and patient-safety consequences; depth depends on role criticality.	binding · Injectables	Checked	ST-07
n447	7.4	Grade A/B staff are trained in aseptic gowning and behaviour; gowning compliance is assessed initially and reassessed at least annually, visually and microbiologically (gloved fingers, forearms, chest, hood/mask/forehead); unsupervised A/B access is limited to people who passed gowning assessment and took part in a successful APS.	binding · Injectables	Checked	ST-08
n448	7.5	Unqualified people should not enter Grade B or Grade A in operation; exceptional entry follows a written procedure, with an authorised person supervising and assessing the impact on area cleanliness; such access is assessed and recorded under the PQS.	binding · Injectables	Checked	ST-10
n449	7.6	A disqualification system removes operators from cleanroom work or unsupervised entry based on ongoing assessment, adverse personnel-monitoring trends, or implication in a failed APS; retraining and requalification are completed before any further aseptic involvement, and for Grade B entry/Grade A intervention requalification should consider participation in a successful APS.	binding · Injectables	Checked	ST-09
n450	7.7; 7.8	Staff report health conditions that could shed abnormal contaminants, with procedures defining actions; staff who have worked with human/animal tissue or other microbial cultures, or on other quality-risk activities, only enter clean areas after documented, effective decontamination/entry procedures.	binding · Injectables	Checked	ST-10
n451	7.9	Personal items (watches, jewellery, make-up, phones) are not allowed in clean areas; manufacturer-supplied electronic devices dedicated to cleanroom use (e.g. tablets) are acceptable if designed for cleaning/disinfection suitable for the grade, and their use and disinfection are covered by the CCS.	binding · Injectables	Checked	ST-13

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n452	7.10; 7.11; 7.12; 7.13; 7.14	Gowning and hand washing follow written procedures; garments suit the grade (Grade B/A: sterile suit, hood enclosing all hair, mask, eye cover, sterile gloves double-layered over sleeves, sterile over-boots; Grade C: covered hair, high-neck suit gathered at wrists, disinfected shoes; Grade D: covered hair, general protective suit; extra items in C/D where CCS requires); garments are visually checked for cleanliness and integrity before and after donning and on exit; sterilised garments/eye covers are checked for sterilisation, hold time and intact packaging; reusable garments are replaced when damaged or at the qualified frequency.	binding · Injectables	Checked	ST-11
n453	7.15; 7.16	Each Grade A/B entry uses fresh sterilised garments (including eye covers and masks) of the right size; the maximum wear time of a sterile gown during a shift is defined in garment qualification; gloves are disinfected regularly during work and damaged garments/gloves are changed immediately.	binding · Injectables	Checked	ST-11
n454	7.17	Reusable cleanroom clothing is laundered in a segregated, qualified process, inspected after washing, and garment management defines a maximum number of laundry and sterilisation cycles.	binding · Injectables	Checked	ST-11
n455	7.18	Non-critical activities in clean areas are minimised, especially during aseptic work; movements are slow and controlled; aseptic technique is maintained at all times; movement near the critical zone is restricted and first-air paths are not obstructed.	binding · Injectables	Checked	ST-13
n456	8.1; 8.2	For terminally sterilised products, components and materials are prepared in at least Grade D; if the product is at high or unusual contamination risk (supports growth, long hold before filling, not mostly in closed vessels) preparation is in at least Grade C, as is preparation of ointments, creams, suspensions and emulsions before terminal sterilisation; primary containers/components are cleaned by validated processes controlling particles, endotoxin and bioburden.	binding · Injectables	Checked	ST-60
n457	8.3; 8.4	Filling of products for terminal sterilisation is done in at least Grade C; where the CCS shows unusual environmental risk (slow filling, wide-necked containers, exposure beyond a few seconds before closing) filling is in Grade A with at least Grade C background.	binding · Injectables	Checked	ST-60
n458	8.5	For terminally sterilised products, bulk solution is passed where possible through a microorganism-retaining filter to lower bioburden and particles before filling, and a maximum permissible time between preparation and filling is set.	binding · Injectables	Checked	ST-60
n459	8.7; 8.8	The aseptic process is clearly defined, its risks assessed and controlled, with CCS-defined acceptance criteria, monitoring and effectiveness review and formally documented residual risks; contamination precautions apply from preparing the aseptic environment through all stages until the product is sealed; particle- and fibre-generating materials are minimised in cleanrooms.	binding · Injectables	Checked	ST-02
n460	8.9; 8.15	Barrier systems, robotics and automation (e.g. dry heat tunnels, automated lyophiliser loading, SIP) should be considered to reduce critical Grade A interventions; aseptic manipulations are minimised by pre-assembled, pre-sterilised equipment and sterilisation in place where feasible.	binding · Injectables	Checked	ST-02
n461	8.11	Where the final formulation cannot be filtered: all product- and component-contact equipment is sterilised before use, all raw materials or intermediates are sterilised and added aseptically, and bulk solutions or intermediates are sterilised.	binding · Injectables	Checked	ST-40
n462	8.12; 8.13	Unwrapping, assembly and preparation of sterilised items with direct or indirect product contact, line set-up and filling are aseptic operations in Grade A with Grade B background (isolator background per 4.20); non-filtered, non-terminally-sterilised exposed ointments, creams, suspensions and emulsions are prepared and filled in Grade A/B.	binding · Injectables	Checked	ST-42
n463	8.14	Aseptic connections are made in Grade A with Grade B background unless they are subsequently sterilised in place or made with intrinsic sterile connection devices designed to limit environmental contamination; aseptic connections are assessed and their effectiveness verified.	binding · Injectables	Checked	ST-42
n464	8.16	An authorised list of allowed, qualified interventions (inherent and corrective) exists, each designed to minimise contamination risk (airflow, critical surfaces, product), using engineering solutions and sterile tools, first evaluated by risk management and APS and kept current.	binding · Injectables	Checked	ST-34
n465	8.16	Non-qualified interventions are used only exceptionally, with the quality unit's authorisation and risk consideration; they are risk-assessed, recorded, fully investigated under the PQS, thoroughly assessed by QA and taken into account at batch disposition.	binding · Injectables	Checked	ST-36
n466	8.17; 9.34	Every intervention and line stoppage is recorded in the batch record with time, duration and the operators involved.	binding · Injectables	Checked	ST-35
n467	8.18(i); 8.18(ii); 8.18(iii)	Validated maximum times apply to: holding between cleaning, drying and sterilisation of equipment/components/containers; holding of sterilised equipment/components/containers before use and during filling/assembly; holding a decontaminated RABS/isolator before use.	binding · Injectables	Checked	ST-37
n468	8.18(iv); 8.18(v)	A validated maximum time applies from start of product preparation to its sterilisation or sterilising filtration and on to the end of aseptic filling, set per product considering composition and storage method; sterilised product also has a maximum hold before filling.	binding · Injectables	Checked	ST-37
n469	8.18(vi); 8.18(vii); 9.36(iii); 9.36(viii)	Aseptic processing time and filling time are each limited to a validated maximum; the APS should be long enough to challenge the process, and should cover maximum permitted holding times of exposed sterile product and equipment.	binding · Injectables	Checked	ST-37
n470	8.19	Aseptic operations, including APS, are regularly observed by people with specific aseptic expertise to verify correct performance and operator behaviour and to correct poor practice.	binding · Injectables	Checked	ST-13
n471	8.20; 8.21	Open primary containers stay under Grade A with the background appropriate to the technology; final containers are closed by validated methods.	binding · Injectables	Checked	ST-55
n472	8.22	For fusion-sealed containers (BFS, FFS, SVP/LVP bags, glass or plastic ampoules), seal-critical parameters are determined, controlled and monitored during operation; glass ampoules, BFS units and fusion-sealed containers of 100 ml or less undergo 100 percent integrity testing by validated methods; over 100 ml, reduced sampling may be accepted only with scientific justification and demonstrated process consistency; visual inspection is not an acceptable integrity test.	binding · Injectables	Checked	ST-64
n473	8.23	Non-fusion closures (e.g. stoppered and crimped vials, syringes) are integrity-tested on samples by validated methods under a scientifically justified sampling plan based on container-closure knowledge, supplier management and component specifications.	binding · Injectables	Checked	ST-64
n474	8.24; 8.25	Containers sealed under vacuum are tested for vacuum retention after a predetermined period before certification and during shelf life; CCI validation considers transport stresses such as decompression and extreme temperatures.	binding · Injectables	Checked	ST-64
n475	8.26; 8.27; 8.29	Particle-generating crimping is separated with extraction; capping of aseptically filled vials is either an aseptic process with sterilised caps or a clean process outside the aseptic area in which vials are protected by Grade A up to leaving the aseptic area and then by Grade A air supply (background at least Grade D) until crimped; manual capping is done in Grade A (isolator or Grade A in Grade B); interventions at capping avoid direct vial contact.	binding · Injectables	Checked	ST-57
n476	8.28	When capping is a clean process under Grade A air, vials with missing or displaced stoppers are rejected before capping using qualified automated stopper-height detection.	binding · Injectables	Checked	ST-57
n477	8.30	Every filled parenteral container is individually inspected for extraneous contamination and defects; defect classes and criticality are set in qualification using patient risk and route of administration; defect types are categorised and batch performance analysed; batches with unusual defect levels versus routine/trend data are investigated; a maintained defect library supports training.	binding · Injectables	Checked	ST-62

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n478	8.30 (last sentences)	Critical defects should not be found in later sampling/inspection of accepted containers; finding one indicates possible failure of the original inspection and triggers an investigation.	binding · Injectables	Checked	ST-63
n479	8.31	Manual inspection runs under controlled illumination and background at a controlled, qualified rate; inspectors are qualified at least annually (with their usual corrective lenses) using defect-library sets and worst-case conditions (time, conveyor speed, container size, fatigue), including eyesight checks; distractions are minimised and frequent breaks taken.	binding · Injectables	Checked	ST-62
n480	8.32	Automated inspection is validated to detect known defects at least as well as manual inspection, and is challenged with representative defect sets before start-up and at regular intervals during the batch.	binding · Injectables	Checked	ST-62
n481	8.33	Inspection results are recorded; defect types and numbers are trended and reject levels trended statistically; adverse trends trigger investigation including impact on product already on the market.	binding · Injectables	Checked	ST-63
n482	8.34; 8.37	Terminal sterilisation is preferred where possible; otherwise post-aseptic terminal heat treatment combined with aseptic processing should be considered; non-pharmacopoeial methods or non-aqueous products need particular attention, with heat the method of choice.	binding · Injectables	Checked	ST-02
n483	8.35; 8.36; 8.40	Sterilisation equipment and cycles are chosen on scientific data; all parameters are defined and critical ones controlled, monitored and recorded; all sterilisation processes are validated (including product composition, storage and maximum time from preparation start to sterilisation) and routine operating parameters and load patterns are followed.	binding · Injectables	Checked	ST-43
n484	8.38; 8.39	Validated loading patterns exist for every sterilisation process (considering maximum and minimum loads) and are periodically revalidated; heat sterilisation cycles are revalidated at least annually for worst-case load patterns, others at a CCS-justified frequency.	binding · Injectables	Checked	ST-43
n485	8.41	Mechanisms detect any sterilisation cycle not conforming to validated parameters; any failed cycle, or one deviating from the validated process (e.g. longer or shorter phases), is investigated.	binding · Injectables	Checked	ST-44
n486	8.42; 8.43	BIs may support validation; they are stored and used per the supplier's instructions; where BIs support validation or routine monitoring (e.g. EO), positive controls are tested for each cycle; BI handling should not contaminate other processes; BI results alone should not override critical parameters; BI suppliers are qualified, transport/storage controlled, and each new BI lot is verified for population, purity and identity (D/Z values may come from the supplier certificate).	binding · Injectables	Checked	ST-44
n487	8.44	Sterilised and non-sterilised products, equipment and components are clearly distinguishable; carriers such as baskets and trays are labelled or electronically tracked with product name, batch number and sterilised/not-sterilised status; process indicators (autoclave tape, irradiation dots) only show exposure, not sterility.	binding · Injectables	Checked	ST-45
n488	8.45	A sterilisation record exists for each run, each cycle has a unique identifier, and conformity of the records is reviewed and approved as part of batch certification/release.	binding · Injectables	Checked	ST-72
n489	8.46; 8.48	Items are sterilised by validated methods and protected against recontamination; if not used immediately they are held in sealed packaging for no longer than an established maximum hold time; sealing precedes sterilisation; packaging is qualified and its sealing validated, including maximum time before sterilisation and sterile shelf life; the integrity of each item's sterile barrier is checked before use. Multi-layer sterile packs may be stored outside cleanrooms if justified.	binding · Injectables	Checked	ST-45
n490	8.47; 8.49	Sealed sterilised items enter Grade A via validated methods (airlocks, pass-throughs, RTP) with disinfection of the outer packaging shown to be effective; non-product-contact items needed for aseptic work that cannot be sterilised undergo validated disinfection and transfer, are protected afterwards, and are included in the EM programme.	binding · Injectables	Checked	ST-46
n491	8.50; 8.51	Each heat sterilisation cycle is recorded (electronically or hardcopy) with accurate, precise equipment; control and monitoring have safeguards/redundancy (e.g. duplex probes on independent control and monitoring systems) to detect non-conforming cycles and abort or fail them; probe positions are set and verified in validation.	binding · Injectables	Checked	ST-44
n492	8.52; 8.53	Sterilising time starts only when the whole load has reached the required temperature (for load-probe-controlled cycles the load probe is within range before the cycle starts); after the high-temperature phase the load is protected from recontamination during cooling and any cooling fluid or gas contacting product or sterilised items is sterilised.	binding · Injectables	Checked	ST-44
n493	8.54	Where parametric release is authorised, a robust, periodically reviewed system covers lifecycle validation and routine process monitoring (see Annex 17).	binding · Injectables	Checked	ST-61
n494	8.55; 8.56; 8.57	Moist-heat sterilisation covers steam and superheated-water (cascade/immersion) systems. Items other than sealed product containers are dry and packed in a barrier allowing air removal and steam penetration; load dryness is confirmed visually as part of cycle acceptance; porous cycles are monitored and recorded by time, temperature and pressure; each item is inspected on unloading for damage, packaging integrity and moisture, and unfit items are removed and investigated.	binding · Injectables	Checked	ST-44
n495	8.58; 8.59	Pre-vacuum autoclaves record chamber-drain temperature throughout sterilisation (load probes optional but control tied to load validation); SIP records temperature at condensate drains; porous-cycle acceptance covers equilibration time, exposure time, pressure-temperature correlation and min/max exposure temperature; fluid cycles use temperature, time and/or F0; critical parameters have defined limits and tolerances in routine cycle acceptance.	binding · Injectables	Checked	ST-44
n496	8.60	Sterilisers with a vacuum phase, or returning to sub-ambient pressure after sterilisation, are leak-tested periodically, normally weekly.	binding · Injectables	Checked	ST-43
n497	8.61	Air removal before and during sterilisation should be assured for processes with air purging (porous loads, lyophiliser chambers): autoclaves run an air-removal test cycle (normally daily) or use an air detector; loads are designed for air removal and free drainage.	binding · Injectables	Checked	ST-43
n498	8.62; 8.64; 8.65	Terminal sterilisation of non-rigid containers (BFS, FFS, bags) avoids distortion through cycle design (pressure, heating/cooling rates, loading); superheated-water fluid-load autoclaves should reach all contact points, are routinely checked for blocked nozzles and debris-free drains, and routine probes correlate with worst-case positions from load mapping.	binding · Injectables	Checked	ST-61
n499	8.63	SIP systems (fixed pipework, vessels, lyophiliser chambers) are validated so all parts are treated; routine monitoring of temperature, pressure and time occurs at locations correlated with the slowest-heating points; after SIP the system stays integral and, where needed, under positive pressure or behind a sterilising vent filter until use.	binding · Injectables	Checked	ST-48
n500	8.66; 8.67	Dry-heat sterilisation/depyrogenation delivers reproducible lethality/endotoxin removal within limits; tunnels maintain pressure differentials and airflow protecting the Grade A zone, supplied through at least HEPA filters integrity-tested periodically (at least twice yearly, text: 'biannually'); tunnel parts touching sterilised components are sterilised or disinfected; critical parameters include belt speed/dwell time, min and max temperatures, heat penetration, heat distribution and airflow/DP profiles.	binding · Injectables	Checked	ST-47
n501	8.68; 8.69	Thermal depyrogenation is validated to give a suitable Fh and at least a 3 log10 endotoxin reduction (which also covers sterilisation); endotoxin-spiked validation containers are representative and fully reconciled, with quantification and recovery demonstrated.	binding · Injectables	Checked	ST-47
n502	8.70	Dry-heat ovens remain at positive pressure to lower-grade areas through sterilisation and post-sterilisation hold (unless packaging integrity is kept), with HEPA-filtered air; critical parameters include temperature, exposure time, chamber overpressure, air speed and quality, heat penetration and distribution, and load pattern including min/max loads.	binding · Injectables	Checked	ST-47
n503	8.71; 8.72	Radiation sterilisation is mainly for heat-sensitive items (ionising radiation per Annex 12), with validation considering product and package density variation; ultraviolet irradiation is not an acceptable sterilisation method.	binding · Injectables	Checked	ST-49

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n504	8.73; 8.74; 8.75	Ethylene oxide is used only when no other method is practicable; validation shows no product damage and that degassing time/conditions reduce residual EO and reaction products to defined limits; direct gas-to-organism contact is essential, so occluded organisms and packaging nature/porosity/quantity matter; loads are pre-conditioned to the required humidity and temperature (using suitable-quality steam) with conditioning time balanced against minimising time before sterilisation.	binding · Injectables	Checked	ST-49
n505	8.76; 8.77; 8.42	Every EO cycle is monitored with suitable BIs in adequate number at validated worst-case locations (with positive controls per cycle); critical parameters considered include EO concentration, pressure, amount of gas, RH, temperature and exposure time.	binding · Injectables	Checked	ST-49
n506	8.78	After EO sterilisation the load is aerated (in chamber, aeration chamber or room) so EO and reaction products desorb to predetermined levels, the aeration phase being validated.	binding · Injectables	Checked	ST-49
n507	8.79; 8.135	Products that cannot be sterilised in their final container are sterile-filtered through a sterile, validated sterilising-grade filter (nominal pore size no more than 0.22 um) and aseptically filled into pre-sterilised containers; the chosen filter should suit the product and match the filter registered in the marketing authorisation.	binding · Injectables	Checked	ST-50
n508	8.80; 8.81; 8.82	Several filtration points (bioburden-reduction prefilters or additional sterilising-grade filters) may be used upstream; a second sterile sterilising-grade filter close to the point of fill should be considered in the CCS; the filter train is justified against product CQAs (fibres, particles, impurities, adsorption, extractables); the system operates within validated parameters, keeps the filtrate sterile, minimises aseptic connections after the final filter, allows cleaning/SIP and permits in-place integrity testing of the final filter before and after filtration, preferably closed.	binding · Injectables	Checked	ST-51
n509	8.83; 8.84; 8.85	Sterile filtration is validated per pharmacopoeia under worst case (grouping justified), with bacterial retention using the product or a justified surrogate and justified challenge organism; validation establishes the integrity-test wetting fluid and specification (with precautions if a non-product fluid is used in situ) and process conditions: pre-filtration hold and bioburden effect, conditioning, maximum filtration/contact time, maximum pressure, flow rate, maximum volume, temperature, and time to filter a known volume with its differential pressure.	binding · Injectables	Checked	ST-51
n510	8.86	Routine controls ensure filtration stays within validated parameters; critical parameter results are included in the batch record, including at least the minimum time to filter a known volume and the differential pressure across the filter; significant differences are documented and investigated.	binding · Injectables	Checked	ST-51
n511	8.87 (PUPSIT)	The sterilised filter assembly is integrity tested before use and after sterilisation (PUPSIT) to detect damage from preparation. Where PUPSIT is not feasible (e.g. very small volumes), an alternative is allowed only after a thorough risk assessment and mitigating controls covering filter sterilisation process control, supply chain (contract steriliser, transport, protective packaging) and product/process knowledge (particle burden that could mask a non-integral filter in post-use testing, upstream clarification).	binding · Injectables	Checked	ST-52
n512	8.87 (post-use)	Every sterilising-grade filter used on a fluid undergoes a validated, non-destructive integrity test after use and before removal from its housing; the test correlates with bacterial retention (e.g. bubble point, diffusive flow, water intrusion, pressure hold).	binding · Injectables	Checked	ST-52
n513	8.88; 8.89; 8.90	Critical sterile gas and air vent filters linked directly to product sterility are integrity-tested after use while still in their housing; non-critical vent filters are confirmed integral at defined intervals; long-duration gas filters are tested at installation and before replacement, with a risk-based maximum duration of use (uses, sterilisation cycles) specified and monitored; gas filters should not become wet.	binding · Injectables	Checked	ST-53
n514	8.91; 8.92	If sterility was validated for a multi-filter system, all filters in it should pass post-use integrity testing. In a redundant set-up (second filter as backup only), the primary is tested post-use; if it passes, testing the backup is unnecessary; if it fails, the backup is tested post-use and an investigation and risk assessment determine why the primary failed.	binding · Injectables	Checked	ST-52
n515	8.93	Bioburden samples are taken from the bulk immediately before the final sterile filtration (before the first filter in a redundant set-up), using sampling systems that do not introduce contamination.	binding · Injectables	Checked	ST-54
n516	8.94; 8.95	Liquid sterilising-grade filters are discarded after one batch and not kept in continuous use beyond one working day unless validated; for CCS-justified, validated campaign use, risks of the duration are assessed, the maximum validated duration documented and enforced with records, and contaminated or defective filters removed.	binding · Injectables	Checked	ST-53
n517	8.96; 8.97	FFS machines meet the environment rules for terminally sterilised (8.3/8.4) or aseptic (8.10) manufacture as applicable; packaging films are controlled against defined specifications (thickness, strength, microbial and particulate contamination, integrity, artwork) with film bioburden/endotoxin sampling defined in the PQS and CCS.	binding · Injectables	Checked	ST-78
n518	8.98; 8.99; 8.100	FFS set-up, filling, sealing and cutting are understood so CPPs are validated, controlled and monitored; product-contact gases (inflation, overlay) are filtered close to point of use and periodically verified; CCS-aligned controls cover critical-zone boundaries, machine and background EM, gowning, integrity testing of filling lines and filters, batch/campaign duration, film decontamination/sterilisation, CIP/SIP, and machine settings and alarm management.	binding · Injectables	Checked	ST-78
n519	8.101; 8.102	FFS critical parameters set in qualification include package dimensions/cutting, forming temperatures (incl. pre-heat and cooling), times and pressures, sealing temperature and its uniformity, sealing time and pressure, environmental and product temperature, batch-specific seal strength and uniformity tests, fill volume/speed/uniformity, printing/embossing settings that protect integrity, and filled-container integrity test methods; they are verified, monitored and recorded during production.	binding · Injectables	Checked	ST-78
n520	8.103; 8.104	FFS procedures describe detecting and fixing forming/sealing problems; rejected units and sealing issues are recorded and investigated; risk-based maintenance includes inspection plans for seal-critical tooling, with quality-relevant findings documented and investigated.	binding · Injectables	Checked	ST-78
n521	8.105; 8.106	BFS for terminally sterilised products is installed in at least Grade D with point-of-fill conditions per 8.3/8.4. For aseptic BFS: shuttle machines need Grade A at the critical zones of parison extrusion, blow-moulding and sealing and a Grade A filling environment at rest and in operation; rotary machines need Grade A inside the parison; the machine sits in at least Grade C provided operators wear Grade A/B clothing, with risk-based personnel monitoring.	binding · Injectables	Checked	ST-77
n522	8.107; 8.108; 8.109	In-operation total particle monitoring of BFS critical zones is not expected (design data should show Grade A would be met); viable EM is risk-based per section 9 and covers all critical processing from equipment assembly onward (sampling the rotary critical zone may be impossible); the EM programme accounts for moving parts, complex airflow, heat output, filter configuration and integrity, and cooling system integrity.	binding · Injectables	Checked	ST-77
n523	8.110; 8.111; 8.112	BFS air/gases touching critical container surfaces during extrusion, forming or sealing are filtered and periodically verified; polymer granulate storage, sampling and distribution prevent particulate/microbial contamination; the extrusion system's sterility-assurance capability is validated and raw polymer bioburden/endotoxin sampling is defined in the PQS and CCS.	binding · Injectables	Checked	ST-77
n524	8.113	BFS interventions that require stopping filling and/or extrusion, moulding and sealing, and any required re-sterilisation of the filling machine, are specified in the filling SOP and simulated in the APS where relevant.	binding · Injectables	Checked	ST-77
n525	8.114; 8.115; 8.116	BFS controls align with the CCS (critical zone, EM, gowning, integrity of filling lines/filters, batch/campaign duration, polymer and extrusion temperature control, CIP/SIP, settings and alarms); BFS CPPs include CIP/SIP of product lines and mandrels, extrusion temperature/speed/throat settings (parison thickness), mould temperatures and cooling rate, preparation/sterilisation of added components, EM and cleaning of critical zones, batch-specific wall-thickness tests, fill volumes/speeds/uniformity, printing/embossing settings, 100 percent CCI methods and parameters, and flash-removal cutter settings; they are verified, monitored and recorded in production.	binding · Injectables	Checked	ST-77
n526	8.117; 8.118; 8.119; 8.120	BFS blowing/forming/sealing issues are detected and fixed per procedure with rejects and sealing issues recorded and investigated; added components (e.g. LVP caps) are decontaminated and added by a clean controlled process	binding · Injectables	Checked	ST-77

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		(aseptic: in Grade A using pre-sterilised components; terminal: validation covers all product pathways including unwetted areas) with validated sealing tests; maintenance covers sealing/integrity/sterility-critical items; moulds are critical equipment and any change triggers a container-integrity assessment and, if indicated, validation.			
n527	8.121	Lyophilisation is a critical step: every activity that could affect sterility of the product or material is to be treated as an extension of aseptic processing of the sterilised product (the Annex uses 'need to', not 'should'). The accompanying design expectation, preventing contamination from filling to the end of lyophilisation with CCS-defined controls, is a 'should'.	binding · Injectables	Checked	ST-59
n528	8.122	Sterilisation of the lyophiliser and associated items (trays, vial support rings) is validated; the hold time between sterilisation and use is challenged in the APS; the lyophiliser is sterilised regularly per design and re-sterilised after maintenance or cleaning; sterilised equipment is protected afterwards.	binding · Injectables	Checked	ST-58
n529	8.123 (applies from 25 Aug 2024)	Lyophilisers and their transfer/loading/unloading areas minimise operator intervention; sterilisation frequency is based on design and contamination risk; lyophilisers loaded or unloaded manually without barrier separation are sterilised before each load; for automated loading/unloading or closed-barrier protection, the frequency is justified and documented in the CCS.	binding · Injectables	Checked	ST-58
n530	8.124	Lyophiliser integrity is maintained after sterilisation and during the cycle; the filter that maintains integrity is sterilised before each use and its integrity test results form part of batch certification; the chamber vacuum/leak test frequency is documented and the maximum permitted air leak rate is specified, with a check at the start of each cycle.	binding · Injectables	Checked	ST-58
n531	8.125	Lyophilisation trays are checked regularly for distortion or damage.	binding · Injectables	Checked	ST-58
n532	8.126	Loading/unloading design specifies and documents the loading pattern; partially closed containers move to the lyophiliser under Grade A at all times with minimal direct intervention (conveyors, clean-air transfer carts, portable UDAF) or in validated closed trays/boxes closed in Grade A and not opened in Grade B; transport and venting should not disturb airflow; unsealed containers stay in Grade A, normally separated from operators by barriers; if stoppers are not fully seated before opening the chamber, product stays in Grade A afterwards; loading/unloading utensils are sterile.	binding · Injectables	Checked	ST-59
n533	8.127; 8.128; 8.129	Closed systems are designed to reduce manual manipulations; product-contact surface sterility of aseptic closed systems is critical (see AN1-200); connections to the sterile pathway after the final filter are designed to be made aseptically (e.g. intrinsic sterile connectors); the integrity of connection components is assured as defined in the CCS, system integrity tests are considered where sterility could be compromised, and supplier assessment covers failure modes.	binding · Injectables	Checked	ST-79
n534	8.130	A closed system's background grade depends on its design and process: for aseptic processing where integrity could be compromised it is located in Grade A; a lower grade is acceptable only if integrity is shown at every use (e.g. pressure testing/monitoring); transfers between classified areas are assessed; if opened (e.g. maintenance) this happens in a suitable classified area (Grade C for terminal, Grade A for aseptic) or is followed by extra cleaning, disinfection and (aseptic) sterilisation.	binding · Injectables	Checked	ST-79
n535	8.131; 8.132	SUS are designed to reduce manipulations and manual complexity; CCS assesses SUS-specific risks: product-surface interaction (adsorption, extractables/leachables), fragility, more manual operations and connections, assembly complexity, pre/post-use filter integrity testing, holes and leaks, compromise when opening outer packaging, and particle contamination.	binding · Injectables	Checked	ST-79
n536	8.133; 8.134	SUS sterilisation is validated without harming performance; supplier assessment (including sterilisation) is critical; for sterile SUS, sterility assurance is verified when qualifying the supplier, and each delivered unit's sterilisation evidence is checked at goods receipt.	binding · Injectables	Checked	ST-79
n537	8.135; 8.136	Adsorption and reactivity of product with contact surfaces are evaluated under process conditions; SUS extractable and leachable profiles and their quality impact are evaluated per component, with leachable studies for high-risk components (absorbing materials, long contact times) and realistic, scientifically based simulated conditions.	binding · Injectables	Checked	ST-79
n538	8.137; 8.138	SUS keep integrity under intended conditions including freeze/thaw and transport (intrinsic sterile connectors, heat- or mechanically sealed, verified to stay integral); risk-based acceptance criteria apply; on receipt each piece is checked against the approved specification and, before use, the outer packaging, label printing and attached documents (certificate of conformance, proof of sterilisation) are visually inspected and the check documented.	binding · Injectables	Checked	ST-79
n539	8.139	Critical manual SUS handling such as assembly and connections is controlled and verified in the APS.	binding · Injectables	Checked	ST-08
n540	9.1; 9.2; 9.3	The environmental and process monitoring programme (total particles, viable EM and personnel monitoring, temperature/RH and other characteristics, and APS for aseptic products) is part of the CCS; no element alone proves asepsis; its data are used for routine batch certification/release (for both aseptic and terminally sterilised products) and for periodic review and investigations.	binding · Injectables	Checked	ST-24
n541	9.4	A documented, risk-assessed EM programme defines sampling locations, frequency, methods and incubation conditions, identifies critical monitoring locations (Grade A, aseptic processing areas and Grade B zones interfacing with Grade A) using process, facility, data, flora and airflow-study knowledge, and is reviewed regularly against trends and the CCS.	binding · Injectables	Checked	ST-18
n542	9.5; 9.16	Cleanrooms, clean-air devices and personnel are routinely monitored in operation throughout all critical stages, including equipment set-up; for Grade A, particle counting spans all critical processing, equipment assembly included.	binding · Injectables	Checked	ST-18
n543	9.6; 9.7; 9.8	Temperature and relative humidity are controlled within ranges suited to product, process and personnel and supporting the cleanliness grade; Grade A monitoring shows aseptic conditions during critical operations at the highest-risk locations with justified device positions; sampling should not contaminate operations.	binding · Injectables	Checked	ST-18
n544	9.9; 9.10; 9.12	Alert levels and action limits are set for viable and total particle results; action limits are at most the Table 5 (particles) and Table 6 (viable) values and may be tighter from trends, process nature or CCS; alert levels come from qualification data and are periodically reviewed; alert levels are set for Grade A (total particles only), B, C and D to detect adverse trends; Grade C/D in-operation limits come from qualification and routine data.	binding · Injectables	Checked	ST-20
n545	9.11	Procedures define trending, covering at least rising numbers of alert/action excursions, consecutive alert excursions, recurring isolated action excursions with a possible common cause (e.g. after planned maintenance), and changes in flora type, counts and predominance, with attention to organisms signalling loss of control or hard to eradicate (spore-formers, moulds).	binding · Injectables	Checked	ST-20
n546	9.13	Action-limit excursions require a root-cause investigation, a product impact assessment that includes batches made between sampling and reporting, and CAPA; alert excursions require assessment and follow-up, considering investigation and/or corrective action.	binding · Injectables	Checked	ST-21
n547	9.15 (note 2)	Occasional isolated 5 um (macro) particle counts in Grade A may be false counts (electronic noise, stray light, coincidence loss), but consecutive or regular low-level counts may signal contamination (filter failure, equipment failure, poor set-up or operating practice) and are investigated.	binding · Injectables	Checked	ST-19
n548	9.17	Grade A is monitored continuously for particles of at least 0.5 and 5 um at a sample flow of at least 28 L/min so that interventions, transient events and deterioration are captured; each result is compared frequently with alert/action levels so excursions are found and handled promptly; alarms sound at alert level; procedures define responses including extra microbial monitoring.	binding · Injectables	Checked	ST-19

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n549	9.18	A similar system is recommended for Grade B, possibly at lower sampling frequency, but frequent and large enough to capture increased contamination and deterioration, with alarms on alert-level exceedance.	binding · Injectables	Checked	ST-19
n550	9.19; 9.20; 9.21	Monitoring system choice considers hazards from materials (live organisms, powders, radiopharmaceuticals); where contaminants could damage counters or pose hazards, the strategy assures classification before and after exposure, with increased viable monitoring and periodic monitoring during simulated operations as defined in the CCS; automated monitoring sample volumes are justified.	binding · Injectables	Checked	ST-18
n551	9.22	During aseptic operations, microbial monitoring is frequent and combines methods (settle plates, volumetric air, glove, gown, surface swabs/contact plates) justified in the CCS without disturbing Grade A/B airflow; room and equipment surfaces are monitored at the end of an operation.	binding · Injectables	Checked	ST-18
n552	9.23	Viable monitoring is also done when normal manufacturing is not running (after disinfection, before start, at batch completion, after shutdowns) and in associated unused rooms to detect contamination incidents; additional locations can verify the effectiveness of corrective actions after an incident.	binding · Injectables	Checked	ST-16
n553	9.24	Grade A viable air monitoring (air sampling or settle plates) is continuous for the full critical processing duration including aseptic set-up; a similar approach is considered for Grade B based on risk; monitoring captures interventions, transients and deterioration without adding intervention risk.	binding · Injectables	Checked	ST-18
n554	9.25	Personnel monitoring locations, types and frequencies are risk-assessed by activity and proximity to critical zones, with periodic in-process sampling; after critical interventions at least gloves (and gown areas if relevant) are sampled; gloves and gown are sampled at each exit from Grade B; after glove sampling following a critical intervention the outer gloves are replaced before continuing, and after gown sampling the gown is replaced before further cleanroom activity.	binding · Injectables	Checked	ST-22
n555	9.26; 9.27	Personnel in Grade A and B are microbiologically monitored, with extra emphasis on gown monitoring for manual operations (e.g. manual compounding or filling) justified in the CCS; where manufacturing staff perform monitoring, the quality unit oversees it regularly.	binding · Injectables	Checked	ST-22
n556	9.28; 9.29	Rapid/alternative microbial methods should be considered and may be adopted once validated as equivalent or better; sampling methods and equipment are understood, with procedures for correct operation and interpretation and recovery-efficiency data.	binding · Injectables	Checked	ST-20
n557	9.30 note (c)	In Grade A any microbial growth results in an investigation.	binding · Injectables	Checked	ST-21
n558	9.31	Microorganisms detected in Grade A and B are identified to species level and their potential impact on product quality for each implicated batch and on the state of control is evaluated; identification in Grade C/D is considered for excursions or indicator organisms (spore-formers, moulds) and often enough to know the typical flora.	binding · Injectables	Checked	ST-21
n559	9.32; 9.35	APS with sterile nutrient media and/or a non-inhibitory surrogate periodically verifies aseptic controls but is not the primary means of validating the aseptic process (which rests on design, PQS adherence, process controls, training and monitoring); media/surrogate mimic risk-relevant product characteristics, with alternative procedures for semi-solids, powders, lyophilised or heat/cold-processed forms; APS should not be used to justify practices with unnecessary contamination risk.	binding · Injectables	Checked	ST-31
n560	9.33	APS mimics routine aseptic manufacture as closely as possible, covering every aseptic step from after sterilisation/decontamination of materials to container sealing, extra aseptic steps for non-filterable products, air instead of inert gas unless anaerobic simulation is intended, a surrogate in the same containers for sterile powder additions, and avoidance of piecemeal unit-operation simulations unless justified and fully covering. Lyophilisation APS covers filling, transport, loading, a representative chamber dwell, unloading and sealing at documented worst case, without boiling or freezing the media, and considering air vacuum break, maximum lyophiliser sterilisation-to-use interval, maximum filtration-to-lyophilisation time, and worst-case quantities (most trays, longest open-door loading).	binding · Injectables	Checked	ST-31
n561	9.34	APS includes the aseptic manipulations and interventions that occur in routine production plus worst cases; inherent and corrective interventions are performed in a manner and frequency similar to routine, and their inclusion and frequency are risk-based.	binding · Injectables	Checked	ST-31
n562	9.36	APS planning considers justified worst-case variables (container size, line speed), representative container/closure sizes with bracketing/matrixing when equivalence is justified, maximum hold times for exposed sterile product and equipment, fill volume enough to contact all surfaces and leave headspace for growth and visible turbidity, air replacing inert gas (with occasional anaerobic runs considered), media supporting pharmacopoeial and local isolates, a justified detection method, sufficient duration to challenge process, intervening operators, shift changes and environment, factors of different/extended shifts (e.g. maximum operator time in the cleanroom), idle interruptions (shift changes, vessel recharging, adding equipment), EM as for routine throughout, campaign start and end risks, and justified end-of-campaign APS only as a supplement.	binding · Injectables	Checked	ST-31
n563	9.37	For sterile active substances, the APS batch is large enough to represent routine operation, simulate worst-case interventions and cover all product-contact surfaces; all simulated materials are microbiologically evaluated and should not compromise recovery.	binding · Injectables	Checked	ST-31
n564	9.38	Initial validation requires at least three consecutive satisfactory APS covering all shifts, repeated after significant changes affecting sterility assurance (HVAC, equipment, process, number of shifts or personnel, major shutdown); periodic APS is normally twice yearly (about every 6 months) for each aseptic process, filling line and shift; each operator takes part in at least one successful APS every year; APS should be considered after the last batch before shutdown, before long inactivity, and before decommissioning or relocation.	binding · Injectables	Checked	ST-29
n565	9.39	For manual aseptic operations (e.g. compounding, filling), each container type, closure and equipment train is initially validated with each operator in at least three consecutive successful APS and revalidated with one APS per operator about every 6 months; the APS batch size mimics routine.	binding · Injectables	Checked	ST-30
n566	9.40	APS unit numbers should suffice to simulate all representative activities, with the justification in the CCS; typically at least 5,000 to 10,000 units; for batches under 5,000 units the APS should contain at least as many containers as the production batch.	binding · Injectables	Checked	ST-31
n567	9.41	APS units are agitated, swirled or inverted before incubation; all integral units are incubated and evaluated, including cosmetic defects and units passed through non-destructive IPCs.	binding · Injectables	Checked	ST-32
n568	9.41 (unit removal)	APS units may be discarded without incubation only when comparable with routine discards and only if production SOPs clearly state that such units must be removed in the same circumstances (intervention type, line location, exact number); never more units may be removed in an APS intervention than in a production run; set-up or mandatory line-clearance units are typically incubated separately and not necessarily part of the acceptance criteria. (One of the few places the Annex uses 'must'.)	binding · Injectables	Checked	ST-32
n569	9.42; 9.43; 9.44; 9.45	Discarded materials that contacted product surfaces (e.g. flushes) are simulated with media and incubated unless shown irrelevant to sterility; units are incubated in clear containers (identical-configuration clear substitutes for amber/opaque, else a validated detection method), promptly and under justified, validated conditions; isolates are identified to species where practical; after incubation, trained and qualified personnel inspect units under suitable conditions, and samples receive positive controls with reference organisms and local isolates.	binding · Injectables	Checked	ST-32
n570	9.46	The target is zero growth; any contaminated unit means the APS has failed. Required follow-up: root-cause investigation; corrective measures; normally at least three consecutive successful repeat APS; a prompt review of all aseptic production records since the last successful APS, including a risk assessment of possible sterility breaches in batches made since then; all unreleased batches included in the investigation scope with release decisions considering	binding · Injectables	Checked	ST-33

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		its outcome; quarantine of all products made on the line after the failure until it is resolved; limits on implicated operators until retrained and requalified; production resumes only after successful revalidation.			
n571	9.47; 9.48	Each APS run is fully documented with a reconciliation of units (filled, incubated, not incubated) and justification for non-incubated units; every intervention is recorded with start and end time and the person involved; all microbial monitoring and other test data go into the APS batch record; an APS is aborted only where written procedures would require the same for commercial lots, and the abort is investigated.	binding · Injectables	Checked	ST-32
n572	9.49	The initial APS validation is repeated when an aseptic process has not run for an extended period, or after changes to process, equipment, procedures or environment that could affect it, or when new containers or container-closure combinations are added.	binding · Injectables	Checked	ST-29
n573	10.1; 10.2	Staff trained and experienced in microbiology and sterility assurance support process design, the EM regime and investigations of microbiological events; specifications for raw materials, components and products include microbial, particulate and endotoxin/pyrogen limits where monitoring or the CCS indicates a need.	binding · Injectables	Checked	ST-07
n574	10.3	Every batch, aseptic or terminally sterilised, has a bioburden assay whose result is considered in final batch review; bioburden limits apply immediately before the final sterilising filter or the terminal sterilisation, linked to the method's efficiency; samples represent worst case (e.g. end of hold time); with overkill terminal cycles, bioburden is monitored at scheduled intervals.	binding · Injectables	Checked	ST-54
n575	10.4	For products with parametric release, a pre-sterilisation bioburden programme on filled units is in place with an assay each batch, worst-case representative sampling locations, identification of organisms found and assessment of their effect on the sterilising process, and endotoxin monitoring where appropriate.	binding · Injectables	Checked	ST-61
n576	10.5	The finished-product sterility test is only the last of a series of critical controls; a pass cannot rescue a batch that failed its design, procedural or validation requirements; the test method is validated for each product.	binding · Injectables	Checked	ST-68
n577	10.6	Sterility testing is done aseptically on samples representative of the whole batch but including those most at risk: for aseptic fills, containers from the beginning and end of the batch plus, based on risk, after critical interventions; for products heat-sterilised in final containers, worst-case load positions (coolest/slowest to heat) of each load; for lyophilised products, samples from different lyophiliser loads; where sub-batches exist (e.g. separate autoclave loads) each sub-batch is sampled and tested separately, with separate testing considered for other tests.	binding · Injectables	Checked	ST-66
n578	10.7; 10.10 (short shelf life)	Where shelf life is too short to await the sterility result (or EM results at manufacture), process design, extra monitoring and/or alternative test methods mitigating the risks are assessed and documented; EM compliance then includes review of the most recent available data, and rapid methods should be considered.	binding · Injectables	Checked	ST-69
n579	10.8; 10.9; 10.11	Decontamination of sterility-sample exteriors (e.g. VHP, UV) should not reduce test sensitivity or sample reliability; product-test media are QC-tested per pharmacopoeia before use; EM and APS media are growth-promotion tested before use with justified reference organisms and representative local isolates, normally by the end user (reliance on supplier/outsourced testing justified, with transport considered); rapid/automated methods are validated for the product or process.	binding · Injectables	Checked	ST-23
n580	10.10	EM data and trend data for the classified areas are reviewed as part of batch certification/release, with a written procedure defining actions for out-of-trend results or limit excursions.	binding · Injectables	Checked	ST-24
n581	Glossary: Inherent / Corrective / Critical intervention; Critical zone; Critical surfaces	Interventions are classified two ways: inherent (integral to set-up, routine operation or monitoring and required by procedure, e.g. aseptic assembly, container replenishment, EM sampling) versus corrective (made to correct or adjust the process, not at a set frequency, e.g. clearing jams, stopping leaks, adjusting sensors, replacing parts); an intervention of either type that enters the critical zone (where product and critical surfaces are exposed) is a critical intervention.	binding · Injectables	Checked	ST-34
n582	Glossary: Grade A air supply; 8.27	Grade A air supply is air through a filter qualified to give Grade A total-particle quality, without a requirement for continuous particle monitoring or Grade A viable limits; it protects fully stoppered, not yet crimped vials.	binding · Injectables	Checked	ST-57
n583	Glossary: Operator; Manual aseptic processing; Closed system; Local isolates	'Operator' includes anyone in the processing operation (set-up, filling, maintenance and other associated personnel); manual aseptic processing covers hand compounding, filling, placing or sealing of open containers of sterile product; a closed system keeps the product unexposed, is sterilised after connections are made (for sterile products), stays closed until the operation ends and excludes RABS/isolators; local isolates are representative site organisms from Grade A/B EM, personnel monitoring or positive sterility tests.	binding · Injectables	Checked	ST-07
n584	4.5-4.9; 4.17; 5.1; 6.5; 6.6	Facility and equipment design expectations (smooth impervious surfaces, no hard-to-clean recesses, low-shedding materials, sealed ceilings, no sinks/drains in Grade A/B, trapped drains elsewhere, observation windows/cameras, up-to-date equipment descriptions and P&IDs, utility installation records, no exposed pipework in cleanrooms) are verified through facility qualification, not batch execution.	binding · Injectables	Checked	discussed in notes of Part C
n585	3.1(v); 2.3; 6.4; 8.33; 9.11	Senior management oversees the state of control; risk-management outcomes are reviewed regularly, during change, for significant emerging problems and in the periodic product quality review; CCS effectiveness forms part of management review; utility, defect and EM data are trended.	binding · Injectables	Checked	ST-73
n586	2.4	Contamination controls for microbial, endotoxin/pyrogen and particle sources are interrelated; although usually assessed, controlled and monitored one by one, their combined effectiveness should be judged together.	binding · Injectables	Checked	ST-70
n587	4.1	Sterile manufacture takes place in cleanrooms entered via change rooms acting as personnel airlocks and via material airlocks, supplied with suitably filtered air and kept to the right cleanliness standard; scientifically justified controls and monitoring evaluate the state of cleanrooms, airlocks and pass-through hatches.	binding · Injectables	Checked	ST-18
n588	4.22	Cleaning ahead of barrier-system bio-decontamination is declared essential, because leftover residues can defeat the decontamination.	binding · Injectables	Checked	ST-76
n589	8.22 (last sentence)	Visual inspection is declared not acceptable as a container-closure integrity test method.	binding · Injectables	Checked	ST-64
n590	8.128 (first sentence)	For closed systems in aseptic processing, keeping every product-contact surface sterile is declared critical.	binding · Injectables	Checked	ST-79
n591	10.5 (second sentence)	A passing sterility test is declared incapable of assuring sterility for a product that did not meet its design, procedural or validation parameters.	binding · Injectables	Checked	ST-68

EU GMP Annex 11

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n147	Annex 11 Principle	Annex 11 covers every computerised system used in GMP-regulated activities. The application must be validated and the IT infrastructure qualified. Replacing a manual operation with a computerised system must not reduce product quality, process control or quality assurance, and must not raise the overall risk of the process.	binding	Checked	DI-03
n148	Annex 11 §1	Risk management must be applied across the system's whole life cycle, weighing patient safety, data integrity and product quality; how far validation and data-integrity controls go must be decided on a justified, documented risk assessment of the system.	binding	Checked	DI-07

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n149	Annex 11 §4.3	A current inventory listing every relevant computerised system and the GMP functions it performs must be available. For critical systems there must also be a current description covering the physical and logical set-up, how data flow, the interfaces to other systems or processes, hardware and software prerequisites, and the security measures.	binding	Checked	DI-06
n150	Annex 11 §4.8	When data are moved into a different format or a different system, the validation must verify that neither their values nor their meaning change during that migration.	binding	Checked	DI-48
n151	Annex 11 §6	Critical data keyed in by hand need an additional accuracy check, either by a second person or by a validated electronic check; risk management must address how critical the data are and what wrong or mistaken entries could cause.	binding	Checked	DI-30
n152	Annex 11 §7.1	Data must be protected against damage by both physical and electronic means; stored data must be verified as accessible, readable and accurate; and access to the data must be assured throughout the retention period.	binding	Checked	DI-44, DI-47
n153	Annex 11 §7.2	All relevant data must be backed up regularly, and backups must be shown to be complete, accurate and restorable - first during validation and then through periodic monitoring.	binding	Checked	DI-45
n154	Annex 11 §8.1; §8.2	The system must be able to produce clear paper copies of data it stores electronically, and for records that support batch release it must be able to print a version indicating whether any value was changed since first entry.	binding	Checked	DI-40
n155	Annex 11 §5	Where the system exchanges data electronically with other systems, suitable built-in checks must ensure the data are entered and processed correctly and securely, so as to reduce risk.	binding	Checked	DI-31
n156	Annex 11 §9	Based on a risk assessment, building into the system an automatically generated audit trail of every GMP-relevant change or deletion should be considered. When GMP-relevant data are changed or deleted, the reason must be documented. Audit trails must be retrievable, convertible into a form people can generally understand, and reviewed at regular intervals. (For an eBMR holding batch-release data the risk assessment will in practice conclude that an audit trail is needed.)	binding	Checked	DI-22
n157	Annex 11 §10	Any change to a computerised system, configuration included, may be made only in a controlled way that follows a defined procedure.	binding	Checked	DI-09
n158	Annex 11 §11	Computerised systems must be evaluated periodically to confirm they are still in a validated state and GMP-compliant; where relevant the review considers the system's current functions, its deviation, incident and problem history, the upgrades made, how it has performed and how reliable and secure it has been, and its validation status.	binding	Checked	DI-11
n159	Annex 11 §12.1-§12.3	Physical and/or logical controls must limit system access to authorised persons (examples include physical keys or badges, individual passwords, biometrics and locking away hardware and storage); how strong the controls are depends on how critical the system is; and granting, changing and withdrawing access rights must be recorded.	binding	Checked	DI-14, DI-16
n160	Annex 11 §12.4	Systems managing data and documents must capture who entered, changed, confirmed or deleted each item of data, and when (date and time).	binding	Checked	DI-22, DI-27
n161	Annex 11 §13	Every incident, including but not limited to system failures and data errors, must be reported and evaluated; for critical incidents the root cause must be determined and must drive corrective and preventive action.	binding	Checked	DI-12
n162	Annex 11 §14	Electronic records may be signed electronically. Such signatures are expected to carry, within the company, the same weight as handwritten ones, to stay permanently tied to the record they sign, and to show the date and time they were applied.	binding	Checked	DI-36, DI-37
n163	Annex 11 §15	If a computerised system records batch certification and release, it must let only Qualified Persons certify batch release, must clearly identify and record who released or certified each batch, and this must be done by electronic signature.	binding	Checked	DI-39
n164	Annex 11 §16	Where computerised systems support critical processes, provisions such as a manual or alternative system must keep those processes going if the system breaks down; how quickly the fallback must be available is set by risk and by the system and process concerned; and the arrangements must be adequately documented and tested.	binding	Checked	DI-43
n165	Annex 11 §17	Data may be archived; archived data must be verified as accessible, readable and intact, and when relevant changes to the system (e.g., hardware or programs) are planned, the ability to retrieve the archived data must be ensured and tested.	binding	Checked	DI-46
n166	Annex 11 §2; §3.1; §3.2; §3.4 (§3.3 is covered by A11-22)	Process owner, system owner, QPs, IT and other relevant staff must cooperate closely, and all personnel need appropriate qualifications, access level and defined responsibilities (§2). Formal agreements stating the third party's responsibilities are required whenever a third party is involved in supplying, setting up, integrating, validating, maintaining (including by remote access), changing or hosting/retaining the system, or in processing data; internal IT departments are to be treated the same way (§3.1). Supplier competence and reliability are key selection factors and the need for an audit is decided by risk (§3.2). Quality-system and audit information on software and system suppliers or developers must be made available to inspectors on request (§3.4).	binding	Checked	DI-08
n167	Annex 11 §4.4; §4.7	User requirements must describe the functions the system needs, rest on a documented risk assessment and GMP-impact analysis, and stay traceable through the life cycle. Validation must evidence appropriate test methods and scenarios, paying special attention to limits on process parameters, limits on data values and how errors are handled; automated test tools and test environments need documented assessments of their adequacy.	binding	Checked	DI-03
n168	12 (Audit trails)	The draft widens and hardens the audit trail rule: wherever users can create, modify or delete GMP data, the system must automatically log each such manual interaction, now including the creation of data and not only changes/deletions. Each entry should identify who acted (with role), what changed (previous and new value), when (with time zone where relevant) and why (the system should prompt for the reason). The audit trail is to be configured and switched on from installation and should not be possible to turn off, edit or delete; commentators note the draft still tolerates such functions being reachable only by administrators independent of GMP work.	draft	Search-based	DI-18, DI-22, DI-23
n169	12 (Audit trails - review)	Per secondary analyses of the draft: audit trail review must be governed by a written procedure, performed by someone independent of the business process being reviewed, documented, and may be done as part of batch record review; discrepancies found must be investigated, resolved and escalated per a defined route; review frequency is risk-based, and QPs must be given access to the relevant audit trail entries.	draft	Search-based	DI-25
n170	(removal of 2011 clause 15) / cross-reference to Annex 16	According to published analyses, the draft no longer carries a stand-alone clause on QP batch certification by electronic signature (that topic is left to Annex 16); instead it requires that QPs can access audit trail entries. Until the draft is final, the 2011 clause 15 still applies.	draft	Search-based	DI-25, DI-39
n171	13 (Electronic signatures)	Per secondary analyses, the draft expects: the signer to re-authenticate when signing; the meaning of the signature (e.g., review, approval) to be clear to the signer at the time of signing; the displayed/printed signature to show the signer's name, role, meaning, and date/time with time zone where applicable; and, for hybrid set-ups where a handwritten signature relates to an electronic record, measures that give high confidence that any later change to the electronic record invalidates the signature. It also references use of external (trust service) e-signature providers.	draft	Search-based	DI-36, DI-37, DI-38
n172	12 (Audit trails - timing of review relative to release)	Secondary summaries of the draft report that review of audit trails should be targeted and risk-based (not line-by-line), done by persons not involved in the reviewed activity, and completed before batch release unless a later review is justified; audit-trail review outcomes that bear directly on product release should be available to the QP when certifying.	draft	Search-based	DI-25
n173	11 (Identity and access management)	The draft replaces the short 2011 security clause with a detailed identity and access management section. Reported expectations: unique personal accounts (shared accounts tolerated only for read-only use, never for creating or changing data); least privilege (no more rights than the job needs); segregation of duties so that people doing GMP	draft	Search-based	DI-14, DI-16, DI-17, DI-19

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		work do not also hold administrator privileges; multi-factor authentication for remote access to critical systems; automatic lock-out after repeated failed log-ins; and controlled, timely handling and review of access rights.			
n174	8 (Alarms)	Alarms were not covered in 2011; the draft adds a dedicated section. Reported expectations: GMP-relevant alarms and their acknowledgements are logged automatically with full details; alarm handling and verification are documented (for example in the batch record) with a log that cannot be erased or deactivated; alarm logs are reviewed periodically for handling and trends. Alarm/event logs are distinct from the audit trail of manual data changes.	draft	Search-based	DI-49
n175	10 (Handling of data)	Reported to cover four themes: verification of input (plausibility checks for manually entered critical data, plus second-person or validated electronic verification); data transfer (preference for validated electronic interfaces over manual transcription, with manual transfer treated as an exception needing extra control); controlled and validated data migration; and risk-based encryption of critical data in storage and in transit.	draft	Search-based	DI-30, DI-48
n176	15 (Security)	Cyber/IT security becomes an explicit GMP expectation. Reported content: firewalls and network segmentation, timely patching, protection against malware, restrictions on removable media, disaster recovery with defined recovery time and recovery point objectives, isolation of unsupported/unpatched platforms, multi-factor authentication for remote access to critical systems, and (reported as draft 15.19) regular penetration testing of critical systems that are exposed to the internet.	draft	Search-based	DI-13
n177	16 (Backup) and 17 (Archiving)	The draft expands both topics: backups must be kept physically and logically separate from the primary data and restore capability must be tested regularly; archiving receives its own, more detailed section than in 2011 (retention, protection and retrievability of archived records).	draft	Search-based	DI-45, DI-46
n178	14 (Periodic review)	Periodic review grows from one sentence in 2011 to a section of about twelve sub-clauses. Secondary summaries mention review of system documentation, maintenance, service agreements, adequacy of backup, disaster recovery and archiving arrangements, data integrity status, changes in regulatory requirements, training effectiveness, and ongoing confirmation of validated status (a one-off qualification is not enough).	draft	Unverified	DI-11
n179	7 (Supplier and service management)	The regulated company stays fully responsible for GMP compliance even when a vendor, cloud/SaaS provider or internal IT runs the system. Suppliers and service providers must be qualified (audit as needed) and covered by written agreements/contracts that fix responsibilities, including data ownership and access, security, communication of security issues, support during inspections, and an exit strategy; complete validation documentation must be accessible for cloud services.	draft	Search-based	DI-08
n193	6 (Requirements)	Reported to contain about six clauses on documenting user requirements and the application configuration, making a traceability matrix mandatory and allowing modern electronic tools to be used to author and maintain requirements; requirements are to be kept current across the lifecycle.	draft	Search-based	DI-06
n194	9 (Qualification and validation) and scope for cloud/service models	Per commentators, the qualification/validation section largely carries over the 2011 expectations, while the draft as a whole extends explicitly to cloud and service-based systems and expects complete validation documentation to be accessible for cloud services; critics note it lacks an explicit formal system-release step.	draft	Search-based	DI-03, DI-10
n200	Annex 11 §3.3; §4.2; §4.5; §4.6	Regulated users must review the documentation that comes with COTS products to confirm it satisfies their user requirements (§3.3). The validation file should contain change-control records (if any) and reports of deviations observed during validation (§4.2). The regulated user must make every reasonable effort to confirm the system was built under a suitable quality management system, and must assess the supplier appropriately (§4.5). For bespoke or customised systems, a process must formally assess and report quality and performance metrics at every life-cycle stage (§4.6).	binding	Checked	DI-03
n226	Draft sections on identity and access management, audit trails (section 12) and electronic signatures (section 13)	The draft (not final as of this research) is much more prescriptive. Per secondary summaries: identity and access management has many sub-clauses, and shared accounts appear limited to read-only use. Audit trails must be always on, capture all changes with metadata and reasons, and be reviewed; critical audit trails are expected to be reviewed with the operation's other records before batch release. E-signatures must be uniquely tied to the signer, the exact content signed and the meaning of the signature, with extra requirements such as re-authentication and rules for hybrid (paper plus electronic) signing. Cybersecurity and periodic review are also covered. Some commentators say MFA is expected for signatures; this is unconfirmed.	draft	Unverified	DI-14, DI-22, DI-25, DI-38
n624	Ch.4 4.29 (personnel matters); Annex 11 §2	Written procedures and associated records are required for personnel matters, including signature lists, GMP and technical training (and checks that it was effective), and gowning and hygiene; Annex 11 §2 requires personnel to have appropriate qualifications for their assigned duties.	binding	Checked	DI-21
n628	Annex 11 §4.1	Validation documentation and reports must cover the relevant steps of the system life cycle, and the manufacturer must be able to defend, from its risk assessment, the validation approach it chose - the standards and protocols used, the acceptance criteria, and the procedures and records produced.	binding	Checked	DI-03

EU GMP Annex 15

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n195	5 (Process validation - general; CPPs/CQAs)	Process validation has to show that the quality attributes and process parameters judged important for a validated state and acceptable quality are met consistently; critical parameters and attributes, and their ranges, come out of development/validation knowledge and govern routine manufacture.	binding	Unverified	PR-08
n196	5 (Ongoing process verification during lifecycle; approx. 5.28-5.32)	Manufacturers must keep monitoring product quality over the lifecycle to show the process stays in a state of control, evaluating relevant trends; ongoing verification runs under an approved protocol or equivalent with a written report, uses statistical tools where suitable to judge variability and capability, feeds the product quality review, looks at gradual changes over time, and has its scope and frequency revisited periodically.	binding	Unverified	PR-09, PR-58
n197	5 (Continuous and hybrid process verification; concurrent validation)	Continuous process verification (e.g., with in-line/at-line PAT and multivariate monitoring) may replace or complement traditional validation. Concurrent validation is only for exceptional, justified cases, and its results and conclusions must be documented and made available to the QP before the batch is certified.	binding	Unverified	PR-09, PR-55
n198	11 (Change control) and 4 (Re-qualification)	Changes that could affect product quality or the validated state (processes, equipment, materials, systems) must go through documented change control within the PQS with impact assessment, and equipment, facilities, utilities and systems must be re-evaluated at suitable intervals to confirm they remain in a state of control.	binding	Unverified	PR-06, PR-23
n199	10 (Cleaning validation - hold times)	Cleaning validation must take into account how long equipment may wait between use and cleaning and between cleaning and next use, which sets validated dirty-hold and clean-hold times.	binding	Unverified	PR-23

EU GMP Annex 16

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n187	General principles / section 1 (certification process)	Before release, the QP must be satisfied that each individual batch was made and tested in line with applicable law, the marketing authorisation (MA) and GMP.	binding	Search-based	PR-04, PR-55, PR-56
n188	1.7 (matters the QP must ensure, directly or by reliance on the PQS)	The QP must be assured (personally or by well-founded, continuously verified reliance on the quality system) that, among other things: production and QC deviations and changes were authorised by the responsible people; required checks and tests were done and production/QC documentation is complete and approved; processes and test	binding	Search-based	PR-53, PR-55

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
		methods are validated and consistent with the MA; all batch-related investigations, including out-of-specification and out-of-trend ones, are complete enough to support certification; and the effect of any open complaints, investigations or recalls has been assessed.			
n189	3 (Handling of unexpected deviations)	A QP may certify a batch affected by an unexpected deviation from MA and/or GMP details in the manufacturing process or analytical controls only if registered specifications are still met, the deviation has been properly investigated with its root cause corrected, and a documented quality-risk assessment supports it (with consideration of whether a variation is needed). Per the EMA Q&A, registered specifications include the in-process, bulk and finished-product specifications in the MA; failing a registered finished-product specification takes the batch outside this route; in-process deviations may be acceptable only where risk assessment shows no impact; the same deviation recurring after discovery is to be handled as a change needing a variation; and the manufacturer must keep a record of batches certified under this provision for management review and product quality review.	guidance	Search-based	PR-08, PR-49
n190	Certification register (section 1)	QP certifications must be entered in a register or equivalent document that is kept current as certifications happen and remains available to competent-authority inspectors for the period national law requires, and at least five years.	binding	Search-based	PR-55
n191	4 (Release of a batch)	A batch may be released to the market only after QP certification; until then it stays at the manufacturing site or moves under quarantine to an approved site. Safeguards must stop uncertified batches reaching saleable stock; they may be physical (segregation, labelling) or electronic (validated computerised systems), and must persist when uncertified stock moves between authorised sites. The way QP certification is notified to the site that transfers stock to saleable status must be set out in a technical agreement and the notification must be formal, unambiguous and follow Chapter 4 documentation rules.	binding	Search-based	DI-39
n192	Partial manufacturing confirmations (section 1 and Appendix I)	Where another site performs only part of the manufacture, a QP at that site must at least confirm that its operations followed GMP and the written agreement, and the certifying QP may rely on such confirmations. Appendix I defines what the confirmation must contain; in summary it identifies the confirming site and QP, ties the statement to the governing technical/quality agreement, states the GMP compliance being confirmed, and is signed and dated.	binding	Search-based	PR-55
n223	Section 1.7 (QP operational responsibilities before certification) incl. records complete and endorsed; deviations/planned changes approved; batch investigation	Before certifying, the QP must be sure (personally or through systems they can justifiably rely on) that: all batch records are complete and endorsed by the right people; any deviations or planned changes have been approved under a defined system, with any change needing a regulatory variation notified or approved; and every investigation connected to the batch, including OOS and OOT, has progressed far enough to support certification. The QP need not re-review every record personally but must have ongoing assurance that the reliance is well founded, and must be able to see non-conformances and changes easily.	binding	Search-based	PR-53, PR-55
n224	Section 3 - handling of unexpected deviations	A batch with an unexpected deviation from the marketing authorisation or GMP (process or analytical methods) may still be certified only if all registered specifications are met, the impact on quality, safety and efficacy has been assessed through QRM and judged acceptable, and the deviation has been investigated with root cause and actions defined. Deviations that keep recurring no longer count as unexpected and cannot be handled this way.	binding	Search-based	PR-49

EU GMP Annex 22 (draft)

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n180	Scope and core expectations (summary)	If AI/ML is used in a critical GMP function, the draft limits it to static, deterministic models and expects a precisely defined intended use, predefined performance/acceptance metrics, representative test data kept independent of training data, explainability and confidence information for outputs, change control, and ongoing performance monitoring; generative AI/LLMs only for non-critical tasks with a human in the loop.	draft	Search-based	PR-60

EU GMP Chapter 4

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n89	Ch.4 4.17(a)-(d); the per-product and per-batch-size rule sits in the unnumbered lead-in to the Manufacturing Formula and Processing Instructions section	For every product and every batch size made, an approved written manufacturing formula and processing instructions must exist. The formula names the product with a product code tied to its specification; gives the dosage form, strength and batch size; lists each starting material and its amount, noting any substance that may be lost during processing; and states the expected final yield with its acceptable range, plus relevant intermediate yields where applicable.	binding	Checked	PR-01, PR-03, PR-31
n90	Ch.4 4.18(a)-(g)	Processing instructions must state where processing takes place and the main equipment; give, or refer to, the methods for readying critical equipment (e.g., cleaning, assembly, calibration, sterilisation); require a pre-start check that equipment and workstation hold no earlier products and no documents or materials unneeded for this process, and that equipment is clean and fit for use; set out detailed ordered steps (material checks, pre-treatment, order of addition, critical parameters such as time and temperature); give in-process control instructions with limits; where needed, define bulk storage requirements (container, labelling, special conditions); and list any special precautions.	binding	Checked	PR-03, PR-08, PR-23, PR-26
n91	Ch.4 4.19(a)-(i)	Approved packaging instructions must exist for each product, pack size and pack type, containing or referencing: product name with bulk and finished batch numbers; dosage form, and strength where applicable; pack content by number, weight or volume; a complete packaging-material list with quantities, sizes, types and specification codes; where appropriate, specimens of printed materials showing where batch number and shelf-life are to be applied; line-clearance checks and a careful pre-start examination of area and equipment, plus other special precautions; a description of the packaging operation including significant ancillary operations and equipment; and in-process controls with sampling instructions and acceptance limits.	binding	Checked	PR-01, PR-03
n92	Ch.4 4.20(a)-(i)	A batch processing record must be kept for every batch processed, derived from the applicable sections of the formula and processing instructions currently approved. It captures: product name and batch number; start, key intermediate stage and completion dates and times; who carried out each significant step and, where appropriate, who checked it; for each starting material its batch and/or QC control number and the quantity actually weighed, plus the lot and quantity of any recovered or reprocessed material that was added; relevant processing operations or events and the main equipment used; in-process control results with the identity of whoever performed them; yields at relevant stages; notes on special problems with signed authorisation of any departure from the formula or instructions; and approval by the person responsible for processing.	binding	Checked	DI-28
n93	Ch.4 4.20 (Note)	Where a process has been validated and is subject to continuous monitoring and control, the batch reports the system generates automatically can be cut down to compliance summaries plus exception and out-of-specification reports.	binding	Checked	PR-52

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n94	Ch.4 4.21(a)-(i)	Every batch or part-batch that is packed needs its own batch packaging record, derived from the applicable parts of the packaging instructions. It holds: product name and batch number; dates and times of packaging; who performed each significant step and, where appropriate, who checked it; identity and conformity checks against the instructions, including in-process results; details of the operations with references to the equipment and lines used; where possible, samples of the printed components actually used, with specimens of the batch code, expiry date and any extra overprint; notes on special problems or unusual events with signed authorisation of departures from the instructions; the amounts and identity of every printed component and of the bulk that were issued, consumed, destroyed or sent back to stock, and the amount of finished product obtained, enough for reconciliation; and approval by the person responsible for packaging.	binding	Checked	DI-28
n95	Ch.4 4.21(h) (final sentence)	The reconciliation data described in 4.21(h) - identity and quantities of printed materials and bulk issued, used, destroyed or returned, and the product quantity obtained - may be left out of the packaging record only where robust electronic controls operate during packaging and the omission is justified.	binding	Checked	PR-39
n96	Ch.4 Principle; 4.1	Documentation may be paper, electronic or photographic and the same rules apply to every medium; the document types and media used must be defined in the QMS and followed. Complex systems must be understood, well documented, validated and adequately controlled. For hybrid and single-medium systems alike, the relationships and controls between masters, official copies, data processing and resulting records must be stated; electronic templates, forms and masters need appropriate controls; and record integrity must be assured for the whole retention period.	binding	Checked	DI-01, DI-43
n97	Ch.4 4.2; 4.3; 4.4	Documents must be carefully designed, drafted, reviewed and issued, and must agree, as appropriate, with the relevant parts of the Product Specification File (investigational products) and of the manufacturing and marketing authorisation dossiers. Making working copies from masters must not be able to introduce errors. Instruction documents must be approved, signed and dated by suitably authorised people and have a defined effective date; every document must have unambiguous content and be uniquely identifiable. Instruction documents should be orderly and easy to check, with style and language suited to their use; SOPs, work instructions and methods specifically must use an imperative, mandatory style.	binding	Checked	PR-02, PR-10
n98	Ch.4 4.5 (periodic review only); superseded-version control rests on 4.3 (effective date) and 4.20/4.21 (records based on currently approved instructions)	Documents in the quality management system must be reviewed regularly and kept current (4.5). Superseded instructions must not be usable for new batches; in the 2011 text this follows from 4.3 (a defined effective date) and 4.20/4.21 (records based on the currently approved instructions), not from 4.5.	binding	Checked	PR-05
n99	Ch.4 4.6; 4.7	Controlled documents should not be handwritten, but documents that call for data entry must leave enough room for the entries; any handwritten entry must be clear, legible and indelible.	binding	Checked	PR-03, PR-11
n100	Ch.4 4.8	Records must be made or completed as each action is taken, and in a manner that keeps every significant manufacturing activity traceable.	binding	Checked	PR-11
n101	Ch.4 4.9 (mandatory reason for GMP-relevant electronic data comes from Annex 11 §9)	Any alteration to an entry must be signed and dated and must leave the original information readable; where appropriate the reason must also be recorded.	binding	Checked	PR-12
n102	Ch.4 4.10	It must be clearly defined which record belongs to each manufacturing activity and where that record is kept, and secure controls - validated where appropriate - must protect record integrity for the entire retention period.	binding	Checked	PR-59
n103	Ch.4 4.11	Batch documentation must be kept for one year after the batch expires or at least five years after QP certification of the batch, whichever is longer. For investigational medicinal products it must be kept at least five years after the last clinical trial using the batch is completed or formally discontinued. Legislation for particular product types (e.g., ATMPs) may require longer periods for some documents.	binding	Checked	PR-59
n104	Ch.4 4.12	Retention of other documentation depends on the business activity it supports. Critical documentation, including raw data (e.g., validation or stability) that supports information in the marketing authorisation, must be kept while the authorisation remains in force. Some of it may be retired once superseded by a complete new data set, with a documented justification that respects batch-record retention; for example, process validation raw data must be kept for no less time than the records of all batches whose release relied on that validation.	binding	Checked	DI-47
n105	Ch.4 4.27	Written release and rejection procedures must exist for materials and products, in particular for QP certification of finished product for sale. The QP must have access to every record, and a system must be in place to highlight special observations and any change to critical data.	binding	Checked	PR-52, PR-55
n106	Ch.4 4.31	Major or critical production and analytical equipment, and rooms where product is processed, need logbooks that record in time order, as appropriate, each use of the area or equipment/method and any calibration, maintenance, cleaning or repair, with dates and the identity of the people who did it.	binding	Checked	DI-28
n107	draft 4.10-4.18 (data governance)	(Draft, not yet binding) Firms must run a data governance system inside the PQS that manages integrity of all GMP data in any format across its whole lifecycle, with controls scaled to risk and supported by suitable expertise and defined ownership.	draft	Search-based	PR-60
n108	draft 4.12 (data lifecycle)	(Draft) Controls must cover each lifecycle stage: creation/recording; processing of raw data into reported (derived) data such that the processing can be reconstructed; review/verification; decisions based on the data; retention, archiving and retrieval; and controlled retirement or destruction.	draft	Search-based	DI-02, DI-33
n109	draft 4.13 (data criticality and data risk)	(Draft) Controls are to be based on two assessments: how critical the data is (its influence on decisions and product quality) and how exposed it is (opportunity for change or deletion and the likelihood that this would be detected).	draft	Search-based	PR-07
n110	draft good documentation practice section (ALCOA++)	(Draft) All GMP records, paper or electronic, are expected to meet the extended data integrity attributes: attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring, available and traceable.	draft	Search-based	DI-02
n111	draft metadata/audit trail provisions and glossary	(Draft) Records include the metadata and audit trail needed to reconstruct their history (who did what, when and why), whatever the medium; raw data (first capture) is defined and must be kept together with the derived data and the steps that produced it.	draft	Search-based	DI-02, DI-22
n112	draft hybrid systems provisions	(Draft) Hybrid paper/electronic arrangements are acknowledged but must be clearly defined, with the interfaces and data flows between formats validated and risk-controlled; commentators report that printing electronic records to sign them by hand, and parallel paper-plus-electronic signatures, are treated as higher risk and to be avoided where possible.	draft	Unverified	DI-43
n113	draft section on signatures in GMP documentation	(Draft) The purpose and control of signatures are spelled out: each signature must make clear who signed and what the signature means (e.g. performed, checked, reviewed, approved); a company signature policy is expected; electronic signatures are fully recognised but must be binding, traceable and stay permanently linked to the signed record for its lifetime.	draft	Search-based	DI-35
n114	draft retention/copies provisions and glossary (true, certified, verified copy)	(Draft) Where the storage medium will not last for the retention period, verified true copies must be made; copies used in place of originals must preserve content and meaning (including metadata) and the verification must be documented; backup, restore and archiving must keep records readable and accessible for the full retention time.	draft	Search-based	DI-41

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n145	Ch.4 4.3; 4.9; 4.13; 4.20(c),(f),(h),(i); 4.21(c),(g),(i); 4.31; e-signature properties Annex 11 §14, QP certification §15	Chapter 4 signature/identification points: instruction documents approved, signed and dated by authorised persons (4.3); specifications authorised and dated (4.13); alterations signed and dated (4.9); processing and packaging records identify who performed each significant step and, where appropriate, who checked it (4.20(c), 4.21(c)), and who performed in-process controls (4.20(f)); departures from instructions carry signed authorisation (4.20(h), 4.21(g)); the responsible person approves the completed processing or packaging record (4.20(i), 4.21(i)); logbooks identify who carried out each operation (4.31). Annex 11 §14, now checked, lets these be electronic signatures that carry the weight of handwritten ones inside the company, stay permanently linked to the record and show date and time; §15 requires QP certification by e-signature.	binding	Checked	DI-24, DI-35
n146	Ch.4 Principle and preamble 'Required GMP documentation' (Records definition); see also 4.28 (distribution records to support recall)	'Written' means recorded or documented on any medium from which data can be rendered human-readable. Records provide evidence of actions taken and, for manufactured batches, a history of each batch including its distribution (4.28 adds that distribution records must allow recall of any batch). Controls must keep documents accurate, intact, available and legible.	binding	Checked	PR-57, PR-59
n181	Data governance and data lifecycle (reported as 4.12 onward)	The draft requires a data governance system, integrated in the PQS and applying whatever the format (paper, electronic, hybrid), covering the whole data lifecycle - reported as creation, processing, verification, decision-making, retention and controlled destruction - and treating metadata and audit trails as part of the record, with defined ownership.	draft	Search-based	DI-02, DI-47
n182	Hybrid systems (reported as 4.82-4.85); electronic signatures	Per secondary summaries, the draft gives hybrid (paper plus electronic) systems specific attention: they must be clearly defined, validated and risk-controlled, with procedures for the hand-offs between manual and computerised parts; the draft also expressly recognises electronic signatures in GMP documentation.	draft	Search-based	DI-35, DI-43
n227	Draft Chapter 4 - data governance, ALCOA++, data lifecycle and metadata, hybrid systems (about draft sections 4.82-4.85), recognition of electronic signatures	The draft replacement for the 2011 Chapter 4 (roughly doubled in length, about half rewritten, per secondary sources) makes ALCOA++ the basis of documentation. It requires data governance within the PQS and lifecycle management of records together with their metadata and audit trails, with named ownership. It sets out risk-based expectations for defining, validating and controlling hybrid (paper plus electronic) systems, and explicitly recognises electronic signatures. The final text and effective date had not been confirmed at the time of this research.	draft	Unverified	DI-01
n616	Ch.4 preamble 'Required GMP documentation' - Records (raw data)	Records include the raw data from which other records are generated. For electronic records the regulated user must define which data count as raw data, and as a minimum every data item on which a quality decision is based must be defined as raw data.	binding	Checked	DI-01
n617	Ch.4 preamble 'Required GMP documentation' - instruction type (manufacturing formulae, processing, packaging and testing instructions)	Manufacturing formulae and processing, packaging and testing instructions must spell out every starting material, item of equipment and computerised system (if any) to be used, and give complete instructions for processing, packaging, sampling and testing; where relevant they must also name the in-process controls and process analytical technology (PAT) to be applied, with acceptance criteria.	binding	Checked	DI-04, DI-10
n618	Ch.4 Principle	Instructions must be error-free and must be available in written, human-readable form.	binding	Checked	PR-02, PR-11
n619	Ch.4 preamble 'Required GMP documentation' - Certificates of Analysis, footnote 1	Where the marketing authorisation provides for it (real-time release testing), certification can be based fully or partly on evaluating summaries and exception reports of real-time batch data from PAT or from process parameters and metrics, instead of end-product test results.	binding	Checked	PR-55
n620	Ch.4 4.13; 4.14; 4.15; 4.16	Approved, dated specifications must exist for starting materials, packaging materials and finished products, and specifications for intermediate and bulk products must be available for critical steps or when such products are bought in or dispatched. Material specifications include how materials are stored and how long they can be held before re-testing; finished-product specifications include the formula, dosage form and pack, storage and handling needs, and shelf-life.	binding	Checked	PR-18, PR-43
n621	Ch.4 4.22; 4.23	Every incoming delivery - starting materials of any kind (bulk, intermediates and finished goods included) and all primary, secondary and printed packaging materials - must be received under a written procedure and recorded. Receipt records must let each received lot be identified (external and internal names or codes, supplier and manufacturer with the manufacturer's batch number, and the internal batch number assigned on receipt) and quantified (total quantity, number of containers and date received), with room for relevant remarks.	binding	Checked	PR-17, PR-57
n622	Ch.4 4.24	Written procedures must govern how starting materials, packaging materials and, as appropriate, other materials are labelled internally, held in quarantine and stored.	binding	Checked	PR-18, PR-45
n623	Ch.4 4.25; 4.26; 4.30	Sampling must follow written procedures that set the method, equipment and amount to take and the precautions against contaminating or degrading the material; testing of materials and products at the different manufacturing stages must follow written procedures describing methods and equipment, and the tests performed must be recorded; and clear operating procedures must be available for major manufacturing and test equipment.	binding	Checked	PR-29
n625	Ch.4 4.29 (validation and qualification, change control, deviations and non-conformances, complaints, recalls, returns, record summaries such as product quality	Policies, procedures, protocols, reports and records of actions or conclusions are required for quality processes that touch batches - among them validation and qualification, change control, investigation of deviations and non-conformances, complaints, recalls and returns - and for record summaries where appropriate, such as the product quality review.	binding	Checked	PR-58
n626	Ch.4 4.29 (environmental monitoring); 4.27 (all records available to the QP)	Written procedures and records are required for environmental monitoring, and every record relevant to a batch must be available to the QP.	binding	Checked	PR-28
n627	Ch.4 4.32	The quality management system must keep a maintained list (inventory) of its documents.	binding	Checked	PR-05

EU GMP Chapter 5

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n115	Ch.5 5.8; 5.44	Yield checks and quantity reconciliations are to be performed as needed to show there are no discrepancies beyond acceptable limits; any significant departure from the expected yield must be recorded and investigated.	binding	Checked	PR-31, PR-53

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n116	Ch.5 5.40	Before a processing operation starts, it must be confirmed that the area and equipment are clean and hold no materials, products, residues or paperwork unrelated to the operation about to begin.	binding	Checked	PR-26
n117	Ch.5 5.9	Different products may only be handled in the same room at the same time, or back-to-back, if this has been shown to create no risk of mix-up or cross-contamination.	binding	Checked	PR-25
n118	Ch.5 5.12; 5.13	During processing, every material, bulk container, major equipment item and, where appropriate, room must carry a label or other identification showing the product or material, strength where relevant, batch number and, where applicable, the production stage. Labels must be unambiguous, easy to read and follow the company's standard layout; colours signalling status (such as quarantine, approved, rejected or cleaned) are an optional extra aid.	binding	Checked	PR-19, PR-22
n119	Ch.5 5.15	Deviating from instructions or procedures is to be kept to a minimum; when a deviation does happen, a competent person must give written approval, with Quality Control involved when relevant.	binding	Checked	PR-13, PR-49
n120	Ch.5 5.37; 5.38; 5.39	Dispensing of starting materials is restricted to designated staff working to a written procedure, so that the right material in the right quantity ends up in clean, correctly labelled containers. An independent second check of each dispensed item and its quantity is required and must be documented. Everything dispensed for one batch is kept as a group and visibly labelled as belonging to that batch.	binding	Checked	PR-19, PR-20, PR-21
n121	Ch.5 5.17; 5.18; 5.19; 5.20; 5.21; 5.22	Contamination of a starting material or product by another material or product must be prevented. The risk of accidental cross-contamination (airborne release such as dust or vapour, biological material, residues left on equipment, operators' clothing) must be assessed, bearing in mind that injectables and long-term therapies are the most sensitive although all products are at risk. Control relies on premises/equipment design (Ch.3) supported by process design and technical/organisational measures, including effective and reproducible cleaning. A QRM process that includes a potency and toxicological evaluation of the products (and considers facility/equipment design, personnel and material flows, microbiological controls, API properties, process, cleaning and analytical capability) decides whether and how far premises, equipment or product-contact parts must be dedicated, or whether a segregated self-contained area is justified, and how extensive the technical and organisational measures must be; Ch.5 gives these measures as examples, not a mandatory set. The measures and their effectiveness must be reviewed periodically under set procedures. Making non-medicinal products in medicinal areas/equipment should normally be avoided and is only acceptable where justified and controlled; technical poisons such as pesticides and herbicides must not be made or stored in medicinal manufacturing/storage areas (unless used to make medicines).	binding	Checked	PR-27
n122	Ch.5 5.49; 5.51	Packaging programmes must be planned to minimise risks of cross-contamination, mix-up or substitution; different products are not packaged close to each other unless physically segregated. The product name and batch number being packed must be displayed at every packaging station or line.	binding	Checked	PR-35
n123	Ch.5 5.50	Before packaging starts, the work area, packaging lines, printers and other equipment must be verified clean and free of any product, material or document from earlier work that is not needed now; this line clearance follows an appropriate checklist.	binding	Checked	PR-34
n124	Ch.5 5.52	On arrival at the packaging area, every product and packaging component must be checked to confirm the right quantity, the right identity and agreement with the packaging instructions.	binding	Checked	PR-33
n125	Ch.5 5.53; 5.54	Containers must be clean when they are filled - the text sets the condition, not an in-house washing step, so supplier-cleaned ready-to-use containers can qualify - and care must be taken to avoid and remove foreign matter such as glass or metal particles. Filling and sealing should normally be followed by labelling as soon as possible; where that is not the case, procedures must prevent mix-ups and wrong labels.	binding	Checked	PR-30, PR-38
n126	Ch.5 5.46; 5.47; 5.48	Printed packaging materials need particular care: they are stored securely so that unauthorised people cannot get to them, and cut labels and other loose printed items are kept and moved in separate closed containers to avoid mix-ups. Issue of packaging materials of any kind (not only printed ones) is limited to authorised staff working to an approved, documented procedure. Each incoming lot or delivery of primary or printed packaging is given its own reference number or identifying mark. Outdated or obsolete primary or printed packaging material must be destroyed and the disposal recorded.	binding	Checked	PR-32
n127	Ch.5 5.55; 5.56	Correct execution of every printing operation (e.g. batch code, expiry date), whether done separately or during packaging, must be checked and recorded, and any hand printing re-checked at regular intervals. Extra care is required with cut labels and with off-line overprinting; roll-fed labels are normally preferred over cut labels to help avoid mix-ups.	binding	Checked	PR-36
n128	Ch.5 5.57; 5.58	The correct operation of electronic code readers, label counters and comparable devices must be checked. Printed or embossed information on packs must be clear and must not easily fade or rub off.	binding	Checked	PR-36, PR-37
n129	Ch.5 5.59	During packaging, on-line checks must at least confirm that finished packs look right and are complete, that the correct product and components are being used and any overprint is right, and that line monitoring devices are working. Units taken off the line as samples must never be put back.	binding	Checked	PR-37
n130	Ch.5 5.60	If product is caught up in an unusual event (e.g. jam, spill, dropped units, suspected mix-up), it may re-enter the process only after it has been specially inspected, the event investigated, and authorised staff have approved; this must be documented in detail.	binding	Checked	PR-37
n131	Ch.5 5.61	Any significant or unusual discrepancy found when reconciling bulk quantity and printed packaging materials against units produced must be investigated and satisfactorily explained before the batch is released.	binding	Checked	PR-39
n132	Ch.5 5.62	When packaging is finished, any unused packaging material already coded with the batch must be destroyed and the destruction recorded; returning uncoded printed materials to stock must follow a documented procedure.	binding	Checked	PR-40
n133	Ch.5 5.43; 5.41	Necessary in-process and environmental controls must be performed and recorded, and intermediate and bulk products must be held under appropriate conditions.	binding	Checked	PR-29
n134	Ch.5 5.66	Rejected materials and products must be clearly marked as rejected and segregated in restricted-access storage. Their fate (return to supplier, reprocessing where appropriate, or destruction) must be approved and documented by authorised people.	binding	Checked	PR-46
n135	Ch.5 5.67	Reprocessing of rejected product is to be exceptional: allowed only when final quality is unaffected, specifications are met, and it follows a defined, authorised procedure after risk evaluation; a record of the reprocessing must be kept.	binding	Checked	PR-47
n136	Ch.5 5.68; 5.69	Adding all or part of earlier batches that meet quality requirements into a later batch of that same product, at a predefined manufacturing stage, needs prior authorisation, must follow a defined procedure after assessing risks (including any effect on shelf life), and must be recorded. QC must decide whether reprocessed product, or product containing recovered material, needs extra testing.	binding	Checked	PR-47
n137	Ch.5 5.70	Market returns that have been outside the manufacturer's control default to destruction unless their quality is not in doubt. Resale, relabelling or use in a later batch is allowed only after QC has critically assessed them under a written procedure, considering the product type, any special storage needs, its condition and history, and how long it has been out. If quality is doubtful the goods must not be reissued or reused (though basic chemical recovery of the active substance may be possible). All actions must be recorded.	binding	Checked	PR-48
n138	Ch.5 5.5; 5.63; 5.64; 5.65	Incoming materials and finished products must be quarantined, physically or administratively, as soon as they are received or produced, and stay so until released for use or distribution. Finished products remain in quarantine, under conditions set by the manufacturer, until final release; the evaluation of product and documentation needed before release is governed by Ch.6. After release they are held as usable stock, again under manufacturer-set storage conditions. (QP certification is required by Directive 2001/83/EC Art. 51 and Annex 16, not by Ch.5.)	binding	Checked	PR-45

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n139	Ch.5 5.10; 5.11	Throughout every processing stage, products and materials must be protected against microbial and other contamination. When dry materials and products are handled, special precautions must stop dust from being generated and spreading, above all for highly active (potent) or sensitising materials.	binding	Checked	PR-27
n140	Ch.5 5.14	Checks must confirm that pipelines and other equipment that transfer product from one area to another are correctly connected.	binding	Checked	PR-26
n141	Ch.5 5.32; 5.33; 5.34 (see also 5.13 and 5.35)	Starting materials in the storage area must be suitably labelled with at least the designated material name (plus the internal code where applicable) and the batch number assigned at receipt, and, where relevant, the lot's QC status (for instance quarantine, under test, released or rejected) and its expiry or retest date; when a fully computerised storage system is used, this information need not all be legible on the label. Procedures or measures must make sure every container holds what it is supposed to hold, and bulk containers that have been sampled must be identified as such. Only starting materials released by QC and still within their retest period may be used.	binding	Checked	PR-17, PR-45
n142	Ch.5 5.1; 5.2	Production must be carried out and overseen by competent staff, and every handling activity for materials and products, from goods receipt through processing and packaging to distribution, must follow written procedures or instructions and be recorded where necessary.	binding	Checked	PR-15
n143	Ch.5 5.23; 5.24; 5.25; 5.26; 5.42	Validation studies must be run according to defined procedures, with results and conclusions recorded. When a new formula or preparation method is introduced, it must be demonstrated fit for routine production: run as defined with the specified materials and equipment, the process must reliably deliver product of the required quality. Significant changes to the process (including equipment or material changes) that may affect product quality or reproducibility must be validated; critical processes must be validated (5.42); and processes and procedures must undergo periodic critical re-validation to confirm they still achieve the intended results.	binding	Checked	PR-06, PR-09
n144	Ch.5 5.71	The manufacturer must promptly tell the marketing authorisation holder about manufacturing constraints that could cause an abnormal restriction of supply, so the MAH can notify the competent authorities.	binding	Checked	discussed in notes of Part B
n629	Ch.5 Principle	Manufacturing must be carried out to clearly defined procedures and in line with GMP so that products reach the required quality, and it must match what the site's manufacturing authorisation and the product's marketing authorisation (MA) allow.	binding	Checked	PR-04
n630	Ch.5 5.3; 5.30	Every incoming consignment must be checked to confirm it matches the order, and containers are cleaned where needed and labelled with the prescribed information. For each delivery of starting material, every container is checked for package integrity (including any tamper-evident seal) and the delivery note, purchase order and supplier labels are checked against each other and against the manufacturer's own approved manufacturer/supplier records; these receiving checks are documented.	binding	Checked	PR-17
n631	Ch.5 5.4	Any damage to containers, or any other problem that could harm a material's quality, must be investigated, recorded and reported to the QC department.	binding	Checked	PR-17
n632	Ch.5 5.6	Bought-in intermediates and bulk products are to be treated at receipt exactly like starting materials.	binding	Checked	PR-17
n633	Ch.5 5.7	All materials and products must be kept under the storage conditions the manufacturer has defined, and stored in an orderly way that allows batches to be kept apart and stock to be rotated.	binding	Checked	PR-18
n634	Ch.5 5.16	Only authorised staff may enter production areas.	binding	Checked	PR-15
n635	Ch.5 5.19; 5.21 (organisational measures on campaign cleaning, cleaning verification, cleaning records and status labels; technical measure on validated clean-i	Cleaning processes must be effective and reproducible (5.19). Among the cross-contamination measures chosen through QRM, 5.21 lists: detailed cleaning records designed to prove cleaning was completed per the approved procedure; status labels showing the cleaning state of rooms and equipment; campaign working ended by cleaning of validated effectiveness; post-campaign cleaning verification for higher-risk products; and automatic clean-in-place of validated effectiveness. Combined with Ch.4 4.31 (logbooks recording cleaning of production equipment and areas), electronic cleaning records and cleaning-status indication are a baseline eBMR/MES function rather than an option.	binding	Checked	PR-24, PR-27
n636	Ch.5 5.21 (organisational measure on recording spills, accidental events and procedural deviations); see also 5.15, 5.22	As one of the QRM-selected organisational controls against cross-contamination, spills, accidental events and departures from procedures are to be recorded.	binding	Checked	PR-27
n637	Ch.5 5.27; 5.29; 5.45 (5.28 noted)	Choosing, qualifying, approving and maintaining suppliers of starting materials, and of primary and printed packaging materials, must be documented within the PQS, with oversight proportionate to each material's risk and supporting evidence kept; where possible, starting materials are bought directly from their manufacturer. For active substances, supply-chain traceability back to the active-substance starting materials must be established, its risks formally assessed and periodically verified, and the traceability records held by the EEA manufacturer or importer; API manufacturers and distributors are audited for GMP/GDP compliance at risk-based intervals. Excipients and their suppliers are controlled on the basis of a formalised risk assessment.	binding	Checked	PR-17, PR-57
n638	Ch.5 5.31	When a single delivery of a material contains several batches, each batch is to be handled on its own for sampling, testing and release.	binding	Checked	PR-17

EMA GMP Q&A

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n225	Q10 section 2.7 - management of outsourced activities and purchased materials; EMA Q&A on data integrity at contract acceptors	The pharmaceutical company remains responsible for controlling outsourced activities and the quality of bought-in materials. It must assess suppliers and contractors before use, set responsibilities in writing (quality agreements), and monitor and review their performance. EU data-integrity expectations add that contract acceptors should apply controls equivalent to the contract giver's, that DI competence is checked before approval and periodically afterwards on a risk basis, and that agreements say how data are transferred and retained.	guidance	Search-based	DI-08

EU GMP Chapter 1

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n214	Q10 3.2.2 and Annex 1; EU GMP Chapter 1 basic GMP requirements on recording and investigating significant deviations, and QP certification before sale	Significant deviations during manufacture must be fully recorded and investigated to find the root cause, with suitable CAPA. Monitoring results and investigation outcomes must be considered when deciding on batch release. A batch must not be sold or supplied until it has been certified (EU: by a Qualified Person) as made and controlled according to its authorisation and GMP. The FDA counterpart is 21 CFR 211.192 (production record review and investigation of unexplained discrepancies, extended to other potentially affected batches).	guidance	Search-based	PR-49, PR-50, PR-53

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n216	Q10 3.2.4 and 4.1 (management review); EU GMP 1.10 product quality review	Management must regularly review process performance and product quality. Inputs include inspection and audit results, periodic quality reviews, complaints and recalls, monitoring conclusions, and how effective changes and CAPAs were. In the EU, the periodic product quality review must cover, among other things, all significant deviations and non-conformances with their investigations and the effectiveness of the resulting CAPAs, plus changes to processes and methods. The FDA counterpart is the annual product review under 21 CFR 211.180(e).	guidance	Search-based	PR-58
n228	1.4(xiv) - level of root cause analysis; justification when human error is concluded	Deviations, suspected defects and other problems need a suitable level of root cause analysis, scaled with QRM. If the true root cause cannot be found, the most likely causes should be identified and dealt with. If human error is blamed, that conclusion must be justified by showing that process, procedural or system causes were considered and not overlooked. CAPA effectiveness should be monitored.	guidance	Search-based	PR-50

ICH Q9(R1) / Q10 / Q12

Ref	Clause	Requirement (paraphrased)	Status	Check	Used in
n204	Section 3 - principles of QRM	QRM rests on two principles. First, risk to quality is judged from scientific knowledge and tied in the end to protecting the patient. Second, the effort, formality and documentation of a QRM exercise should match the level of risk.	guidance	Search-based	PR-07
n205	Section 5.1 - formality in QRM	Formality is a spectrum, not a yes/no choice. How formal a QRM activity should be depends on the uncertainty (how little is known about hazards and harms), the importance of the decision for quality, and the complexity of the process or subject. More uncertainty, importance or complexity calls for a more formal, structured and documented approach.	guidance	Search-based	PR-06, PR-07
n206	Section 5.2 - risk-based decision-making; section 5.3 - subjectivity in QRM	The revision clarifies risk-based decision-making (decisions informed by knowledge about the risk) and asks firms to reduce subjectivity in risk assessments. The means are well-defined scoring scales, evidence and data rather than opinion, and suitable team input.	guidance	Search-based	PR-07
n207	Section 4.6 - risk review (text unchanged from 2005 Q9)	QRM outputs should be reviewed again as new knowledge and experience build up, at a frequency set by the level of risk. Once a QRM process exists, it should be revisited after planned events (for example product reviews, audits or inspections, change control) and unplanned events (for example root causes found in failure investigations, recalls) that could affect the original decision.	guidance	Search-based	PR-07
n208	Section 6.1 - QRM for product availability risks arising from quality/manufacturing issues	The revision asks firms to treat quality and manufacturing problems that threaten supply (and so harm patients through shortages) as risks to manage. Examples are process drift, poor process capability, unreliable equipment or suppliers, and changes that could interrupt supply. Changes and events should be judged on both quality impact and availability impact.	guidance	Search-based	PR-49
n209	Section 1 (Introduction) - digitalisation and emerging technologies; Annex II applications	Digital and emerging technologies can lower risk when they are fit for their intended use, but they can also bring new risks that need control. QRM should therefore be applied to how computerised systems, advanced data-analysis methods and advanced production or control methods are designed, validated and transferred.	guidance	Search-based	DI-03, DI-07
n210	Annex II - QRM applied to production (in-process sampling and testing, validation, production planning) and to integrated quality management	Q9 lists places where QRM helps. They include deciding how often and how extensively to perform in-process tests (for example reduced testing where control is proven, or use of PAT supporting real-time release), planning production (including sequential or concurrent campaigns), deciding what documentation and training are needed, and handling quality defects, periodic review and change control.	guidance	Search-based	PR-07, PR-29
n211	Section 4.3 risk assessment and definitions (hazard, harm, risk); emphasis on root cause	The revision draws a clearer line between a hazard (a potential source of harm), the harm itself (damage to health, including from loss of quality or availability) and the risk (the combination of how likely the harm is and how severe). It also stresses root cause analysis in investigations.	guidance	Search-based	PR-07, PR-50
n212	Section 3.2.1 - process performance and product quality monitoring system	Firms need a monitoring system, designed with QRM, that can measure and analyse the parameters and attributes named in the control strategy. It must show whether the process is still in a state of control, find sources of variation, and take in quality feedback from inside and outside the firm, such as complaints, rejects, non-conformances, recalls, deviations, audits and inspection findings. That knowledge then improves process understanding.	guidance	Search-based	PR-58
n213	Section 3.2.2 - corrective action and preventive action (CAPA) system	CAPAs should come out of investigations of complaints, rejects, non-conformances, recalls, deviations, audits, inspection findings and monitoring trends. Investigations should follow a structured method aimed at finding root cause, with effort and documentation in proportion to risk (per Q9). CAPA should lead to better products, processes and understanding, and its effectiveness should be evaluated.	guidance	Search-based	PR-51
n215	Section 3.2.3 - change management system (with Table III across lifecycle)	Changes should be assessed with QRM, with effort and formality proportionate to risk. Each proposed change is judged against the marketing authorisation (including any design space) and current product and process knowledge, and the firm decides whether a regulatory submission is needed. People with the right expertise (development, manufacturing, quality, regulatory, medical as relevant) review it. After implementation, the firm checks that the change met its goals and did not harm product quality.	guidance	Search-based	PR-06
n217	Section 1.6.1 - knowledge management (enabler); section 3.1 - lifecycle goals incl. technology transfer and product discontinuation	Product and process knowledge should be managed across the lifecycle: gathered, analysed, stored and shared, including knowledge from development, technology transfer, commercial manufacture and discontinuation. Knowledge management and QRM are the two enablers of the PQS. At discontinuation, activities such as keeping documentation and samples and continuing necessary monitoring must still be managed.	guidance	Search-based	PR-05
n218	Chapter 3 - Established Conditions (ECs) and reporting categories	ECs are the legally binding elements of the approved application considered necessary to assure product quality, such as certain process parameters and ranges, equipment principles, controls and specifications. Changing an EC must be reported to the regulator in a category set by risk: prior approval, moderate-risk notification or low-risk notification. Elements that are not ECs can be changed under the firm's PQS alone, without a filing. In the EU, ECs are handled through the variations framework.	guidance	Search-based	PR-04, PR-56
n219	Chapter 4 - Post-Approval Change Management Protocol (PACMP); Chapter 5 - Product Lifecycle Management (PLCM) document	A PACMP is a protocol agreed with the regulator in advance. It sets out a planned change, the studies and acceptance criteria that will support it, and the lower reporting category that applies once the criteria are met. The PLCM document is a central summary of the product's ECs, their reporting categories, any PACMPs and planned post-approval commitments. In the EU both are covered by the Variations Guidelines applying from 15 Jan 2026.	guidance	Search-based	PR-04
n220	Chapter 6 - PQS and change management across the supply chain and product lifecycle	An effective Q10-style PQS across the whole supply chain is a precondition for using Q12 tools. All CMC changes are managed within the PQS, and knowledge must flow between the marketing authorisation holder and every manufacturing site and supplier involved. Holders of referenced information, such as master files, must tell the MAH about relevant changes so the MAH can judge their regulatory impact.	guidance	Search-based	PR-06

Reviewer sign-off

I have reviewed the controls in this pack and recorded my decisions against each control.

Reviewer name	
Qualification / role	
Organisation	
Parts reviewed	☐ A ☐ B ☐ C ☐ D ☐ Appendix
Signature	
Date	

General comments